# On the equation $n'' = n$ and a problem of Erdős and Barbeau on products of prime–reciprocal sums

Vico Bonfioli*

Version 1.76.1 18 October 2026

**Abstract**

Erdős Problem #307 (posed by Barbeau, 1976; recorded in Erdős–Graham, 1980) asks whether there exist two finite sets $P, Q$ of primes with $\left(\sum_{p\in P}\frac{1}{p}\right)\left(\sum_{q\in Q}\frac{1}{q}\right) = 1$. Writing $a = \prod P$, $b = \prod Q$, a solution exists if and only if the *arithmetic derivative* has a two–cycle $a' = b$, $b' = a$ with $a \neq b$; equivalently iff $n'' = n$ has a solution that is not a fixed point $p^p$. That reformulation, and the equivalence with the derivative, are due to Ufnarovski–Åhlander [8], whose Conjecture 4 asserts in exactly these terms that $\left(\sum 1/p_i\right)\left(\sum 1/q_j\right) = 1$ has no solution in distinct primes. Our starting point is that #307 *is* that conjecture, an identification neither literature records, which ties the problem to the Ufnarovski–Åhlander dynamics: their conjecture that every $n'' = n$–cycle has period one would force #307 to have answer NO, while a YES would refute it. Exploiting the equivalence we prove a *rigidity lemma* (each prime–reciprocal sum is automatically in lowest terms, so $N_P = D_Q$, $N_Q = D_P$, and $P \cap Q = \varnothing$) and a quantitative *barrier*: building on Rosen's bound $|P \cup Q| \geq 59$, any solution has $\min(\prod P, \prod Q) \geq 2.09 \times 10^{56}$, for *both* members. This is the first valid unconditional bound of its kind. The prior claims are Kovič's: his Proposition 17 is sound but conditional on the smaller member being odd, while Proposition 18 (the only unconditional cardinality and size bound in the literature, giving $r + s \geq 34$ and $\max\{m, n\} \geq \prod_{i\leq 17} p_i$) rests on an invalid step, deriving $p_1 q_s < rs$ by multiplying an upper bound against a lower one, where the available products give only $p_1 q_1 < rs < p_r q_s$ and no cardinality bound at all; we give an explicit model and machine–check both points. Against the strongest surviving unconditional bound, that paper's computer search $x > 10^4$, the gain is 52 orders of magnitude. A finite verification over the complete list of 49,961 admissible 59–prime supports then closes the 59–prime level: any solution in fact has $|P \cup Q| \geq 61$ and $\min(\prod P, \prod Q) > 5.89 \times 10^{58}$: level 60 is empty, closing both its sectors, the pair sector by a mass bound on the free tail with no factorisation, and the single–tail sector by an exact identity decided over 7,713 admissible bases with no factorisation either. We add a parity dichotomy (a hypothetical all–odd cycle would need $\geq 1412$ primes and have $\min(\prod P, \prod Q) > 10^{2519}$), a proof that longer cycles are strictly harder ($k = 3$ already forces 361,139 primes), and a function–field theorem: over $\mathbb{F}_q[t]$ the derivative has *no* cycles, so the obstruction over $\mathbb{Z}$ is purely archimedean. A local congruence anatomy of the cycle equation, the Erdős–Wintner fact that the abundance density 0.0420 is a genuine constant, and a Thābit/Euler–type constructive calculus (the *Frame Rule* and a divisor–trick) that faithfully parametrises solutions but, by an exact identity $\Delta_0 = M_0 N(1 - s_{M_0} s_N)$, provably does *not* reduce the difficulty below #307 itself (the natural "make $\Delta_0$ small" objective is #307 in disguise). No fixed–shape family can help either: for families of *any* growth rate, not merely polynomial ones, a product identity holding infinitely often forces every member constant, which closes the Thābit/Euler programme (primes $3 \cdot 2^n - 1$ and relatives) that is the classical technique for problems of this kind. Recast as a bilinear breeder it yields a conditional

*vicobonfioli@gmail.com. See the acknowledgments for a candid note on the use of AI assistance in this work.

reduction (H) $\Rightarrow$YES to an explicit prime–density hypothesis, which a quantitative ledger shows is not a feasible finite search. On the counting side we prove that both Pythagorean layers have density zero unconditionally *on the squarefree stratum*, which is where every tool in the proof lives and where a #307 solution lies by rigidity; the unrestricted statement is not proved. This comes with a rate, $\ll N(\log\log\log N)^{2+\varepsilon}(\log\log N)^{-1}$ for every $\varepsilon > 0$, which rests on a cancellation estimate uniform for moduli up to $(\log\log N)^2/(\log\log\log N)^{2+\varepsilon}$; the far stronger $\ll N(\log\log N)^{1/2+\delta}(\log N)^{-1/4}$ also holds, by estimating the pretentious distance without expanding in characters, which is uniform to power–of–log moduli. The sharp exponent $N^{1/2}$ is not reached either, and we record three distinct reasons: the multiplicative decomposition scatters the cancellation across frequencies, a bilinear decomposition is unavailable for a signed non-multiplicative summand, and the natural large-sieve reformulation is circular at its principal term. The 34 reciprocity-immune families at level 60 are upgraded from probable to proved by Atkin–Morain certificates, archived for independent verification. The rigidity result, the barrier, and the arithmetic core of the counting argument are formalised in Lean 4 (mathlib), extremality proved rather than assumed; the only non–logical input is a verified evaluation of the first 59 primes. The problem remains open. The heuristic case for existence is weaker than it appears: the constant in the expected-count heuristic cannot be measured even in principle, since the barrier places the entire relevant region above $10^{112.9}$ while the largest value of $\sigma(a)\sigma(a')$ observed below $10^7$ is 0.5535.

# 1 Introduction

Let $P, Q$ be finite, nonempty sets of (distinct) primes. Erdős Problem #307 [11], attributed to Barbeau's 1976 "Computer challenge corner" problem #477 [2] and listed in Erdős–Graham [3], asks:

> do there exist finite sets of primes $P, Q$ with $\Big(\sum_{p\in P}\frac{1}{p}\Big)\Big(\sum_{q\in Q}\frac{1}{q}\Big) = 1$ ?

The problem is verifiable in the sense that a single example would settle it. No example is known. The published partial facts are a remark of J. Rosen on MathOverflow [10], that any solution needs at least 59 primes in $P \cup Q$, and the problem page [11], which asserts $|P \cup Q| \geq 60$.

The relation between the two is worth stating precisely, since this paper proves the second. The page's justification reads, in full, that $P$ and $Q$ are disjoint and $\sum_{p\in P\cup Q} 1/p \geq 2$, "whence $|P \cup Q| \geq 60$". That implication does not hold as displayed: the reciprocals of the first 59 primes already sum to $2.00235 > 2$, so a mass of 2 is attainable on 59 primes and the stated hypotheses yield $|P \cup Q| \geq 59$, which is Rosen's bound. Reaching 60 requires excluding the 49,961 admissible 59–element supports one at a time, which is what Proposition 4.5 does and what `erdos307_sixty` machine-checks. We therefore claim no priority for the *statement* $|P \cup Q| \geq 60$, which predates this work on the problem page, and we do claim the proof. The barrier in the form current at the time is carried upstream in Google DeepMind's `formal-conjectures` as `FormalConjectures/ErdosProblems/307.lean`, where it appears as `erdos_307.barrier` with a `formal_proof using lean4` attribution to `Closed.lean` of this development at a pinned commit; that snapshot states $59 \leq |P \cup Q|$ and $(\prod P)^2 \geq 4 \times 10^{112}$, and the bounds here are since stronger. The known "solutions" of Cambie and others solve only the *weakened* version in which 1 is permitted as an element, e.g. $(1 + \frac{1}{5})(\frac{1}{2} + \frac{1}{3}) = 1$; these do not bear on the prime version.

Expanding the product exhibits #307 as a rigidly shaped Egyptian fraction for 1: $1 = \sum_{p\in P}\sum_{q\in Q} 1/(pq)$, whose denominators are the *rectangular grid* $P \times Q$ of semiprimes; a *rank–one*, multiplicatively separable representation. The unrestricted companion (representing 1 by reciprocals of distinct two–prime–factor numbers, with no product structure imposed) is by contrast *solved*, and comprehensively so: every natural number is a sum of distinct semiprime unit fractions [26], and every

$a/b$ with $b$ squarefree above an explicit threshold is likewise representable, so the $\omega = 2$ problem is settled well past the single value 1. Johnson (1978) gave a 48–term decomposition, Watanabe [5] 47–term ones, and Graham's survey [4] records exactly this contrast, noting after Barbeau that the product form "is not known." Thus #307 is precisely the rank–one restriction of a solved problem. Nor is the density–zero nature of the prime denominators the obstruction in itself: Bloom [7] represents every positive rational by distinct unit fractions drawn from the (equally sparse) shifted primes $\{p-1\}$. What resists is the rank–one rigidity (the demand that the representation *factor* as a product of two prime–reciprocal sums) which the barrier (Theorem 4.3) shows costs at least 59 primes, hence of order 870 semiprime terms against Johnson's 47.

Our starting point is an identification of #307 with a question about the *arithmetic derivative* $n \mapsto n'$, the unique map with $0' = 1' = 0$, $p' = 1$ for primes $p$, and the Leibniz rule $(ab)' = a'b + ab'$. (The map goes back to J. Mingot Shelly (1911) and appeared as a 1950 Putnam problem; Barbeau [1] gave the first systematic study, with the modern account in Ufnarovski–Åhlander [8] and OEIS [12].) For $n = \prod_i p_i^{e_i}$ one has $n' = n \sum_i e_i/p_i$. We show (Section 3) that #307 is *equivalent* to the existence of a two–cycle of $n \mapsto n'$ that is not a fixed point. The two–cycle reformulation itself is not new, and it is older than is generally recorded. Ufnarovski–Åhlander [8] prove that a period–two point of $n \mapsto n'$ has squarefree members with disjoint supports, and deduce that the period–two case of their Conjecture 3 is equivalent to their Conjecture 4: $\big(\sum_{i\le k} 1/p_i\big)\big(\sum_{j\le l} 1/q_j\big) = 1$ has no solution in distinct primes. That is #307, in the derivative literature, in 2003. It was restated "in disguise" by Seva [21] in 2019 (via $Q \mapsto Q\sum_{p\mid Q} 1/p$, which agrees with the arithmetic derivative on squarefree integers, where the question lives), who is the first to name it as #307, and obtained independently by Bado [28]; see Section 14. What is new here is the identification of the two problems, #307 is Ufnarovski–Åhlander's Conjecture 4, and the resulting two–way bridge, importing their period–one conjecture and exporting our bounds. Remarkably, Barbeau authored foundational papers on both sides (1961 and 1976 [1, 2]).

The bridge is productive in both directions.

- *Importing* from the derivative literature: the Ufnarovski–Åhlander theorem that any two–cycle has squarefree members with disjoint supports [8] makes the "finite sets of distinct primes" hypothesis of #307 automatic, and links #307 to the Ufnarovski–Åhlander conjecture that all derivative cycles have period one.

- *Exporting* to that literature: our barrier (Theorem 4.3) is, we believe, the first valid unconditional bound of its kind. Kovič [9] states three bounds on an $n'' = n$ solution. A computer search gives $x > 10^4$, unconditionally. Proposition 17 gives $> 3 \cdot 5 \cdots 29 \approx 3.23 \times 10^9$, but only when the *smaller* member is odd; Kovič notes explicitly that no estimate follows otherwise. Proposition 18 claims the unconditional $r + s \ge 34$ and $\max\{m, n\} \ge \prod_{i=1}^{17} p_i \approx 1.92 \times 10^{21}$; its proof does not establish them, for the reason set out in Section 14. Against the strongest *valid unconditional* bound, $10^4$, our $\min(\prod P, \prod Q) \ge 2.09 \times 10^{56}$ is a gain of 52 orders of magnitude; against the conditional Proposition 17 it is 47, in the case where that proposition applies. On cardinality there is no prior bound to improve: $|U| \ge 60$ appears to be the first, and it re-derives Kovič's claimed $r + s \ge 34$ as a corollary by an unrelated route.

**Results and organisation.** Section 2 proves the Rigidity Lemma (Theorem 2.1): the fraction $\sum_{p\in S} 1/p$ is automatically in lowest terms, which forces, for any solution, the exact identities $N_P = D_Q$, $N_Q = D_P$, disjointness $P \cap Q = \varnothing$, and $D_Q = s\, D_P$ where $s = \sum_{p\in P} 1/p$. The disjointness core was anticipated by R. Israel [10]; the full rigidity is what powers everything later. Section 3 establishes the bridge. Section 4 proves the barrier, the parity dichotomy, and that

longer cycles are strictly harder (so two–cycles are extremal). Section 6 gives a local congruence anatomy, a heuristic model, and a proof (via Erdős–Wintner) that the relevant abundance density is a genuine constant. Section 7 adds a function–field lens: over $\mathbb{F}_q[t]$ the derivative has *no* cycles, so the obstruction in $\mathbb{Z}$ is purely archimedean. Section 8 develops the Frame Rule and divisor–trick, and, crucially, records the exact identity that renders the natural constructive objective circular. Section 9 recasts the rule as a bilinear breeder, records a conditional reduction (H) $\Rightarrow$ YES to an explicit prime–density hypothesis, and shows quantitatively why (H) does not amount to a feasible finite search. Section 10 documents the computations, and Section 11 the Lean 4 formalisation. We are explicit throughout about what is proved, what is heuristic, and what is borrowed; the problem remains open.

**Notation.** For a finite set $S$ of distinct primes put

$$D_S \coloneqq \prod_{p\in S} p, \qquad N_S \coloneqq \sum_{p\in S} \frac{D_S}{p}, \qquad \text{so} \qquad \sum_{p\in S} \frac{1}{p} = \frac{N_S}{D_S}.$$

We call $N_S$ the *cofactor sum*. For a solution we write $a = D_P$, $b = D_Q$, $s = N_P/D_P = \sum_{p\in P} 1/p$ and $t = N_Q/D_Q = \sum_{q\in Q} 1/q$, so $st = 1$, and (without loss of generality) $s > 1 > t$. We write $\omega(n)$ for the number of distinct primes dividing $n$.

## 2 The Rigidity Lemma

**Lemma 2.1** (Rigidity / lowest terms)**.** *For a finite set $S$ of distinct primes,* $\gcd(N_S, D_S) = 1$. *Equivalently, $\sum_{p\in S} 1/p$ is already in lowest terms, and $v_p\big(\sum_{r\in S} 1/r\big) = -1$ for every $p \in S$.*

*Proof.* The prime divisors of $D_S = \prod_{r\in S} r$ are exactly the elements of $S$, so it suffices to show $p \nmid N_S$ for each $p \in S$. Fix $p \in S$ and reduce $N_S = \sum_{r\in S} D_S/r$ modulo $p$. For $r \neq p$ the term $D_S/r = \prod_{u\in S\setminus\{r\}} u$ still contains the factor $p$, so $p \mid D_S/r$. The single surviving term is $D_S/p = \prod_{u\in S\setminus\{p\}} u$, whence

$$N_S \equiv \prod_{u\in S\setminus\{p\}} u \pmod{p}.$$

Each $u$ in this product is a prime distinct from $p$, so $p \nmid u$; as $p$ is prime it does not divide the product. Thus $N_S \not\equiv 0 \pmod{p}$, i.e. $p \nmid N_S$. As $p$ ranged over all prime divisors of $D_S$, $\gcd(N_S, D_S) = 1$. The valuation statement is the same computation: $v_p(N_S/D_S) = v_p(N_S) - v_p(D_S) = 0 - 1 = -1$. □

**Remark 2.2.** Distinctness and primality are both essential: $v_p$ of the cofactor of a repeated prime need not be $-1$, and the final step uses Euclid's lemma. The case $|S| = 1$ is consistent ($N_{\{p\}} = 1$).

**Theorem 2.3** (Solution structure)**.** *Suppose $P, Q$ are finite sets of distinct primes with* $\dfrac{N_P}{D_P} \cdot \dfrac{N_Q}{D_Q} = 1$. *Then*

$$N_P = D_Q, \qquad N_Q = D_P, \qquad P \cap Q = \varnothing,$$

*$N_P$ is squarefree with prime support exactly $Q$ (and $N_Q$ squarefree with support $P$), and $D_Q = s\, D_P$ where $s = N_P/D_P$.*

*Proof.* Cross-multiplying, $N_P N_Q = D_P D_Q$. By Lemma 2.1 both $N_P/D_P$ and $N_Q/D_Q$ are in lowest terms. Since $N_Q/D_Q = D_P/N_P$ and the lowest–terms representative of a positive rational is unique, comparing $N_Q/D_Q$ with $D_P/N_P$ gives $N_Q = D_P$ and $D_Q = N_P$.

*Disjointness* (the Israel mechanism [10]). If $r \in P \cap Q$ then $r \mid D_P$ and, by $N_P = D_Q = \prod Q$, also $r \mid N_P$, but $\gcd(N_P, D_P) = 1$ forces $r \nmid N_P$, a contradiction. (Equivalently $v_r(s) = v_r(t) = -1$, so $v_r(st) = -2 \neq 0 = v_r(1)$.) Hence $P \cap Q = \varnothing$.

Finally $N_P = D_Q = \prod_{q \in Q} q$ is a product of distinct primes, hence squarefree with prime support $Q$; symmetrically for $N_Q$. And $s\, D_P = (N_P/D_P)D_P = N_P = D_Q$. □

The disjointness in Theorem 2.3 means a solution is genuinely a partition of $U \coloneqq P \cup Q$, and $D_P D_Q = \prod_{p \in U} p$. The valuation form $v_p(\sum 1/p_i) = -1$ underlying disjointness is due to R. Israel [10], and appears also in Seva [21]; the cross–matching form ($N_P = D_Q$, $N_Q = D_P$) was obtained independently and concurrently by Bado [28, Thm. A] (Section 14).

## 3 The bridge to the arithmetic derivative

**Lemma 3.1.** *For squarefree $m$, the arithmetic derivative equals the cofactor sum: $m' = \sum_{p \mid m} m/p = N_{\{p:\, p \mid m\}}$. Moreover $\gcd(m, m') = 1$.*

*Proof.* For $m = \prod_{p \mid m} p$ squarefree, $m' = m \sum_{p \mid m} 1/p = \sum_{p \mid m} m/p$, which is exactly the cofactor sum of the prime set of $m$. Coprimality is Lemma 2.1. □

**Theorem 3.2** (Bridge)**.** *The following are equivalent.*

(1) *Erdős #307 has a solution: finite prime sets $P, Q$ with $\big(\sum_{p \in P} 1/p\big)\big(\sum_{q \in Q} 1/q\big) = 1$.*

(2) *The arithmetic derivative has a two–cycle: integers $a \neq b$ with $a' = b$ and $b' = a$.*

(3) *The equation $n'' = n$ has a solution $n$ that is not a fixed point (i.e. $n' \neq n$, so $n$ is not of the form $p^p$).*

(*The implication* (2) $\Rightarrow$ (1) *uses the Ufnarovski–Åhlander squarefreeness theorem for cycles; all other implications, and everything used in the barrier of Section* 4*, are unconditional; see Remark* 3.3.)

*Proof.* (1) $\Rightarrow$ (2). Let $a = D_P, b = D_Q$. By Lemma 3.1, $a' = N_P$ and $b' = N_Q$, and by Theorem 2.3, $N_P = D_Q = b$ and $N_Q = D_P = a$. Thus $a' = b$, $b' = a$, and $a \neq b$ since $s \neq 1$ (a prime–reciprocal sum is never 1, being $N/D$ in lowest terms with $D > 1$).

(2) $\Rightarrow$ (1). Let $a' = b, b' = a$ with $a \neq b$. By the Ufnarovski–Åhlander analysis of two–cycles (cf. [8, 9]), both $a$ and $b$ are squarefree; their prime supports $P, Q$ are then disjoint since $\gcd(a, b) = \gcd(a, a') = 1$. By Lemma 3.1, $a' = N_P$ and $b' = N_Q$; the cycle gives $N_P = b = D_Q$ and $N_Q = a = D_P$, so $\big(\sum_{p \in P} 1/p\big)\big(\sum_{q \in Q} 1/q\big) = \frac{N_P}{D_P}\frac{N_Q}{D_Q} = \frac{D_Q}{D_P}\frac{D_P}{D_Q} = 1$.

(2) $\Leftrightarrow$ (3). A two–cycle $a \neq b$ gives $a'' = a$ with $a' = b \neq a$, so $a$ is a non–fixed solution of $n'' = n$. Conversely a solution of $n'' = n$ with $n' \neq n$ yields the two–cycle $(n, n')$. The fixed points of $n \mapsto n'$ are exactly the numbers $p^p$ [1, 8]. □

**Remark 3.3** (Status of the cited input)**.** Part (2) $\Rightarrow$ (1) uses that a derivative two–cycle has squarefree, support–disjoint members. Squarefreeness is the content of the Ufnarovski–Åhlander analysis of cycles (a non–squarefree $n$ has $p \mid \gcd(n, n')$ for some $p$, and along $n \mapsto n'$ such common factors propagate, precluding return); disjointness then follows from $\gcd(a, a') = 1$. We use it as a cited theorem. Even *without* it, the implication (1) $\Rightarrow$ (2) and the entire barrier below are unconditional, because Rigidity already delivers squarefree, disjoint $P, Q$ from a #307 solution.

Formalised. `Erdos307.bridge_forward` is (1) $\Rightarrow$ (2) and cites nothing; the distinctness $a \neq b$ is *derived* there, not assumed (`dprod_ne_dprod`). Support–disjointness is available separately as

`disjoint_primeFactors_of_cycle`, also from Rigidity; it is not consumed by `bridge_forward`, which does not need it. `bridge_backward` is (2) ⇒ (1) with the Ufnarovski–Åhlander squarefreeness appearing as an explicit hypothesis, and `two_cycle_iff_double_fixed` is (2) ⇔ (3) for an arbitrary self–map, on *no* axioms at all. 0 `sorry` (`lean/Erdos307/Bridge.lean`).

**Corollary 3.4.** *If the Ufnarovski–Åhlander conjecture holds in the weak form "every cycle of $n \mapsto n'$ has period one," then Erdős #307 has answer* NO. *Conversely, a* YES *to #307 exhibits a period–two cycle and refutes that statement.*

# 4 The barrier

**Lemma 4.1** (Strict AM–GM)**.** *If $s, t > 0$ with $st = 1$ and $s \neq 1$, then $s + t > 2$.*

*Proof.* $s + t - 2 = s + 1/s - 2 = (s-1)^2/s > 0$ since $s \neq 1$. □

**Lemma 4.2** (The number 59, after Rosen [10])**.** *Any finite set $U$ of distinct primes with $\sum_{p\in U} 1/p > 2$ has $|U| \geq 59$. Moreover, among $k$–element prime sets, $\sum 1/p$ is maximised by the $k$ smallest primes.*

*Proof.* Replacing any prime of $U$ by a smaller prime not in $U$ increases $\sum 1/p$, so the maximum over $k$–element sets is at the first $k$ primes. The reciprocal sum of the first 58 primes is $1.99874\ldots < 2$ and of the first 59 primes is $2.00235\ldots > 2$, so a sum exceeding 2 needs at least 59 primes. □

**Theorem 4.3** (Barrier)**.** *For any solution $(P, Q)$ of #307, with $U = P \cup Q$:*

(a) $|U| \geq 59$ *and* $\displaystyle\prod_{p\in U} p \geq \Pi_{59} \coloneqq \prod_{i=1}^{59} p_i \approx 8.77 \times 10^{112}$;

(b) $\min(\prod P, \prod Q) \geq \sqrt{\Pi_{59}/T_{59}} = 2.0929\ldots \times 10^{56}$, *where* $T_{59} = \sum_{i=1}^{59} 1/p_i = 2.00235\ldots$. *In particular both $\prod P$ and $\prod Q$ exceed $2.09 \times 10^{56}$.*

*Consequently every $n'' = n$ solution has both members $> 2.09 \times 10^{56}$. The strongest valid unconditional bound previously available is the computer search $x > 10^4$ of [9], so this is a gain of 52 orders of magnitude; against the conditional Proposition 17 of that paper ($3.23 \times 10^9$, when the smaller member is odd) it is 47. The unconditional $r + s \geq 34$ and $\max\{m, n\} \geq 1.92 \times 10^{21}$ claimed in Proposition 18 of [9] are not established by their proof (Section 14); the bound above does not rely on them and in fact implies the first.*

*Proof.* (a) By $st = 1$, $s \neq 1$ and Lemma 4.1, $\sum_{p\in U} 1/p = s + t > 2$, so $|U| \geq 59$ by Lemma 4.2; since $P, Q$ are disjoint (Theorem 2.3), $\prod_{p\in U} p = D_P D_Q$, and a product of $\geq 59$ distinct primes is at least the product of the 59 smallest, $\Pi_{59}$.

(b) Take $D_P \leq D_Q$ to be the minimal side. By Theorem 2.3, $D_Q = s\, D_P$ with $s > 1$, hence

$$s\, D_P^2 \;=\; D_P\, D_Q \;=\; \prod_{p\in U} p \;=:\; R, \qquad \text{so} \qquad D_P^2 = \frac{R}{s}.$$

Since $s \leq s + t = T(U) \coloneqq \sum_{p\in U} 1/p$ (as $t > 0$),

$$D_P^2 = \frac{R}{s} \;\geq\; \frac{R}{T(U)}.$$

It remains to bound $R/T(U)$ below over all admissible $U$. Among $k$–element prime sets the quotient $\big(\prod U\big)/\big(\sum_{p\in U} 1/p\big)$ is minimised by the $k$ smallest primes (smallest numerator and, by Lemma 4.2, largest denominator simultaneously), and passing from 59 to more primes only increases it (the product grows multiplicatively while the reciprocal sum grows by a single small term). Hence the global minimum over admissible $U$ is at $U = \{p_1, \dots, p_{59}\}$, giving $R/T(U) \geq \Pi_{59}/T_{59}$. Therefore

$$D_P^2 \;\geq\; \frac{\Pi_{59}}{T_{59}} \;=\; \frac{\Pi_{59}^2}{N_{59}} \;=\; 4.3806\ldots \times 10^{112}, \qquad D_P \;\geq\; 2.0929\ldots \times 10^{56},$$

where $N_{59} = T_{59}\,\Pi_{59}$ is the integer cofactor sum of the first 59 primes. □

**Remark 4.4** (Why the *minimum* is bounded, not only the maximum)**.** The product identity $D_P D_Q \geq \Pi_{59}$ alone gives only $\max(\prod P, \prod Q) \geq \sqrt{\Pi_{59}} \approx 2.96 \times 10^{56}$. The stronger statement (b), a lower bound on the *smaller* side, genuinely uses rigidity: it is the identity $D_Q = s\,D_P$ (equivalently $D_P^2 = R/s$) together with $s \leq T(U)$ that converts the product bound into a bound on $D_P$. We verified numerically that $\min_U R(U)/T(U)$ is attained exactly at the first 59 primes (the minimising configuration is unique), so the constant $2.09 \times 10^{56}$ is sharp for this method.

**Proposition 4.5** (Closing the 59–prime level)**.** *No solution of #307 has $|P \cup Q| = 59$. Hence $|P \cup Q| \geq 60$, $\prod_{p\in U} p \;\geq\; \Pi_{60} > 2.46 \times 10^{115}$, and*

$$\min\big(\textstyle\prod P, \prod Q\big) \;\geq\; \sqrt{\Pi_{60}/T_{60}} \;=\; 3.5053\ldots \times 10^{57}.$$

*Proof (finite verification).* Let $U = P \cup Q$ with $|U| = 59$; as in Theorem 4.3(a), $T(U) > 2$. The candidate list is finite and complete: (i) every prime $p \leq 167$ lies in $U$; omitting $p$ caps $T(U)$ at $T_{60} - 1/p < 2$, since the 59 largest prime reciprocals avoiding $p$ are those of the first 60 primes less $p$, and $1/p > T_{60} - 2 = 0.00590\ldots$ exactly for $p \leq 167$; (ii) every element $p' \geq 281$ satisfies $T_{58} + 1/p' > 2$ (the other 58 reciprocals sum to at most $T_{58}$), so $p' \leq 787$. Thus $U$ consists of the 39 primes up to 167 plus twenty primes from $(167, 787]$, and exactly 49,961 such sets have $T(U) > 2$. For each, a solution supported on $U$ forces both $N' + 2N = (a+b)^2$ and $N' - 2N = (a-b)^2$ to be perfect squares, where $N = \prod U$ and $N' = a^2 + b^2$ (rigidity; the $k = 0$ case of Proposition 14.2); one integer test that simultaneously excludes all $2^{58}$ splits of $U$. Exact–integer verification shows $N' + 2N$ is a non–square for *every* one of the 49,961 candidates, by two independent implementations with different enumeration orders and prunings (`code/close59.py` and `code/hunt/close59.rs` in the repository). The companion condition is checked separately and agrees: $N' - 2N$ is a non–square for every one of the same 49,961 supports (`code/close59_minus.rs`). That check is split–free in a second way, since $N' - 2N$ is the CRT combination of the residues $N/\ell \bmod \ell$: reducing $N' = \sum_{\ell \mid N} N/\ell$ modulo $\ell$ leaves $N/\ell$, and $2N \equiv 0$, so $N' - 2N \equiv N/\ell \pmod{\ell}$ at every $\ell \mid N$, and $0 \leq N' - 2N < N$ here because $T(U) < 3$. Hence no 59–element support closes, and $|U| \geq 60$. The bound on the minimal side repeats the proof of Theorem 4.3(b) with the minimum of $R/T$ now over $|U| \geq 60$, attained at the first 60 primes: $D_P^2 \geq \Pi_{60}/T_{60} = \Pi_{60}^2/N_{60}$, where $N_{60}$ is the cofactor sum of the first 60 primes. □

**Corollary 4.6** (The coprime variant inherits the closed level)**.** *Consider the weakened variant in which the elements of $P$ and $Q$ are merely pairwise coprime integers $\geq 2$ (the open form of the variant, $1 \notin P \cup Q$). Any solution has $|P \cup Q| \geq 60$ elements, and $\min(\prod P, \prod Q) > 3.50 \times 10^{57}$.*

*Proof.* Rigidity holds verbatim for pairwise–coprime families (the cofactor–sum argument uses only coprimality), so again $T(U) = s + t > 2$. The least prime factors $\ell(e)$ of the elements are distinct, and $\sum_e 1/e \;\leq\; \sum_e 1/\ell(e)$. Suppose $|U| = 59$ and some element $e$ is not prime, so $e \geq \ell(e)^2$.

(Formalised: `Erdos307.coprime_loss` is the per–element bound, `loss_antitone` its monotonicity, and `loss_at_277` the numeric consequence (`lean/Erdos307/Coprime60.lean`); the slack itself is the level–59 census.) If $\ell(e) \le 277$, the loss is $1/\ell - 1/e \ge (\ell-1)/\ell^2 \ge 276/277^2 = 0.00359\ldots$, exceeding the maximal available slack $T_{59} - 2 = 0.00235\ldots$; if $\ell(e) \ge 281$, then $e \ge 281^2$ while the other 58 reciprocals sum to at most $T_{58}$, giving $T(U) \le T_{58} + 281^{-2} = 1.99875\ldots < 2$. Either way $T(U) < 2$, a contradiction: at 59 elements every element is prime, and Proposition 4.5 applies verbatim. The product bound transfers because $\prod_e e \ge \prod_e \ell(e)$ and $T(U) \le \sum_e 1/\ell(e)$, so the extremal ratio bound at the first 60 primes still applies. □

The next bound is what Corollary 4.11 consumes: for a support written as a base together with one further prime, the two square conditions bound that prime in terms of the base alone. It is stated here in the form the level theory uses; its role in the level–by–level exclusion programme is taken up in the companion.

**Proposition 4.7** (The Pythagorean tail prime is bounded)**.** *Let $S$ be a finite set of primes, $D = \prod S$, $N = \sum_{p\in S} D/p$, $A = N + 2D$, $B = N - 2D$, and suppose $N > 2D$. Let $q$ be a prime with $q \nmid 2D$ for which $Aq + D$ and $Bq + D$ are both perfect squares. Then there are integers $m, c$ with $mc = 4D$ and*

$$q^2m^2 - 4Nq + c^2 - 4D = 0, \qquad \big(qm^2 - 2N\big)^2 = 4\big(AB + Dm^2\big),$$

*so that $q = 2(N \pm k)/m^2$ with $k^2 = AB + Dm^2$ and $m \mid 4D$. Moreover $m$ is* even*, say $m = 2\alpha$ with $\alpha \mid D$ and $\beta = D/\alpha$, and then $q$ is a root of*

$$\alpha^2q^2 - Nq + (\beta^2 - D) = 0, \qquad \text{so} \qquad q = \frac{N + \sqrt{N^2 - 4D^2 + 4\alpha^2 D}}{2\alpha^2},$$

*a quantity strictly decreasing in $\alpha$. Hence the maximum is at $\alpha = 1$ and*

$$q \;\le\; \tfrac{1}{2}\Big(N + \sqrt{N^2 - 4D^2 + 4D}\Big) \;=\; tD + O(1), \qquad t = \tfrac{1}{2}\big(\sigma + \sqrt{\sigma^2 - 4}\big), \quad \sigma = \sigma(S) = N/D.$$

*The largest admissible tail is therefore the base times the very ratio $b/a$ that the mass forces. At level* 59 *this is* $9.207 \times 10^{112} = 1.0497\,D$*, a factor* 7.63 *below* $4N$*.*

*Proof.* Write $x^2 = Aq + D$ and $y^2 = Bq + D$ with $x, y \ge 0$. Since $A - B = 4D$ *exactly*, $x^2 - y^2 = (A - B)q = 4Dq$, that is $(x + y)(x - y) = 4Dq$. The prime $q$ divides exactly one of the two factors: were it to divide both it would divide $2x$ and $2y$, hence $x$ and $y$ as $q$ is odd, hence $q^2 \mid 4Dq$ and $q \mid 4D$, against $q \nmid 2D$. Write the factor divisible by $q$ as $qm$ and the other as $c$, so that $mc = 4D$ and, in either case, $2x = qm + c$. Multiplying by $m$,

$$2xm \;=\; qm^2 + mc \;=\; qm^2 + 4D,$$

and squaring against $x^2 = (N + 2D)q + D$ gives $4m^2\big((N + 2D)q + D\big) = q^2m^4 + 8Dqm^2 + 16D^2$. Substituting $16D^2 = m^2c^2$ and cancelling $m^2 > 0$ leaves $q^2m^2 - 4Nq + c^2 - 4D = 0$; completing the square and using $AB = N^2 - 4D^2$ gives the second display. For the parity, $x + y$ and $x - y$ differ by $2y$ and so have equal parity; were both odd their product would be odd, whereas it is $4Dq$. Hence both are even, and as $q$ is odd, $m$ is even. Finally $m^2 \ge 4$ and $c^2 \ge 0$ give $4Nq = q^2m^2 + c^2 - 4D \ge 4q^2 - 4D$, that is $Nq \ge q^2 - D$; if $q \ge N + 1$ then $q^2 - Nq - D \ge q - D > 0$ because $D < N < q$, a contradiction, so $q \le N$. (Using only $m \ge 1$ here gives $q \le 4N$; the parity removes the factor four.) For the sharp form, $m = 2\alpha$ and $c = 2\beta$ with $\alpha\beta = D$ turn $q^2m^2 - 4Nq + c^2 - 4D = 0$ into $\alpha^2q^2 - Nq + \beta^2 - D = 0$, whose larger root is $\big(N + g(u)\big)/(2u)$

with $u = \alpha^2$ and $g(u)^2 = N^2 - 4D^2 + 4uD$. Its derivative in $u$ has the sign of $2uD - g(N+g)$, and $4uD = g^2 - N^2 + 4D^2$ turns that into $-\big((g+N)^2 - 4D^2\big)/2$, negative since $N > 2D$. So the root decreases in $\alpha$ and is largest at $\alpha = 1$. □

**Proposition 4.8** (The level–by–level programme terminates at 60)**.** *Let $U$ be a finite set of primes with $\sum_{p\in U} 1/p > 2$ and let $B$ be arbitrary. Then there is a prime $q > B$ with $q \notin U$ such that $U \cup \{q\}$ again has $\sum 1/p > 2$. Consequently, for every $n > 59$ there are infinitely many $n$–element prime supports of mass exceeding 2.*

*Proof.* The primes are infinite, so choose a prime $q$ exceeding both $B$ and $\max U$; then $q \notin U$ and $\sum_{p\in U\cup\{q\}} 1/p = \sum_{p\in U} 1/p + 1/q > 2$. Iterating from the first 59 primes, whose reciprocal sum is $N_{59}/P_{59} = 2.00235\ldots > 2$, produces admissible supports of every cardinality above 59, each containing a prime beyond any prescribed bound. □

**Remark 4.9** (What the level barrier costs, and what it does not)**.** Proposition 4.5 closes level 59 by enumerating its 49,961 admissible supports, and Proposition 5.2 together with the search of Remark 5.19 closes level 60 by adjoining one prime to each. The obvious continuation is to enumerate the admissible 60–element supports and repeat. Proposition 4.8 says there is no such enumeration *of supports*.

The finiteness at level 59 is an accident of one inequality being tight: 59 primes reach mass 2 only just, at $2.00235\ldots$, so demanding mass $> 2$ from exactly 59 primes pins them near the smallest 59. Allow a sixtieth and the first 59 already carry the mass alone, leaving the sixtieth entirely free. The mass condition, having been met, constrains nothing further. The next proposition shows that the *solutions* are nevertheless finite at every level: the haystack is infinite, the needles are not.

**Proposition 4.10** (Each level carries only finitely many solutions)**.** *Fix cardinalities $m, n \geq 0$ and rationals $c_1, c_2 \geq 0$. The equation*

$$\Big(c_1 + \sum_{a\in A} \frac{1}{a}\Big)\Big(c_2 + \sum_{b\in B} \frac{1}{b}\Big) = 1$$

*has only finitely many solutions in sets $A, B$ of integers $\geq 2$ with $|A| = m$, $|B| = n$; in particular, for every $(m, n)$ there are only finitely many pairs of prime sets with $\sigma(P)\sigma(Q) = 1$, $|P| = m$, $|Q| = n$, and the bound is effective.* 0 `sorry` (`Erdos307.sols_finite`, `erdos307_level_finite`).

*The statement is the analogue, for this pair of equations, of the finiteness known for relatively prime amicable pairs: Artjuhov and Borho proved that for each fixed $K$ there are only finitely many coprime amicable pairs $(M, N)$ with $\omega(MN) = K$, and Pollack [13] gives the explicit $MN < (2K)^{2^{K^2}}$. The two settings share their shape, a coprime pair each of which is a divisor sum of the other, and both bounds are of tower type: at $K = 60$ Pollack's is $(120)^{2^{3600}}$, so neither side of the analogy yields an enumeration. The proof below is independent of that literature and is included because it is effective and short.*

*Proof.* Induct on $m + n$; the constants absorb removed elements. Write $u, v$ for the two sums and $\varepsilon = 1 - c_1c_2$. Expanding, any solution satisfies $c_1v + u(c_2 + v) = \varepsilon$, and the left side is strictly positive for $m + n \geq 1$ (if $m \geq 1$ then $u > 0$ and $c_2 + v > 0$, else the product is 0; if $m = 0$ then $c_1(c_2 + v) = 1$ forces $c_1 > 0$, $v > 0$), so $\varepsilon > 0$ is fixed by $(c_1, c_2)$ alone. One of the two terms is at least $\varepsilon/2$.

If $u(c_2 + v) \geq \varepsilon/2$: every $b \in B$ has $1/b \leq 1/2$, so $v \leq n/2$, whence $u \geq \varepsilon/(2c_2 + n)$; since $u$ is a sum of $m$ terms, some $a_0 \in A$ has $1/a_0 \geq u/m$, that is

$$a_0 \;\leq\; \frac{m\,(2c_2 + n)}{\varepsilon}.$$

Moving $a_0$ into the constant, $(A \setminus \{a_0\}, B)$ solves the equation with $(m-1, n, c_1 + 1/a_0, c_2)$, and the map is injective for fixed $a_0$. If instead $c_1 v \geq \varepsilon/2$, then symmetrically some $b_0 \leq 2c_1 n/\varepsilon$ moves into $c_2$. Either way the solution set injects into finitely many strictly smaller problems, and the induction closes. The bounds compound: each step divides by an $\varepsilon$ whose denominator is at most the product of the elements already fixed, so the resulting height bound is doubly exponential in $m+n$; crude, but effective. □

**Corollary 4.11** (The largest element of a support is explicitly bounded)**.** *Let $U$ be the support of a two–cycle, $q = \max U$, $S = U \setminus \{q\}$, $D = \prod S$ and $N = \sum_{p \in S} D/p$. Then*

$$N < 2D \implies q \leq \frac{D}{2D - N}, \qquad N \geq 2D \implies q \leq \tfrac{1}{2}\Big(N + \sqrt{N^2 - 4D^2 + 4D}\Big).$$

*In both regimes $q$ is bounded by an explicit function of $S$ alone, so the enumeration at each level is a finite tree whose branching is known in advance.* 0 ***sorry*** *for the mass regime* (`Erdos307.max_le_of_mass_deficit`, `max_le_of_mass_deficit'`).

*Proof.* The mass of a two–cycle support is $\geq 2\,(\sigma(P)+\sigma(Q) \geq 2\sqrt{\sigma(P)\sigma(Q)} = 2)$, so $N/D + 1/q \geq 2$. If $N < 2D$ then $1/q \geq 2 - N/D > 0$ and the first bound follows. If $N \geq 2D$ then $D < N$ and $q \nmid 2D$, and both $Aq + D$ and $Bq + D$ are squares, being $D(U) + 2\prod U = (\prod P + \prod Q)^2$ and $D(U) - 2\prod U = (\prod P - \prod Q)^2$ under $\prod U = qD$ and $D(U) = D + qN$; the second bound is then Proposition 4.7 at $\alpha = 1$, the discriminant being nonnegative because $N \geq 2D$. □

**Proposition 4.12** (The mass ladder: the 59 smallest elements are bounded at every level)**.** *Let $U$ be a set of $k$ primes with $\sum_{p \in U} 1/p \geq 2$, written $u_1 < \cdots < u_k$. Then for every $j$ with $\sum_{i<j} 1/u_i < 2$,*

$$u_j \;\leq\; \frac{k-j+1}{2 - \sum_{i<j} 1/u_i} \;\leq\; \frac{k-j+1}{2 - T_{j-1}}.$$

*The hypothesis holds for all $j \leq 59$, since $T_{58} = 1.99874\ldots < 2$. In particular, at every level $k \geq 59$,*

$$u_{59} \;\leq\; \frac{k-58}{2 - T_{58}} \;=\; 793.67\ldots \times (k - 58),$$

*so $u_1, \ldots, u_{59}$ are bounded by a quantity linear in $k$; only $u_{60}, \ldots, u_k$ escape the mass, and those are covered by Corollary 4.11.* 0 ***sorry*** (`Erdos307.mass_ladder_step`, `mass_ladder_step'`).

*Proof.* The $k - j + 1$ elements $u_j, \ldots, u_k$ carry reciprocal mass at least $r_j = 2 - \sum_{i<j} 1/u_i$, and each is at least $u_j$, so $r_j \leq (k-j+1)/u_j$, which is the first bound. For the second, $u_i \geq p_i$ gives $\sum_{i<j} 1/u_i \leq T_{j-1}$, and $r_j > 0$ exactly when $\sum_{i<j} 1/u_i < 2$, which $T_{j-1} \leq T_{58} < 2$ guarantees for $j \leq 59$. □

*What this gives at the first open level.* At level 61 the ladder reads $u_1 \leq 30.5$, $u_{10} \leq 103.7$, $u_{30} \leq 201.1$, $u_{58} \leq 808.0$, $u_{59} \leq 2381.0$: the 59 smallest elements of any level–61 support lie among the first 353 primes, so positions $1, \ldots, 59$ range over at most $\binom{353}{59} \approx 10^{67.9}$ configurations, with $u_{60}$ and $u_{61}$ then confined by Corollary 4.11. This replaces the doubly exponential height of Proposition 4.10 at this level by an explicit and much smaller count. It is of course still far beyond enumeration; the point is that the first open level now has a stated size rather than an abstract finiteness.

Two consistency checks. At level 59 the ladder gives $u_{59} \leq 793.67\ldots$, and `Erdos307.elt_le_795` bounds the elements of a 59–element support by 795 by an unrelated argument; the same value appears again as the mass regime of Corollary 4.11 at the base of the first 58 primes. Falsification: over 5,781 supports of mass $\geq 2$ generated by perturbing initial segments, 0 violations of the ladder, and over 5,885 supports 0 violations of $u_{59} \leq (k-58)/(2 - T_{58})$ (`code/mass_ladder.gp`).

**Corollary 4.13** (An explicit height at the first open level)**.** *Every solution of #307 at level* 61 *has* $\prod(P \cup Q) < 10^{567}$.

*Proof.* Write $U = S \cup \{u_{60}, u_{61}\}$ with $S$ the 59 smallest elements, $D = \prod S$.

If $\sigma(S) \geq 2$ then $S$ is one of the 49,961 admissible 59–element supports of Corollary 16.14, whose products satisfy $\log_{10} D \leq 113.397$ (`code/maxprod59.rs`). Splitting $S = P_0 \sqcup Q_0$ and writing $M = \prod P_0$, $N = \prod Q_0$, $M' = D(P_0)$, $N' = D(Q_0)$, Theorem 8.1 determines the two remaining primes: $u = (MN' + N^2)/\Delta$ and $v = (M'N + M^2)/\Delta$ with $\Delta = MN - M'N'$. Here $\Delta = D\big(1 - \sigma(P_0)\sigma(Q_0)\big)$ is a positive integer, since $\sigma(P_0)\sigma(Q_0) < \sigma(P)\sigma(Q) = 1$, so $\Delta \geq 1$; and $MN = D$ exactly, so $MN' = D\,\sigma(Q_0) \leq tD$ with $t = 1.10200\ldots$ the window root at $T_{61}$. Hence $u, v \leq D^2 + tD$, giving $\log_{10} u, \log_{10} v \leq 226.79$ and $\log_{10} \prod U \leq 113.397 + 2(226.79) = 566.98$. If instead the two new primes lie on the same side, say $a = Muv$ and $b = N$ with $M = \prod P_0$ and $N = \prod Q_0$ (the other orientation is the same with the sides exchanged), then $a' = b$ and $b' = a$ read $M'uv + M(u + v) = N$ and $N' = Muv$, so $M^2(u + v) = MN - M'N' = \Delta$, which is again positive, and $uv = N'/M \leq N' \leq \sigma(S)\, D < 2.0024\, D$, whence $\log_{10} \prod U \leq 2(113.397) + 0.31 < 567$ (`Erdos307.same_side_delta`).

If $\sigma(S) < 2$ then Proposition 4.12 bounds every element of $S$ and a branch–and–bound over the admissible chains gives $\log_{10} D \leq 113.526$ (`code/ladder_maxprod.rs`). The ladder extends one step, $u_{60} \leq 2/(2 - \sigma(S)) \leq 2D$ since $2 - \sigma(S) \geq 1/D$, and Corollary 4.11 then gives $u_{61} \leq tD\, u_{60}$, so $\log_{10} \prod U \leq 454.75$. The first case dominates. □

*What the height means.* Proposition 4.10 makes every level finite by an argument using only the two cardinalities, and its height compounds doubly exponentially: at level 61 it is a tower, not a number one can write. Corollary 4.13 replaces it by $10^{567}$, because it uses the two square conditions and the cycle equations rather than the sizes alone. Together with the barrier $\prod(P \cup Q) \geq (2.09 \times 10^{56})^2 = 10^{112.6}$ this confines level 61 to the window

$$10^{112.6} \;\leq\; \prod(P \cup Q) \;<\; 10^{567}.$$

Two remarks on what this is and is not. It is not an enumeration: $10^{567}$ is far past any search, and the reduction that matters is not the height but Theorem 8.1, which turns the two unknown primes into a *computed* pair once the split of the bottom 59 is fixed, so the free parameters at level 61 are the support and the split, not the primes. And it is not a route to the problem, since the level index remains unbounded; it is a statement about one level.

*What the window does not do.* It is worth recording what fails, since the natural guess is wrong. One might hope the mass window pins all but the top few elements of a support, leaving a bounded search. It does not, and the pinning weakens as the level rises: a prime $p$ is forced into a $k$–element support exactly when $1/p > T_{k+1} - 2$, which forces 39 primes at level 59, 27 at level 60 and only 21 (all $p \leq 73$) at level 61, leaving 40 of the 61 elements unconstrained by mass. The bound of Corollary 4.11 comes from the square conditions, not from the mass, and this is why: past level 59 the mass condition, once met, constrains nothing further. Consistency check: at the base of the first 58 primes the mass regime gives $q \leq 793.67\ldots$, while `Erdos307.elt_le_795` bounds the elements of a 59–element support by 795 through an unrelated argument (`code/level_dichotomy.gp`; 400 bases, 0 failures). At level 61 the second regime applies at the extremal base and gives $q \leq 1.1020\, D$ with $\log_{10} D = 117.84$.

*The level–by–level programme survives its own barrier.* Together with Proposition 4.8 this splits the situation cleanly: at every level above 59 the admissible supports are infinite, and the solutions are finite with an effective height bound. Restricted to any fixed level, #307 is therefore *decidable*: a terminating computation either exhibits the finitely many solutions or certifies that the level is

empty. What is genuinely unbounded in the problem is only the level itself, so #307 is equivalent to the halting of one explicit search, which is the precise content of the problem page's VERIFIABLE tag. We record the honest scale: the bound of Proposition 4.10 at level 61 is astronomically beyond enumeration, so decidability in principle does not yield a computation in practice; the value of the proposition is structural. Validated on the inhomogeneous case $c_1 = 1$, where solutions are the known 1–containing coprime examples: the enumeration recovers $(\{5\}, \{2, 3\})$, $(\{41\}, \{2, 3, 7\})$, $(\{1805\}, \{2, 3, 7, 43\})$ and their congeners, every one containing an element within the proven bound (`code/level_finite_check.gp`).

Two consequences, and the second is the one that matters. The searches of Remark 5.19, however far they are pushed in $\omega(m)$, bear on level 60 alone; widening them is not a route upward. And any further progress on #307 must be *non–enumerative*: exhausting supports one level at a time is not merely expensive beyond level 60, it is impossible, so the method behind every unconditional exclusion in this note is now spent. That is a no–go, not a resource limit.

Formalised in `lean/Erdos307/LevelBarrier.lean` (`admissible_extends`, `admissible_of_card_add`, `level_enumeration_terminates`) on the three standard axioms, with a non–vacuity witness exhibiting the first 59 primes as a set the hypotheses apply to.

**Theorem 4.14** (Parity dichotomy)**.** *Let $a' = b, b' = a$ be a derivative two–cycle (so $a, b$ squarefree, $\gcd(a, b) = 1$). Then exactly one of the following holds.*

(i) Mixed parity. *Exactly one of $a, b$ is even; say $a = 2k$ with $k$ odd. Then $b = a'$ is odd and $\omega(b)$ is even.*

(ii) All odd. *Both $a, b$ are odd. Then $P \cup Q$ avoids 2, so reaching reciprocal sum $> 2$ needs $\geq 1412$ odd primes, and $\min(a, b) > 10^{2519}$. Moreover $|P|$ and $|Q|$ are both odd.*

*Proof.* They cannot both be even, as $\gcd(a, b) = \gcd(a, a') = 1$. In case (i), write $a = 2k$, $k$ odd squarefree; $b = a' = k + 2k'$ is odd. For odd squarefree $b = \prod_i q_i$ one has $b' = \sum_i b/q_i \equiv \omega(b) \pmod 2$ (each $b/q_i$ is odd). Since $b' = a$ is even, $\omega(b) \equiv 0 \pmod 2$. In case (ii) the same congruence gives the cardinalities: $b = a' \equiv \omega(a) \pmod 2$ with $b$ odd forces $\omega(a) = |P|$ odd, and symmetrically $|Q|$ is odd. (The same conclusion follows from Proposition 2.3 of [22] applied to the expansion $1 = \sum_{p \in P, q \in Q} 1/(pq)$, whose rank is $|P|\,|Q|$, with $n = 2$; it does not improve $|P \cup Q| \geq 1412$, since $|P| + |Q|$ is even either way.) The union of supports omits 2; the reciprocal sum of the first 1411 odd primes is $< 2$ while that of the first 1412 exceeds 2, so $|P \cup Q| \geq 1412$ and $\prod_{p \in P \cup Q} p \geq 10^{5040}$ (product of the 1412 smallest odd primes, by direct computation). Applying the same rigidity/ratio argument as in Theorem 4.3(b) (now with $2 \notin U$) gives $\min(a, b) \geq \sqrt{R/T(U)} > 10^{2519}$. The all–odd case is therefore vastly more constrained than the mixed case and, heuristically, empty. □

**Proposition 4.15** (Longer cycles are harder; two–cycles are extremal)**.** *Every member of a $k$–cycle of the arithmetic derivative is squarefree, and for $k \leq 3$ the members are pairwise coprime (each pair is consecutive). Their masses $s(a_i) = a_{i+1}/a_i$ multiply to 1, so by AM–GM $\sum_i s(a_i) \geq k$; for $k \leq 3$ this is the union's reciprocal sum, whence the support contains at least $m_k$ primes, where $m_k$ is least with $\sum_{i \leq m_k} 1/p_i \geq k$. Now $m_2 = 59$ and $m_3 = 361{,}139$ (last prime 5,195,977), so a three–cycle has at least 361,139 primes in its support and product exceeding $10^{2.26 \times 10^6}$ (hence its largest member exceeds $10^{752{,}194}$); by Mertens $m_k$ grows doubly exponentially. Two–cycles are thus the extremal, and only practically relevant; case for that measure. For $k \geq 4$ non–consecutive members may share primes, so the displayed bound is one on the* multiset *reciprocal sum; the union's own sum is bounded in Proposition* 4.16.

**Proposition 4.16** (The barrier is uniform in the period)**.** *Let $a_1 \to a_2 \to \cdots \to a_k \to a_1$ be a cycle of the arithmetic derivative with $k \geq 2$ and $a_i \neq a_{i+1}$, and let $U = \bigcup_i \operatorname{supp}(a_i)$. Then*

$$\sum_{p \in U} \frac{1}{p} \;\geq\; \frac{k}{\lfloor k/2 \rfloor},$$

*so $|U| \geq m(k)$, the least $n$ with $T_n \geq k/\lfloor k/2 \rfloor$, and $\prod_{p \in U} p \geq \Pi_{m(k)}$. Since $k/\lfloor k/2 \rfloor \geq 2$ always, with equality exactly for even $k$,*

$$|U| \;\geq\; 59 \quad \textit{and} \quad \prod_{p \in U} p \;>\; 8.77 \times 10^{112}$$

*for a cycle of* every *period.*

*Proof.* Rigidity gives $\gcd(a_i, a_i') = 1$, and $a_{i+1} = a_i'$, so $a_i$ and $a_{i+1}$ are coprime. Hence for each prime $p$ the set $S_p = \{i : p \mid a_i\}$ contains no two cyclically consecutive indices, i.e. it is an independent set in the cycle graph $C_k$, and therefore $|S_p| \leq \lfloor k/2 \rfloor$. Consequently

$$\sum_i \sigma(a_i) \;=\; \sum_{p \in U} \frac{|S_p|}{p} \;\leq\; \left\lfloor \tfrac{k}{2} \right\rfloor \sum_{p \in U} \frac{1}{p}.$$

The masses satisfy $\prod_i \sigma(a_i) = \prod_i a_{i+1}/a_i = 1$, so $\sum_i \sigma(a_i) \geq k$ by AM–GM. Combining the two displays gives the claim, and $T$ is increasing. □

The bound is uniform rather than growing, and it is the union support that a search or a sieve sees. Odd periods are genuinely harder, even ones are not:

| $k$ | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|
| $k/\lfloor k/2 \rfloor$ | 2 | 3 | 2 | 5/2 | 2 | 7/3 | 2 | 9/4 |
| $m(k)$ | 59 | 361,139 | 59 | 1,413 | 59 | 400 | 59 | 231 |

so $m(9) < m(7) < m(5)$: the sequence is not monotone in $k$, and $m(k) = 59$ *exactly* for every even $k$, not merely in the limit. The consequence worth naming is that Theorem 4.3 is not a statement about period two at all: every nontrivial cycle of the arithmetic derivative, of any length, carries at least 59 primes and a product past $8.77 \times 10^{112}$. The barrier therefore bounds the whole of Ufnarovski–Åhlander's Conjecture 3 and not merely its period–two case, which is #307.

**Corollary 4.17** (What a bounded sweep rules out, for every period at once)**.** *Let $a_1 \to \cdots \to a_k \to a_1$ be a cycle with every member at most $X$. Then*

$$k \;\geq\; \frac{\log_{10} \Pi_{59}}{\log_{10} X} \;=\; \frac{112.94\ldots}{\log_{10} X}.$$

*Proof.* Proposition 4.16 gives $|U| \geq 59$ for a cycle of any period, so $\prod_{p \in U} p \geq \Pi_{59} > 8.77 \times 10^{112}$. Every prime of $U$ divides some member, so $\prod_{p \in U} p$ divides $\prod_i a_i \leq X^k$. □

| $X$ | $10^6$ | $10^7$ | $10^9$ | $10^{12}$ | $10^{18}$ | $10^{30}$ |
|---|---|---|---|---|---|---|
| rules out every $k \leq$ | 18 | 16 | 12 | 9 | 6 | 3 |

The sweep of Proposition 16.7, now carried to $10^8$, therefore already excludes every cycle of period at most 14 with all members in range, and not merely period two. This is the practical form of Proposition 4.16: because the barrier does not weaken with the period, a single bounded search settles a whole initial segment of Ufnarovski–Åhlander's Conjecture 3 at once, whereas the period–two reading of the same computation settles only $k = 2$.

**Proposition 4.18** (The barrier as a function of period and smallest prime)**.** *Let $a_1 \to \cdots \to a_k \to a_1$ be a cycle with $k \geq 2$, let $U$ be the union of the supports, and suppose $\min U \geq P$. With $c(k) = k/\lfloor k/2 \rfloor$, let $M(k, P)$ be the least $m$ for which the $m$ smallest primes $\geq P$ have reciprocal sum at least $c(k)$. Then $|U| \geq M(k, P)$, and $\prod_{p \in U} p$ is at least the product of those $m$ primes.*

*Proof.* Proposition 4.16 gives $\sum_{p \in U} 1/p \geq c(k)$, and among sets of $m$ primes all $\geq P$ the reciprocal sum is maximised by the $m$ smallest such primes. □

| $M(k,P)$ | $k=2$ | $k=3$ | $k=5$ | $k=7$ | $k=9$ |
|---|---|---|---|---|---|
| $P=2$ | 59 | 361,139 | 1,413 | 400 | 231 |
| $P=3$ | 1,412 | — | 361,138 | 40,260 | 15,473 |
| $P=5$ | 40,259 | — | — | — | 1,266,094 |
| $P=7$ | 588,500 | — | — | — | — |

(dashes exceed $\pi(4 \times 10^7)$; `code/kcycle_uniform.gp`). The family contains three results proved elsewhere in this note as special cases, and reproduces each of them exactly: $M(2,2) = 59$ is Theorem 4.3; $M(2,3) = 1412$, with product $10^{5040.1}$ and hence $\min(\prod P, \prod Q) > 10^{2519}$, is the all–odd branch of Theorem 4.14; and $M(3,2) = 361{,}139$ is Proposition 4.15. That three independent derivations land on the same integers is the check the family deserves.

Two readings. The parity dichotomy is not special to period two: an all–odd cycle of *any* even period needs 1,412 primes and a product past $10^{5040}$, and of period 9 still needs 15,473. And the family supplies, in valid form, the conditional statements that Proposition 18 of [9] asserts but does not establish: for $\min U \geq 3$ that proposition claims $r + s \geq 57$ where the true bound is $M(2,3) = 1{,}412$, and for $\min U \geq 5$ it claims $r + s \geq 110$ where the true bound is $M(2,5) = 40{,}259$, larger by factors of 24.8 and 366.

## 5 The certificate programme and its anatomy

The barrier of Section 4 bounds a solution; it does not exclude one. This section records the programme that attempted the exclusion level by level, its completeness, and the point at which it provably stops. Nothing here is used by the results of the core note.

**Remark 5.1** (Verification tier, and why the ladder stops here)**.** Theorem 4.3 is machine–checked in Lean 4, and so is Proposition 4.5: the theorem `erdos307_sixty` formalises the entire argument; the forced–primes and element bounds, the Pythagorean plus–square, and a pruned depth–first search over the pool whose *completeness is proved, not assumed* (`dfs_sound`), the search itself running under the kernel; on top of the two independent unverified programs. The level cannot be climbed further by finite means: already at $|U| = 60$ the family $U = \{p_1, \ldots, p_{59}\} \cup \{q\}$ qualifies for *every* prime $q > 277$, and on it the two square conditions read $x^2 = Aq + \Pi_{59}$, $y^2 = Bq + \Pi_{59}$ with $A, B = N_{59} \pm 2\Pi_{59}$, i.e. the Pell–type system $Bx^2 - Ay^2 = -4\Pi_{59}^2$ with 113–digit coefficients. We verified that no accessible modulus obstructs it (mod 32, and all odd $\ell < 4000$: the conditions pin $q \equiv 5 \pmod 8$ and thin each $\ell \mid \Pi_{59}$ to about half of its residue classes and each $\ell \nmid \Pi_{59}$ to about a quarter, but never to zero), consistent with the local–solubility theme of Section 6. The obstruction, however, lives at the primes *of* $A$ and $B$, and it is detectable *without factoring*: Proposition 5.2 below closes this family for *every* $q$ by a single Jacobi–symbol computation. (An exact sweep to $q \leq 10^{12}$, `code/hunt/tail_sweep.rs`, $1.25 \times 10^{11}$ candidates, none with $Aq + \Pi_{59}$ even a single square, which corroborates it independently.) The tail family also exposes the self–similarity of the

ladder: writing $a = qm$, so that $m\,b = \Pi_{59}$ partitions the first 59 primes, its two closing conditions collapse to the exact equation

$$D(m)\,D(b) + m^2 \;=\; \Pi_{59}$$

together with the divisibility $m \mid D(b)$ (then $q = D(b)/m$, required prime); verified exhaustively on a 12–prime toy universe. Equivalently, the deficit $1 - \sigma(m)\sigma(b)$ must equal $m^2/\Pi_{59}$ *exactly*: the circularity $\Delta_0 = M_0 N(1 - s_{M_0} s_N)$ of Section 8 in its purest form, with the deviation forced to be the perfect square $m^2$. Closing even this one family is thus a $2^{58}$–fold exact–coincidence search of the same type as #307 itself (expected count $\sim 2^{58}/\Pi_{59} \approx 10^{-95}$): each rung of the ladder past 60 contains the whole problem. The one case with *no* free large prime (a 60–set drawn entirely from small primes) is instead a finite search, and empty in the minimal range: a direct both–squares enumeration of all 60–prime sets with $T > 2$ supported in the first 70 primes (4,011,289 sets; `code/hunt/close60.rs`) finds none passing even the plus–square, so no level–60 solution has all its primes $\leq 349$. (The first 68 primes give 1,258,448 sets, also empty, and reproduce independently.) By Corollary 15.5 this verdict is decisive rather than provisional: at these masses a support passing the plus– and minus–squares, with the parity of Proposition 15.1(2), *is* a two–cycle, so a survivor would have been a solution and not a candidate. The enumeration extends with compute.

**Proposition 5.2** (The canonical tail family is empty: a reciprocity kill)**.** *No solution of #307 has support $U = \{p_1, \ldots, p_{59}\} \cup \{q\}$, for any prime $q$.*

*Proof.* A solution supported on $U$ forces $x^2 = Aq + \Pi_{59}$ with $x = a + b$ and $A = N_{59} + 2\Pi_{59}$ (Remark 5.1). Every prime $\ell \mid A$ exceeds 277: for $\ell \leq 277$ one has $\ell \mid \Pi_{59}$, so $\ell \mid A$ would force $\ell \mid N_{59}$, contradicting $\gcd(N_{59}, \Pi_{59}) = 1$ (rigidity). Hence $\ell \nmid \Pi_{59}$, and reduction mod $\ell$ gives $x^2 \equiv \Pi_{59} \pmod{\ell}$, requiring the Legendre symbol $(\Pi_{59} \mid \ell) \neq -1$. But the Jacobi symbol satisfies

$$\left(\tfrac{\Pi_{59}}{A}\right) \;=\; -1$$

(a finite gcd–speed computation, script `code/tailkill.py`; no factorisation of the 114–digit $A$ is needed), so some $\ell \mid A$ of odd multiplicity has $(\Pi_{59} \mid \ell) = -1$. The family is empty, for every $q$.

The companion symbol at $B = N_{59} - 2\Pi_{59}$ equals $-1$ as well; necessarily: for any admissible base $S$ (writing $D = D_S$, $N = N_S$, $A_S, B_S = N \pm 2D$) one has $A_S \equiv B_S \pmod 8$ (as $4D \equiv 8 \bmod 16$) and $A_S \equiv B_S \equiv N \pmod p$ for every odd $p \mid D$, while the quadratic–reciprocity sign difference carries the exponent $\frac{p-1}{2} \cdot \frac{A_S - B_S}{2}$ with $\frac{A_S - B_S}{2} = 2D$ even; hence $(D \mid A_S) = (D \mid B_S)$ always; one binary certificate per tail family. Running it over all 49,961 admissible bases of Proposition 4.5 proves 31,219 of the one–new–prime families at level 60 empty (62.5%, curiously near 5/8); the remaining 18,742 are Jacobi–inconclusive: their $-1$–Legendre primes, if any, occur to an even count, visible only through a factorisation of $A_S$. A factoring attack on those (trial division and Brent–Pollard $\rho$ over a Montgomery core, `code/hunt/survivor_kill.rs`; the found factors are kept as a pairwise–coprime basis, so a kill certificate (an odd–multiplicity basis piece $P$ with $(D_S \mid P) = -1$) needs no primality testing at all), pushed to $\rho$–budget $10^6$ with a Pollard $p - 1$ stage, closes a further 15,264: in total 46,483 of the 49,961 one–new–prime families at level 60 are proven empty (93.0%). The 3,478 still open (3,444 of them once Proposition 5.5 has emptied the 34 immune ones) resist all of this for a structural reason, not a want of effort: each is Baillie–PSW *composite* yet has no prime factor below $\sim 10^{12}$, so its $A_S$ (or $B_S$) is a *balanced* semiprime of two $\sim$57–digit primes, splitting one is factoring at cryptographic scale. Two cheap tests confirm no shortcut survives: a product–tree batch–GCD over all 37,484 values $A_S, B_S$ finds only shared primes below $10^9$ (already found by $\rho$, so *no* free kills), and Baillie–PSW leaves exactly 34 families in which $A_S$ and $B_S$ are both *probable* primes; these are immune *conditionally on that primality* (a prime $A_S$ with $(D_S \mid A_S) = +1$ has no inert factor, hence no congruence obstruction for any $q$). The conditionality

is not pedantry and not removable by more of the same computation: Baillie–PSW is rigorous only in its *composite* verdicts, so the 46,483 kills are unconditional while the 34 immunities are not. Upgrading them would require primality *certificates* for thirty-four 114–digit integers (ECPP or a Pocklington factorisation of $A_S-1$). The elementary route does not reach: immunity needs *both* $A_S$ and $B_S$ prime, and partial factorisation of $n-1$ (trial division to $3\times10^6$ together with the fully split part) leaves $\log F/\log n$ below $1/3$ on at least one side of every one of the 34, so neither Pocklington ($F>\sqrt{n}$) nor Brillhart–Lehmer–Selfridge ($F>n^{1/3}$) applies to any family: 0 of 34 are certifiable this way (`code/immune_certify.py`). This is now settled. Running the Adleman–Pomerance–Rumely –Cohen–Lenstra test on all 68 integers proves every one of them prime in 8 seconds of wall clock, so the 34 immunities upgrade from [C] to [P] and are *unconditional* (`code/immune_prove.gp`, PARI/GP 2.17.4, whose `isprime` is a proof and not a probable–prime test). The run reproduces the whole chain independently: 49,961 admissible bases, 31,219 Jacobi kills, and, among the 81 bases whose $A_S$ and $B_S$ are both prime, exactly the 34 with $(D_S \mid A_S)=+1$; the other 47 carry $(D_S \mid A_S)=-1$ and were already killed. The primality is moreover backed by *independently checkable* evidence rather than a verdict: `code/immune_certify.gp` emits Atkin–Morain ECPP certificates for all 68 integers, archived at `data/certs/immune_ecpp.txt` (512 KB) so a third party can check them without trusting the prover. That file is a `primecertexport` dump and is not machine–readable: PARI's `read` parses its header as a polynomial, so `primecertisvalid` cannot consume it. The working check is `code/immune_cert_verify.gp`, which extracts the 68 asserted subjects and re–proves each by ECPP in 1.3 seconds: 68/68 prime, 0 failures. The structural half of the campaign is now formal: the identity $(D_S \mid A_S)=(D_S \mid B_S)$, which is what makes one binary test decide a family rather than two, is proved in Lean 4 as `Erdos307.jacobiSym_A_eq_B` (`lean/Erdos307/Campaign.lean`), on the three standard axioms and with no `native_decide`; it follows from periodicity, $A_S-B_S=4D_S$ being exactly the period of $J(a\mid\cdot)$. The primality itself is not formalised and cannot presently be: mathlib's only route to a large prime is `lucas_primality`, which needs a complete factorisation of $n-1$, and that is precisely what the 0 of 34 above rules out. Note the two counts answer different questions and both are correct: 81 is the number of prime pairs, 34 the number of *immune* families among them. A complete kill/immune split of the residual is in any case beyond feasible computation, and the honest state of the single–tail sector is 46,483 *empty* [P] *and* 3,478 *open*, the latter including the 34 immune ones, whose primality the APRCL run above makes unconditional [P]; Proposition 5.5 then empties those 34, leaving 3,444. □

**Proposition 5.3** (Sector decomposition of level 60)**.** *Let $U$ be a 60–element prime support with $T(U)>2$. Then exactly one of the following holds:* (i) $T(U)-2\ge 1/\max U$*, and $U=S\cup\{q\}$ for an admissible base $S$ (one of the* 49,961 *of Proposition 4.5); the* single–tail sector*, or* (ii) $T(U)-2<1/\max U$*, and, writing $p>q$ for the two largest elements and $W$ for the rest, $T(W\cup\{p\})<2$ and $T(W\cup\{q\})<2$; the* pair sector*, whose members escape every single–tail family.*

*Proof.* $U$ lies in the tail family of $U\setminus\{r\}$ iff $T(U)-1/r\ge 2$, which is easiest for $r=\max U$; (i) is that case ($T=2$ being impossible by rigidity, the base is admissible). In case (ii) no removal reaches 2, and a fortiori the two stated 59–subsets stay below 2. □

**Remark 5.4** (The campaign; completeness of the certificate; the arity barrier)**.** The reciprocity certificate captures *every* $q$–independent congruence obstruction of a tail family: at a prime $\ell\mid A_SB_S$ the system degenerates to the rank–one congruences $x^2\equiv D_S$ or $y^2\equiv D_S \pmod\ell$, while at $\ell\nmid 2A_SB_S$ the conditions define conics in $(x,q)$ or $(y,q)$ with $\sim\ell$ points, deleting at most half of the residue classes of $q$ and never all (the 2–adic layer pins $q\equiv 5 \bmod 8$ without emptiness); by CRT and Dirichlet, admissible $q$ survive any finite set of such conditions. Hence a tail family admits a congruence kill iff $(D_S\mid\ell)=-1$ for some $\ell\mid A_SB_S$, which the Jacobi symbol detects

for 31,219 bases, and partial factorisation (`code/hunt/survivor_kill.rs`) for a further 15,264: in total 46,483 of the 49,961 single–tail families at level 60 are proven empty (93.0%) and 3,478 remaining open. Of the latter, 34 have $A_S$ and $B_S$ both prime and are emptied by Proposition 5.5; the other 3,444 carry a balanced ∼57–digit semiprime and are factoring–hard. The 34 immune families are not a bottleneck; they are the tractable ones, and they are empty. Immunity means $A_S$ is prime, and that is exactly what makes the family decidable directly.

**Proposition 5.5** (Immune families are decided in polynomial time)**.** *Let $S$ be a level–60 base with $A_S = N_S + 2D_S$ prime and $(D_S \mid A_S) = +1$. Then a solution in the tail family $S \cup \{q\}$ forces $x^2 = A_S q + D_S$ with $x$ one of the two square roots of $D_S$ modulo $A_S$, so the family has at most two candidate tail primes, computable by Tonelli–Shanks. All 34 immune families at level 60 are empty.*

*Proof.* Write $\sigma = N_S/D_S$, so $A_S = (\sigma+2)D_S$. Proposition 4.7 bounds $q \le tD_S$ with $t = \frac{1}{2}(\sigma + \sqrt{\sigma^2-4})$, whence $x^2 = A_S q + D_S \le \big((\sigma+2)t+1\big)D_S^2$. At level 60 one has $\sigma > 2$ and $\sigma$ close to 2, so $(\sigma+2)t + 1 < 5.202$ and $x < 2.281\, D_S < (\sigma+2)D_S = A_S$; over all 49,961 admissible bases the ratio $x/A_S$ is at most 0.56982. Hence $x$ lies in $[0, A_S)$ and, $A_S$ being prime, is one of the two roots of $X^2 \equiv D_S$. Each root gives one candidate $q = (x^2 - D_S)/A_S$, to be tested for integrality, primality and the companion condition that $B_S q + D_S$ be a square. Carrying this out on the 34 families, whose $A_S$ and $B_S$ are unconditionally prime by the APRCL and ECPP runs recorded above, every candidate fails (`code/immune_decide.gp`), either because $q = (x^2 - D_S)/A_S$ is not a prime exceeding 787 or because $B_S q + D_S$ is not a square; the conclusion is therefore [P], not [C]. The script re–derives the 49,961 bases, the 31,219 Jacobi kills and the 34 immune families from scratch before deciding them. □

Primality of $A_S$ plays no role in the bound $x < A_S$, only in the ease of extracting the roots, so the argument is general.

**Proposition 5.6** (The split sieve)**.** *Let $S$ be a finite set of primes, $D = \prod S$, and let $M$ be a finite set of primes disjoint from $S$. Suppose the pair $a = \prod(T \cup M)$, $b = \prod(S \setminus T)$ with $T \subseteq S$ is a derivative two–cycle. Then*

$$\prod T \;\Big|\; \partial\big(\textstyle\prod(S \setminus T)\big), \qquad \Big(\prod T\Big)^2 \partial(\textstyle\prod M) + \partial\big(\prod T\big)\,\partial\big(\prod(S \setminus T)\big) = D.$$

*The second identity carries no reference to the tail; the first is its reduction modulo $\prod T$, Theorem 2.3 removing the factor $\partial(\prod T)$, and is therefore a consequence of it rather than an additional hypothesis. Writing $\alpha = \prod T$ and $\beta = \prod(S \setminus T)$, the divisibility is by Lemma 6.1 the system of congruences $\sum_{p \in S\setminus T} p^{-1} \equiv 0 \pmod r$, one for each $r \in T$. Each odd $r$ costs $1/r$, while the congruence at 2 holds exactly when $|T| \equiv |S| \pmod 2$, so the expected number of surviving $T$ of size $j$ is $m_j = e_j + [\, j \equiv |S| \ (2)\,]\, e_{j-1}$, where $e_j$ is the $j$-th elementary symmetric function of the $1/p$, $p \in S$ odd. Neither statement involves $A_S$, so the criterion applies to the families whose $A_S$ is composite, which Proposition 5.5 cannot reach, and to the pair sector of Proposition 5.3(ii), where $|M| = 2$ and $\partial(\prod M) = p + q$. Two consequences are recorded. Since $\partial(\alpha)\partial(\beta) > 0$ for $T$ and $S \setminus T$ both nonempty, the identity forces $\alpha^2 < D$. And for $X$ a set of odd primes $\partial(\prod X) \equiv |X| \pmod 2$, so $2 \in T$ forces $|T| \equiv |S| \pmod 2$. 0 `sorry` (`Erdos307.split_criterion_gen`, `split_snd_gen`, `split_snd_of_criterion`, `csum_odd_card`; `lean/Erdos307/SplitSieve.lean`).*

*Proof.* Leibniz on the disjoint union gives $\partial(\prod(T \cup M)) = \prod M \cdot \partial(\prod T) + \prod T \cdot \partial(\prod M)$. The equation $\partial(b) = a$ reads $\partial(\beta) = \alpha \prod M$, which is the divisibility. The equation $\partial(a) = b$ reads $\beta = \prod M \cdot \partial(\alpha) + \alpha\, \partial(\prod M)$; multiplying by $\alpha$, replacing $\alpha \prod M$ by $\partial(\beta)$ and using $\alpha\beta = D$ gives

the identity. For the converse direction of the first claim, reduce the identity modulo $\alpha$: both $\alpha^2\partial(\prod M)$ and $D$ vanish, so $\alpha \mid \partial(\alpha)\partial(\beta)$, and $\gcd(\alpha, \partial(\alpha)) = 1$ by Theorem 2.3. □

**Corollary 5.7** (The single–tail criterion is exact)**.** *Let $M = \{q\}$ with $q \geq 3$ a prime not in $S$, and $\alpha^2 < D$. Then $a = q\alpha$, $b = \beta$ is a two–cycle if and only if $\alpha^2 + \partial(\alpha)\partial(\beta) = D$ and $q = \partial(\beta)/\alpha$. If $|S|$ is odd, $2 \in S$ and $\Sigma = \sum_{p\in S} 1/p < 2.01$, the identity holds if and only if $D \mid \alpha^2+\partial(\alpha)\partial(\beta)$, that is, if and only if $\sum_{p\in S\setminus T} p^{-1} \equiv 0 \pmod r$ for every $r \in T$ and $\alpha^2+\partial(\alpha)\,\beta/r \equiv 0 \pmod r$ for every $r \in S \setminus T$; and every two–cycle has $|T|$ odd.* 0 ***sorry*** *for the algebraic core* (`Erdos307.split_cycle_of_identity`, `split_card_odd`; `lean/Erdos307/SplitExact.lean`).

*Proof.* The forward direction is Proposition 5.6. Conversely, the identity reduced modulo $\alpha$ gives $\alpha \mid \partial(\beta)$ as in its proof; with $q = \partial(\beta)/\alpha$ this is $\partial(b) = a$, and substituting $\partial(\beta) = q\alpha$ into $\alpha\beta - \alpha^2 = \partial(\alpha)\partial(\beta)$ and cancelling $\alpha$ gives $\beta = \alpha + q\,\partial(\alpha) = \partial(q\alpha)$, which is $\partial(a) = b$. For the divisibility, put $N = \alpha^2 + \partial(\alpha)\partial(\beta)$; then $N/D = \alpha^2/D + \sigma(T)\sigma(S \setminus T) < 1 + \Sigma^2/4 < 2.02$, and $N = 2D$ would give $\alpha^2/D \geq 2 - \Sigma^2/4$, hence $q = \sigma(S \setminus T)D/\alpha^2 \leq \Sigma/(2 - \Sigma^2/4) < 3$. The local form reduces $N$ modulo each prime of $S$, using $\partial(\beta) \equiv \beta/r \pmod r$ for $r \in S \setminus T$. For the parity: if $2 \in T$ this is Proposition 5.6; if $2 \notin T$, then $\alpha$ and $q$ are odd and $\partial(\alpha) \equiv |T| \pmod 2$, so $\beta = \alpha + q\,\partial(\alpha) \equiv 1 + |T|$ is even only for $|T|$ odd. □

The 58 character conditions of the tail at the odd primes of $S$ add nothing to the square of Proposition 5.8: $\partial(D) \equiv D/p \pmod p$ turns $x^2 = A_S q + D_S$ into $x^2 \equiv q\,D_S/p$, a nonzero square modulo $p$, which is the condition $(q \mid p) = ((D_S/p) \mid p)$.

*What the sieve decides.* The criterion was applied to every level–60 single–tail base surviving the certificate of Proposition 5.2, 18,742 of them, a superset of the 3,444 open families, over all $T$ with $|T| \leq 6$: about $9 \times 10^{11}$ splits, no factorisation, and no two–cycle. The observed survivor count is 5.20 per base against the 5.26 of the model $m_j$, which matches size by size (`code/split_sieve.rs`, `code/gen_bases.rs`). On the 7,713 bases that also survive trial division to $10^6$, the pass was extended to $|T| \leq 8$: $2.0 \times 10^{13}$ splits, with 37 survivors at $|T| = 7$ against 39.1 predicted, none at $|T| = 8$ against 0.51, and again no two–cycle. The range $9 \leq |T| \leq 34$, where the model expects 0.57 survivors over all 7,713 bases, is not decided by enumeration, since $\alpha^2 < D$ bounds $|T|$ by 34, so the sieve reduces the level–60 single–tail question without closing it.

**Proposition 5.8** (Single–tail families are decided by factoring)**.** *Let $S$ be a set of primes containing 2, with $D_S = \prod S$ squarefree, and let $q$ be a prime completing $S$ to a solution support. Then $A_S$ is odd and coprime to $D_S$, and $x = a + b$ satisfies $x^2 = A_S q + D_S$ with $0 < x < A_S$. Hence $x$ is one of the $2^{\omega(A_S)}$ square roots of $D_S$ modulo $A_S$, each giving a single candidate $q = (x^2 - D_S)/A_S$, and the family is decided by one factorisation of $A_S$ together with $O(2^{\omega(A_S)})$ primality tests.*

*Proof.* Since $2 \in S$ and $D_S$ is squarefree, exactly one term of $N_S = \sum_{p\in S} D_S/p$ is odd, so $N_S$ is odd and $A_S = N_S + 2D_S$ is odd. If $p \in S$ divided $A_S$ it would divide $N_S$, but $N_S \equiv D_S/p \not\equiv 0 \pmod p$; as the primes of $D_S$ are exactly $S$, $\gcd(A_S, D_S) = 1$. For the bound, write $\sigma = N_S/D_S$ and $A_S = (\sigma+2)D_S$. Proposition 4.7 gives $q \leq tD_S$ with $t = \frac{1}{2}(\sigma + \sqrt{\sigma^2 - 4})$, so $x^2 \leq \big((\sigma+2)t+1\big)D_S^2$. Across all 49,961 admissible level–60 bases $\sigma \in (2, 2.00236]$, whence $(\sigma + 2)t + 1 < 5.202$ and $x < 2.281\,D_S$, while $A_S = (\sigma + 2)D_S > 4\,D_S$; the ratio $x/A_S$ is at most 0.56982 over the whole set (`code/lvl60check.gp`). Thus $x < A_S$. When instead $\sigma \leq 2$, as for the pair–sector bases of Proposition 5.3(ii), Corollary 4.11 bounds the tail by $q \leq D_S/(2D_S - N_S) = D_S/|B_S|$, so $x^2 = A_S q + D_S \leq A_S D_S/|B_S| + D_S < A_S^2$ because $|B_S| \geq 1$ and $A_S > D_S$; the conclusion $x < A_S$ therefore holds in both regimes. Being odd and coprime to $D_S$, $A_S$ admits exactly $2^{\omega(A_S)}$ square roots of $D_S$ when $D_S$ is a residue modulo every prime power dividing it, and none otherwise; these

are computed from the factorisation by Tonelli–Shanks, Hensel lifting at any repeated factor, and the Chinese remainder theorem. □

This replaces a $2^{58}$ search per family by a single factorisation, and the integer to be factored is $A_S \approx 4D_S$, of 114 decimal digits, not the 230–digit product the divisor search ranges over. It also supersedes the reciprocity route rather than extending it: a congruence kill decides at best half of the split families, since for $A = pq$ with $(D_S \mid A) = +1$ the two Legendre symbols are either both $+1$, when no kill exists, or both $-1$, when one does, whereas the root enumeration decides every family whose $A_S$ is factored. One cheap idea deserves to be recorded as closed, because the level–60 bases are far from random: all 49,961 share the 39 forced primes below 167, so one may hope that two of the $A_S$ share a prime factor and split each other for free. A batch GCD in the sense of Bernstein (one product tree over the hard cofactors, then a remainder tree) tests this for every pair at once in seconds. Over the 7,713 families that survive the Jacobi certificate and trial division to $10^6$, a superset of the 3,478, exactly two of the 6,972 composite cofactors of $A_S$ share a factor, and both shared factors have seven digits, so they lie barely past the trial–division bound and are found instantly by any factoring method. The structure the bases share is in $D_S$ and does not propagate to $A_S = N_S + 2D_S$: at the wall the residual behaves as a set of pairwise coprime balanced semiprimes, and no amount of shared structure among the bases is going to split them. (`code/batchgcd.gp`)

Level 60 is therefore not an exponential obstruction but a factoring project: the 34 with $A_S$ prime are already done, and 3,444 integers of 114 digits remain. A 114–digit general number field sieve is routine, costing core–days apiece rather than the $3.2 \times 10^7$ core–years the divisor search of Proposition 4.7 would demand, a reduction by six orders of magnitude. It should be said plainly that this does not by itself deliver $|P \cup Q| \geq 61$. The single–tail sector is only one of the two cases of Proposition 5.3; the other, the pair sector, is empty by Proposition 5.12, without factoring. What has changed is the character of the obstruction: level 60 is no longer exponential in $|S|$, but linear in the number of families, each at the cost of one factorisation.

Those 3,478 are expected to be empty, and by a wide margin: evaluating the expected count of Remark 5.16 over them, with $\kappa_S \leq 1$ and the tail sum bounded by $\log\log(tD_S)$, gives at most $5.5 \times 10^{-108}$, the bases having $D_S$ of 113 digits. The divergence of the global count comes from large $a$ and not from this level, and the first solution the heuristic locates sits at $10^{116.7}$ or beyond. So the factoring wall guards the verification of something the heuristic already asserts, rather than a plausible solution. The weapon is intrinsically *arity–one*, and provably fails on the pair sector. With $s = p + q$, $t = pq$, a pair–sector member requires the two simultaneous squares $x^2 = A_W t + D_W s$ and $y^2 = B_W t + D_W s$ (the would–be third condition $s^2 - 4t = \square$ is automatic, being $(p - q)^2$), together with the reciprocity–linked constraints $(D_W p \mid q) = (D_W q \mid p) = 1 \pmod q$, a solution with $q \mid a$ has $N_U \equiv D_W p$, forcing $D_W p$ to be a square). At a prime $\ell \mid A_W$ the condition $x^2 \equiv D_W s \pmod \ell$ constrains only $s$, which the free second prime absorbs; the pair system is in fact locally soluble at *every* modulus (a theorem, Proposition 5.11 below, corroborated by a direct scan over all $\ell \leq 600$) so, unlike the single–tail family, the pair sector admits *no* congruence obstruction at all. It is a genuine two–parameter surface per base, #307 in miniature (Remark 5.15); reciprocity cannot touch it, and only a direct both–squares search or a descent argument can.

The sector is nevertheless an explicit finite object, and small. A level–60 support $U$ is covered by the one–new–prime campaign exactly when $T(U \setminus \{\max U\}) > 2$, the bases of Proposition 4.5 being the 59–prime sets of mass exceeding 2; the complement is $U = R \cup \{m\}$ with $|R| = 59$, $m = \max U$ and $T(R) \leq 2 < T(R) + 1/m$. The mass ladder at $k = 60$ puts every element of $R$ below $793.67 \times 2 = 1587.4$, so the complement is a search over 59–subsets of the 250 primes under 1588 with $T(R) \in (2 - 1/\max R,\ 2]$. There are exactly 18,234,653 of them, the largest $\max R$

being 1583 (`code/pairsector_count.rs`, corroborated by an independent binned counting DP in `code/pairsector_countdp.rs`). Each is an arity–*one* family in $m$; Proposition 5.11 concerns only the family with both new primes free.

That the sector is finite at all rests on $T(R) \neq 2$: at $T(R) = 2$ every $m$ would satisfy $T(R) + 1/m > 2$ and the tail would be unbounded. It cannot occur. If $2 \notin R$ then $D$ is odd and each $D/p$ is odd, so $N = \sum_{p|D} D/p$ is a sum of 59 odd terms and is odd; if $2 \in R$ then $D/2$ is odd and the other 58 cofactors are even, so $N$ is again odd. Since $2D$ is even, $N \neq 2D$ (`Erdos307.mass_ne_two`; `code/pairsector_parity.gp`).

Every base is decided by Proposition 5.12, which tests its tails one by one and needs no certificate (`code/pairsector_kill.rs` in campaign mode reproduces the 49,961, 31,219 and 18,742 above).

**Proposition 5.9** (Level 60 is empty)**.** *No solution of #307 has $|P \cup Q| = 60$. Consequently every solution has $|P \cup Q| \geq 61$.*

*Proof.* By Proposition 5.3 a 60–element support of mass above 2 lies in the single–tail sector or the pair sector, and the latter is empty by Proposition 5.12. In the single–tail sector write $a = q\alpha$, $b = \beta$, $\alpha\beta = D = \prod S$, $\rho = \alpha^2/D < 1$. For $\rho \leq 2^{-40}$ the identity of Proposition 5.6 gives $\sigma(T)\big(\Sigma - \sigma(T)\big) = 1 - \rho$, which confines $\sigma(T)$ to a window of width about $\rho/\sqrt{\Sigma^2 - 4}$ around a root of $X(\Sigma - X) = 1$; every $T$ in the window, at every $|T|$, is enumerated by a meet–in–the–middle over four quarters of $S$ and decided exactly. For $\rho > 2^{-40}$ the tail is bounded, $q = \sigma(S \setminus T)/\rho < \Sigma\, 2^{40}$, and the square of Proposition 5.8 fixes $(q \mid p) = ((D/p) \mid p)$ at the 58 odd $p \in S$; every $q$ in range is sieved by these conditions and each survivor decided by the two squares. Over the 7,713 bases, a superset of the 3,444 open families, the first search examines $4.50 \times 10^{11}$ candidates and finds no $T$ satisfying the identity, and the second examines $2.27 \times 10^{13}$ candidates and leaves no $q$ making both squares perfect. The last claim is Proposition 5.3 with the barrier of Theorem 4.3. □

The computations are `code/splitmitm/split_mitm.rs` and `code/qsieve/q_sieve.rs`, the latter reproduced base by base by the independent `code/qsieve/q_sieve2.rs`, with identical survivor lists (all empty) on every base. Both are audited: `code/splitmitm/stageA_exact.gp` re-decides every reported split by the identity directly and confirms exactly one completed pass per base with no candidate misreported; `code/qsieve/q_exact.gp` does the same for the tail sieve; and `split_mitm --audit` plants 100 genuine two–cycle–shaped splits across 5 open bases and recovers all 100. The assembly from level 60 to the bound 61 is formalised (`Erdos307.erdos307_sixtyone_of_level60`). The level–60 closure is formalised at a second tier, in directories that the main library does not import (`lean/Erdos307/Computational`, `lean/QNative`), where each compiled evaluation adds one `native_decide` axiom beside the three standard ones. The reduction is proved in the kernel: a tail cycle falls in the regime $\alpha^2 2^{40} \leq D$ or satisfies $q\,\alpha^2 \leq N$ (`tail_split_core`), the two squares force $q \equiv A(1 - D) \pmod 8$ and $(q \mid \ell) = ((D/\ell) \mid \ell)$ at every odd $\ell \in S$ (`two_adic_class`, `tail_legendre`), and a divisor $d \mid A$ with Jacobi symbol $(D \mid d) = -1$ kills a base with no factoring (`kill_of_divisor`). The compiled evaluations are: the pair sector is empty (`pair_sector_empty`); every one of the 49,961 window bases is killed by a Jacobi symbol or is one of 3,698 listed survivors (`killCheck_run`); the $q$–sieve returns `true` on all 3,698 (chunk theorems `chunk_0`–`chunk_36`). The barrier $|P \cup Q| \geq 61$ is then a theorem (`Erdos307.erdos307_sixtyone_of_stageA`) modulo one named statement, `StageAList`: no survivor base carries a cycle with $\alpha^2 2^{40} \leq D$. Stage A is not formalised; it was recomputed by `code/splitmitm/stageA2.rs`, an independent program (interleaved halves, a bucketed hash join, the identity tested modulo two 31–bit primes with the restriction $|T|$ odd), on the 3,698 survivors: $1.05 \times 10^{11}$ candidates in the lower window, no identity hit, and a candidate count at least that of the original run on every base. A rerun of the factoring search found 15,044 kills and 3,698 survivors (the search is not deterministic across runs; the figures

above, 15,264 and 3,478, came from an earlier run); all 3,698 lie among the 7,713 bases on which Stage A and the sieve ran.

**Corollary 5.10** (Sharpened bound)**.** *Every solution has* $\prod_{p\in U} p \geq \Pi_{61} > 6.97 \times 10^{117}$ *and* $\min(\prod P, \prod Q) \geq \sqrt{\Pi_{61}/T_{61}} = 5.8918\ldots \times 10^{58}$.

*Proof.* Repeats Proposition 4.5's bound on the minimal side with the minimum of $R/T$ now over $|U| \geq 61$, attained at the first 61 primes. □

**Proposition 5.11** (The pair sector admits no congruence kill)**.** *Let $W$ be any pair–sector base and $A = A_W$, $B = B_W$, $D = D_W$, so that a pair–sector solution requires $x^2 = A\,\alpha\beta + D(\alpha+\beta)$ and $y^2 = B\,\alpha\beta + D(\alpha+\beta)$ with $\alpha, \beta$ the two new primes. For every prime power $\ell^j$ this system has a solution in units $\alpha, \beta$ and integers $x, y$ at a nonsingular point. Consequently, by CRT and Dirichlet, no finite system of congruence conditions empties any pair–sector family: the arity barrier of Remark 5.4 is a theorem, and the certificate programme provably ends at arity one.*

Formalised, in part. ***Erdos307.no_pinning_isUnit*** *is the step that makes the tail–kill mechanism unavailable,* ***collapse_identity*** *is regime* (2)*'s observation that the two square conditions are one,* ***regime_one_first*** *and* ***regime_one_second*** *are the explicit unit of regime* (1)*, and* ***targets_agree_mod_eight*** *is the mod–8 collapse of regime* (3)*. The lift from $\ell$ to $\ell^j$ that regime* (1) *needs is* ***square_lifts_to_prime_powers****, proved from* ***hensel_all_powers****: if $\ell$ is an odd prime not dividing $c$ and $c$ is a square modulo $\ell$, it is a square modulo every power of $\ell$. Regime* (2) *needs no citation to Weil's bound. Its two steps are* ***Erdos307.pairlocal_inner_sum****, the evaluation of the inner sum of $S_{fg}$ as $-\chi\big((A\alpha+D)(B\alpha+D)\big)$ by the quadratic complete sum of Proposition 5.17, with its side condition discharged as $4D^2\alpha^2 \neq 0$; and* ***pairlocal_count_pos****, the arithmetic turning the resulting $O(\ell)$ bounds into a positive count from $\ell \geq 17$. 0* ***sorry*** (`lean/Erdos307/PairLocal.lean`, `lean/Erdos307/PairLocalCount.lean`)*. Checked directly: over* 872 *random bases with $A - B = 4D$ and every odd $\ell \leq 400$, the count of admissible $(\alpha, \beta)$ was positive in every case and tracks $(\ell-1)^2/4$; the crude bounds $|S_f|, |S_g| \leq 2\ell$, $|S_{fg}| \leq 3\ell$ held in all* 409 *cases tested* (`code/pairlocal_count.gp`).

*Proof.* First, no $\ell$ pins either target to a constant: that needs $\ell \mid A$ (or $B$) *and* $\ell \mid D$, whence $\ell \mid N$, contradicting $\gcd(N, D) = 1$ (rigidity); so the tail–kill mechanism is unavailable, and solubility is decided in three regimes.

*(1) $\ell$ odd, $\ell \mid D$.* This covers every odd $\ell \leq 103$: at level 60, omitting a prime $p$ caps $T(U)$ at $T_{61} - 1/p$, and $1/p > T_{61} - 2 = 0.00944\ldots$ exactly for $p \leq 103$, so all such primes lie in $U$, hence in $W$ (the two adjoined primes are the largest). Take $\beta = 1$, $x = 1$, $\alpha = (1-D)/(A+D) \bmod \ell^j$, a unit since $A + D \equiv N \not\equiv 0 \pmod{\ell}$: the first equation holds by construction, and the second target $(B+D)\alpha + D \equiv N \cdot N^{-1} = 1 \pmod{\ell}$ is a unit congruent to a square, so $y$ exists modulo $\ell^j$.

*(2) $\ell$ odd, $\ell \nmid 2D$* (in particular every $\ell \geq 107$ outside $W$). The two conditions are secretly *one*: setting $m = (x^2 - y^2)/4D$ and $u = (x^2 - Am)/D$, the relation $A - B = 4D$ gives the polynomial identity $Bm + Du = y^2$. Only *existence* is wanted, and counting the pairs $(\alpha, \beta)$ directly settles it with no curve and no Weil bound. Write $\chi$ for the quadratic character, $f = A\alpha\beta + D(\alpha+\beta)$ and $g = B\alpha\beta + D(\alpha+\beta)$. Summing $\frac{1}{4}(1+\chi(f))(1+\chi(g))$ over $\alpha, \beta \in \mathbb{F}_\ell^\times$ and discarding the at most $2(\ell-1)$ pairs with $fg = 0$,

$$4\,\#\{\text{both nonzero squares}\} \;\geq\; (\ell-1)^2 + S_f + S_g + S_{fg} - 8(\ell-1).$$

For fixed $\alpha$ each target is *linear* in $\beta$, so the inner sums of $S_f$ and $S_g$ vanish by the linear evaluation above, leaving $|S_f|, |S_g| \leq 2\ell$. And $fg$ is a *product of two linear forms* in $\beta$, so its inner sum is the quadratic evaluation, whose side condition $ad \neq bc$ is here $D\alpha(A-B)\alpha = 4D^2\alpha^2 \neq 0$, exactly the hypotheses $\ell \nmid 2D$ and $\alpha \neq 0$; summing the resulting $-\chi\big((A\alpha+D)(B\alpha+D)\big)$ over $\alpha$ is that same

evaluation again, so $|S_{fg}| \le 3\ell$. Hence the count is positive as soon as $(\ell-1)^2 - 7\ell - 8(\ell-1) > 0$, i.e. from $\ell \ge 17$, far below the $\ell \ge 107$ in play. The point is nonsingular ($\alpha\beta \neq 0$; the Jacobian contains $\mathrm{diag}(2x, 2y)$), so it lifts to every $\ell^j$.

*(3)* $\ell = 2$. Both targets are odd and agree modulo 8 (as $\alpha\beta$ is odd, $D(\alpha+\beta) \equiv 0 \bmod 4$, and $(A-B)\alpha\beta = 4D\alpha\beta \equiv 0 \bmod 8$), and a finite table over ($A \bmod 8$, $D \bmod 8$) shows that for every base some achievable pair $(\alpha\beta, \alpha+\beta) \bmod 8$ makes the common target $\equiv 1 \pmod 8$; a 2–adic square, soluble modulo $2^j$ for every $j$.

All three regimes, the collapse identity, the threshold $1/103 > T_{61} - 2 > 1/107$, and the mod–32 layer were verified computationally on three bases and every $\ell \le 600$ (`code/pairlocal.py`; the measured density of admissible $(x, y)$ in regime (2) is 0.497–0.503 against the predicted $\frac{1}{2}$). □

**Proposition 5.12** (The pair sector of level 60 is empty)**.** *No* 60*–element support in the pair sector of Proposition 5.3 carries a two–cycle.*

*Proof.* Write the support as $U = R \cup \{m\}$ with $m = \max U$ and $D = \prod R$, and let $N = ab = Dm$ with $a' = b$, $b' = a$, where $a, b$ are coprime and squarefree (Theorem 2.3). By Proposition 5.3(ii), $T(R) < 2 < T(R) + 1/m$, that is

$$\max R < m < \frac{1}{2 - T(R)}.$$

Now $N' = a'b + ab' = a^2 + b^2$. Let $p \in R$ be odd. It divides exactly one of $a, b$, so $N'$ is a nonzero square modulo $p$. Since $p \mid D$ and $p \nmid m$, $N' = D'm + D \equiv (D/p)\, m \pmod p$, hence

$$\Big(\frac{m}{p}\Big) = \Big(\frac{D/p}{p}\Big).$$

Every base contains 2, since 59 odd primes have reciprocal sum below 1.51, so these are 58 conditions. For none of the 18,234,653 bases $R$ of Remark 5.4 does an odd $m$ in the displayed range satisfy all 58: the ranges are short, $1/(2 - T(R)) < 1.14 \times 10^8$, and the 2,143,165,628 pairs $(R, m)$ are checked by table lookup with no survivor. □

The check is `code/pairsector_close.rs` (one thread, about four CPU–minutes). It decides sector membership and the range of $m$ in exact integer arithmetic, with 0 ambiguous cases, recovers a planted tail and the count 18,234,653, and agrees with an independent implementation on every count. The algebraic core is formalised (`Erdos307.pair_squares`, `pair_tail_bound`, `pair_char_square`, `pair_legendre`; `lean/Erdos307/PairClosed.lean`). The finite check itself is also formalised, at a tier that adds one compiled evaluation (`native_decide`): `Erdos307.pair_sector_empty` in `lean/Erdos307/Computational/`, which is not imported by the main library. Proposition 5.11 is not contradicted: it concerns the family with both new primes free, and here the sector's own mass condition bounds the one tail that remains. Level 60 is thereby reduced to the 3,444 open single–tail families.

**Remark 5.13** (Anatomy of the certificate; the empirical 5/8 law)**.** Writing $D = 2D_1$, the reciprocity signs in the certificate cancel ($\frac{A-1}{2} + \frac{N-1}{2} = N - 1 + 2D_1$ is even), leaving the clean product

$$\big(\tfrac{D}{A}\big) \;=\; \big(\tfrac{2}{A}\big)\big(\tfrac{D_1}{N}\big), \qquad \big(\tfrac{2}{A}\big) = +1 \iff N \equiv 3, 5 \pmod 8,$$

and one layer down $(N \mid D_1) = \prod_{p \mid D_1} \big(\frac{D/p}{p}\big)$, since $N \equiv D/p \pmod p$: the cofactor structure of rigidity recurs inside its own certificate. This product evaluates in closed form: with $D/p = 2D_1/p$

and quadratic reciprocity,

$$(N \mid D_1) = (2 \mid D_1) \prod_{\{p,q\}\subset D_1} (-1)^{\frac{p-1}{2}\frac{q-1}{2}} = (-1)^{s+\binom{t}{2}}, \quad s = \#\{p \mid D_1 : p \equiv 3,5\ (8)\},\ t = \#\{p \mid D_1 : p \equiv 3\ (4)\},$$

a Rédei genus character of $D_1$, *independent of* $N$. This is forced, not merely observed: $(N \mid D_1) = \prod_{p|D_1}(N \mid p)$, and cofactor survival gives $N \equiv D/p = 2D_1/p \pmod p$, so $(N \mid p) = (2 \mid p)\,(D_1/p \mid p)$. The first factors multiply to $(-1)^s$ since $(2 \mid p) = -1$ exactly for $p \equiv 3,5 \pmod 8$; and, each unordered pair $\{p,q\}$ occurring once in each of the two cofactors,

$$\prod_{p|D_1}\Big(\tfrac{D_1/p}{p}\Big) = \prod_{\{p,q\}\subset D_1}\Big(\tfrac{q}{p}\Big)\Big(\tfrac{p}{q}\Big) = \prod_{\{p,q\}\subset D_1}(-1)^{\frac{p-1}{2}\frac{q-1}{2}} = (-1)^{\binom{t}{2}},$$

by quadratic reciprocity, since $\frac{p-1}{2}$ is odd exactly for $p \equiv 3 \pmod 4$. (Checked independently, prime by prime, on 20,000 bases: 0 violations.) The inner symbol is thus not a coin but a genus invariant; its empirical fairness (49.9%, 50.0% over the 49,961 bases) is that character's equidistribution; and $(D_1 \mid N)$ follows by the reciprocity sign $(-1)^{\frac{D_1-1}{2}\frac{N-1}{2}}$, hence depends on $N$ only through $N \bmod 4$. In fact the entire certificate closes into a finite functional. Let $v(S) = (n_1, n_3, n_5, n_7) \bmod 4$ count the odd primes of $S$ in the residue classes $1,3,5,7$ modulo 8. Every factor above is a function of $v$ alone: $s \equiv n_3 + n_5$, $t = n_3 + n_7$ (with $\binom{t}{2}$ needing only $t \bmod 4$), $D_1 \equiv 3^{n_3}5^{n_5}7^{n_7} \pmod 8$, $S(D_1) \equiv (n_1+n_5)-(n_3+n_7) \pmod 4$, and (by the mod–8 law of Proposition 15.13, $D_1' \equiv D_1 S(D_1) \pmod 8$) also $N = D_1 + 2D_1' \equiv D_1 + 2\big(D_1 S(D_1) \bmod 4\big) \pmod 8$. Hence

$$(D \mid A) \;=\; K\big(v(S)\big)\,:$$

*the kill is decided by sixteen residue counts modulo* 4. The realized vectors are exactly the coset $n_1 + n_3 + n_5 + n_7 \equiv 58 \equiv 2 \pmod 4$ (all 64 of its classes occur), $K$ is constant on each, and the functional reproduces the direct 113–digit Jacobi computation on every one of the 49,961 bases (`code/kill_classvector.py`). The five–eighths law is then an exact finite statement, and its mechanism is the *reciprocity switch*: for $N \equiv 1 \pmod 4$ the reciprocity sign is inert and the kill reduces to the genus coin; exact conditional rate $12{,}699/24{,}975 = 0.5085$; while for $N \equiv 3 \pmod 4$ the sign couples the parity of $t$ into the product and the rate jumps to $18{,}520/24{,}986 = 0.7412$ (by class modulo 8: 0.509, 0.741, 0.508, 0.741 at $N \equiv 1,3,5,7$; the 3/4 classes are $N \equiv 3 \pmod 4$, not the $(2 \mid A)$ classes). The blend $\frac{1}{2}\cdot\frac{24975}{49961} + \frac{3}{4}\cdot\frac{24986}{49961} = 0.62503$ against the exact $31{,}219/49{,}961 = 0.62487$ is the whole of the "coincidence". The correlation is an ensemble property, a random–subset proxy gives ≈0.45/0.55; and the functional yields no second certificate: it recomputes the same symbol.

**Proposition 5.14** (Why a one–sided mass argument cannot reach the barrier)**.** *The threshold $\sigma(U) = t + 1/t$ of Proposition 5.18 makes the barrier equivalent to a lower bound $|\sigma(a) - 1| > c$ with $c$ absolute, and it is worth recording what such a bound cannot be proved from. The primary pseudoperfect numbers satisfy the one–sided equation $n' = n - 1$ exactly, so along them $|\sigma(n) - 1| = 1/n$ tends to 0: the eight known members give $1/n$ from $\frac{1}{2}$ down to $1.18 \times 10^{-31}$. They are solutions of the coprime relaxation (Corollary 12.7, the case $u = 1$). Hence no argument that uses only $a' = b$ with $b$ squarefree and coprime to $a$ can bound $|\sigma(a) - 1|$ below by an absolute constant, since the same argument would prove the corresponding statement for the relaxation, which is false. Any such bound must consume the second equation $b' = a$. This is why the mass arguments of Section 4, and the Sylvester–type extremal bounds, reproduce* 60 *and stop: they are one–sided, and $1/a$ is the most a one–sided argument can give.*

**Remark 5.15** (The kill–program is one–sided)**.** Every kill is a proof that no solution hides in that family, but if #307 has a solution, the solution's own family is unkillable, so no accumulation of kills can overshoot the truth: the program's asymptote *is* the existence question. Together with the self–similarity of Remark 5.1 (each rung past 60 contains the whole problem), this locates the campaign precisely: it corners where a solution could live, one certified region at a time, and can terminate only if #307 has answer NO, which the heuristics disfavour.

**Remark 5.16** (The reciprocity kills are the local factors of the expected count)**.** The divergent naive count of Section 6 and the reciprocity sieve of Proposition 5.2 are the *same* computation. Decompose the expected number of solutions over tail families $S \cup \{q\}$: the summand for a base $S$ is $\kappa_S \left(D_S\sqrt{\text{mass}^2-4}\right)^{-1} \sum_q q^{-1}$, where $\kappa_S = \prod_\ell$(local solubility density at $\ell$) is exactly the reciprocity data. A reciprocity–killed base carries an inert $\ell \mid A_S$ with $(D_S \mid \ell) = -1$, so $N_S + 2D_S \equiv D_S \pmod{\ell}$ is a non–residue for *every* $q$, forcing $\kappa_S = 0$; the family drops from the count entirely. An immune base ($A_S, B_S$ both prime, both residues) has no inert prime, so $\kappa_S > 0$ and, treating $q$ as ranging freely with density $1/q$, the summand *looks* like a divergent $\sum_q q^{-1}$ (verified on an explicit 114–digit immune base: local density 1 at every tested prime, partial sums growing without bound). That model overstates the space: Proposition 4.7 bounds $q$ itself, so the true sum is finite, and Proposition 5.8 identifies it exactly with the roots of $D_S$ modulo $A_S$, of which there are only two once $A_S$ is prime. The immune part of the divergence was therefore never genuine, only an artifact of summing a local density over an unbounded range that Proposition 4.7 had already closed; the kills remove ∼93% of the count's mass, and Proposition 5.5 removes the rest of the 34 conditionally immune families' contribution unconditionally, both candidate values of $q$ failing outright, and the pair sector is empty by Proposition 5.12. The level–60 part of the count's "weakly favours YES" signal is therefore zero; the count itself, which ignores the $N(r) \le 1$ rigidity, is unaffected.

**Proposition 5.17** (Local completeness of the reciprocity sieve)**.** *For an admissible base $S$ and an odd prime $\ell$, the admissible tail residues $Q_\ell = \{q \bmod \ell : (A_S q + D_S \mid \ell) \neq -1$ and $(B_S q + D_S \mid \ell) \neq -1\}$ fall into exactly three regimes.* (i) *$\ell \mid D_S$, which covers every $\ell \le 167$ uniformly over the* 49,961 *admissible bases, since $1/p \ge (T_{59}-2)+\frac{1}{281}$ forces every prime $p \le 167$ into every base: then $A_S \equiv B_S \equiv N_S \not\equiv 0 \pmod{\ell}$, the two conditions collapse to the single condition $(N_S q \mid \ell) \neq -1$, and $|Q_\ell| = (\ell+1)/2 \ni 0$.* (ii) *$\ell \mid A_S B_S$: one condition is the constant $(D_S \mid \ell)$, so $Q_\ell = \emptyset$ iff $(D_S \mid \ell) = -1$ (exactly the certificate of Proposition 5.2) and otherwise $|Q_\ell| \ge (\ell-1)/2$.* (iii) *$\ell \nmid A_S B_S D_S$ (hence $\ell \ge 173$): the complete character sums give $|Q_\ell| \ge (\ell-3)/4 - 2 \ge 40$ (the linear sums vanish, the quadratic sum is $-\chi(A_S B_S)$, boundary terms $\le 2$).* (*Regimes* (i) *and* (ii) *are formalised:* `Erdos307.regime_one_collapse`, `regime_one_same_argument` *and* `regime_two_constant` (`lean/Erdos307/LocalComplete.lean`). *Regime* (iii)*'s counting step is* `regime_three_count`*, and its two complete character–sum hypotheses are now theorems rather than citations: $\sum_q \chi(aq+b) = 0$ for $a \neq 0$ is* `Erdos307.sum_quadraticChar_affine`*, the reindexing $q \mapsto aq+b$ followed by* `quadraticChar_sum_zero`*; and $\sum_q \chi((aq+b)(cq+d)) = -\chi(ac)$ for $ad \neq bc$ is* `Erdos307.sum_quadraticChar_quad` *proved by factoring out $\chi(ac)$, translating to $\sum_u \chi\big(u(u+e)\big)$ with $e \neq 0$, writing $u(u+e) = u^2(1+e/u)$ for $u \neq 0$ so that the character value is $\chi(1+e/u)$, and reindexing by the involution $u \mapsto e/u$ of the nonzero elements to reach $\sum_{t \neq 0} \chi(1+t) = 0 - \chi(1) = -1$. Both hold over any finite field of odd characteristic.* 0 `sorry` (`lean/Erdos307/CompleteSums.lean`). *Checked against direct evaluation for all odd primes $\ell \le 59$: the linear sum over every $(a,b)$ with $a \neq 0$, and the quadratic sum over all* 33,008,952 *quadruples with $ad \neq bc$,* 0 *violations* (`code/complete_sums_check.gp`).) *Consequently $Q_\ell \neq \emptyset$ forces $|Q_\ell| \ge 2$ at every modulus, the surviving local density is positive (e.g. $\prod_{\ell \le 300} |Q_\ell|/\ell \approx 2.2 \times 10^{-19} > 0$ on the canonical base), and by CRT* no finite system of congruences (no covering) can annihilate an unkilled family*: the binary certificate of Proposition 5.2 is the complete local theory of the tail sector, there is no "second kill layer," and the one–sidedness of*

*Remark 5.15 is structural rather than an artefact of method. (All three regimes verified exhaustively on the canonical base: collapse and $|Q_\ell| = (\ell+1)/2$ at $\ell \in \{3, 5, 7, 101, 277\}$; the Weil floor at every prime $281 \le \ell \le 600$; $A_S, B_S$ carry no factor below $10^6$, consistent with the balanced–semiprime wall of Remark 5.4.)*

**Proposition 5.18** (What a barrier improvement must supply)**.** *Let $(a, b)$ be a solution with $a < b$ and put $t = b/a$, so that $t = \sigma(a)$ and, by $\sigma(a)\sigma(b) = 1$,*

$$\sigma(U) \;=\; \sigma(a) + \sigma(b) \;=\; t + \tfrac{1}{t}.$$

*For $n$ with $T_n > 2$ let $t_n > 1$ be the root of $t + 1/t = T_n$. Since $\sigma(U) \le T_{|U|}$,*

$$\sigma(a) > t_n \implies |U| \;\ge\; n + 1,$$

*and this is the only route by which a mass argument can raise the barrier. Numerically*

| $n$ | 59 | 60 | 61 | 62 | 63 | 64 |
|---|---|---|---|---|---|---|
| $T_n$ | 2.0023502 | 2.0059089 | 2.0094424 | 2.0128554 | 2.0161127 | 2.0193282 |
| $t_n$ | 1.0496677 | 1.0798804 | 1.1020081 | 1.1199914 | 1.1352477 | 1.1490254 |

So the barrier level is decided entirely by how close $\sigma(a)$ may lie to 1, and each further level demands a definite gap: $|U| \ge 61$ needs $\sigma(a) > 1.0799$, a full 8% above 1.

That is why the mass mechanism is exhausted here rather than merely difficult. The only unconditional lower bound on the gap is $\sigma(a) - 1 = (a' - a)/a \ge 1/a$, which is worthless at $a > 10^{56}$; and $\sigma$ genuinely comes close to 1 from above on very few primes, the Sylvester-type set $\{2, 3, 7, 41\}$ giving $\sigma = 1.00058\ldots$ with four. Indeed no bound in $\omega(a)$ alone can exist: the greedy Sylvester construction gives, for each $k$, squarefree $a$ with $\omega(a) = k$ and $0 < |\sigma(a) - 1|$ doubly exponentially small in $k$, from below $1 - \sigma = 7.8 \times 10^{-592}$ at $k = 12$, and from above $\sigma - 1 = 4.6 \times 10^{-591}$ at the same $k$ (`code/sigma_gap.gp`). So $|\sigma(a) - 1| \ge f(\omega(a))$ is false for every positive $f$, and the unconditional $1/a$ is of the right shape. Nothing in the cycle equations forbids $a' - a = 1$. A barrier past 60 must therefore come from a mechanism that bounds $\sigma(a)$ away from 1 without appealing to the size of $U$, which is what Theorem 4.3 does in the opposite direction, and we know of none.

The same reduction locates the extreme case. Writing $c = b - a$, the identity $\sigma(U) - 2 = (t-1)^2/t = c^2/D$ is the minus layer $D' - 2D = c^2$, and $\gcd(c, D) = \gcd(b - a, ab) = 1$ because $\gcd(a, b) = 1$. Since an admissible support contains every prime $\le 167$ (Remark 5.1), $c$ is divisible by none of them, so $c = 1$ or $c \ge 173$. The branch $c = 1$ is $D' = 2D + 1$, equivalently $a' = a + 1$ and $(a + 1)' = a$; it is the closest a solution could come to $\sigma(a) = 1$, and it is exactly the configuration a mass argument cannot reach. Below $3 \times 10^6$ the equation $a' = a + 1$ has only the solutions 30, 858, 1722 and 66,198, and none extends: for the first three $a + 1$ is prime, so $(a + 1)' = 1$, and for the last $a + 1$ is not squarefree. The branch is not excluded, only unsearched, since the barrier puts it above $10^{56}$. The level–59 computation that gives $|U| \ge 60$ is, in these terms, an exhaustive verification that $\sigma(a) > t_{59}$, and Proposition 4.8 says the same verification is unavailable one level up.

**Remark 5.19** (The tail is not a Lehmer question)**.** Remark 16.24 names the exponential Pell orbit of Remark 5.28 as the only live branch, and it is natural to read the surviving question as the primality of a term $q_n$ along it. That reading does not hold. The orbit is genuinely infinite and both square conditions do hold along all of it; what fails is that no term past $N$ can be prime, by Proposition 4.7. The single identity responsible is $A - B = 4D$, used elsewhere only to form

the Pell system and never to factor $4Dq$. *The threshold in largest–prime coordinates, and which sectors it leaves alive.* Remark 5.1 already writes a cycle as $a = qm$ with $q$ the largest prime; Bado [28] makes the same reduction exact and independent of the level. With $q = P^+(ab)$, $q \mid a$, write $a = qd$, $b = e$: then $e = d + qd'$, $e' = qd$, and $q = (e - d)/d'$. Here $\sigma(a) = 1/q + \sigma(d)$, so $t$ (or $1/t$, according to which side carries $q$; $t + 1/t$ is symmetric and the reading below is unaffected) is $\sigma(d)$ up to $1/q$, and Proposition 5.18 becomes a statement about the *sector* $d$. It is very uneven. (The rows below take $q$ astronomically large, which Proposition 4.5 supplies once $d$ is small: at $q = 281$ the first row would read 1295.)

| $d$ | $\sigma(d)$ | $\sigma + 1/\sigma$ | $\lvert U\rvert \geq$ |
|---|---|---|---|
| 2 | 0.500000 | 2.500000 | 1413 |
| $2 \cdot 3$ | 0.833333 | 2.033333 | 69 |
| $2 \cdot 3 \cdot 7$ | 0.976190 | 2.000581 | 59 |
| $2 \cdot 3 \cdot 7 \cdot 43$ | 0.999446 | 2.000000 | 59 |
| $2 \cdot 3 \cdot 11 \cdot 23 \cdot 31$ | 0.999979 | 2.000000 | 59 |
| 3 | 0.333333 | 3.333333 | $\sim 1.1 \times 10^8$ |

The last row is a Mertens estimate, the exact $K$ being past $\pi(10^6)$. Inverting $t + 1/t \leq T_K$ in the same convention gives the windows: $|U| \leq 60$ forces $\sigma(d) \in [0.9260, 1.0799]$, the $t_{60}$ of the table above, and $|U| \leq 68$ forces $\sigma(d) \in [0.8372, 1.1945]$ (`code/sector_window.gp`). So a sector can be cheap only if $\sigma(d)$ sits within a few per cent of 1. The primary pseudoperfect numbers, having $\sigma(n) = 1 - 1/n$ exactly, are among those extremal sectors (they are not the only ones, the Sylvester–type $\{2, 3, 7, 41\}$ above being another) and by Corollary 12.7 they are the $u = 1$ solutions of #307's coprime relaxation, so Erdős #313 sits at the floor of this barrier as well.

The reduction is not new to the tail theory: it *is* the parametrisation of Proposition 4.7, with $\alpha = d$ and $\beta = e$. Substituting $e = d + qd'$ and $e' = qd$ into $\alpha^2 q^2 - Nq + (\beta^2 - D)$ at $\alpha = d$ gives $e(e - d - qd') = 0$ identically (`code/alpha_is_d.gp`; the discriminant identity is $4d^2$ times the same vanishing, not a second fact). Two things follow. Everything proved about tail families is a statement about sectors: the reciprocity kill of Proposition 5.2, and the factoring wall behind the families it leaves open. And the sector $d = 2$ is $\alpha = 2$, which the table puts at $|U| \geq 1413$, so the level–60 programme never meets it. Negatively, the reduction does not shorten that residual: the unknown is the divisor $\alpha \mid D$, of which there are $2^{59}$, and the only $\alpha$–independent certificate is the Jacobi one already used.

**Remark 5.20** (The sector barrier sharpened by $d'$, and a second derivation of 60)**.** The sector reading above uses only the mass. It can be sharpened, and the sharpening is what Bado [6] exploits at $d = 42$. Since $e = d + qd'$, one has $\gcd(e, d) = \gcd(qd', d) = 1$ and $\gcd(e, d') = \gcd(d, d') = 1$, so

$$\operatorname{supp}(e) \text{ avoids } \operatorname{supp}(d) \cup \operatorname{supp}(d') \qquad \text{in every sector,}$$

and the mass $\sigma(e)$, which is $1/\sigma(d)$ to many places, must be carried by primes drawn from a thinned set. Two further identities come with it, both immediate from $e' = qd$: $de - d'e' = d^2$ and $\sum_{r \mid e} 1/r = d/d' - d^2/(d'e)$; and the associated defect $\Delta_d(R) = dR - d'R'$ satisfies $\Delta_d(Rr) = r\Delta_d(R) - d'R$ with terminal value $\Delta_d(e) = d^2$ (`code/sector_general.gp`, symbolic in $d, d', q, R, R'$). When $d$ is primary pseudoperfect, $d' = d - 1$; that is the only reason those sectors look distinguished.

Running the resulting bound sector by sector (mass $d/d'$ from primes avoiding $\operatorname{supp}(d) \cup \operatorname{supp}(d')$, together with the parity constraint that $d$ even forces $e$ odd and $\omega(e)$ even) gives

$$\min_d \big(1 + \omega(d) + \omega(e)\big) \;=\; 60,$$

over all sectors $d$ built from primes up to 47, attained at $d = 15470 = 2 \cdot 5 \cdot 7 \cdot 13 \cdot 17$ and at many others, among them the primary pseudoperfect $47058 = 2 \cdot 3 \cdot 11 \cdot 23 \cdot 31$ (`code/sector_dprime.gp`). This is Theorem 4.3's constant recovered by a route sharing nothing with its proof: that one is an exhaustive computation over the 49,961 admissible 59–prime supports, this one a largest–prime descent and a reciprocal–mass count. It does not improve the barrier, it bottoms out at exactly 60, but two independent derivations of the same constant are worth recording, and the sectors where it is tight are where any improvement must act.

At $d = 42$ the mass bound alone gives $\omega(e) \geq 62$ and $|U| \geq 66$; [6] obtains $\omega(e) \geq 64$ and $|U| \geq 68$ by adjoining the residue conditions $\sum_{r|e} r^{-1} \equiv 0 \pmod 3$ and $\pmod 7$ and $\prod_{r|e} r \equiv 1 \pmod{41}$ and maximising over admissible supports. We have re-run that certificate independently, obtaining the same maximum and the same 51,281 reachable states. The residue sieve cannot be pushed further as a sieve: the complete system of local conditions is the sector equation itself. For each $r \mid e$, reducing $e' = qd$ modulo $r$ gives $\prod_{s|e,\, s\neq r} s \equiv -d^2/d' \pmod r$, one condition per prime of $e$; conversely these, with $e \equiv d \pmod{d'}$ and $1 < \sigma(e) < 2$, force $e' = qd$ exactly, both sides lying in $(e, 2e)$. Bado's conditions modulo 3 and 7 are the sums of these; the maximiser certifying his bound violates them at five of its six smallest primes, and adjoining the three at $5, 11, 13$ to the dynamic programme lowers the 64–prime maximum from 1.03032 to 1.02853, still above $42/41$ (`code/sector42_znam_violations.gp`, `code/sector42_znam_dp.rs`). The next step is therefore not a sieve but an exact enumeration, and it goes through.

**Proposition 5.21** (The $d = 42$ sector at 64 primes)**.** *No solution of #307 has $a = 42q$ with $q$ prime and $\omega(b) = 64$. Hence in that sector $\omega(b) \geq 66$ and $|P \cup Q| \geq 70$.*

*Proof (finite verification).* Write $e = b$, $T = 42/41$, and let $A_1 < A_2 < \cdots$ be the primes other than $2, 3, 7, 41$, $S_j = \sum_{i \leq j} 1/A_i$. By the sector identity, $\sigma(e) = T - 1764/(41e)$, and $e > 3.5 \times 10^{57}$ by Theorem 4.3, so $T - 10^{-55} < \sigma(e) < T$. Exactly, $A_{64} = 337$, $A_{197} = 1229$, $A_{198} = 1231$, and $S_{61} + 3/1231 < T$ (`code/sector42_k64.gp`). If $m$ of the 64 primes of $e$ exceed 1229 then $\sigma(e) \leq S_{64-m} + m/1231$, which for $m \geq 3$ is below $S_{61} + 3/1231 < T - 10^{-55}$; so $m \leq 2$. Next, if $\mathrm{supp}(e) = S \cup \{\ell\}$ with $R = \prod S$, the identity $42e - 41e' = 1764$ reads $\ell\, \Delta_{42}(R) = 41R + 1764$, so $\Delta_{42}(R) > 0$ and $\ell$ is the integer $(41R + 1764)/\Delta_{42}(R)$: the last prime is determined by the others.

*Case $m \leq 1$.* Take $\ell$ the prime exceeding 1229 if $m = 1$, else the largest prime of $e$, so $\ell \geq A_{64}$. The other 63 primes $S$ lie among $A_1, \ldots, A_{197}$ with $\sigma(S) = \sigma(e) - 1/\ell \in [T - 1/A_{64} - 10^{-55}, T)$. The enumeration tests 36,901,596,763 sets, every such $S$ among them; for none is $(41R + 1764)/\Delta_{42}(R)$ an integer.

*Case $m = 2$.* Let $\ell_2 \leq \ell_1$ be the two primes exceeding 1229, $S$ the other 62, and $\delta = T - \sigma(S)$. Then $1/\ell_1 + 1/\ell_2 \in (\delta - 10^{-55}, \delta)$, so $\delta < 2/1231$ and $\ell_2 \in (1/\delta,\ 2/\delta + 1)$. The enumeration tests 166,213 sets, every such $S$ among them; the least $\delta$ among them is $1.539 \times 10^{-6}$, so every $\ell_2$ range is finite (largest upper bound 1,299,387). For each $S$ and each prime $\ell_2$ in its range, $S \cup \{\ell_2\}$ determines $\ell_1$ as above; it is never an integer.

The enumeration prunes on the mass in floating point with a margin of $10^{-9}$ on both ends of each window, so no set in a window is lost; the integrality test is a double–double prefilter with a conservative band followed by exact integer arithmetic (7,454,365 and 2,495 exact tests respectively), the integer layer checked against PARI to the digit (`code/sector42_k64.rs`, two runs with independent big–integer implementations). The non–enumerative half of the argument (the forced–prime identity and its converse, the sign condition, the case–split inequality and the parity step) is formalised in `lean/Erdos307/Sector42.lean`, with the two finite computations appearing as explicit hypotheses of the concluding statement. So $\omega(e) \neq 64$. By [6], $\omega(e) \geq 64$ and $\omega(e)$ is even ($e$ is odd and $e' = 42q$ is even); hence $\omega(e) \geq 66$. Finally $q \notin \{2, 3, 7\}$ and $q \nmid e$, since $\gcd(a, b) = 1$ and $a = 42q$, so the four listed primes are distinct and $|P \cup Q| = |\{2, 3, 7, q\}| + \omega(e) \geq 70$. □

**Proposition 5.22** (The primary pseudoperfect sectors)**.** *Let $d$ be squarefree and* $\operatorname{supp}(e) = S \cup \{\ell\}$ *with $R = \prod S$. The sector equation $de - d'e' = d^2$ reads*

$$\ell\big(dR - d'R'\big) \;=\; d'R + d^2,$$

*so the last prime is determined by the others in every sector, not only at $d = 42$. Running the resulting enumeration at the first admissible $\omega(e)$ above the mass floor gives, in each case, no solution:*

| $d$ | $\omega(e)$ | *sets* | $\lvert P \cup Q\rvert$ *before* | *after* |
|---|---|---|---|---|
| 42 | 62 | 175 | 66 | 68 |
| 47058 | 54 | 41,654 | 60 | 62 |
| 2214502422 | 70 | 11,646 | 77 | 79 |

(`code/sector_kexclude.rs`; the identity is checked symbolically and in exact rationals in `code/sector_kexclude.g`.) *At each $\omega(e)$ in the table the mass bound $S_{\omega-1} + 1/A_{N+1} < d/d'$ excludes even one prime beyond the truncation, so there a single phase is a complete argument. At $d = 47058$ this raises a primary pseudoperfect sector off the global barrier: that sector sat at exactly 60, and $\lvert P \cup Q\rvert \geq 62$ there. With Proposition 5.21 the $d = 42$ sector stands at $\lvert P \cup Q\rvert \geq 70$.*

*The condition that makes a phase count sufficient is worth isolating, because it prices every rung of the ladder in advance and because a phase–1 enumeration that returns nothing proves nothing without it.*

**Proposition 5.23** (When the sector enumeration is complete)**.** *Let $d$ be squarefree, $d' = \sum_{p \mid d} d/p$, $t = d/d'$, and let $A_1 < A_2 < \cdots$ be the primes dividing neither $d$ nor $d'$, with $S_m = \sum_{i \leq m} 1/A_i$. Fix a truncation $N$ and a target $\omega(e) = k$. If $j$ of the primes of $e$ exceed $A_N$ then*

$$\sigma(e) \;\leq\; S_{k-j} + \frac{j}{A_{N+1}},$$

*so whenever $S_{k-j} + j/A_{N+1} < t$ no admissible $e$ has exactly $j$ primes beyond the truncation. Writing $J = J(k, N)$ for the least such $j$, every admissible $e$ has at most $J - 1$ primes beyond $A_N$, and the phases $1, \ldots, J-1$ exhaust the possibilities: the enumeration is complete.*

*Proof.* The $k - j$ primes of $e$ not exceeding $A_N$ lie among $A_1, \ldots, A_N$, so their reciprocals sum to at most that of the $k - j$ smallest, namely $S_{k-j}$; each of the remaining $j$ is at least $A_{N+1}$ and contributes at most $1/A_{N+1}$. A sector solution requires $\sigma(e) \geq t$, which the displayed bound then contradicts. Since the bound is decreasing in $j$ once $S_{k-j}$ falls below $t$, the values $j \geq J$ are all excluded together. □

*At the floor $k = K(d)$ one has $J = 1$, and there phase 1 alone is a complete argument, which is the case the table above uses; at $k = K(d) + 2$ one has $J = 3$, and there phases 1 and 2 together are complete. Both correspondences are verified directly at $d = 47058$, at $k = 54$ and $k = 56$ respectively. Since only those two phases are implemented, the reachable rungs are those with $J \leq 3$, and $J$ is what prices the ladder:*

| $d$ | $\omega(d)$ | $K(d)$ | $k$ | $J$ | $\lvert P \cup Q\rvert$ *if excluded* |
|---|---|---|---|---|---|
| 42 | 3 | 62 | 62 | 1 | 68 |
| 42 | 3 | 62 | 64 | 3 | 70 |
| 42 | 3 | 62 | 66 | 5 | 72 |
| 47058 | 5 | 54 | 54 | 1 | 62 |
| 47058 | 5 | 54 | 56 | 3 | 64 |
| 47058 | 5 | 54 | 58 | 6 | 66 |
| 2214502422 | 6 | 70 | 70 | 1 | 79 |
| 2214502422 | 6 | 70 | 72 | 3 | 81 |

*at $N = 400$ (`code/sector_phasecount.gp`). The truncation is worth minimising rather than maximising: $J$ is what must stay small, and at $d = 2214502422$ with $k = 72$ it equals 3 for every $N \geq 300$ and 4 below that, so $N = 300$ is the smallest complete choice and the one that leaves the fewest subsets to enumerate. The value of $J$ grows quickly with $k$, and raising the truncation barely helps, since $j/A_{N+1}$ falls only as fast as $A_{N+1}$ grows: at $d = 42$ the rung $k = 66$ needs $J = 5$ at every $N$ from 400 to 3000. That is what stops the $d = 42$ sector at $|P \cup Q| \geq 70$, and it is a limitation of the implemented phases rather than of the method.*

*The next step at $d = 47058$ costs more, because the single–phase condition fails there. With $N = 400$ and $A_{N+1} = 2791$ one has $S_{55} + 1/A_{N+1} = 1.00629$ and $S_{54} + 2/A_{N+1} = 1.00309$, both above $d/d' = 1.0000213$, so one and two primes beyond the truncation must each be enumerated; but $S_{53} + 3/A_{N+1} = 0.99984 < d/d'$, so three or more cannot occur and the two phases together are complete. Both are empty: phase 1 enumerates $4.387 \times 10^{11}$ leaves and 88,472,925 exact checks, phase 2 a further 373,644 leaves and 59,276 exact checks, and neither yields an admissible forced prime (`code/sector_kexclude.rs`, 12,658 s on nine cores). Hence $\omega(e) \neq 56$. Since $d$ is even, $e = d + qd'$ is odd and $e' = dq$ is even, so Proposition 5.25 forces $\omega(e)$ even; with $\omega(e) \geq 56$ already in hand this gives $\omega(e) \geq 58$ and*

$$|P \cup Q| \;=\; 1 + \omega(d) + \omega(e) \;\geq\; 1 + 5 + 58 \;=\; 64$$

*at $d = 47058$. The truncation is what stops the ladder here: at $\omega(e) = 57$ even four primes beyond $A_N$ remain possible, so the next rung needs a larger $N$ rather than more time at this one.*

**Remark 5.24.** The $d = 42$ row recovers [6]'s $\omega(e) \geq 64$ by a route with nothing in common with its proof: that one maximises mass over a 51,281-state residue automaton, this one enumerates 175 prime sets and tests an integrality condition. Two independent derivations of the same bound.

Running the same exclusion across sectors rather than up the ladder is what the identity really buys. Among all sectors $d$ that are squarefree with $\omega(d) \leq 12$ and every prime factor at most 71, exactly 16,234 have mass floor $|P \cup Q| = 60$; in every one of them the floor value of $\omega(e)$ is excluded, the largest enumeration being 41,654 sets (`code/sector_at60.gp`, `code/sector_kexclude.rs`). The counts 250, 1,533 and 16,234 are reproduced from scratch by `code/sector_at60_count.rs`. Determining them requires the complete factorisation of $d'$, not merely its small factors: a prime of $d'$ that goes unfound is never matched in the exclusion set, so the primes available to $e$ are overcounted, $K(d)$ comes out too small and the floor too low, which is the direction in which a sector at 60 could be misread as lying below 60 and skipped. Since $d'$ passes $10^{14}$ already at $\omega(d) = 10$ and routinely carries a large prime factor, the enumerator factors it with Pollard rho under a Miller–Rabin test, and refuses outright when $d'$ would exceed the width its modular arithmetic is exact to. So

$$|P \cup Q| = 60 \text{ is impossible in every such sector,}$$

and any support of size 60 must have $\omega(d) \geq 13$ or a sector prime exceeding 71. This does not move Theorem 4.3, whose 60 is proved by a different route and over all supports; what it does is remove the entire low–$\omega$, small–prime part of the sector decomposition from the level–60 frontier.

The sectors that could reach 60 form a *finite* set, which is worth recording because the sector decomposition appears to range over unboundedly many $d$. At level 60 the support is a 59–prime base together with one tail prime, and a base element $x$ satisfies $T_{58} + 1/x \geq 2$, so $x \leq 1/(2 - T_{58}) = 793.6\ldots$; every prime up to 167 lies in the base, and $d$ is a subset of the 138 primes at most 787. So the question "which sectors admit $|P \cup Q| = 60$" is a finite one. It is not, however, a small one: the count of sectors whose mass floor is exactly 60 rises from 250 over primes at most 47

with $\omega(d) \le 8$ to 1,533 over primes at most 59 with $\omega(d) \le 10$ and 16,234 over primes at most 71 with $\omega(d) \le 12$ (all cleared, each family a subset of the next), and it does not saturate: raising the cap one step further, to primes at most 89 at the same $\omega(d) \le 12$, gives 125,813, a factor of 7.75 for one more prime in the pool. Those are counted, not cleared, and are recorded here only as the next term of the growth. Holding $\omega(d) \le 8$ and letting the prime bound $B$ grow, the count runs 26, 83, 250, 588, 1762, 5172 at $B = 31, 41, 47, 59, 71, 89$; holding $\omega(d) \le 6$ it runs 19, 34, 53, 77, 107, 166, 217 at $B = 31, 41, 47, 59, 71, 89, 107$. Against $\pi(B)$ the fitted exponent grows with $\omega$, from about 2.7 at $\omega = 6$ to about 6.8 at $\omega = 8$, so the count behaves like a binomial $\binom{\pi(B)}{\omega}$. With the true bounds $B = 787$ and $\omega(d) \le 58$ the family is finite but of order $\binom{138}{58}$, and enumerating it is not possible. Clearing sectors one at a time therefore strengthens the statement at each step and cannot reach the barrier. Finiteness here converts the sector programme into a computation of the same order as the one it was meant to replace.

Across every sector enumerated so far the floor value of $\omega(e)$ is not attained, and that is a falsifiable statement in its own right.

**Proposition 5.25** (Odd sectors keep the prime 2 only when $\omega(d)$ is odd)**.** *Let $d$ be odd and squarefree. Then $d' \equiv \omega(d) \pmod 2$. Consequently, if $\omega(d)$ is even then $2 \mid d'$, so $2 \mid dd'$ and $2 \notin \operatorname{supp}(e)$ by Remark 5.20; the mass $d/d'$ must then be carried entirely by odd primes dividing neither $d$ nor $d'$.*

*Proof.* For odd squarefree $d$ each cofactor $d/p$ is a product of odd primes, hence odd, so $d' = \sum_{p \mid d} d/p$ is a sum of $\omega(d)$ odd terms and has the parity of $\omega(d)$. Verified with no violation on the 81,053 odd squarefree $d < 2 \times 10^5$. □

The cost of losing 2 is not marginal, because $\frac{1}{2}$ is the largest single contribution available: at $d = 15015 = 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$, where $\omega(d) = 5$ is odd and 2 survives, $K(d) = 127$, while at $d = 255255 = 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17$, where $\omega(d) = 6$ is even and 2 is lost, $K(d) = 1535$. Hence a sector with $d$ odd and $\omega(d)$ even has mass floor far above 60, and every sector of mass floor 60 with $d$ odd has $\omega(d)$ odd. That is what the enumeration shows: of the 16,234 sectors of mass floor 60 over primes at most 71, the 4,846 with $d$ odd have $\omega(d) \in \{9, 11\}$ and none has $\omega(d)$ even, although nothing in the enumeration imposes it.

**Conjecture 5.26** (The floor gap)**.** *For every squarefree $d$, no two–cycle in the sector $d$ has $\omega(e)$ equal to the mass floor $K(d)$, the least $k$ with $S_k(d) \ge d/d'$.*

The evidence is 5,680 sectors with no attained floor: the 1,533 of mass floor 60 over primes at most 59, a further 3,600 of the family over primes at most 71, the three sectors of Proposition 5.22, and 480 sectors of mass floor 61 through 68 over primes at most 47, the last of these a range disjoint from all the others and tested after the statement was fixed. Its consequence is that the barrier would move: the minimum of $1 + \omega(d) + K(d)$ over sectors is 60, attained at $d = 2 \cdot 3 \cdot 7 \cdot 23 \cdot 41$ and checked over 109,293 sectors, so with the conjecture and the parity of Proposition 5.22 every sector would give $|P \cup Q| \ge 62$. The evidence must be read with its power, which is negligible. The integrality $\Delta_d(R) \mid d'R + d^2$ has probability of order $1/\Delta_d(R)$, and $\Delta_d(R)$ is of the size of a product of some sixty primes; over the 5,680 sectors and their some $10^4$ candidate sets each, the expected number of attained floors is of order $10^{-92}$ whether the conjecture is true or false. An enumeration of this kind therefore cannot support it, and no feasible enlargement would. Nor is the floor structurally distinguished: the rate at which candidates reach the exact divisibility test matches the width of the numerical prefilter at the floor and above it alike, the ratios of observed to expected being 0.72 and 0.86 at the floor for $d = 47058$ and $d = 2214502422$ and 1.01 at $\omega(e) = 64$ for $d = 42$, so the deficit at the floor is the small number of candidate sets and nothing else. The

conjecture is recorded because it is what every enumeration performed here verifies and because its consequence is sharp, not because the enumerations are evidence for it.

The ladder does not continue cheaply. In every sector the first rung above the mass floor is small, the three counts above, and the next requires the two-phase machinery of Proposition 5.21, because the truncation bound no longer excludes primes beyond $A_N$: at $d = 42$ that step cost $3.7 \times 10^{10}$ sets, and at $d = 47058$ the analogous $\omega(e) = 56$ is $4.4 \times 10^{11}$, not run. The cheapness of the first rung and the cost of the second is the shape of the method, not an accident of $d = 42$.

**Remark 5.27.** The same argument decides $\omega(e) = 66$ in principle, but with up to four primes beyond the truncation the nested ranges multiply, and the enumeration is no longer a twenty–minute job; we have not run it. What the proposition shows is where the sector programme actually stops: not at the residue sieve, whose limit is 68, but at the cost of an exact enumeration in which the last one or two primes are forced by the rest. The two–large–prime case, a factoring problem in general $((\Delta\ell_1 - 41R)(\Delta\ell_2 - 41R) = 1681R^2 + 1764\Delta)$, is finite here only because the smaller of the two is trapped in $(1/\delta, 2/\delta)$ and $\delta$ is bounded below over a finite list.

Three consequences, in increasing order of what they cost.

*The gate is finite, not Lehmer.* The tail primes of a base lie in the explicit finite set $\{2(N \pm k)/m^2 : m \mid 4D,\ k^2 = AB + Dm^2\}$. So the second of the three gates is a search over the divisors of $4D$, not a primality question along an exponential sequence, and the classification of the residual difficulty as Lehmer–class rather than Schinzel–class does not survive. Checked sound *and complete* against exhaustive search on 1200 bases with zero mismatches, and stepped out along a real orbit, in `code/tailbound.gp`: on the base $D = 1$, $N = 7$ (bound $N = 7$) the orbit gives $q_1 = 7$ prime, then $q_2 = 741895$, $q_3 = 76921173511$, and every later term composite, with both square conditions holding throughout.

*The count reverses sign, and the search has now been run.* The expected number of admissible $m$ is $\sum_{m|4D}(AB + Dm^2)^{-1/2} \approx D^{-1/2}\sum_{m|4D} m^{-1}$, which on the first immune base is $2.4 \times 10^{-56}$ and below $10^{-38}$ summed over all 34 immune families, the opposite of the "weakly favours YES" of Remark 5.16 on exactly the families that reading was about. That prediction is now checked. The full divisor space is $4 \cdot 2^{58} \approx 1.15 \times 10^{18}$ per family and is not enumerable, but because the hit probability falls like $1/(m\sqrt{D})$ the expected count is carried by small $m$: restricting to $\omega(m) \le 8$ leaves 99.9998% of the expected count. Over all 34 families that slice is $3.0777 \times 10^{11}$ candidates, and

$$\text{it contains no } m \text{ with } AB + Dm^2 \text{ a perfect square}$$

(4,933 survive the modular cascade, exact `issquare` kills all of them; the earlier $\omega(m) \le 7$ pass, $4.7081 \times 10^{10}$ candidates and 771 survivors, agrees).

The same search has been run over *every* admissible base, which matters because Proposition 4.7 requires no primality of $A_S$ or $B_S$, only the algebra. The 34 immune families are the ones left after the Jacobi kill and the factoring attack, and their immunity is conditional on a Baillie–PSW verdict; the other 3,444 of the 3,478 open families are open precisely because $A_S$ or $B_S$ is a balanced semiprime nobody can split. The tail search does not care. Over all 49,961 admissible bases at $\omega(m) \le 4$, which is 99.2% of the expected count per base, the run is $9.1296 \times 10^{10}$ candidates, 1,469 survive the modular cascade, and exact `issquare` kills all 1,469. Every survivor was additionally checked to satisfy $m \mid 4D$, the hypothesis the proposition needs. So

$$\text{no admissible base admits a Pythagorean tail prime with } \omega(m) \le 4,$$

unconditionally and with no primality input. Where Proposition 5.2 leaves 3,478 of the one–new–prime families at level 60 open, of which 34 only conditionally immune, this leaves none open on

the stated slice. It is a slice result and not a proof: the untested remainder is 0.8% of the expected count per base, and $\omega(m) \geq 5$ is untested. The run is `code/tailsearch.rs` (a cascade of quadratic–residue filters modulo primes coprime to $2D$, 771 survivors) with every survivor then tested exactly by `issquare` in `code/tailsearch_verify.gp` (0 squares). A positive control recovers the known solutions of the toy base $D = 1$, $N = 7$, and an exhaustive exact run at $\omega(m) \leq 2$ agrees with the filter, so the cascade is not silently over–rejecting.

It is worth recording what the slice covers in the other measure, since the two are thirteen orders apart. The parametrisation runs over $m \mid 4D$ with $m$ even, so $m = 2\alpha$ with $\alpha \mid D$, and at level 59 that is $2^{59} = 5.7646 \times 10^{17}$ divisors. Splitting on the parity of $\alpha$, the divisors with $\omega(m) \leq 4$ number $2\sum_{j\leq 3}\binom{58}{j} = 65{,}136$, and those with $\omega(m) \leq 8$ number $2\sum_{j\leq 7}\binom{58}{j} = 692{,}376{,}800$. As fractions of the space the bound $q \leq N$ actually proves finite, these are

$$1.130 \times 10^{-13} \qquad \text{and} \qquad 1.201 \times 10^{-9},$$

against the 99.2% and 99.9998% of the *expected count* quoted above. Both figures are correct and they measure different things: the expected count concentrates on small $\omega$, the hit probability falling like $1/(m\sqrt{D})$, so a slice can carry almost all of the expectation while carrying almost none of the space. Only the second pair of numbers is a statement about what has been checked. This is VERIFIED on the stated slice, not on the full divisor set: the residual 0.0018% of the expected count is untested.

The filter primes must be coprime to $2D$, which is itself a structural fact rather than a convenience. Since $D$ contains every prime up to 167, for any $\ell \mid D$

$$AB + Dm^2 - N^2 = D\big(m^2 - 4D\big) \equiv 0 \pmod{\ell},$$

so the value is the square $N^2$ modulo $\ell$ for *every* $m$, and as $\gcd(D, N) = 1$ Hensel lifts that root to every power of $\ell$. The tail search therefore has no local obstruction at any prime dividing $D$, the analogue of Proposition 5.17 for this formulation; the classical 64/63/65/11 pre–filter for squares rejects nothing here, measured at a 75% pass rate modulo 64 and 100% modulo 63. Formalised as `tail_value_mod_dvd` and `hensel_lift_identity`; the induction from the step to all powers is routine and is not formalised.

*The paper's own height estimate now says the same thing.* Remark 5.28 puts the least point of a nonempty fiber at height $\exp\big(10^{112}\big)$ on regulator grounds, so its $q \approx x^2/A$ carries some $10^{112}$ digits, against a bound of 115. Two independent estimates, one from divisor counting and one from the class–number formula, agree that no immune family admits a Pythagorean tail prime. Neither is a proof: the bound $q \leq N$ is proved, the emptiness is HEURISTIC.

Formalised in `lean/Erdos307/TailBound.lean` (`tail_factor`, `tail_quadratic`, `tail_discriminant`, `tail_bound`, `tail_finite`) on the three standard axioms, with a non–vacuity witness at the base above.

**Remark 5.28** (The immune signal and its Pell orbit)**.** The orbit below is real, but primality along it is not the residual question, since Proposition 4.7 bounds the tail prime by $N$.

For a reciprocity–immune base the Pell system of Remark 5.1, $B_S x^2 - A_S y^2 = -4D_S^2$, is locally soluble everywhere ($\kappa_S > 0$), and a Pythagorean pair in the tail family $S \cup \{q\}$ is *exactly* a solution with $q = (x^2 - D_S)/A_S = (y^2 - D_S)/B_S$ a positive prime. Its solutions grow like $\varepsilon^n$ in the fundamental unit of the order of discriminant $4A_SB_S$ (224 digits, $A_SB_S$ a nonsquare, on an explicit immune base). Writing $x_{n+1} = tx_n - x_{n-1}$ with $x_n = (\alpha^n + \beta^n)/2$ and $\alpha\beta = 1$, the squares $x_n^2$ have characteristic roots $\alpha^2, \beta^2, 1$ and the constant shift is absorbed by the root 1, so

$$q_n = \frac{x_n^2 - D_S}{A_S} \qquad \text{satisfies} \qquad q_{n+3} = (t^2-1)q_{n+2} - (t^2-1)q_{n+1} + q_n,$$

a non–degenerate ternary linear recurrence (verified on four Pell parameters). A reciprocity–killed base is the complementary degeneration: there the same system is locally *insoluble* ($\kappa_S = 0$), so no candidate exists at all, and the dichotomy of Proposition 5.2 is Pell–solubility versus Pell–insolubility.

The return is milder than "$k = 0$" suggests: on the Pythagorean locus the deviation $k = (a' - b)/a$ is an *integer* (the identity $v(D(u) - v) = u(u - D(v))$ with $\gcd(u, v) = 1$), forced into $\{0, -1\}$ by the near–balanced immune ratio, so $k = 0$ is a sign choice rather than a measure–zero coincidence, and the naive Wieferich shortcut that would pin $q$ is vacuous, since $x^2 \equiv D_S \pmod q$ holds identically from $x^2 = A_S q + D_S$.

Neither square condition carries the difficulty alone. Imposing only the minus–square, $d^2 = B_S q + D_S$, the admissible $d$ satisfy $d^2 \equiv D_S \pmod{B_S}$, and along such a class $d = d_0 + tB_S$ one has

$$q(t) \;=\; \frac{d_0^2 - D_S}{B_S} \;+\; 2d_0\, t \;+\; B_S\, t^2,$$

a quadratic in $t$ (toy bases: $3 + 54t + 173t^2$, $7 + 238t + 1693t^2$, $2 + 446t + 17357t^2$): a Bunyakovsky question, of the same class as the $p = 5$ family of Proposition 15.14(iii), and conjecturally routine. It is the *conjunction* of the two squares that replaces the quadratic parameter by a fundamental unit and turns the orbit exponential.

*Where the reading broke.* That exponential growth was taken to place the question in the Lehmer class, on two grounds: no algebraic factorisation is available, $D_S$ being squarefree so that $x^2 - D_S$ is irreducible, and no covering congruence can kill the family, by Proposition 5.17. Both observations are correct, and neither is the relevant one. The factorisation that does exist is not of $x^2 - D_S$ but of $x^2 - y^2 = 4D_S q$, available because $A_S - B_S = 4D_S$ *exactly*: the identity was used here only to build the Pell system, never multiplicatively. Proposition 4.7 draws it, and the ternary recurrence above then supplies no prime term past $q \le N$.

Three nested gates separate an immune family from a solution: fiber nonemptiness, a tail prime, and the $k = 0$ return. The first is a class–group condition, since Hasse supplies only rational points, and it is quantitative: immunity makes $A_S B_S$ a product of two distinct primes, hence squarefree, so the conic lives in the near–maximal order of $\mathbb{Q}(\sqrt{A_S B_S})$ of discriminant $\sim A_S B_S \approx 10^{224}$, and the class–number formula puts the expected regulator at $R = \sqrt{\Delta}\, L(1, \chi)/(2h) \approx 10^{112}$; barring an anomalously large class number, the *least* point of a nonempty fiber sits at height $\exp\big(10^{112}\big)$. That estimate is used in Remark 5.19 as independent corroboration that the second gate is empty. A direct sweep to $q \le 10^{12}$ was structurally incapable of reaching even the first gate.

**Remark 5.29** (The affine sieve does not reach a rank–one orbit)**.** The modern instrument for primes in orbits is the affine sieve of Bourgain, Gamburd and Sarnak [46], which produces $r$–almost–primes in orbits of thin groups. It is the one method designed for exactly the shape of question Remark 5.28 isolates, so its failure here is worth recording precisely rather than left implicit.

The affine sieve needs a spectral gap: the congruence quotients of the acting group must form an expander family, and that is what supplies the level of distribution the sieve consumes. Here the $q_n$ run along a Pell orbit, so the acting group is the cyclic group generated by the fundamental unit of the order of discriminant $4A_S B_S$. Cyclic groups are abelian, hence amenable, and have no spectral gap; expansion is precisely the property an abelian group lacks. The sieve therefore returns nothing, and it returns nothing on $2^n - 1$ for the same reason. The obstruction is rank, not thinness as such: non–elementary thin orbits, Apollonian packings among them, do yield almost–primes.

The natural repair fails for a reason worth stating too. One may embed the rank–one orbit in a larger thin group whose orbit does expand, and sieve there. What the affine sieve then delivers is infinitely many almost–primes in the *larger* orbit, inside which our sequence has relative density

zero; the sieve does not localise to a prescribed sub–orbit. Expansion is bought at the cost of losing the sequence.

This meets the unit–rank trichotomy of Remark 7.13 from a second direction. There, rank zero gave a classical gate and denied a continuous family, positive rank the reverse. Here, rank one is exactly the cell in which the affine sieve has no traction. Stated as a single missing property: what is wanted is a substitute for expansion at rank one, and that is the Mersenne problem itself, not a lemma about #307.

**Remark 5.30** (The gap to the record, measured)**.** It is worth stating how far the required statement is from the best that is known, because the distance is not one of degree.

Write a sequence's *density exponent* $\theta$ for $\#\{n \le X\} \asymp X^{\theta+o(1)}$. The sparsest sets currently proved to contain infinitely many primes are the values of binary forms: $a^2+b^4$ at $\theta = 3/4$, by Friedlander and Iwaniec [41], and $a^3+2b^3$ at $\theta = 2/3$, by Heath-Brown [42], the latter being the sparsest polynomial form known to capture its primes. Our $q_n$ grow like $\varepsilon^{2n}$, so $\#\{n : q_n \le X\} \asymp \log X$ and

$$\theta = 0.$$

The distance from 2/3 to 0 is categorical rather than quantitative. Both records are bilinear arguments: they run a sieve of dimension one and close it with a Type II estimate obtained by averaging over a set of size a positive power of $X$. Every gain in such an argument is a gain of a power of $X$, and against $\log X$ terms a power of $X$ gains nothing; below $\theta = 1/2$ the standard sieve axioms already fail, and at $\theta = 0$ there is no set left to average over. This is the same wall as Remark 5.29 approached from the analytic rather than the spectral side, and it is why every sequence of exponent zero, Mersenne, Fibonacci, Lucas and ours alike, is open together.

The structure our sequence does carry is the standard one and confers nothing. From $A_S q_n = x_n^2 - D_S$ one gets $x_n^2 \equiv D_S$ for every prime $p \mid q_n$ not dividing $D_S$, so all prime factors of $q_n$ lie in the density–$\frac{1}{2}$ set where $D_S$ is a quadratic residue. That constrains the factorisations without excluding any, which is exactly the position the primitive divisor theorems already left us in: divisors, never primality.

**Remark 5.31** (The residue, atomised, and where its difficulty comes from)**.** Decomposing the question into independently known statements locates the obstruction on a single atom and identifies its source.

Two atoms carry theorems. Every $q_n$ beyond the thirtieth has a primitive prime divisor [49]; and the greatest prime factor satisfies $P^+(q_n) > n \exp\big(\log n/(104 \log\log n)\big)$, by Stewart [43], which settled questions of Schinzel and Erdős and was the first general advance on Bang and Carmichael. Every other atom, that $q_n$ be squarefree infinitely often, that $\omega(q_n)$ be bounded infinitely often, that $q_n$ be a $P_k$, and primality itself, is open.

The chain that would close the crux is $P^+$ large, hence $\omega$ small, hence prime. It breaks at the first link, and by a measurable amount. Stewart's bound certifies $\log P^+ \ge (1+o(1)) \log n$ against $\log q_n \approx 2n \log \varepsilon$, so the fraction of the term's size that is certified is of order $\log n/n$: 10.6% at $n = 10$, 0.32% at $n = 10^3$, $4 \times 10^{-4}$% at $n = 10^6$. Primality is the fraction 1. No sharpening of the constant 104 changes this, since $n^{1+o(1)}$ is the shape linear forms in logarithms produce. The Subspace Theorem is the other available instrument and does not help: it is stronger under weaker hypotheses but ineffective [44], and an ineffective statement cannot exhibit the single prime term a construction needs.

The source is visible in the comparison with the polynomial case. For $n^2+1$ the greatest prime factor exceeds $n^{1.3}$ infinitely often [45], that is $q^{0.65}$, a *positive proportion* of the term's size, and on such sequences almost-primality is available too. For an exponentially parametrised sequence

only $(\log q)^{1+o(1)}$ is available, a vanishing proportion. The gap between the two regimes is exactly the gap between polynomial and exponential parametrisation, and Proposition 8.6 proves #307 admits no polynomial one. So the obstruction recorded in Remark 5.29 as a question of rank, in Remark 5.30 as one of density exponent, and here as one of $P^+$, is a single obstruction with three faces, and its source is Proposition 8.6: forced exponentiality is what places the residue in the class where nothing of the required strength is known.

Two things sharpen that attribution, and both make the obstruction less incidental rather than more. First, the boundary is exactly where Proposition 8.6 sits. The fraction of a term's size that Stewart's bound certifies is $\log P^+/\log q_n \approx \log n/\log q_n$, so it stays bounded below precisely when $\log q_n \ll \log n$, that is when $q_n$ grows polynomially: degree $d$ gives the constant $1/d$. Every super-polynomial rate sends it to 0, and not only the exponential one: $\exp(\sqrt{n})$ and $n^{\log n}$ do as well. So polynomial growth is not one convenient case among several, it is the whole of the favourable region, and it is exactly what Proposition 8.6 excludes.

Second, the hypotheses of Proposition 8.6 are worth testing, since a gap there would dissolve everything above. It assumes a *fixed* number of polynomials, and its proof uses that finiteness when it splits $\sum_i 1/p_i$ into constants plus a remainder tending to 0. A family whose support grows with the parameter is therefore not covered. That gap is real and useless: a growing support means a growing number of simultaneous primality conditions, which is a harder demand than one, not an easier one, and it does not return the family to the polynomial regime. The proposition forbids the case that would help and leaves uncovered only cases that would not.

**Remark 5.32** (The squarefree atom, and what it reduces to)**.** Among the four open atoms one is structurally easier, and it is worth saying why and how far it gets. Squarefreeness of $q_n$ asks only for an *upper* bound on a count of bad events, never for the production of a prime, so it is the one atom not subject to the parity obstruction that blocks bounded $\omega$, $P_k$, and primality.

The argument it wants is a union bound. For a prime $p \nmid 2A_S D_S$, $p^2 \mid q_n$ forces $x_n^2 \equiv D_S \pmod{p^2}$; the Pell orbit is purely periodic modulo $p^2$ with period $\pi(p^2)$, there are at most two square roots of $D_S$, and each is met a bounded number of times per period, so the density of bad $n$ attached to $p$ is at most $C/\pi(p^2)$. Where $\pi(p^2) = p\,\pi(p)$, which is the generic behaviour, and $\pi(p) \asymp p$, this is $\asymp C/p^2$, and the union bound closes as soon as $C\sum_p p^{-2} < 1$. The constant is comfortable: $\sum_p p^{-2} = 0.4522474\ldots$, so any $C < 2.21$ suffices, and a positive proportion of the $q_n$ would be squarefree, hence infinitely many.

What is missing is the hypothesis $\pi(p^2) = p\,\pi(p)$. Its failure is precisely a Wall–Sun–Sun prime for the sequence, and there the density is $\asymp 1/p$ rather than $\asymp 1/p^2$, whose sum diverges and destroys the bound. No such prime is known for any classical sequence and their finiteness is unproved, so the atom does not close: it *reduces* to a Wall–Sun–Sun statement. That is a sharper position than "open", since it names the obstruction and shows it is a single hypothesis rather than a programme, but it is a reduction and not a proof.

The unconditional pieces are formalised. `Erdos307.primeSquareSum_certificate` gives $\sum_p p^{-2} < \frac{1}{2}$ in the finite form used above, the first six primes together with the bound $\frac{1}{2}\sum_{m\ge 16} m^{-2} < \frac{1}{30}$ for the odd primes beyond; `good_density_pos` and `squarefree_positive_density_of_bound` are the union bound and the conditional implication. All carry 0 `sorry` and the three standard axioms (`lean/Erdos307/Squarefree.lean`).

Two further points fix the standing of this atom, and both are negative. First, on the immune bases the argument is blocked at its *lowest* rung as well as its highest. The density attached to a single small $p$ is $\#\{n \bmod \pi(p^2) : p^2 \mid q_n\}/\pi(p^2)$, read off the Pell orbit modulo $p^2$, and that needs the fundamental solution, the object Remark 16.3 (i) places at $10^{24.7}$ operations. For a Lucas sequence with a known fundamental unit the small-prime rung is a finite computation; here it is

not. The ladder is therefore blocked at both ends, by non-computability below and by Wall–Sun–Sun above. Second, and more to the point, squarefreeness is not what the construction needs: Remark 5.28 requires $q$ to be *prime*, and a squarefree $q$ with several prime factors is useless there, so even the dream lemma of this atom would leave #307 exactly where it stands. Of the four atoms this is the easiest, and it is also the only one whose resolution would not advance the problem.

# 6 Local anatomy and a heuristic model

**Lemma 6.1** (Local anatomy)**.** *Let $a$ be squarefree and $q$ a prime with $q \nmid a$. Then $q \mid a'$ if and only if $\sum_{p|a} p^{-1} \equiv 0 \pmod q$, where $p^{-1}$ is the inverse of $p$ modulo $q$. If $q \mid a$, then $q \nmid a'$.*

Formalised. *`Erdos307.anatomy_iff`, with the factorisation $a' \equiv a \sum_{p|a} p^{-1}$ proved as `csum_eq_mul_recipSum` rather than assumed; the second half is `anatomy_on_support` and is Rigidity* (`lean/Erdos307/Frame.lean`).

*Proof.* $a' = a \sum_{p|a} 1/p$. If $q \nmid a$ then $q \mid a'$ iff $q \mid a \sum_{p|a} p^{-1}$ iff $\sum_{p|a} p^{-1} \equiv 0 \pmod q$. If $q \mid a$, then by Lemma 2.1 applied to the prime set of $a$, $q \nmid a'$. □

Thus the cycle equation $a' = b$, $b' = a$ factors, prime by prime, into $\omega(a)+\omega(b)$ local congruences of exactly the shape that governs amicable–pair heuristics: for each $q \mid b$ one needs $\sum_{p|a} p^{-1} \equiv 0 \pmod q$, and symmetrically. These local conditions, however, are individually *empty of obstruction*:

**Proposition 6.2** (No single–modulus congruence obstruction)**.** *For every modulus $m$ with $2 \le m \le 210$, the reduced cross–match system $N_P \equiv D_Q$, $N_Q \equiv D_P \pmod m$ is realisable by disjoint squarefree prime sets $P, Q$, in both structural branches: $\gcd(m, D_P D_Q) = 1$, and the branch in which a prime divisor of $m$ lies in $P \cup Q$. Hence no individual modulus in this range can rule out a solution of #307.*

*Construction.* Write $\Sigma_S \equiv \sum_{p \in S} p^{-1} \pmod m$, so $N_S \equiv D_S \Sigma_S$. The reduced system is $\Sigma_P \equiv D_Q D_P^{-1}$, $\Sigma_Q \equiv D_P D_Q^{-1} \pmod m$; its only coupling is $\Sigma_P \Sigma_Q \equiv 1$, automatic, with $D_P, D_Q$ otherwise free. By Dirichlet each unit class mod $m$ contains primes, fixing $D_P, D_Q$; appending a prime $r \equiv 1 \pmod m$ leaves $D_S$ fixed while sending $N_S \mapsto r N_S + D_S \equiv N_S + D_S$, i.e. incrementing $\Sigma_S$ by 1, so (when $\gcd(D_S, m) = 1$) every target class is reached with disjoint sets. In the branch $\ell \mid m$, $\ell \in P$, reduction mod $\ell$ collapses the system to $N_Q \equiv D_P \equiv 0 \pmod \ell$, met by inverse–paired residues in $Q$; composite $m$ needs no separate gluing, the increment acting directly mod $m$. We verified both branches for all $2 \le m \le 210$ (for prime and prime–power moduli the inside branch requires the directed construction above; seeding a prime divisor of $m$ into $P$, a blind search missing these as a search–power artefact, not an arithmetic obstruction). □

**Remark 6.3** (Precise scope)**.** This excludes only a *single fixed–modulus* congruence disproof. The witnesses are $m$–dependent (the modulus–6 witness already fails modulo 210), and a single $(P, Q)$ solving the system for *all* $m$ simultaneously is, for bounded integers, equivalent to the integer identities $N_P = D_Q$, $N_Q = D_P$ (that is, to #307 itself) so per–modulus solvability is *no* evidence for existence. Nor does it exclude other finite or local negative arguments (size/Mertens bounds, density, magnitude–congruence hybrids, or a Hasse–type failure across infinitely many moduli); indeed our own $|U| \ge 59$ and $2.09 \times 10^{56}$ barriers are finite–data results fully consistent with it. Together with Theorem 7.1 (every non–archimedean shadow trivially negative for degree reasons) it brackets the problem: no single–modulus obstruction over $\mathbb{Z}$, fatal non–archimedean analogues over $\mathbb{F}_q[t]$, the difficulty living at the archimedean place.

**A first–order count.** Model a candidate squarefree $a$ as "$P$–abundant" ($\sum_{p|a} 1/p > 1$) and ask for $b = a'$ to itself be $Q$–abundant with $\sum_{q|b} 1/q = 1/s$. Two empirical inputs (Section 10) calibrate this:

- the density of $P$–abundant $n$ is $\delta = 0.0420$ (and $\approx 0.0176$ among squarefree $n$); a theorem, not merely an observation (Proposition 6.5);
- for a qualifying squarefree $a$, $b = a'$ is squarefree about 95% of the time, but its mass $\sum_{q|b} 1/q$ has mean $\approx 0.047$ against the required $\approx 0.934$; the empirical tail probability $\Pr[\sum_{q|b} 1/q \geq 1/s]$ is 0 in 40,920 samples at these scales (Section 10).

The gap is structural: at small scale a $P$–abundant $a$ hogs the cheap mass–carrying small primes, and $b = a'$, being coprime to $a$, is both too small and too coprime to assemble reciprocal mass $1/s$. The extremal case is stark: for the primorial $a = \prod_{i \leq k} p_i$, though $\sum_{p|a} 1/p$ rises through 2 at $k = 59$, the cofactor sum $a'$ is essentially prime, with $\sum_{q|a'} 1/q < 0.2$ for every $k < 100$ (and only 0.0014 at $k = 59$): the family that *maximises* the forward mass is the most starved on the return. Equivalently, for squarefree $n$ with $n'$ squarefree, $n'' = n\,\sigma(n)\,\sigma(n')$, so the two–step return ratio $n''/n = \sigma(n)\sigma(n')$ is the #307 product itself; over *all* such $n < 10^6$ it stays strictly below 1, never exceeding 0.54 (maximal at $n = 969969 = 3 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19$), so $D$ is two–step contracting throughout the tested range and must reverse to unit ratio for a solution to exist. The first scale at which $b$ could plausibly carry the required mass is of the order of the barrier ($\sim 10^{56}$); this dynamical mass–gap is *consistent with* the proved barrier rather than an independent derivation of it (the empirical tail probability 0 at small scale is equally consistent with NO and with YES–beyond–the–barrier). A naive "expected number of solutions" built from the local congruences (one factor $\sim 1/q$ per condition) diverges like a constant times $\log X$; this is a divergent naive count of exactly the type known to be unreliable, so we treat it only as very weak, non–rigorous suggestive evidence that solutions could exist at or beyond the barrier.

**Remark 6.4** (Calibration)**.** We present the count of the previous paragraph as a heuristic, not a theorem. Heuristics of amicable–pair type are known to be delicate; we report it as weak evidence, consistent with the divergence of the refined expected count, that the answer to #307 is plausibly YES (and hence that the period–one form of the Ufnarovski–Åhlander conjecture is plausibly false).

*A cautionary refinement.* One can try to sharpen this into a quantitative rate using a Kubilius/Erdős–Wintner model, stratified over which small primes lie in the support of $a$ versus remain available to $b$. Two lessons emerge, both methodological. First, a plain Monte–Carlo estimate *undercounts* the rate by some $10^{61}$: the expectation is dominated by rare, balanced splits of the small primes that sampling almost never sees, so a naive estimate of this kind is misleadingly small and only an explicit stratified sum corrects it. Second, and this is why we record no numerical prediction, the stratified estimate is itself unstable: refining the stratification from the first 8 to the first 14 primes drops the rate by some 17 orders of magnitude and pushes the implied first occurrence from $\sim 10^{10^{38}}$ out past $\sim 10^{10^{55}}$, with no sign of convergence. The only robust conclusion is negative: every version places a hypothetical first solution *doubly exponentially* beyond any conceivable search, consistent with the constructive deficit of Section 9; the model does not reliably decide existence in either direction.

**Proposition 6.5** (The abundance density is a theorem)**.** *The additive function $f(n) = \sum_{p|n} 1/p$ satisfies the Erdős–Wintner three–series criterion [19] (here $\sum 1/p^2$ and $\sum 1/p^3$ converge), so $f$ has a limiting distribution and the density $\delta = \lim_{x\to\infty} \frac{1}{x}\#\{n \leq x : f(n) > 1\}$ exists, equal to*

$\Pr\big[\sum_p X_p/p > 1\big]$ *for independent* $X_p \sim \mathrm{Bernoulli}(1/p)$. *Moreover the constant carries a* certified *enclosure:*

$$0.04202 \;\le\; \delta \;\le\; 0.04223.$$

*Writing* $S_1 = \sum_{p \le P_1} X_p/p$, $P_1 = 3 \times 10^4$, *and* $T \ge 0$ *for the tail, monotone upper and lower cdf envelopes of* $S_1$ *are propagated through the Bernoulli convolution on a grid of step* $10^{-7}$ *with the shift* $1/p$ *rounded down for the upper and up for the lower envelope (which preserves them, cdfs being nondecreasing) so* $\delta \ge 1 - U(1)$ *costs nothing (the tail is nonnegative), while* $\delta \le 1 - L(1 - a) + \Pr[T \ge a]$ *with the explicit Chernoff bound* $\Pr[T \ge a] \le \exp(1.36 - P_1 a)$, $a = 10^{-3}$ *(from* $\log \mathbb{E}\, e^{\lambda T} \le \lambda s_2 + 0.72\lambda^2 s_3$ *at* $\lambda = P_1$, *using* $s_2 < 1/P_1$, $s_3 < 1/(2P_1^2)$ *and* $e^x - 1 \le x + 0.72x^2$ *on* $[0, 1]$*); float slack* $10^{-9}$ *exceeds the accumulated rounding by nine orders, and the machinery reproduces an exact four–prime reference to* $2 \times 10^{-16}$ *(*`code/delta_cert.py`*). The point value* $0.0420$*, at the bottom of the enclosure, agrees to four places with the sieve to* $2 \times 10^7$*; the upper edge is conservative, carried almost entirely by the* $a$*–window. As for the density of abundant integers, no closed form is expected.*

*Two dynamical probes: no second obstruction.* Two measurements sharpen the picture beyond the first–order mass count, and both cut towards rather than against a solution. *(i) Image density.* A two–cycle needs *both* members in the image $D(\mathbb{N})$ (each is the other's derivative), so one might fear a second obstruction; that the high–mass numbers a cycle requires are seldom derivatives. The opposite holds: of the integers below $10^7$, 68.3% lie in $D(\mathbb{N})$, and the fraction *rises* with mass, from 61.7% at $\sum_{p|n} 1/p < 0.2$ to 78.2% at $\sum_{p|n} 1/p > 1$ and 80% beyond. The high–mass numbers are thus over–represented among derivatives; the second cycle condition is comfortably met. *(ii) The barrier is where the balanced mass–product reaches* 1. Set $r(n) = \big(\sum_{p|n} 1/p\big)\big(\sum_{q|n'} 1/q\big) = D^2(n)/n$, so $r = 1$ is exactly the cycle condition. Exhaustively over $n \le X$ the maximum of $r$ climbs $0.50 \to 0.62$ as $X$ runs $10^3 \to 10^9$ (with no exact cycle, as the barrier demands), with no ceiling; the climb follows the balanced split $r \approx (S(k)/2)^2$, $S(k) = \sum_{i \le k} 1/p_i$, which reaches 1 precisely when $S(k) = 2$, i.e. $k \approx 59$. The dynamical trajectory thus reproduces the $|U| \ge 59$ barrier from the opposite direction. Neither probe decides existence; both *remove* hypothetical obstructions, leaving the exact archimedean closure as the sole gate.

*Why the analytic method is starved:* $N(r) \le 1$. The rank–one rigidity (Section 13) has a sharp consequence for the circle–method machinery that resolves the *unrestricted* Egyptian–fraction problem. For a rational $r$ let $N(r) = \#\{\, P : \sum_{p \in P} 1/p = r \,\}$ count the finite prime sets with reciprocal sum $r$. Then

$$N(r) \le 1 :$$

by the Rigidity Lemma (Theorem 2.1) the sum $\sum_{p \in P} 1/p = N_P/D_P$ is already in lowest terms, so the reduced denominator of $r$ equals $D_P = \prod P$ and $P$ is exactly its set of prime divisors; uniquely determined ($N(r) = 1$ precisely when that denominator is squarefree and the numerator of $r$ is its cofactor sum). The circle method needs the opposite: a target hit by a positive proportion of subsets, so a main term dominates. This is exactly what Bloom's shift to $\{p-1\}$ manufactures, $p-1$ being composite and divisor–rich, a rational is the reciprocal sum of *many* subsets of $\{1/(p-1)\}$ (already among the first fourteen shifted primes the multiplicity reaches 3 and grows), whence [7] represents every positive rational there. Over the primes themselves $N(r) \le 1$ leaves nothing to average: the analytic engine that reaches $\{p - h\}$ is provably starved on $\{p\}$, and this is the method–level reason #307 (the rank–one, exact–prime case) lies beyond it.

## 7 A function–field lens: the obstruction is archimedean

Define an arithmetic derivative on $\mathbb{F}_q[t]$ by $P' = 1$ for monic irreducibles and the Leibniz rule, so $f' = \sum_i e_i\, f/P_i$ for $f = \prod_i P_i^{e_i}$ (the exponents $e_i$ read modulo the characteristic).

**Theorem 7.1.** *Over $\mathbb{F}_q[t]$ this derivative has no two–cycles and no nonzero fixed points. Indeed $\deg f' \le \deg f - 1$ for every nonconstant $f$. The analogues of the integer fixed points $p^p$ also vanish: $(P^p)' = p\,P^{p-1}P' = 0$ in characteristic $p$. The same holds verbatim for every twisted variant ($P' = u_P \in \mathbb{F}_q^\times$, Leibniz).*

*Proof.* Each term $e_i\, f/P_i$ has degree $\deg f - \deg P_i \le \deg f - 1$, so $\deg f' \le \deg f - 1$. A two–cycle $f' = g$, $g' = f$ would force $\deg g \le \deg f - 1$ and $\deg f \le \deg g - 1$, hence $\deg f \le \deg f - 2$; a fixed point $f' = f$ is excluded likewise. (Verified exhaustively over $\mathbb{F}_2$ to degree 7, $\mathbb{F}_3$ to degree 5, and $\mathbb{F}_5$ to degree 4; twisted variants over $\mathbb{F}_2$ and $\mathbb{F}_5$ on small supports through $2.3 \times 10^6$ twist configurations, none closing.) □

Formalised. `Erdos307.ff_no_cycle` is the theorem, on the single hypothesis $\deg f' + 1 \le \deg f$; `deg_drop_of_terms` is that hypothesis from the term bound, and `no_fixed_point_of_drop` excludes fixed points. The argument is stated on the degrees alone, so it holds verbatim for every twisted variant, the twist multiplying each term by a unit (`lean/Erdos307/FunctionField.lean`).

**Corollary 7.2** (The function–field Erdős–Barbeau equation is insoluble)**.** *For every finite field $\mathbb{F}_q$, all nonempty finite sets $P, Q$ of monic irreducibles, and all unit weights $u_\pi, v_\rho \in \mathbb{F}_q^\times$,*

$$\Big(\sum_{\pi\in P} \frac{u_\pi}{\pi}\Big)\Big(\sum_{\rho\in Q} \frac{v_\rho}{\rho}\Big) \neq 1.$$

*Proof.* Write $f = \prod P$, $g = \prod Q$, $D_u(f) = \sum_\pi u_\pi f/\pi$; the equation reads $D_u(f)\, D_v(g) = fg$. If some $\pi \in P \cap Q$, then $\pi^2 \mid fg$ while $D_u(f) \equiv u_\pi f/\pi \not\equiv 0$ and $D_v(g) \equiv v_\pi g/\pi \not\equiv 0 \pmod\pi$, so $\pi \nmid D_u(f)D_v(g)$; impossible; hence $\gcd(f,g) = 1$. The same congruence gives $\gcd(f, D_u(f)) = \gcd(g, D_v(g)) = 1$, so $g \mid D_u(f)$ and $f \mid D_v(g)$, forcing $D_u(f) = c\,g$, $D_v(g) = c^{-1}f$ with $c \in \mathbb{F}_q^\times$: a twisted two–cycle, excluded by the degree drop. The weighted product equation is thus insoluble over every $\mathbb{F}_q(t)$; it *is* soluble over the unit–rich number rings of Theorem 7.10, since any twisted cycle gives $(D_u(a)/a)(D_v(b)/b) = (b/a)(a/b) = 1$, and over $\mathbb{Q}$, where the admissible weights are $\pm1$, it is #307 itself with the signed barrier of Remark 7.3, open beyond $2.09\times10^{56}$. The three regimes (non–archimedean monotonicity (closed, negative), unit–broken monotonicity (closed, positive), ordered–archimedean (open, barred)) now have theorems on two of three doors. □

Over $\mathbb{Z}$, by contrast, $n' = n\sum_i e_i/p_i$ can exceed $n$ precisely because the archimedean absolute value lets a sum of strictly smaller terms outgrow the original; the "mass $> 1$" phenomenon $\sum 1/p > 1$, which has no degree–theoretic analogue. Theorem 7.1 thus localises the entire existence question to the archimedean place: every non–archimedean shadow of #307 is trivially negative. This is consistent with (and partly explains) both the census finding of Section 10 and Proposition 6.2 (no single modulus obstructs the cross–match system in the tested range); the difficulty is not local. Any eventual proof must therefore distinguish $\mathbb{Z}$ from $\mathbb{F}_q[t]$; over the latter the decisive tools are the degree and the Mason–Stothers/Wronskian inequality [20], neither of which has a $\mathbb{Z}$–analogue. The conjectural analogue, *abc*, does not fill the gap, and quantifiably so. Every cycle inherits, from $N' = a^2 + b^2$ with $N = ab$, the relation $(a-b)^2 + 4ab = (a+b)^2$ on coprime squarefree members, but this triple is *abc*–unexceptional, of quality $q = \log(a+b)^2/\log\operatorname{rad}\big((a-b)\cdot ab\cdot(a+b)\big) \approx \frac{1}{2}$ (median 0.543, never above 0.74 across $6 \times 10^4$ coprime squarefree pairs; the only $q > 1$ cases

are degenerate smooth–gap pairs such as $(5,3)$, vanishingly far below the $10^{56}$ scale a solution must occupy). Granting *abc* in its strong explicit form would therefore yield no bound here: the function–field collapse rests on an inequality genuinely *absent* over $\mathbb{Z}$, not merely unproven, and the size mismatch is by a factor of two in the exponent. #307 lives in the *abc*–comfortable zone.

*The free grading, and what evaluation destroys.* Theorem 7.1 has a structural reading. In the free Leibniz world (the polynomial ring $\mathbb{Z}[x_p]$ with $x_p' = 1$) the derivative of a squarefree monomial of degree $n$ is the elementary symmetric form $e_{n-1}$, homogeneous of degree $n-1$: the formal derivative strictly lowers the grading, cycles are impossible, and over $\mathbb{F}_q[t]$ the degree realises this grading faithfully. Over $\mathbb{Z}$ the evaluation $x_p \mapsto p$ destroys homogeneity, and the data show it destroys it completely: for random squarefree $a$ of fixed size, $\omega(a')$ is governed by the size of $a'$ alone (Erdős–Kac), flat in $\omega(a)$ within each parity class (measured: mean $\omega(a') = 2.7$–$3.0$ as $\omega(a)$ runs over $3, 5, 7$, and $3.6$–$4.0$ over $4, 6, 8$, the offset being the parity anatomy of Theorem 4.14). A two–cycle is therefore a doubly anti–typical event: an additively assembled integer must carry multiplicative rank far in the Erdős–Kac tail *and* on an exactly prescribed support; the two independent prices whose product is the heuristic rarity of Section 6. The deeper additive–multiplicative content collapses, on inspection, into the $S$–unit form of Proposition 7.6: dividing the cycle equation by $D_Q$ exhibits it as an $n$–term unit equation; one more coordinate system in which the same exactness reappears unchanged.

**Remark 7.3** (The obstruction is *ordered*–archimedean)**.** The barrier is finer than "archimedean": its engine is AM–GM on real, *positive* masses, i.e. the ordering of $\mathbb{R}$. Over $\mathbb{Z}[i]$ (a PID, so the Rigidity Lemma and cross–matching survive verbatim, with cycles defined up to units), the masses $\sum 1/\pi$ are *complex*: phases can align so that $\left|\sum 1/\pi\right| > 1$ with as few as three Gaussian primes (e.g. $\frac{1}{1+i} + \frac{1}{2+i} + \frac{1}{2-i} = 1.3 - 0.5i$), and the 59–prime bound has no analogue. The ordered–archimedean thesis thus predicts that small derivative two–cycles *may* exist over $\mathbb{Z}[i]$ where they are barred over $\mathbb{Z}$, and the prediction is confirmed. Two facts make the contrast precise. *First*, phases do not help over $\mathbb{Z}$: for any *signed* derivative ($p' = \pm 1$, Leibniz) a two–cycle still satisfies $|s||t| = 1$ with $|s| \le \sigma(a)$, $|t| \le \sigma(b)$ and disjoint supports (the rigidity proof is unchanged), so $\sigma(a) + \sigma(b) \ge 2$ and the full 59–prime, $2.09 \times 10^{56}$ barrier applies to every choice of signs: real phases cannot cancel. *Second*, fourth roots of unity suffice instantly: Leibniz derivatives on $\mathbb{Z}[i]$ with unit prime–values ($\pi' \in \{\pm 1, \pm i\}$) possess genuine two–cycles at norm $\sim 10^4$. The smallest:

$$a = (1+i)(2+7i)(6+11i) = -129 - i, \qquad b = 3\,(2+i)(1+2i)(6+5i) = -75 + 90i,$$

with $a' = b$ under the *standard* values $\pi' = 1$ on the primes of $a$, and $b' = a$ exactly under $\pi' = i$ on $3$, $2+i$, $1+2i$ and $\pi' = 1$ on $6+5i$; seven prime classes ($N(a) = 16{,}642$, $N(b) = 13{,}725$) against the fifty–nine that $\mathbb{Z}$ demands. Exhaustive search finds exactly sixteen such cycles up to conjugation and direction having a member of norm $\le 2 \times 10^7$ with $\omega \le 5$ on the enumerated side ($\omega$–profiles $(3,4)$, $(4,4)$, $(5,4)$, and even $(4,6)$) and the population grows steadily (4 below $3 \times 10^5$, 16 below $2 \times 10^7$), consistent with an infinite, slowly growing family; in six of the sixteen one side requires no twist at all. All sixteen were verified exactly, by independent recomputation; no twisted *fixed* point ($a' = ua$) exists in the same range. The strict two–sided normalisation $\pi' = 1$ (first–quadrant representatives) has no cycle up to units (nor any fixed point, zero–derivative, or unit–derivative element) with $N(a) \le 2 \times 10^9$ ($9.4 \times 10^8$ candidates, partners by full factorisation); this is the expected order rather than a separate rigidity, a single fixed phase system being heuristically $4^{\omega-1}$–fold rarer than the union of all systems. The contrast with $\mathbb{Z}$ survives this accounting: over $\mathbb{Z}$ *every* one of the $2^{\omega-1}$ sign systems is barred by the theorem above, while over $\mathbb{Z}[i]$ the union of phase systems is demonstrably populated from norm $1.7 \times 10^4$ on. The barrier of #307 is thus localised precisely: it is the impossibility of phase cancellation in the ordered field $\mathbb{R}$, where the only

available phases $\pm 1$ provably gain nothing; while the moment fourth roots of unity are admitted, the analogous existence problem closes at seven primes. The cycles found come in conjugate pairs (supports not conjugation–stable); a conjugation–*stable* Gaussian cycle would descend to rational data through the norm form $a^2 + b^2$, i.e. precisely the Pythagorean layer of Proposition 14.2; now a concrete target rather than a speculation.

**Remark 7.4** (The descended operator; the symmetric sector)**.** Conjugation–stable supports admit an exact descent to $\mathbb{Z}$. Such a squarefree Gaussian integer is an associate of an odd squarefree rational $m$ (split primes entering as whole conjugate pairs; the $(1+i)$–sector is empty by a short parity argument), and a conjugation–equivariant unit assignment makes the twisted derivative rational:

$$D(m) \;=\; \sum_{\substack{p \mid m \\ p \equiv 1\,(4)}} t_p \,\frac{m}{p} \;+\; \sum_{\substack{q \mid m \\ q \equiv 3\,(4)}} \varepsilon_q \,\frac{m}{q}, \qquad t_p \in \{\pm 2x_p, \pm 2y_p\}, \quad \varepsilon_q = \pm 1,$$

where $p = x_p^2 + y_p^2$; a Leibniz derivative on odd squarefree integers whose prime values have size $2\sqrt{p}$ rather than 1, and whose masses $\sim 2/\sqrt{p}$ reach 2 with four primes, so no 59–prime barrier exists for it. A two–cycle of $D$ is exactly a conjugation–stable Gaussian cycle and would plant the phenomenon on rational soil. Parity constrains the hunt: $D(m)$ is odd iff $m$ carries an odd number of primes $\equiv 3 \pmod 4$, so every member must contain such a prime. One level deeper, since $q^{-1} \equiv q \pmod 8$ for odd $q$, the descended operator obeys $D(m) \equiv m\,W(m) \pmod 8$ with $W(m) = \sum_{p \mid m} t_p\, p$, so a two–cycle forces $W(m) \equiv W(m') \equiv mm' \pmod 8$; a mod–8 constraint on the sign assignment that prunes the $4^{\#\mathrm{split}} 2^{\#\mathrm{inert}}$ phase enumeration severalfold in future searches (verified on $2 \times 10^5$ random assignments; `code/peeling_groupoid.py`). An exhaustive search over members $m \le 10^9$ (Gaussian norm $10^{18}$, $\omega \le 8$, $1.4 \times 10^9$ closure tests) found no cycle, no fixed point $|D(m)| = m$, and no vanishing $D(m) = 0$. The search has since been extended two decades further in $m$, on a box narrower in the other parameters: $\omega \in [4, 6]$ and odd primes below 600, with $m$ running over $[10^9, 10^{10}]$ and then $[10^{10}, 10^{11}]$. Both tiers are exhaustive over their 5,778 prime pairs and return nothing: 10,004,024 members and $6.94 \times 10^8$ closure tests on the first, 22,218,430 members and $1.82 \times 10^9$ closure tests on the second, so $3.22 \times 10^7$ members and $2.52 \times 10^9$ closure tests in all, with no two–cycle at any point (`code/hunt/descended_hunt2.rs`, `code/hunt/run_tier2_par.sh`). The tier boxes are not nested in the one above, being narrower in $\omega$ and in the prime range while wider in $m$, so the two counts complement rather than supersede one another; and the tiers test closure only, not the fixed–point and vanishing conditions recorded above. We claim no obstruction from this: equivariance prunes the available phases fourfold per split prime, and a phase count makes emptiness at this depth the expected order. The descent target remains open over $\mathbb{Z}[i]$, but not in the real–quadratic world, where the corresponding sector contains a fully rational pair (Remark 7.5). Finally, the descended coefficients are classical objects: at a split prime $p$ the four values $\{\pm 2x_p, \pm 2y_p\}$ are exactly the Frobenius traces of the congruent–number curve $y^2 = x^3 - x$ (CM by $\mathbb{Z}[i]$) and of its twists ($a_p \in \{\pm 2x_p, \pm 2y_p\}$ verified by point counting for all $p < 120$) so the conjugation–stable sector is the arithmetic derivative twisted by the Hecke eigensystem of $\mathbb{Q}(i)$, and its mass statistics are governed unconditionally by Hecke's equidistribution theorem: the first point of contact between #307 and automorphic forms. Concretely: over random squarefree $m$ the three–series conditions converge (the variance term is $\sum 2/p^2$), so the descended mass $D(m)/m$ has an Erdős–Wintner *limiting distribution* whose prime components are arcsine laws scaled by $1/\sqrt{p}$; the required input, equidistribution of the Hecke angles, is visible exactly (Kolmogorov–Smirnov distance 0.0041 from uniform over the 8,977 split primes below $2 \times 10^5$, well inside the 95% band 0.0144).

**Remark 7.5** (The wall is the orderability of $\mathbb{Q}$)**.** The cycles over $\mathbb{Z}[i]$ permit a dissection of the barrier that no amount of theory over $\mathbb{Z}$ could provide. The entire structural chain of this note (rigidity, cross–matching, the Pythagorean identity, the split discriminant) transfers verbatim to twisted derivatives over $\mathbb{Z}[i]$: Leibniz holds for disjoint supports, so a cycle $(a, b)$ has $D(N) = a^2+b^2$ for $N = ab$, with $D(N)-2N = (a-b)^2$ and $2N-D(N) = \big(i(a-b)\big)^2$ (all verified exactly on the cycles found). The last equation exposes the wall: over $\mathbb{Z}[i]$, where $-1 = i^2$ is a square, *both* discriminant factors are squares automatically, and the plus/minus distinction of Proposition 15.12 (which over $\mathbb{Z}$ carries the entire obstruction, via $(a-b)^2 \geq 0 \Rightarrow \sigma(N) \geq 2$) has no content. The unique non–transferring link is the sign–definiteness of squares: the barrier is, in the end, the *formal reality* of $\mathbb{Q}$ in the sense of Artin–Schreier. (The two Artin–Schreier theorems thus bracket the problem: their characteristic–2 map builds the tower of Proposition 15.7, their theory of orderable fields names the wall.) This classifies where the mechanism can operate at all: it needs an ordering making squares nonnegative (formal reality, hence a real embedding) *and* bounded twisting phases (finite unit group); by Dirichlet's unit theorem, unit rank $r_1 + r_2 - 1 = 0$ with $r_1 \geq 1$ forces degree one. $\mathbb{Q}$ *is the only number field over which the mass barrier can operate*: over imaginary quadratic fields squares lose sign–definiteness (over $\mathbb{Z}[i]$ even $-1$ is a square), and over every other field the unit rank is positive and twisted masses are unbounded. This is a classification of the wall's habitat, not an existence claim elsewhere, but over $\mathbb{Z}[i]$ existence is no longer a claim: it is a list of sixteen. The ladder of imaginary quadratic rings confirms the mechanism quantitatively. Over $\mathbb{Z}[\sqrt{-2}]$, whose only units are $\pm 1$, precisely the twisting freedom that provably gains nothing over $\mathbb{Z}$; sign–twisted cycles nevertheless exist from norm 11,462 on ($a = -90 - 41\sqrt{-2}$, $b = 75 - 45\sqrt{-2}$, seven prime classes): the complex embedding alone carries the phenomenon. Over the Eisenstein ring $\mathbb{Z}[\omega]$, with six units, cycles appear at norm 2,964 with *six* prime classes ($a = -40 + 22\omega$, $b = 56 - 21\omega$); the smallest derivative two–cycle exhibited in any ring, and the population at fixed height grows with the unit group: at norm $\leq 2 \times 10^7$ the censuses read 14, 56, and 278 twisted–cycle instances for unit groups of order 2, 4, 6 (a factor $\sim$4–5 per two extra units, as a phase count predicts) while the strict canonical systems of all three rings remain empty to norm $2 \times 10^9$ ($3 \times 10^9$ candidates in total, partners by full factorisation). Two further objects appeared over $\mathbb{Z}[\sqrt{-2}]$: a composite with *unit* derivative ($a' \sim 1$; impossible over $\mathbb{Z}$, where $n' = 1$ characterises the primes), and an *anti–holomorphic* cycle: $a = -3163-1474\sqrt{-2}$ satisfies $a' = -\bar{a}$ exactly, under a single conjugation–equivariant sign assignment closing both directions ($N(a) = N(b) = 14{,}349{,}921$, five prime classes each side). The equation $a' = u\bar{a}$ is vacuous over $\mathbb{Z}$ only in its *one–dimensional* reading, where it says $n' = \pm n$. In the reading that matches the problem it is not vacuous but central: by Proposition 15.2 the Pythagorean layer over $\mathbb{Z}$ *is* the anti–holomorphic equation $\mathcal{D}z = \mu\bar{z}$ for the coordinatewise derivative on $z = a + bi$, and #307 is exactly its unit–multiplier case. The ring ladder and the main problem are therefore instances of one equation rather than analogues; what singles out $\mathbb{Z}$ is the extra constraint $\operatorname{Im}\mu = 1$, which confines the multiplier to a line meeting $\mathbb{Z}[i]^\times$ in the single point $\mu = i$. That the anti–holomorphic sector is thin is uniform across the ladder: over $\mathbb{Z}[\sqrt{-2}]$ the cycle above is still the *only* one after widening the census simultaneously in all three directions; prime–norm $\leq 2 \times 10^4$, $\omega \leq 8$, product–norm $\leq 10^{10}$ (a factor 700 past the cycle itself), $8.09 \times 10^8$ supports, `code/hunt/antiholo_hunt.rs`, and over $\mathbb{Z}$ the sector is empty below $10^{112}$ by Corollary 14.3. We record the plateau as consistent with genuine finiteness, not as evidence for it: a random model for $a' = -\bar{a}$ predicts a count growing like $\log X$, which the tested range cannot separate from a constant. All cycles were verified exactly by independent recomputation. Finally, the second pillar was tested in isolation: over $\mathbb{Z}[\sqrt{2}]$ (*totally real*, so squares are sign–definite in both embeddings, but with infinite unit group $\pm(1+\sqrt{2})^{\mathbb{Z}}$) twisted cycles appear already at norm 882: with unit twists of height $\leq 2$, $a = 21\sqrt{2}$ (classes $\sqrt{2}$, $-1+2\sqrt{2}$, $1+2\sqrt{2}$, 3) cycles exactly with

$b = 51 + 13\sqrt{2}$, of norm 2263 and only *two* prime classes (against the fifty–nine that $\mathbb{Z}$ demands) and the composite $369 - 15\sqrt{2}$ has derivative exactly $\varepsilon^6 = 99 + 70\sqrt{2}$, a unit. Both pillars of the classification are thus load–bearing, and each is now confirmed by explicit counterexample on the soil where it alone fails: drop formal reality and cycles appear (imaginary quadratic rings, norms $10^3$–$10^4$); keep formal reality but drop unit finiteness and cycles appear (real quadratic, norm 882, the smallest in any ring); over $\mathbb{Q}$, where both hold, the barrier stands at $10^{112}$ by theorem. The bracketing of $\mathbb{Q}$ is complete from both axes. The census sharpens it: at norm $\leq 2 \times 10^7$ the real–quadratic ring carries 30,204 twisted–cycle instances against 14, 56, 278 for the imaginary rings with $2, 4, 6$ units infinite unit rank dominates by two orders of magnitude, and contains both the *minimal* architecture, $a = \sqrt{2}\,(5 - \sqrt{2})$ cycling with $b = 7$ at norms $(46, 49)$, two prime classes per member (the rational prime 7 being composite in the ring), the smallest derivative cycle found in any ring; and a *fully rational* pair:

$$D(3707) = 1547, \qquad D(1547) = 3707 \qquad (3707 = 11 \cdot 337, \quad 1547 = 7 \cdot 13 \cdot 17),$$

exactly, with unit heights $\leq 2$ on both sides, verified independently: two ordinary integers in a derivative two–cycle. The descent phenomenon that is empty over $\mathbb{Z}[i]$ to norm $10^{18}$ is realised over $\mathbb{Z}[\sqrt{2}]$ at four digits. The canonical system of $\mathbb{Z}[\sqrt{2}]$ remains cycle–free to norm $2 \times 10^9$ ($7.7 \times 10^8$ candidates), now with seven unit–derivative composites.

The cycle phenomenon over rings with units is governed by the theory of unit equations, which supplies its rigidity theorem.

**Proposition 7.6** ($S$–unit rigidity of twisted cycles)**.** *Let $\mathcal{O}$ be the ring of integers of a number field and $S$ a finite set of primes of $\mathcal{O}$. If $(a, b)$ is a twisted two–cycle with* $\operatorname{supp}(ab) \subseteq S$*, then dividing $D(a) = b$ by $b$ exhibits a solution of*

$$x_1 + \cdots + x_{\omega(a)} = 1, \qquad x_i = u_i \,\frac{a/\pi_i}{b},$$

*in the $S$–unit group of the field, and likewise for $b$. By the theorem of Evertse–Schlickewei–Schmidt [47] (cf. [40]), such an equation has only finitely many nondegenerate solutions; hence* every fixed support carries at most finitely many twisted cycles free of vanishing subsums*, and the species can be infinite only through growing supports; precisely the behaviour of every census above. (Vanishing subsums correspond to partial zero–derivative patterns; none occurred in any search. The relation was verified exactly on the $\mathbb{Z}[\sqrt{2}]$ specimen: the four $S$–unit terms of $a = 21\sqrt{2}$ sum to 1.)*

**Remark 7.7** (Galois folding, and the thinness of symmetric sectors)**.** Over a quadratic field with automorphism $\sigma$, a $\sigma$–equivariant choice of unit values ($u_{\pi^\sigma} = u_\pi^\sigma$) makes $D$ commute with conjugation, so the *single* equation $D(a) = a^\sigma$ already implies that $(a, a^\sigma)$ is a twisted two–cycle (checked symbolically and on 1,988 random instances). In the two–prime case the partner is forced: $\rho = -\big(\bar{u}v\,\pi + \bar{v}\,(\pi^\sigma)^2\big)/\big(N(u) - N(\pi)\big)$, with integrality a congruence and the entire residue of the problem the primality of $|N(\rho)|$ along Pell–exponentially growing sequences; Lehmer territory rather than Bunyakovsky. Whether a folded sector is starved or rich is decided by its degrees of freedom. When the fold leaves no slack the sector is thin: the folded *two*–prime family over $\mathbb{Z}[\sqrt{2}]$ is empty in the tested box (36,000 parameter triples, none even integral), the conjugation–stable sector over $\mathbb{Z}[i]$ is empty to Gaussian norm $10^{18}$, and the anti–holomorphic cycle over $\mathbb{Z}[\sqrt{-2}]$ is unique in its census. But with *three* primes the folded equation $u_1\pi_2\pi_3 + u_2\pi_1\pi_3 + u_3\pi_1\pi_2 = (\pi_1\pi_2\pi_3)^\sigma$ carries three unit unknowns against two real equations, and this underdetermined system is *populated*: an exhaustive box ($p_i \leq 137$, unit heights $\leq 4$) contains twenty–four solutions over eleven prime triples, the smallest being

$$a = (3 + \sqrt{2})(5 - 2\sqrt{2})(5 - \sqrt{2}) = 57 - 16\sqrt{2}, \qquad D(a) = a^\sigma = 57 + 16\sqrt{2},$$

with unit values $\varepsilon^2$, $\varepsilon$, $-\varepsilon^{-1}$: *the derivative conjugates the element*, at norm 2737 on both sides, the equivariant return being automatic (verified exactly). Existence of twisted cycles over the four rings tested is a theorem by exhibition; for *every* ring with infinite unit group it is now reduced to the abundance of lattice points on these underdetermined unit systems (measured plentiful already at $k = 3$) a geometry–of–numbers question rather than a primality one.

**Remark 7.8** (The forced third prime: a Schinzel–type reduction over $\mathbb{Z}[\sqrt{2}]$, unblocked)**.** Within that abundance the arithmetic is sharper than free lattice–counting. Fixing $\pi_1, \pi_2$ and the units $u_1, u_2, u_3$, the folded equation rearranges to $\pi_3\,(u_1\pi_2+u_2\pi_1)-(\pi_1\pi_2)^\sigma\pi_3^\sigma = -u_3\pi_1\pi_2$, a $2\times 2$ integer linear system for $\pi_3 = (x, y)$ whose determinant is $N(u_1\pi_2 + u_2\pi_1) - N(\pi_1\pi_2)$ (an identity, verified on all 168 solutions of the box). So $\pi_3$ is *forced* by $(\pi_1, \pi_2, u_1, u_2, u_3)$, and infinitude of twisted cycles is exactly the primality of this forced $\pi_3$ for infinitely many such data a Schinzel/Bunyakovsky–type condition over $\mathbb{Z}[\sqrt{2}]$. Crucially it is *not* the regime of Proposition 8.6: the exponential unit freedom $u_i = \pm\varepsilon^{k_i}$ keeps the family non–polynomial yet populated, so the polynomial–rigidity that forbids reducing the *rational* #307 to such a hypothesis does not apply here. A Schinzel–type theorem over $\mathbb{Z}[\sqrt{2}]$ would render twisted infinitude unconditional, whereas none touches #307 itself; the twisted problem sits strictly closer to the reach of current sieve theory than its rational shadow. Concretely the twisted solution set is *abundant* at accessible scale; thousands of cycles at modest norm (Proposition 7.11), and the count of distinct prime–norm signatures grows roughly in proportion to the norm bound in our census (e.g. $6, 12, 20$ at bounds $40, 80, 160$, unit heights saturating); whereas the rational Pythagorean layer is *empty* below $10^{112}$ (Corollary 14.3). The twisted and rational problems thus differ not in the difficulty of the search but in whether solutions exist to be found at all: for the twist they are plentiful and the only question is their primality, exactly the Schinzel–shaped residue above. That residue is a *pair*, not a triple or a Lehmer sequence: fixing $\pi_1$ and bounding the units, the cycle count still grows with $N(\pi_2)$ (verified: $33, 48, 60$ for $\pi_1 = 3+\sqrt{2}$ at $N(\pi_2) \le 60, 120, 240$), so $\pi_1$'s primality is free and only the *simultaneous* primality of $\pi_2$ and the forced $\pi_3$ remains. Setting up the sieve pins the difficulty precisely. Writing $C = u_1\pi_2 + u_2\pi_1$, $E' = (\pi_1\pi_2)^\sigma$, $F = -u_3\pi_1\pi_2$, the forced third prime is $\pi_3 = (\bar{C}F + E'\bar{F})/\det$ with $\det = N(C) - N(E')$ a binary quadratic form in $(c, d) = \pi_2$. The first condition is the clean form $N(\pi_2) = c^2 - 2d^2$; the second, however, is *not* a second form but the rational condition $N(\pi_3) = N(G)/\det^2$ ($G = \bar{C}F + E'\bar{F}$ quadratic, $N(G)$ quartic, $\gcd(N(G), \det) = 1$), evaluated on the integral locus $\det \mid G$, and $N(\pi_3)$ is unbounded, spiking as $\det \to 0$ (values to $\sim 10^{11}$ already at $N(\pi_2) \le 2 \times 10^4$). On this *frozen* $(\pi_1, \text{units})$ slice the question is primality along the folded conic bundle, and the slice is thin: its unimodular fibres $\det = \pm 1$ are *empty* by a genus obstruction (there $\det = \pm 1$ would need $x^2 - 2y^2 \in \{-55, -43\}$, both non–residues). But the thinness is an artefact of freezing exactly the parameters that carry the abundance. The *full* twisted variety $D(a) = a^\sigma$ ($\pi_1, \pi_2, \pi_3$ and their units all free) is a cubic threefold whose solution count grows with positive exponent; a positive–density attack must run there, not on a fixed fibre, so the fixed–fibre conic sieve is a closed dead end. The honest placement is then: the reachable object is the full threefold at the Heath–Brown $x^3 + 2y^3$ frontier, compounded by a *coupled* second primality ($N(\pi_2)$ and $N(\pi_3)$ simultaneously prime); the two–sparse–conditions/parity barrier where the circle method loses minor–arc cancellation. This is strictly below the exponential/Lehmer tier of the immune families (Remark 5.28, where no sieve reaches at all), but a generation beyond present technology; it is where genuinely new analytic input for the twist would first have to bite.

The natural attempt to *prove* that infinitude (forcing one prime to be the conjugate of another so that a single free prime remains, whence Dirichlet would suffice) is blocked, and the block is a clean structural law.

**Proposition 7.9** (No conjugate prime pair in a twisted cycle)**.** *Let $D$ be a Leibniz derivative with unit prime–values over a quadratic ring with automorphism $\sigma$, and $a$ a squarefree twisted cycle $D(a) = a^\sigma$. Then no two prime factors of $a$ are $\sigma$–conjugate. Equivalently a twisted cycle's support is disjoint from its conjugate; it is never conjugation–stable, which is exactly why the exhibited $\mathbb{Z}[i]$ cycles are asymmetric, coming in conjugate pairs of* distinct *supports rather than on one symmetric support. This* separates *two problems easily conflated: the twisted equality $D(a) = a^\sigma$ treated here, and the symmetric target of Remark 7.4, which is a genuine two–cycle $D(m) = m'$ ($m \neq m'$, both conjugation–fixed) of the* descended *rational operator; not a twisted equality. The present proposition bars the first from the symmetric locus; it does not bear on the second, which remains genuinely open (the exhaustive emptiness there, Remark 7.4, is unexplained by it).*

*Proof.* Suppose a split prime $\pi$ and its distinct conjugate $\pi^\sigma$ both divide $a$. Reducing $D(a) = \sum_i u_i\,(a/\pi_i)$ modulo $\pi$ kills every term but the one at $\pi_i = \pi$, so $D(a) \equiv u_\pi\,(a/\pi) \not\equiv 0 \pmod{\pi}$ ($u_\pi$ a unit and $\pi \nmid a/\pi$ by squarefreeness). Yet $a^\sigma = \prod_i \pi_i^\sigma$ carries the factor $(\pi^\sigma)^\sigma = \pi$, so $a^\sigma \equiv 0 \pmod{\pi}$; hence $D(a) \neq a^\sigma$. An exhaustive $\mathbb{Z}[\sqrt{2}]$ search confirms no cycle carries a conjugate pair. □

We codify what is theorem and what remains open in this circle.

**Theorem 7.10** (Existence of twisted derivative two–cycles)**.** *Leibniz derivatives with unit prime–values admit genuine two–cycles over $\mathbb{Z}[i]$, $\mathbb{Z}[\sqrt{-2}]$, $\mathbb{Z}[\omega]$, and $\mathbb{Z}[\sqrt{2}]$. The proof is by exhibited certificate, each verified exactly by independent recomputation: the smallest instances are, in order of norm, $\big(\sqrt{2}(5-\sqrt{2}),\,7\big)$ over $\mathbb{Z}[\sqrt{2}]$ at norms $(46, 49)$; the Eisenstein cycle at norm 2,964; the sign–twisted cycle over $\mathbb{Z}[\sqrt{-2}]$ at norm 11,462, and the cycle over $\mathbb{Z}[i]$ at norm 16,642; together with the rational pair $D(3707) = 1547$, $D(1547) = 3707$ over $\mathbb{Z}[\sqrt{2}]$.*

**Proposition 7.11** (Geometry of the solution set)**.** *Over $\mathbb{Z}[\sqrt{2}]$, consider the Galois–folded system $u_1\pi_2\pi_3 + u_2\pi_1\pi_3 + u_3\pi_1\pi_2 = (\pi_1\pi_2\pi_3)^\sigma$ of Remark 7.7.* (i) *For* every *triple of split primes the system is solvable with the $u_i$ on the real unit hyperbola $\{|st| = 1\}$: solving the second embedding for $u_2$ and letting the third coordinate tend to zero, the first–embedding residual sweeps $\pm\infty$ while the remaining terms stay bounded, so a root exists by continuity; the real solution set is a nonempty curve (verified numerically on 200 random triples, all solvable). The volume of the solution variety never vanishes.* (ii) *By Proposition 7.6 each fixed support carries finitely many integral solutions, so infinitude can only come from growing supports.* (iii) *The measured integral hit rates are 11 of 455 prime triples in the smallest box, and 30,204 cycles of norm $\leq 2\times10^7$ at twist height $\leq 2$.* (iv) *What remains open is precisely whether the discrete unit lattice meets the real curve in integer points for infinitely many supports, inhomogeneous Diophantine approximation on exponentially parametrised curves. Two shortcuts are closed: bounded–unit polynomial families are obstructed (the rational component of the return forces unit coefficients to track growing prime coordinates, which bounded units cannot), and even the cleanest forward family ($D\big(\sqrt{2}(x-\sqrt{2})\big) = x+2$ identically in $x$, supplying a candidate cycle whenever $x^2-2$ and $x+2$ are prime) has sporadic returns: the closing two–term unit equation, effectively finite for each $x$, is solvable at $x = 5$ and provably insolvable at $x = 15$ and $x = 21$. Those failures are, however, exactly the rigid case: a* prime *return $x \pm 2$ leaves the closing system no slack. The identity has four unit branches $D\big(\sqrt{2}(x\pm\sqrt{2})\big) = x \pm 2$ (independent signs, twists 1 and $\pm\varepsilon^{\pm1}$), and when $x \pm 2$ is composite squarefree the closing system gains one unit unknown per prime class against the same two equations, and does close: $x = 107, 121, 387, 1693$ yield cycles with rational partners $b = 105, 119, 385, 1695$, each verified exactly. Infinitude along this family is thereby reduced to Bunyakovsky for $x^2 - 2$ together with recurring composite–return closure: the softest form the gap takes anywhere in this paper. Even the primality half of that reduction can be removed. By the*

*half–dimensional sieve [16, 17] (applied to* $(2y+1)^2-2=4y^2+4y-1$ *to keep* $x$ *odd),* $x^2-2$ *is a* $P_2$ *infinitely often; every prime factor of* $x^2-2$ *has* 2 *as a quadratic residue and therefore splits in* $\mathbb{Z}[\sqrt{2}]$*, so* $\sqrt{2}(x-\sqrt{2})$ *is infinitely often squarefree and all–split with* $\omega\in\{2,3\}$*. The candidate supports thus exist* unconditionally*; all that remains open on this route is whether the unit gates open along that supply infinitely often. Exactness migrates; it does not disappear.*

**Proposition 7.12** (No vanishing derivatives, no transport, and the price of exactness)**.** (i) *No squarefree* $z$ *with* $\omega(z)\geq 1$ *has* $D(z)=0$*, for any twisted derivative over any of the rings considered (indeed over any UFD): modulo a prime* $\pi\mid z$ *every cofactor term but one vanishes, leaving* $D(z)\equiv u_\pi\, z/\pi\not\equiv 0 \pmod{\pi}$*. The zero–derivative counters of every census above are therefore theorems, not observations.* (ii) *Consequently cycles cannot be transported multiplicatively: for squarefree* $c$ *coprime to* $ab$*, Leibniz gives* $D(ca)=D(c)\,a+c\,D(a)$*, so* $(ca,cb)$ *is a cycle exactly when* $D(c)=0$*; impossible. Cycles do not breed; each one requires fresh exactness.* (iii) *The two cycle equations can be bought by construction, at a conserved price. Buying* neither *leaves the generic stratum: two exact unit conditions (*30,204 *specimens). Buying* one *is free in two dual ways; forward,* $\pi\coloneqq q-\varepsilon^m\sqrt{2}$ *makes* $D(\sqrt{2}\,\pi)=q$ *an identity (the four–branch family above is the case* $m=\pm1$*); return,* $\pi\coloneqq(\rho+\varepsilon^{-1}\rho^\sigma)/\sqrt{2}=(c+2d)-d\sqrt{2}$ *for* $\rho=c+d\sqrt{2}$ *closes* $D(b)=\sqrt{2}\,\pi$ *identically for every associate* $b$ *of* $\rho\rho^\sigma$*, and the residue is one primality,* $|N(\pi)|$*, plus one exact unit gate, measured at* 2 *of* 50 *pairs with both primalities in the smallest box. The two hits re–derive, with no search, precisely the two census minimal–shape cycles with prime rational partner: norms* $(46,49)$ *and* $(206,49)$*. Buying* both *collapses the parameters to unit–indexed quadratics in* $q$ *with a square–discriminant condition; the resulting* $q=7$ *line carries the known cycles at* $m=1$ *and* $m=-2$ *(discriminants* 25 *and* 81*) together with one Galois image, and every other prime–norm rung out to* $|m|\leq 14$ *fails its return, verified exactly. Each purchased identity is repaid as a sparser arithmetic condition elsewhere: the hardness of exactness is conserved, never destroyed.*

**Remark 7.13** (The two faces of the gap: unit rank dichotomises the obstruction)**.** The invariant that governs abundance in Remark 7.5 (the unit rank $r=r_1+r_2-1$ of Dirichlet) also governs the *type* of the residual open problem, and the two types are mutually exclusive. When $r=0$ (the rings $\mathbb{Z}$, $\mathbb{Z}[i]$, $\mathbb{Z}[\sqrt{-2}]$, $\mathbb{Z}[\omega]$) the unit group is finite, so a given pair of supports admits only $|\mathcal{O}^\times|$ twist assignments and hence finitely many candidate cycles: the solution locus is a set of *isolated points*, zero–dimensional, and infinitude can arise only from infinitely many *distinct* supports independently passing a closing test; a Bateman–Horn/Schinzel–type recurrence over prime tuples, with no continuous family to deform along. The constituent primalities are individually classical (each support prime is a value of a binary norm–form, so by Fermat–Dirichlet (and, for $\mathbb{Z}[i]$, by the Green–Sawhney theorem with *both* coordinates prime) there is no shortage of supports); it is their *simultaneous* closure that is not classical. This was tested directly: the minimal shape $a=(1+i)\pi$ over $\mathbb{Z}[i]$ yields *no* cycle for any Gaussian prime $\pi$ of norm $\leq 4000$ under any of the sixteen twists, and the sixteen genuine cycles are irregular mixtures of inert and split classes ($\omega=3,4,5$) with no one–parameter family among them; the signature of isolated points, not a curve. When $r\geq 1$ (every real quadratic ring, e.g. $\mathbb{Z}[\sqrt{2}]$), Proposition 7.11 makes the real solution locus a positive–dimensional curve in each support, but the unit lattice is an infinite rank–$r$ discrete set, and infinitude requires it to meet that curve in *integer* points infinitely often; inhomogeneous Diophantine approximation on an exponentially parametrised curve. No unit rank delivers both a classical gate and a continuous family: rank zero gives the former and denies the latter, positive rank the reverse. This is why no single ring lets the infinitude be *finished* by present methods; a structural account of the obstruction's universality, not a barrier theorem. It also locates $\mathbb{Z}$ exactly: the unique ring carrying *both* the rank–zero sporadicity *and*, by formal reality, the mass barrier; the worst cell of the trichotomy of Corollary 7.2, which is precisely why #307 is the hard representative

of its family.

## 8 A Thābit–type rule and its honest limits

The bridge recasts the search for a solution as the search for a derivative two–cycle, and the derivative's Leibniz structure makes the cycle equations *linear* in the "closing" primes. This yields a constructive calculus analogous to Thābit ibn Qurra's rule for amicable pairs and to Euler's divisor method.

**Theorem 8.1** (Frame Rule)**.** *Let $M, N$ be coprime squarefree integers with derivatives $M', N'$, and set $\Delta := MN - M'N'$. Suppose*

$$p = \frac{MN' + N^2}{\Delta}, \qquad q = \frac{M'N + M^2}{\Delta}$$

*are integers, both prime, each coprime to $MN$, with $p \neq q$. Then $a := Mp$, $b := Nq$ form a derivative two–cycle $a' = b$, $b' = a$; equivalently $(P, Q)$ with $P$ the prime set of $Mp$ and $Q$ that of $Nq$ solves Erdős #307.*

Formalised. `Erdos307.frame_cycle` *verifies both cycle equations from the two divisibility hypotheses, and* `frame_solve` *runs the derivation backwards, so the two quotients are* forced *rather than guessed and the rule is an equivalence. The Leibniz step $a' = M'p + M$ is* `csum_insert_prime`*. What no algebra reaches is the primality of the two quotients* (`lean/Erdos307/Frame.lean`).

*Proof.* With $p$ prime and $p \nmid M$, $a = Mp$ is squarefree and $a' = M'p + M$; similarly $b' = N'q + N$. The cycle conditions $a' = b$, $b' = a$ read

$$M'p + M = Nq, \qquad N'q + N = Mp,$$

a linear system in $(p, q)$ with determinant $-(MN - M'N') = -\Delta$. Solving gives exactly $p = (MN' + N^2)/\Delta$ and $q = (M'N + M^2)/\Delta$; substituting back verifies both equations identically. (Both identities were checked symbolically.) □

**Remark 8.2** (Linear relations and a factorised form of $\Delta$)**.** For integers $p, q$ satisfying the two displayed quotients one has, by direct substitution,

$$Mp - N'q = N, \qquad Nq - M'p = M, \qquad (\Delta p - N^2)(\Delta q - M^2) = MN\,M'N'.$$

(Proved by hand; the first two are the cycle equations themselves, the third is the product of $\Delta p - N^2 = MN'$ and $\Delta q - M^2 = M'N$.) Two consequences. A prime dividing both $p$ and $q$ divides $M$ and $N$, so $\gcd(p, q) = 1$; and $\gcd(p, N) \mid M$, $\gcd(q, M) \mid N$, so $\gcd(p, N) = \gcd(q, M) = 1$ without further hypothesis. Of the clause "each coprime to $MN$" in Theorem 8.1, only $p \nmid M$ and $q \nmid N$ are therefore independent conditions. In a census of all coprime squarefree $M, N \leq 400$ (GP, observed) the value $\Delta = 1$ occurs only for $M = N = 1$, the smallest other value is $\Delta = 5$ (at $\{2\}, \{3\}$), and each of the ten integral frames found has $\min(p, q) = 1$. This is a finite census and settles nothing for larger bases.

A two–primes–per–side variant ("Rule 2") is obtained identically: with $a = Mp_1p_2$, $b = Nq_1q_2$ and elementary symmetric functions $e_1, e_2, f_1, f_2$, the cycle equations are linear in the $f_i$,

$$f_2 = \frac{Me_1 + M'e_2}{N}, \qquad f_1 = \frac{MNe_2 - MN'e_1 - M'N'e_2}{N^2},$$

and a solution is realised whenever $f_1^2 - 4f_2$ is a perfect square with prime roots.

**An Euler-style divisor trick.** Fix $N$ and let $M = M_0 r$ with $r$ a new prime, so $M' = M_0' r + M_0$. Put $\Delta_0 := M_0 N - M_0' N'$.

**Proposition 8.3** (Divisor trick)**.** *With the above notation and* $p_{\text{num}} := MN' + N^2$*:*

(a) $\Delta(r) := MN - M'N' = r\,\Delta_0 - M_0 N'$ (linear in $r$)*;*

(b) $\Delta_0\, p_{\text{num}} - M_0 N'\,\Delta(r) = K$ *where* $K := \Delta_0 N^2 + (M_0 N')^2$ *is* constant in $r$*;*

(c) *consequently* $\Delta(r) \mid p_{\text{num}} \Rightarrow \Delta(r) \mid K$*, and the converse holds provided* $\gcd(\Delta(r), \Delta_0) = 1$ (*equivalently* $\gcd(M_0 N', \Delta_0) = 1$*, since* $\Delta(r) \equiv -M_0 N' \bmod \Delta_0$)*;*

(d) *each positive divisor* $d \mid K$ *with* $d \equiv -M_0 N' \pmod{\Delta_0}$ *yields a candidate* $r = (d + M_0 N')/\Delta_0$ *and recovers* $\Delta(r) = d$*; this enumeration is complete (it captures every genuine solution). The resulting* $p = p_{\text{num}}/d$ *is integral when* $\gcd(M_0 N', \Delta_0) = 1$*, and otherwise must be checked.*

*Proof.* (a) and (b) are direct expansions (verified symbolically; $\partial K/\partial r = 0$). (c): from (b), $\Delta(r) \mid p_{\text{num}} \Rightarrow \Delta(r) \mid \Delta_0 p_{\text{num}} = M_0 N' \Delta(r) + K \Rightarrow \Delta(r) \mid K$. For the converse, $\Delta(r) \mid K$ gives $\Delta(r) \mid \Delta_0 p_{\text{num}}$, whence $\Delta(r) \mid p_{\text{num}}$ iff $\gcd(\Delta(r), \Delta_0) = 1$. (d) inverts (a). Excluded degeneracies: $\Delta_0 = 0$ ($r$ undefined) and $K \le 0$. □

Thus, exactly as in Euler's amicable–pair machinery, one factors the single number $K$ once and reads off candidates from its divisors lying in a fixed residue class modulo $\Delta_0$; only the primality of $r, p, q$ then remains. Two caveats deserve emphasis: the converse in (c) and the automatic integrality of $p$ require $\gcd(M_0 N', \Delta_0) = 1$ (it fails, e.g., for $M_0 = 2$, $N = 19 \cdot 47$, $r = 5$, where $\Delta(r) \mid K$ but $\Delta(r) \nmid p_{\text{num}}$), and the count of admissible classes is governed by $\Delta_0$, not merely by $\tau(K)$.

When the closing side carries a *single* new prime, the rule collapses to a normal form that separates the two layers of the problem.

**Proposition 8.4** (One–prime normal form; the exactness stratum)**.** (a) *Erdős #307 has a solution if and only if there exist squarefree* $M, N$ *with*

$$MN - M'N' = M^2,$$

*such that* $p := N'/M$ *is a prime not dividing* $M$*. Indeed the equation forces* $M \mid M'N'$*, hence* $M \mid N'$ *by* $\gcd(M, M') = 1$*, and* $N = M + M'p$*; then* $a = Mp$*,* $b = N$ *satisfy* $a' = M'p + M = N$ *and* $b' = N' = Mp = a$ *(coprimality and* $a \neq b$ *are automatic). Conversely every solution* $(a, b)$ *and every prime* $p \mid a$ *yield such a pair with* $M = a/p$*, so each solution carries* $\omega(a)$ *witnesses. The problem thus splits into an* exactness *layer (the Diophantine equation) and a single* primality *layer (one prime value of the determined quantity* $N'/M$*).* (b) The exactness layer alone carries the full barrier. *Call* $(M, N)$ *a* relaxed witness *if* $MN - M'N' = M^2$ *with* $p = N'/M$ *composite. The identity* $\big(\sigma(M) + \frac{1}{p}\big)\sigma(N) = \frac{N}{Mp} \cdot \frac{Mp}{N} = 1$ *holds for* every *witness, so by strict AM–GM* ($\sigma(N) = 1$ *being impossible)* $\sigma(M) + \sigma(N) > 2 - \frac{1}{p}$*. Rigidity forces three structural facts. The supports of* $M$ *and* $N$ *are* automatically *disjoint: a common prime* $q$ *would divide* $N' = Mp$ *through* $q \mid M$*, against* $\gcd(N, N') = 1$*. Likewise* $\gcd(p, MN) = 1$*: a prime shared by* $p$ *and* $M$ *propagates to* $N = M + M'p$ *and lands in the previous case, while one shared by* $p$ *and* $N$ *divides* $N'$*. And if* $p$ *is even, both members are odd:* $2 \mid N' = Mp$ *forces* $N$ *odd by the same lemma, while* $M$ *even would make* $N = M + M'p$ *even. The union of supports therefore avoids every prime divisor of* $p$*, and for even* $p$ *avoids* $2$*; exact greedy computation then gives, for* every *composite* $p$*,*

$$\omega(MN) \;\ge\; 59, \qquad MN \;>\; 8.77 \times 10^{112}$$

*the cycle barrier itself; with strictly stronger floors at every finite $p$: $\omega \geq 194$, $MN > 10^{497}$ at $p = 9$; $\omega \geq 230$, $MN > 10^{609}$ at $p = 4$; the even–$p$ floors rising towards the all–odd bound $10^{5040}$, and the infimum $8.77 \times 10^{112}$ approached only as $p \to \infty$ through composites free of prime factors $\leq 277$. Deleting primality thus lowers the barrier by* nothing*: the 59–prime, $10^{112}$ wall is carried entirely by the exactness equation, and the primality of $N'/M$ is a separate demand on top of it. (No witness, prime or composite, exists with $M \leq 120$, $p \leq 2000$, exhaustively.)*

**The decisive obstruction: a circular objective.** The heuristic yield of the divisor trick from one frame is

$$\mathbb{E}[\text{cycles}] \;\approx\; A \cdot \frac{1}{\ln r \,\ln p \,\ln q}, \qquad A = \#\{d \mid K:\; d \equiv -M_0 N' \pmod{\Delta_0}\} \;\approx\; \frac{\tau(K)}{\Delta_0},$$

the last step being an equidistribution heuristic. To make $\mathbb{E} \geq 1$ at the barrier scale ($1/\ln^3 \approx 5 \times 10^{-7}$) one needs $A \gtrsim 2 \times 10^6$ admissible divisors per frame, hence $\Delta_0 = O(1)$ with $\tau(K) \sim 10^6$. The natural objective is therefore "engineer frames with $\Delta_0$ small." But here is the obstruction, an exact identity:

$$\boxed{\Delta_0 \;=\; M_0 N\big(1 - s_{M_0}\, s_N\big), \qquad s_{M_0} = \sum_{p \mid M_0} \frac{1}{p}, \quad s_N = \sum_{p \mid N} \frac{1}{p}.}$$

So $\Delta_0$ is small *relative to* $M_0 N$ exactly when $s_{M_0} s_N \approx 1$ (which is the Erdős #307 condition for the frames themselves) and $\Delta_0 = O(1)$ in absolute terms at scale requires $s_{M_0} s_N = 1 - O(1/M_0 N)$, i.e. a reciprocal–product within $O(1/M_0 N)$ of 1, *strictly harder* than #307. The frame–engineering objective is thus circular: the Frame Rule is a faithful *parametrisation* of solutions, not a reduction in difficulty.

**Remark 8.5** (Honest status of the constructive program)**.** The Frame Rule (Theorem 8.1) and divisor trick (Proposition 8.3) are exact theorems; they make the existence of a solution *falsifiable in principle by finite computation per frame*, and they place #307 in the same constructive lineage as Thābit's and Euler's rules. They do not, as they stand, lower the difficulty below #307 itself: by the boxed identity the bottleneck (small $\Delta_0$) is the original problem in disguise, and a complete small–frame sweep ($2.2 \times 10^5$ frames) produces zero cycles with total expected count $< 1$, as the barrier demands. We regard the calculus as an exact parametrisation of solutions whose central obstruction we have now identified precisely; not a pathway that lowers the difficulty, and not evidence that a solution is within reach.

**Proposition 8.6** (No polynomial families)**.** *Let $p_1(t), \ldots, p_k(t)$, $q_1(t), \ldots, q_l(t)$ be integer polynomials, each taking* positive *prime values on a common infinite branch $t \to +\infty$ (a fixed–shape Schinzel–type family), such that $\big(\sum_i 1/p_i(t)\big)\big(\sum_j 1/q_j(t)\big) = 1$ for infinitely many $t$. Then every $p_i$ and $q_j$ is constant: the family is a single fixed solution. Consequently no polynomial parametrisation can reduce #307 to Bunyakovsky, Schinzel's Hypothesis H, or Bateman–Horn.*

Formalised. `Erdos307.const_prod_eq_one_of_tendsto` *is the limit step,* `eps_eq_zero_of_identity` *the vanishing step, and* `nonconstant_empty` *the conclusion that both index sets of nonconstant members are empty;* `nopoly` *chains them. The polynomial input enters only as the two hypotheses it supplies, that each nonconstant member contributes a strictly positive reciprocal and that the remainders tend to* 0*, so what is imported is visible.* 0 `sorry`*, three standard axioms* (`lean/Erdos307/NoPoly.lean`).

*Proof.* Holding at infinitely many $t$, the relation holds identically as rational functions. A nonconstant $p_i$ taking infinitely many prime (hence positive) values has positive leading coefficient on

an infinite branch, so $p_i(t) \to +\infty$ there; split each side into constants and nonconstant members, $\sum_i 1/p_i = A_P + \varepsilon_P(t)$ with $\varepsilon_P(t) \to 0^+$. Letting $t \to \infty$ in the identity gives $A_P A_Q = 1$ (in particular both sides contain constants), and subtracting, $A_P\varepsilon_Q(t) + A_Q\varepsilon_P(t) + \varepsilon_P(t)\varepsilon_Q(t) = 0$ for all large $t$. Every term on the left is nonnegative, and strictly positive as soon as one nonconstant member exists; hence there are none. □

# 9 A breeder form and the feasibility of certificate search

Allowing *two* free primes on one side turns the divisor trick into a cleaner bilinear form, Euler's amicable–pair method, in the shape used by te Riele's breeding algorithms [14], and lets us state precisely the strongest honest conclusion: a *conditional* reduction, which nonetheless does not become a feasible finite search.

**Proposition 9.1** (Bilinear breeder form)**.** *Fix coprime squarefree $M_0, N$ and seek a two–cycle $a = M_0 rp$, $b = Nq$ with $r, p, q$ distinct primes coprime to $M_0 N$. Put $G := NM_0 - N'M_0'$ $(= \Delta_0)$, $c := M_0 N'$, and $K := GN^2 + c^2$. Then the cycle equations $a' = b$, $b' = a$ are equivalent to*

$$(Gr - c)(Gp - c) = K, \qquad q = \frac{M_0 rp - N}{N'}.$$

*Proof.* Expanding, $(Gr - c)(Gp - c) - K = -N'G\,(a' - b)$ (verified symbolically), so the bilinear equation is equivalent to $a' = b$; the second cycle equation $b' = a$ then determines $q$ as stated. □

**Proposition 9.2** (The breeder is integral)**.** *Keep the notation of Proposition* 9.1 *and let $G \neq 0$. Then*

$$(Gr - c)(Gp - c) = K \iff G\,rp - c(r + p) = N^2,$$

*and every solution of that equation has*

$$N \mid a', \qquad q = \frac{M_0 rp - N}{N'} = \frac{a'}{N} \in \mathbb{Z}.$$

*Proof.* Expanding, $(Gr - c)(Gp - c) - c^2 = G\big(G\,rp - c(r + p)\big)$ while $K - c^2 = GN^2$, so the two equations differ by the factor $G \neq 0$. Substituting $G = NM_0 - N'M_0'$, $c = M_0 N'$, $a = M_0 rp$ and $a' = M_0' rp + M_0(r + p)$ into $G\,rp - c(r + p) = N^2$ gives $NM_0 rp - N'M_0' rp - M_0 N'(r + p) = N^2$, that is

$$N\,(a - N) \;=\; N'\big(M_0' rp + M_0(r + p)\big) \;=\; N'a'.$$

Hence $N \mid N'a'$, and $N$ squarefree gives $\gcd(N, N') = 1$, so $N \mid a'$ and $q = a'/N$. □

This is worth isolating because the displayed $q = (M_0 rp - N)/N'$ is a quotient by a quantity of the size of the barrier, and one might expect a candidate to be usable only when $N'$ happens to divide $M_0 rp - N$, a further demand of probability about $1/N'$, which would leave the construction hopeless for reasons having nothing to do with primality. There is no such demand. The expectation $\mathbb{E} \approx \big(\tau(K)/G\big)(\ln r \ln p \ln q)^{-1}$ is therefore the correct count: the breeder loses candidates only to primality, never to integrality. A scan of 51,132 frames (`code/breeder_integral.gp`) produced 9 admissible candidates, every one with integral $q$. Formalised in `lean/Erdos307/Breeder.lean`.

Each factorisation $K = d \cdot (K/d)$ with $d \equiv -c \pmod{G}$ proposes a candidate $(r, p) = \big((d + c)/G, (K/d + c)/G\big)$; a solution is any such candidate whose $r, p, q$ are all prime. A dual, partition–side form is occasionally useful:

**Proposition 9.3** (Subset form)**.** *For a support $U$ with $D = \prod_{p\in U} p$ and $\Sigma = \sum_{p\in U} D/p$, a partition $U = P \sqcup Q$ solves #307 iff $S = P$ satisfies $A(\Sigma - A) = D^2$, where $A := \sum_{p\in S} D/p$. Moreover $A(\Sigma - A) - D^2 = -(1 - s_P s_Q)\, D^2$.*

(Both identities verified symbolically and on $2 \times 10^4$ random supports.) The last identity says the "analytic" residual $1 - s_P s_Q$ and the "arithmetic" residual $A(\Sigma - A) - D^2$ are one equation up to the factor $D^2$: the two obvious steering handles carry no independent information.

**A conditional reduction.** Several ingredients of a construction are unconditional. Greedy Egyptian–fraction choice gives disjoint $P, Q$ with $0 < 1 - s_P s_Q = O(1/T)$ (solvability to any precision); a prime $r$ nearest $M_0 N'/\Delta_0$ gives $G \le \Delta_0^{0.475}(M_0 N')^{0.525}$ unconditionally (Baker–Harman–Pintz [15]), and each small prime $\ell$ can be forced to divide $K$ by one congruence on a free prime slot of $N$ (Dirichlet then supplies the prime). Granting these, #307 reduces to a single prime–density statement,

(H) among the admissible candidates so produced, some triple $(r, p, q)$ is prime,

and (H) $\Rightarrow$ #307 is YES. This (H) is a Hardy–Littlewood/Bateman–Horn–type assertion for an explicit, computable, *non–polynomial* family; we know of no named conjecture implying it. That is not an accident of our construction:

**Proposition 9.4** (No fixed–shape family, of any growth rate)**.** *Let $P(t) = \{p_1(t), \dots, p_k(t)\}$ and $Q(t) = \{q_1(t), \dots, q_l(t)\}$ be families of integers of* any *shape (no polynomial, algebraic or analytic hypothesis is imposed) indexed by $t \in \mathbb{N}$, with $k, l$ fixed. Suppose each member takes positive prime values, each nonconstant member tends to $+\infty$, and $\big(\sum_i 1/p_i(t)\big)\big(\sum_j 1/q_j(t)\big) = 1$ for* infinitely many *$t$. Then every member is constant.*

*Proof.* Let $\varphi$ enumerate, in increasing order, the $t$ at which the relation holds; $\varphi$ is strictly monotone, so $\varphi(n) \to \infty$. Split each side into its constant members, contributing $A_P$ and $A_Q$, and its nonconstant ones, contributing $\varepsilon_P(t), \varepsilon_Q(t) > 0$ which tend to 0 because each nonconstant member tends to $+\infty$ and there are finitely many. Along $\varphi$ the relation holds at every index and the remainders still tend to 0, so Proposition 8.6's two steps apply verbatim: the limit gives $A_P A_Q = 1$, and subtracting leaves $A_P\varepsilon_Q + A_Q\varepsilon_P + \varepsilon_P\varepsilon_Q = 0$ with every term nonnegative and some term strictly positive if a nonconstant member exists. □

The proof of Proposition 8.6 uses polynomiality in exactly one place, to pass from "holding at infinitely many $t$" to "holding identically", and restricting to the subsequence removes that need. What the argument actually consumes is three hypotheses (finitely many members, positive values, nonconstant members growing) and none of them is about shape.

The strengthening is not cosmetic. Excluding polynomial families rules out a reduction of #307 to Bunyakovsky, Schinzel or Bateman–Horn, but it says nothing about *exponential* families, and those are what the classical technique for problems of this kind actually uses: Thābit's rule for amicable pairs runs on primes $3\cdot 2^n - 1$, $3\cdot 2^{n-1} - 1$, $9\cdot 2^{2n-1} - 1$, and Euler's generalisation is of the same shape. A member $3\cdot 2^n - 1$ is positive and tends to infinity, so it satisfies the hypotheses above and the family is excluded. Proposition 9.4 therefore closes the whole Thābit/Euler programme for #307, not merely its polynomial shadow: no fixed–shape rule of any growth rate can generate solutions, and the constructive calculus of Section 8 is not failing for want of a cleverer ansatz.

Two limits on the statement should be read with it. The hypothesis that the families have *fixed* cardinality is used, in the splitting into constants plus a remainder tending to 0, and a family

whose support grows with $t$ is not covered; that gap is real, and harmless, since a growing support is a growing number of simultaneous primality demands rather than a weaker one. And positivity is a genuine hypothesis, not a consequence, for the reason recorded in Remark 9.5. Formalised as `Erdos307.no_family_of_any_shape` on three standard axioms, in the same shape as Proposition 8.6: the abstract core is proved, and the two facts the family supplies, that each nonconstant member contributes a strictly positive reciprocal and that the remainders tend to 0, enter as hypotheses rather than being derived, so what is imported stays visible. The theorem's statement quantifies over arbitrary index types and arbitrary $f_P : \mathbb{N} \to \iota \to \mathbb{R}$; no polynomial occurs anywhere in it.

**Remark 9.5** (Positivity is a hypothesis, not a consequence)**.** It is not enough to ask only that each polynomial take prime values for infinitely many integers $t$, without requiring the values to be positive on a common branch. Without positivity the statement is *false*: take $P = (2, 2, t, -t)$ and $Q = (2, 2)$. Each member takes prime values for infinitely many integers, and

$$\left(\tfrac{1}{2} + \tfrac{1}{2} + \tfrac{1}{t} - \tfrac{1}{t}\right)\left(\tfrac{1}{2} + \tfrac{1}{2}\right) = 1$$

identically, with two nonconstant members present. The proof's step "every term on the left is nonnegative" silently assumes positivity, which is exactly the missing hypothesis; with it, as now stated, the proof is correct and the counterexample is excluded.

The formalisation carries the repaired statement, not the printed one: `Erdos307.nonconstant_empty` takes $0 < f_P(i)$ for every nonconstant member as an explicit hypothesis, so the Lean development was already stating the corrected proposition. This is the one place in the paper where formalisation caught a missing hypothesis before review did.

This upgrades the earlier computational observation (that the breeder family is not polynomially parametrised) to a theorem for *all* fixed–shape families: the prime–density hypothesis behind #307 is irreducibly non–polynomial, which is why no standard conjecture applies.

**Proposition 9.6** (Near–miss frames are extremal for $G$)**.** *Let $a$ be squarefree with $a'$ squarefree, and take the frame $(M_0, N) = (a, a')$, so that $M_0' = a'$ and $N' = a''$. Then*

$$G \;=\; NM_0 - N'M_0' \;=\; a'\,(a - a''),$$

*and since $a'' \neq a$ unless #307 is solved, $|a - a''| \geq 1$ and therefore $|G| \geq a'$.*

*Proof.* Direct substitution; the inequality is $|a - a''| \geq 1$ for the nonzero integer $a - a''$. □

The point is not that the bound is large but that it runs the wrong way. Writing $r = a''/a$ for the near–miss ratio, $G = aa'(1 - r)$, so a frame built from a good near–miss has $1 - r$ small and $aa'$ correspondingly large, and the two cancel *exactly*, because $a(1 - r) = a - a''$ is an integer and a nonzero one. Improving the near–miss therefore does not shrink $G$; it drives $a'$ up and $G$ with it. The witness of Proposition 16.7, with $|1 - r| = 1.65 \times 10^{-13}$, gives $|G| \geq a' > 10^{112}$, against the $\log_{10} G \approx 13$ that the Baker–Harman–Pintz route reaches at the barrier. So the natural reading of "steer $\Delta_0$ small by steering $s_{M_0} s_N$ to 1" is not merely circular in the sense of Section 8: on near–miss frames it is inverted.

Small $|G|$ does occur, but only on small frames: over 1,143,888 coprime squarefree frames with $M_0, N \leq 2000$ the minimum is $|G| = 5$, at $(M_0, N) = (2, 3)$. That is the supply–demand conflict in raw form, and it can be measured. Binning by $|G|$ and taking the best frame in each bin (`code/breeder_supply.gp`) gives

| $\log_{10} \lvert G \rvert$ | 0.5 | 1.5 | 2.5 | 3.5 | 4.5 | 5.0 |
|---|---|---|---|---|---|---|
| $\max \tau(K)/\lvert G \rvert$ | 0.600 | 0.313 | 0.214 | 0.053 | 0.0065 | 0.0030 |

The ratio is largest on the smallest frames, where it is already below 1, and decays monotonically from there. Since a productive frame needs $\tau(K)/|G|$ well above 1, no frame in the range supplies one, and the trend is against scale rather than with it.

**The descent, and why it cannot be iterated.** Adjoining a prime $w$ to $M_0$ acts on $G$ *linearly*: with $M_0 \mapsto M_0 w$ one has $M_0' \mapsto M_0' w + M_0$ and hence

$$G \;\longmapsto\; wG - M_0 N', \qquad \text{so} \qquad |G_{\text{new}}| = |G| \cdot |w - w^*|, \qquad w^* := \frac{M_0 N'}{G} = \frac{\sigma(N)}{\varepsilon},$$

writing $\varepsilon = 1 - \sigma(M_0)\sigma(N) = G/(M_0 N)$. Since $G = 0$ is #307, this is a descent towards a solution, and on the $\varepsilon$ scale it is quadratic: $\varepsilon_{\text{new}} = \varepsilon^2 \theta / \sigma(N)$ with $\theta = w - w^*$. It is a Newton iteration for #307.

Three facts close it. First, it cannot start below the barrier: $w^* = \sigma(N)/\varepsilon \geq 2$ needs $\varepsilon \leq \sigma(N)/2$, i.e. $\sigma(M_0)\sigma(N)$ near 1, which by AM–GM needs $\sigma(M_0 N) \geq 2$ and so $\omega(M_0 N) \geq 59$. Over 4,284,896 coprime squarefree frames with $M_0 \leq 5000$, $N \leq 3000$ the smallest $|\varepsilon|$ is 0.470, and $w^* < 2$ throughout.

Second, only $w = \lfloor w^* \rfloor$ or $\lceil w^* \rceil$ can shrink $|G|$, since $|G_{\text{new}}| < |G|$ demands $|w - w^*| < 1$. So a step descends only when one of those two integers is prime, with probability $\approx 2/\log w^*$.

Third, and decisively, a successful step squares the difficulty. From $w^* = M_0 N'/G$,

$$w^*_{\text{new}} = \frac{M_0 w N'}{G\theta} = w^* \cdot \frac{w}{\theta} \;\approx\; \frac{(w^*)^2}{\theta},$$

so $\log w^*$ more than doubles at every step while $\log|G|$ falls by only $|\log\theta| \approx 0.69$. The gain is linear in the number of steps $k$ and the cost is quadratic in the exponent: $|G|$ falls like $2^{-k}$, whereas the probability of surviving $k$ steps is $\prod_{j<k} 2/\log w_j^* \asymp 2^{-k^2/2}$. Starting from the greedy frame $N = 30$, $M_0 = \prod_{7 \leq p \leq 271} p$, where $\log|G| = 249$ and $w^* = 428.2$ (`code/breeder_descent.gp`):

| paths available | $10^2$ | $10^4$ | $10^6$ | $10^8$ | $10^{10}$ | $10^{12}$ |
|---|---|---|---|---|---|---|
| reachable depth $k$ | 3 | 5 | 6 | 7 | 8 | 8 |
| $\lvert G\rvert$ shrinks by | 8 | 32 | 64 | 128 | 256 | 256 |

against the factor $10^{108}$ required to reach $G = 0$. Even $10^{12}$ independent descent paths buy eight steps and a factor 256. This is the quantitative form of the circularity of Section 8: the Newton iteration for #307 exists, is exact, and converges quadratically, and each step costs a prime whose size is the square of the last one's.

**Why** (H) **is not a finite search.** It is tempting to read (H) as "run a certificate hunt that is expected to succeed." It is not, for a quantitative reason. The expected number of certificates from one frame is $\mathbb{E} \approx \big(\tau(K)/G\big) \cdot (\ln r \, \ln p \, \ln q)^{-1}$, and *both* factors are pinned by the geometry.

- *Demand.* $G = \Delta_0 = M_0 N(1 - s_{M_0} s_N)$ exactly, so $G$ is small only in the circular regime $s_{M_0} s_N \approx 1$; over $2 \times 10^4$ random coprime squarefree frames the ratio $G/(M_0 N)$ has median 1 and best value $\approx 10^{-0.23}$, and the honest Baker–Harman–Pintz reduction still leaves $\log_{10} G \approx 13$ at the barrier.

- *Supply.* $\tau(K)$ is bounded by the *size* of $K = GN^2 + c^2$ ($\geq c^2$), which is an *output* of the frame, not a free dial. The maximal order of the divisor function [18, Thm. 317] caps $\log_2 \tau(K)$ at $\approx 36$ for $K \leq 10^{54}$ (the barrier geometry) and at $\approx 112$ even for $K \leq 10^{240}$; reaching $\tau(K) = 2^{120}$ needs $K \gtrsim 10^{263}$, i.e. frame scale far beyond the barrier.

Enlarging a frame grows $K \propto N^2$ but gains divisors only at rate $\log 2/\log\log K < 1$ per log–unit of $K$, while it grows $G$ at rate 1 per log–unit of $N$; so $\tau(K)/G$ cannot be driven up. Optimising $\mathbb{E}$ over all $(M_0, N)$ splits with geometry–consistent $K$ and the maximal–order $\tau$, and granting the most favourable demand and singular–series constants, yields $\log_{10}\mathbb{E} \le -4$ at the barrier and *decreasing* with scale; empirically 385 factorable frames realise a median of 0 and a maximum of 1 admissible divisor, against the $\sim 10^5$ a single productive frame would need. A self–consistent accounting; capping the number $w$ of forced prime factors of $K$ by the frame's own residue freedom (counted generously), sharpens this to $\log_{10}\mathbb{E} \le -8$ at $B = 56$, falling to $\approx -260$ at $B = 3000$: the deficit is *scale–invariant*, so no choice of scale, slot count, or budget split reaches expectation 1. (The construction is even self–similar: engineering $K$'s algebraic; e.g. Gaussian–integer; factorisation to supply divisors reduces to prescribing $N$ and $N'$ simultaneously, an arithmetic–antiderivative inverse problem of the same difficulty class as #307.) The breeder thus makes #307 conditional on (H) but does *not* render it a feasible finite computation: the supply–demand conflict through $K = GN^2 + c^2$ is exactly the circularity of Section 8 in quantitative form. There is also a structural reason the analogy with amicable pairs was bound to fall short: te Riele's breeding amplifies a *known seed* pair into new ones, whereas the barrier (Theorem 4.3) forbids any seed below $2.09 \times 10^{56}$ the method imports a precondition the problem cannot supply.

## 10 Computations

All verdicts use exact integer/rational arithmetic; floating point is used only for pre–screening.

- *Exhaustive non–existence below* $10^8$. Enumerating all prime sets $P$ with $\sum 1/p > 1$ and $\prod P \le 10^8$ (forcing $Q$ as the prime support of $N_P$ and testing $N_Q = D_P$): 1,075,419 candidate sets, no solution. Subsumed by Theorem 4.3 but an independent sanity check.

- *Derivative sieve.* No solution of $n'' = n$ with $n' \ne n$ and both members $\le 2 \times 10^7$ (smallest–prime–factor sieve of the derivative).

- *Function field.* The arithmetic derivative on $\mathbb{F}_q[t]$ has no two–cycles and no nonzero fixed points, verified exhaustively over $\mathbb{F}_2$ (degree $\le 7$), $\mathbb{F}_3$ ($\le 5$), $\mathbb{F}_5$ ($\le 4$), confirming Theorem 7.1.

- *Density.* The sieve gives density 0.042029 to $2 \times 10^7$ (0.017623 among squarefree); the independent Erdős–Wintner convolution over primes $\le 10^5$ gives $\delta = 0.04202$, agreeing to four places (Proposition 6.5).

- *Conditional mass.* Of 43,087 qualifying squarefree $a$ sampled, $b = a'$ is squarefree 95.0% of the time; on the resulting 40,920 squarefree $b$ the mean reciprocal mass is 0.047 vs. the required $1/s \approx 0.934$, with empirical tail probability $\Pr[\sum_{q|b} 1/q \ge 1/s] = 0/40{,}920$.

- *Frame/ledger sweep.* Over $2.2 \times 10^5$ factorable small frames: the divisor–trick identities hold on every instance; the smallest $\Delta_0$ observed is 22; the equidistribution $A \approx \tau(K)/\Delta_0$ holds only in aggregate ($\sum A = 127$ vs. $\sum \tau(K)/\Delta_0 = 39.8$) and fails per frame ($A = 0$ for 99.95% of frames); total expected cycles $< 1$; zero cycles, as the barrier requires.

- *$\mu$–Sondow line census.* Enumerating the lines $n' = n \pm d$ (squarefree members coprime to $d$) finds no adjacent opposite–sign pair at additive distance $d$: exhaustively to $2 \times 10^7$ for $|d| \le 200$, and along the small–$|\delta|$ genealogy tree out to members of more than a thousand digits. Every $\delta = \pm 1$ member encountered is a known Giuga or primary pseudoperfect number; the closest opposite–sign approach recorded is $|y - (x + d)| = 11$ (at $d = 1$, 30 vs. 42, and $d = 37$, 210 vs. 258), attained only among small members; the two trees do not come close at height.

Code and logs accompany this note.

## 11 Formalisation in Lean 4

The rigidity and barrier results are formalised against mathlib (Lean 4, toolchain `v4.30.0`). The following are machine–checked and `sorry`–free, depending only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`:

- Lemma 2.1 (`rigidity_coprime`) and Theorem 2.3 (`solution_structure`);
- Lemma 4.1 (`reciprocal_sum_gt_two`); the exact thresholds $\sum_{\text{first } 58} 1/p < 2 < \sum_{\text{first } 59} 1/p$ grounding $|U| \geq 59$ (`recipSum58_lt_two`, `recipSum59_gt_two`); the numeric core $\Pi_{59}^2 \geq (4 \times 10^{112})N_{59}$ (`barrier_numeric`), and the algebraic core of the barrier (`barrier`), giving $D_P^2 \geq 4 \times 10^{112}$;
- Theorem 16.29 (`log_lt_half_sub`, `delta_decreasing`, `two_le_add_of_mul_eq_one`) and Theorem 16.33 (`criterion_fails_of_pos`, `criterion_fails_of_nonpos`, `no_lambda_works`, `lyapunov_criterion_fa` neither of which uses `native_decide`;
- Proposition 16.36 (`half_works`, `no_f_above_two`, `mass_sum_threshold`) and Proposition 16.37 (`exists_f_of_injective`, `increment_iff`, `no_two_cycle_of_decreasing`), which together machine-check the closure of the sign branch in Theorem 16.31;
- Proposition 16.22 (`exists_block_sum_near`, `infimum_zero`);
- Theorem 16.31(1) in its mechanism (`Erdos307.Congruential`) and Proposition 17.21 (`live_slot_threshold`, `threshold_bracket`).

The single non–elementary ingredient, that the $k$ smallest primes minimise $\prod$ and maximise $\sum 1/p$ among $k$–element prime sets, is isolated as an explicit hypothesis of `barrier` (the ratio bound $R/T(U) \geq \Pi_{59}/T_{59}$), so the dependency is transparent rather than hidden. `Erdos307.Extremal` proves it (`prod_first_primes_le`, `recipSum_le_first_primes`, `hRatio_of_extremal`), which is why Theorem 4.3 appears above in the hypothesis-free form `erdos307_barrier_closed`.

On the trust boundary: there is none beyond the logic. Every declaration in the development, without exception, depends only on `propext`, `Classical.choice` and `Quot.sound`; there is no `native_decide` site and no `ofReduceBool`.

The one that resisted was the pruned search behind Proposition 4.5. `dfs` compares rationals, and the Lean kernel cannot reduce a `Rat` comparison at all, since `Nat.gcd` is well-founded recursion rather than a GMP primitive; `decide` does not merely run slowly on it, it fails to evaluate. Three changes move the execution into the kernel. Scaling by $M = \prod(\texttt{forced39} \cup \texttt{pool99})$, a 327–digit integer, clears every denominator *exactly*: each $1/p$ becomes $M/p$ and thr becomes the integer $M \cdot \text{thr}$, so the scaled search branches identically and the soundness theorem `dfs_sound` is reused unchanged. The leaf test replaces `Nat.sqrt` on a 130–digit number, some hundreds of structural steps, by a residue certificate: eleven moduli $\{8, 19, 167, 127, 149, 67, 59, 101, 73, 191, 41\}$, with their quadratic residues stored as bitmasks, cover all 49,961 supports, and a non–residue is a proof of non–squareness (`lean/Erdos307/NonSquare.lean`). Finally the product and cofactor sum are carried down the recursion rather than rebuilt at each leaf. The kernel then runs the whole search in about a minute (`dfsA_run`, `dfs_of_dfsA`).

Proposition 4.5's product bounds are proved in this paper and not in Lean: `erdos307_sixty` establishes $|P \cup Q| \geq 60$ alone.

## 12 Why every known coprime example uses 1

Corollary 4.6 shows the weakened variant is no smaller than #307 itself: with $1 \notin P \cup Q$ a solution needs 60 pairwise coprime elements, and at 59 every element is forced prime. What that leaves unexplained is the shape of the examples that *are* known, $1 = (1 + \frac{1}{5})(\frac{1}{2} + \frac{1}{3})$, and the family generated by primary pseudoperfect numbers such as $1806 = 2 \cdot 3 \cdot 7 \cdot 43$ giving $1 = (1 + \frac{1}{1805})(\frac{1}{2} + \frac{1}{3} + \frac{1}{7} + \frac{1}{43})$. Every one of them uses the element 1. That is not an accident of search.

**Proposition 12.1** (No coprime family has reciprocal sum 1)**.** *Let $A$ be a finite set of pairwise coprime integers $\geq 2$. Then $\sum_{a \in A} 1/a \neq 1$.* 0 *sorry* (`Erdos307.Coprime.sum_inv_ne_one`).

*Proof.* Suppose $\sum_{a \in A} 1/a = 1$ and fix $a \in A$; put $R = \prod_{b \in A,\, b \neq a} b$. Multiplying by $R$ gives $R/a + \sum_{b \neq a} R/b = R$. Each term of the sum is an integer, since $b \mid R$ for $b \neq a$, and so is $R$; hence $R/a \in \mathbb{Z}$, i.e. $a \mid R$. But $a$ is coprime to every $b \neq a$, so $\gcd(a, R) = 1$ and $a = 1$, contradicting $a \geq 2$. □

**Remark 12.2.** Every known example of the weakened variant has one factor equal to $1 + 1/n$ and the other equal to $n/(n+1)$, i.e. one side is a coprime family whose reciprocal sum is $1 - 1/(n+1)$ and which is completed to 1 only by adjoining the element 1. Proposition 12.1 says the adjunction is unavoidable: the element 1 contributes a full unit to the sum while dividing nothing, and it is the only element that can, because for every other element the divisibility above closes. So the known examples are not near misses for the open form, they are on the other side of a divisibility, and no refinement of them approaches $1 \notin P \cup Q$.

The rigidity step that Corollary 4.6 inherits from the prime case is the same divisibility in cofactor form: for a coprime family $A$ with product $\alpha$, the numerator $M = \sum_{a \in A} \alpha/a$ satisfies $\gcd(M, a) = 1$ for every $a \in A$, because every term but $\alpha/a$ is divisible by $a$ while $\alpha/a = \prod_{b \neq a} b$ is coprime to it. This is `Erdos307.Coprime.coprime_numerator`, and it closes the one step of Corollary 4.6's proof that was previously carried in prose.

### 12.1 A rejection sieve that factors nothing

The obvious search for a 1–free coprime example fixes one side and reads off the other: a coprime family $A$ with product $\alpha$ determines its partner $\beta = \sum_{a \in A} \alpha/a$, and $B$ must then be a coprime factorisation of $\beta$ with $\sum_{b \in B} 1/b = \alpha/\beta$. The recorded obstacle is that testing a candidate appears to require factorising $\beta$, which at the sizes Corollary 4.6 forces is out of reach. It does not.

**Proposition 12.3** (A necessary condition, computable in small modular arithmetic)**.** *Let $A$ be a coprime family with $\alpha = \prod A$, $x = \sum_{a \in A} 1/a$ and $\beta = \sum_{a \in A} \alpha/a$. If some coprime family $B$ satisfies $\left(\sum_A 1/a\right)\left(\sum_B 1/b\right) = 1$, then*

$$y_{\max}(\beta) \;:=\; \sum_{p^e \,\|\, \beta} \frac{1}{p^e} \;\geq\; \frac{1}{x}.$$

*Moreover $y_{\max}(\beta)$ is computable without forming or factorising $\beta$: for every prime power $p^e$ with $p \nmid \alpha$, Lemma 17.1 gives $\beta \equiv \alpha \sum_{a \in A} a^{-1} \pmod{p^e}$, so each $v_p(\beta)$ is decided by arithmetic modulo $p^e$ alone.*

*Proof.* Write $y = \sum_B 1/b = N/\gamma$ with $\gamma = \prod B$ and $N = \sum_{b \in B} \gamma/b$. As in Remark 12.2, $\gcd(\beta, \alpha) = \gcd(N, \gamma) = 1$, so $xy = 1$ reads $\beta N = \alpha\gamma$ with $\beta \mid \gamma$ and $N \mid \alpha$, forcing $\gamma = \beta$ and $N = \alpha$. Hence $B$ is a coprime factorisation of $\beta$ and $\sum_B 1/b = \alpha/\beta = 1/x$. Every coprime

factorisation is obtained from the factorisation into prime powers by merging blocks, and merging strictly decreases the reciprocal sum, since $1/(mn) < 1/m + 1/n$ for $m, n \geq 2$ (0 `sorry`, `Erdos307.Coprime.merge_loses`). Hence the prime–power factorisation is the maximiser and $1/x = \sum_B 1/b \leq y_{\max}(\beta)$. The congruence is Lemma 17.1 with $r = p^e$, legitimate because $p \nmid \alpha$. □

The test is therefore free: primes above a trial bound $Z$ contribute below $1/Z$ apiece, so trial moduli up to $10^5$ already pin $y_{\max}$ to within $10^{-5}$, using no integer larger than $10^{10}$. Verified against direct multiprecision evaluation of $v_p(\beta)$ over 200 random families and all $p \leq 500$: 13,065 valuations, 0 mismatches (`code/beta_smoothness.rs`, checked by the script in the same file's header).

What the sieve says is stark. Over 5,272 splits of the first 60 primes lying in the mass window $|xy - 1| \leq 10^{-2}$, the mean of $y_{\max}(\beta)$ is 0.1795 and its maximum is 0.8371, against a requirement of $1/x \approx 0.95$: *not one candidate survives* (`code/beta_smoothness.rs`). The mechanism is a tension between the two constraints. Reaching $x \approx 1$ obliges $A$ to hold small primes; but $\gcd(\alpha, \beta) = 1$ bars every prime of $A$ from dividing $\beta$, and $y_{\max}$ is carried almost entirely by the small primes. The mass window and the smoothness requirement pull on the same scarce resource.

The same tension defeats the complementary route, which uses Lemma 17.1 in the forward direction: $b \mid \beta$ is equivalent to $\sum_{a \in A} a^{-1} \equiv 0 \pmod{b}$, a subset–sum condition on $A$ that again needs no factorisation, so one may try to *impose* the divisibilities $\beta$ must satisfy. Annealing on the forced divisor $D(A) = \prod\{b \in B : b \mid \beta\}$ over ground sets of 60 to 70 primes reaches $\log_{10} D \approx 10$ against $\log_{10} \beta \approx 50$–$75$ in a window $|xy - 1| \leq 10^{-2}$, and degrades to $\log_{10} D \approx 3$ as the window tightens to $10^{-7}$, below which no split is found at all (`code/split_forcing.rs`). Since a random split already satisfies $\sum_{b \in B} 1/b \approx 1$ of the congruences by chance, the search inside the mass window is performing at the random baseline: the freedom the congruence route needs is exactly the freedom the mass window removes.

### 12.2 Candidates that are smooth by construction

Both failures above share an order of operations: fix a ground set $U$, split it, and hope $\beta$ comes out smooth. Reversing the order removes the obstruction. Choose the small primes $B_0$ that are to divide $\beta$, draw $A$ from primes above $\max B_0$, and *impose* $\sum_{a \in A} a^{-1} \equiv 0 \pmod{q}$ for each $q \in B_0$. Two points make this work where the earlier routes did not. First, the conditions are congruences on $A$ alone, so they can be solved exactly rather than sampled, we use meet–in–the–middle over disjoint swap pairs drawn from adjacent primes, so the mass moves by less than $10^{-4}$. Second, and decisively, $B$ is no longer the complement of a fixed ground set but whatever factors $\beta$; the rigid window $x + y = T(U)$ of Section 12.1 therefore does not apply, and the only surviving requirement is $x > 1$.

Two obstructions appear and both are removable. Every pool prime is odd, so $a^{-1} \equiv 1 \pmod{2}$ and the residue mod 2 is pinned to $|A| \bmod 2$: no swap can move it, and $|A|$ must be even. And forcing $q \mid \beta$ does not prevent $q^2 \mid \beta$, in which case the block contributes $1/q^2$ in place of $1/q$, at $B_0 = \{2, \ldots, 41\}$ an unconstrained solution came out with $2^5 \parallel \beta$ and $3^3 \parallel \beta$, which alone costs 0.76 and fails the sieve. Imposing $q \parallel \beta$, tested by the same identity taken modulo $q^2$, repairs it.

**Proposition 12.4** (The necessary condition is constructible)**.** *For every $B_0$ among the first 8 to 12 primes there are coprime families $A$ of primes above $\max B_0$ with $x = \sum_A 1/a > 1$ such that each $q \in B_0$ divides $\beta$ exactly once. Such $A$ satisfy $y_{\max}(\beta) \geq \sum_{q \in B_0} 1/q > 1 > 1/x$, hence pass Proposition 12.3.*

Verified in exact multiprecision arithmetic at $B_0 = \{2, \dots, 37\}$: $|A| = 3290$, largest member 31,063, $\log_{10} \alpha = \log_{10} \beta = 13{,}163.1$, $\gcd(\alpha, \beta) = 1$, every forced prime exactly dividing $\beta$, and

$$y_{\max}(\beta) = 1.5927\ldots \;\geq\; 1/x = 0.99597\ldots$$

(`code/smooth_by_construction.rs`). Against 0 of 5,272 for the sampled search, the construction passes at every depth from 8 to 12; depth 13 needs modulus $3.0 \times 10^{14}$ and more swap pairs than the $2^{24}$ table cap admits.

**Remark 12.5.** The two routes are worth separating. The *factorisation* obstacle is genuinely gone, both tests run in arithmetic modulo $10^{10}$ and never form $\beta$, and Proposition 12.3 is a sound, essentially free first pass that any future search should run. What replaces it is a harder obstacle: inside the mass window the candidates are not merely hard to test, they are uniformly far from the target, by a margin of 0.77 in $y_{\max}$ at the observed best. A search for a 1–free coprime example must therefore generate candidates that are smooth *by construction* rather than filtered for smoothness after the fact. Section 12.2 gives such a construction. What it does not give is a test of sufficiency: deciding whether some coprime factorisation of $\beta$ sums to exactly $\alpha/\beta$ needs the factorisation of $\beta$, whose unfactored part in the verified instance has 13,151 digits. Factoring returns, but only at the last step, and it no longer blocks the generation of candidates.

### 12.3 The two–member side, and what every known example really is

The searches above treat both sides symmetrically. They need not be. Suppose one side has exactly two members, $A = \{u, v\}$. Then $\alpha = uv$ and $\beta = u + v$, so $u$ and $v$ are the roots of $z^2 - \beta z + \alpha = 0$: *$A$ is recovered from $(\alpha, \beta)$ by a quadratic, with no factorisation at all.* The roots are automatically coprime, since $\gcd(u, v)$ divides $\gcd(u + v, uv) = \gcd(\beta, \alpha) = 1$.

**Proposition 12.6** (The two–member reduction)**.** *Let $B$ be a coprime family, $\beta = \prod B$, and let $u \geq 1$ be an integer with $u < \beta$. Then $\{u, \beta - u\}$ and $B$ satisfy $\big(\sum_A 1/a\big)\big(\sum_B 1/b\big) = 1$ if and only if*

$$D(B) \;=\; u(\beta - u), \qquad \textit{that is} \qquad \sum_{b \in B} \frac{1}{b} = \frac{u(\beta - u)}{\beta}.$$

0 *sorry* (`Erdos307.Coprime.two_element_identity`).

*Proof.* $\sum_{a \in A} 1/a = 1/u + 1/(\beta - u) = \beta/\big(u(\beta - u)\big)$, and $\sum_B 1/b = D(B)/\beta$. The product is $D(B)/\big(u(\beta - u)\big)$, which is 1 exactly when $D(B) = u(\beta - u)$. □

The value of $u$ decides everything, and it identifies the known examples.

$u = 1$. The condition is $D(B) = \beta - 1$, which is precisely the *primary pseudoperfect* condition, and $A = \{1, \beta - 1\}$. This is $(1 + \frac{1}{5})(\frac{1}{2} + \frac{1}{3})$ from $\beta = 6$ and $(1 + \frac{1}{1805})(\frac{1}{2} + \frac{1}{3} + \frac{1}{7} + \frac{1}{43})$ from $\beta = 1806$; the Sylvester products $2, 6, 42, 1806$ are pseudoperfect because Sylvester's recursion maintains $\beta - D(B) = 1$ identically. So the whole recorded family of solutions is the case $u = 1$, and it contains the member 1 for the structural reason that $u$ *is* that member.

This is an equivalence, not merely a supply of examples, and it links two of Erdős's problems. That a primary pseudoperfect number yields a solution of the relaxation is an observation of Cambie on the problem page; Proposition 12.6 supplies the converse, since it derives $D(B) = u(\beta - u)$ as a *criterion*.

**Corollary 12.7** (#307's relaxation at $u = 1$ is #313)**.** *The two–member solutions of the coprime relaxation of #307 that contain the element* 1 *are exactly the primary pseudoperfect numbers:* $\big(\{1, \beta - 1\}, B\big)$ *solves the relaxation if and only if* $\beta = \prod B$ *satisfies* $\sum_{q|\beta} 1/q + 1/\beta = 1$*. Consequently Erdős Problem #313, the infinitude of primary pseudoperfect numbers [27], is equivalent to the statement that the relaxation of #307 has infinitely many two–member solutions containing* 1*.*

*Proof.* Proposition 12.6 at $u = 1$ reads $D(B) = \beta - 1$, and dividing by $\beta$ gives $\sum_{q|\beta} 1/q = 1 - 1/\beta$, which is the primary pseudoperfect condition. Both directions are the same computation. □

The identity $D(B) = u(\beta - u)$ also names the right generalisation. Writing $\partial$ for the arithmetic derivative, [25] calls $(R, c)$ a *port* and says a squarefree $B$ *fills* it when $cB - R\,\partial(B) = 1$; the case $R = 1$, $c = 1$ is #313. Our condition is $u\beta - \partial(\beta) = u^2$, that is, the port $(1, u)$ filled at value $u^2$ rather than 1. The 1–free case of the relaxation is therefore a port–filling problem one step outside the range that #313 occupies, which is where any machinery developed for #313 would have to be pushed.

**Proposition 12.8** (Only the value one sustains itself)**.** *For squarefree* $B$ *and* $c \geq 1$ *put* $\Delta_c(B) = cB - \partial(B)$*. Then* $\gcd(\Delta_c(B), B) = 1$*, and appending a prime* $q \nmid B$ *gives* $\Delta_c(qB) = q\,\Delta_c(B) - B$*. Consequently a step returns to the same value,* $\Delta_c(qB) = \Delta_c(B)$*, only when* $\Delta_c(B) = 1$*, and then* $q = B + 1$*.* 0 ***sorry*** (`Erdos307.port_fixed_value`)*.*

*Proof.* A prime $p \mid B$ dividing $\Delta_c(B)$ would divide $\partial(B)$, contradicting Theorem 2.3; so $\gcd(\Delta_c(B), B) = 1$. The recursion is $\partial(qB) = B + q\,\partial(B)$. Fixing the value gives $\Delta_c(B)(q - 1) = B$, so $\Delta_c(B) \mid B$, and coprimality forces $\Delta_c(B) = 1$. □

**Proposition 12.9** (The ten–factor primary pseudoperfect number does not extend by one or two primes)**.** *Let* $N_{10} = N_9(N_9 + 1) = 359793511891993165345874739057735720 06$ *be the primary pseudoperfect number with ten prime factors of [25]. There is no prime* $p \nmid N_{10}$ *with* $N_{10}p$ *primary pseudoperfect, and no pair of distinct primes* $p, q \nmid N_{10}$ *with* $N_{10}pq$ *primary pseudoperfect.* 0 ***sorry*** (`Erdos307.N10_no_one_prime`, `N10_no_two_prime`; `lean/Erdos307/PPNInherit.lean`)*.*

*Proof.* The idea is that inheritance is a factorisation problem rather than a search: the new primes are pinned to a divisor pair of one fixed integer, so exhausting that integer's divisors decides the question. Write $K = N_{10}$, so $K' = K - 1$. By Leibniz, $(Kp)' = K + p(K - 1)$, and $(Kp)' = Kp - 1$ forces $p = K + 1$, which is divisible by 7. For two primes, $(Kpq)' = Kpq - 1$ is equivalent to $(p-K)(q-K) = K^2+1$. Both factors have the same sign; if both are negative then $(K-p)(K-q) \leq (K-2)^2 < K^2 + 1$, so both are positive and $d = \min(p, q) - K$ is a divisor of

$$K^2+1 = 21807157 \cdot 480382349 \cdot P_{60}, \qquad P_{60} = 123572138719194583969192220095883252267503088389616114960309$$

all three factors prime. Hence $d \in \{1,\ 21807157,\ 480382349,\ 21807157 \cdot 480382349\}$, and $K + d$ is divisible by $7, 7, 5, 2141$ respectively, so it is not prime. The primality of $P_{60}$ and of $N_9+1$ is certified by Pratt trees whose leaves lie below $10^9$ (`lean/Erdos307/PrattN10.lean`, `code/pratt_n10.gp`). □

The proposition is the frontier case of a routine that is validated on known values before it is used there: the same criterion, run on every primary pseudoperfect number with at most ten prime factors, recovers 42 from 2, 1806 from 6, and the eight–factor $2214502422 \cdot 2217342227 \cdot 1729101023519$ from 2214502422, and returns nothing at $N_9$ or $N_{10}$ (`code/ppn_inherit.py`). It bears on the conditional infinitude theorem of [25], whose hypothesis splits a terminal prime into five: at the current frontier the one– and two–prime steps are exhausted rather than unsearched.

Three primes would require factoring $K^2x^2 + x - K$, about 150 digits, for each candidate prime $x > K$.

*Why #313 has a family and the* 1*–free case has none.* Proposition 12.8 explains the asymmetry between Corollary 12.7's two sides. At $u = 1$ the target value is 1, which is a fixed point of the port step: from $\Delta = 1$ the choice $q = B + 1$ returns $\Delta = 1$, and iterating from $B = 2$ gives $2, 6, 42, 1806, \ldots$, Sylvester's sequence, whose partial products are exactly the known primary pseudoperfect numbers. The family costs nothing because the recursion sits still.

At $u \geq 2$ the target is $u^2$, and by the proposition *no* value other than 1 is self-sustaining, so there is no orbit to sit on: every step must be steered, and the deficit $\Delta$ wanders. This is the structural content of the failed descent recorded earlier, over 399 seeds the recursion $\Delta \mapsto q\Delta - B$ never reached a divisor of $B + 4$, the smallest $\Delta$ attained being $2.7 \times 10^6$ (`code/two_side_descent.gp`). The search was not unlucky. Verified: 23,916 pairs $(B, c)$ with $\gcd(\Delta_c(B), B) = 1$ and 0 failures, 14,920 checks of the recursion with 0 failures, and among all self-sustaining steps found in a sweep, every one had $\Delta = 1$ (`code/port_recursion.gp`).

The consequence for transfer is specific. Machinery built for #313 targets the value 1 and may use its stability; anything of that kind does not carry to $u \geq 2$, where the stability is provably absent. What does carry is the recursion itself, which is value–blind.

Proposition 12.8 is sharp at period one only. A two–step orbit $\Delta \mapsto q_1\Delta - B \mapsto \Delta$ is possible above the value 1, and requires exactly $\Delta \mid q_1 + q_2$ (0 `sorry`, `Erdos307.port_period_two`); at $c = 1$ such orbits occur, for instance $B = 33$ with $\Delta : 19 \mapsto 5 \mapsto 19$ at $q_1 = 2$, $q_2 = 17$. They are single events rather than families, since returning a second time needs a fresh pair: the condition ties the primes to $B$. In the regime that the 1–free case actually requires, $c = 2$ with $B$ and both primes odd, a sweep of 162,110 states found none (`code/port_orbit.gp`).

**Proposition 12.10** (The barrier survives relaxing the equations to divisibilities)**.** *Let $P, Q$ be disjoint finite sets of primes and suppose only that*

$$\prod Q \;\mid\; D(P) \qquad \text{and} \qquad \prod P \;\mid\; D(Q),$$

*in place of the two equalities a two–cycle requires. Then $\sigma(P)\sigma(Q) \geq 1$, hence $\sigma(P) + \sigma(Q) \geq 2$, and therefore $|P \cup Q| \geq 60$ and $\min(\prod P, \prod Q) \geq 2.09 \times 10^{56}$ as in Theorem 4.3.* 0 ***sorry*** (`Erdos307.mass_from_divisibility`).

*Proof.* Divisibility between positive integers gives $\prod Q \leq D(P) = \sigma(P) \prod P$ and $\prod P \leq D(Q) = \sigma(Q) \prod Q$. Multiplying and cancelling the positive quantity $\prod P \prod Q$ leaves $1 \leq \sigma(P)\sigma(Q)$, and the arithmetic–geometric mean inequality gives $\sigma(P) + \sigma(Q) \geq 2$, which is the only input Theorem 4.3 consumes. □

*What the relaxation localises.* The relaxation is worth recording because it is natural to expect the mass condition to be an artefact of the *exact* equations, so that weakening them would open room below level 60. It does not: the divisibilities alone already force the mass, because a divisibility between positive integers is an inequality, and the two inequalities point in opposite directions. Consequently the difficulty in #307 is not located in the passage from "$\prod Q$ divides $D(P)$" to "$\prod Q$ equals $D(P)$"; both carry the same barrier.

At $|P| = |Q| = 2$ the relaxed system has no solutions at all among the primes below 283 (2,925,810 pairs, 0 solutions, `code/relax_divisibility.gp`), and Proposition 12.10 says why: four primes cannot carry mass 2. A random search for relaxed solutions is therefore vacuous rather than evidential, and the proposition is checked instead on the arithmetic it actually asserts, over 200,000 random positive quadruples with 0 counterexamples.

**Corollary 12.11** (The balanced splits are exactly the forbidden ones)**.** *In the situation of Corollary 4.13, a split $S = P_0 \sqcup Q_0$ can support a solution only if $\sigma(P_0)\sigma(Q_0) < 1$, hence only if $\sigma(P_0)$ lies strictly* outside *the base's own mass window $[1/t_0, t_0]$, where $t_0 + 1/t_0 = \sigma(S)$. In particular the mass–balanced splits, $\sigma(P_0) \approx \sigma(S)/2$, are precisely excluded.* 0 ***sorry*** (`Erdos307.split_outside_window`).

*Proof.* Theorem 8.1 gives $\alpha = (MN' + N^2)/\Delta$ with $\Delta = D(1 - \sigma(P_0)\sigma(Q_0))$, and If the two new primes lie on the same side, then $a = Muv$, $b = N$ and the equations $a' = b$, $b' = a$ give $M^2(u + v) = MN - M'N' = \Delta$ (`Erdos307.same_side_delta_pos`), so $\Delta > 0$ again. $\alpha > 0$ forces $\Delta > 0$. With $\sigma(P_0) + \sigma(Q_0) = \sigma(S)$ fixed, the product is the downward parabola $z \mapsto z(\sigma(S) - z)$, which exceeds 1 exactly between the roots of $z^2 - \sigma(S)z + 1$ and is maximal at the midpoint. □

*Why level* 61 *is still not enumerable.* This is the complement of Proposition 16.12, which places the mass of a *solution* inside the window of the full support; here the split of the *base* must sit outside its own. It is a genuine constraint and it points a search away from where one would naturally begin, but it does not make level 61 enumerable, and it is worth recording the arithmetic rather than leaving the size vague. Sampling $2 \times 10^6$ splits of the first 59 primes, 89.4% satisfy $\sigma(P_0)\sigma(Q_0) < 1$, so the condition removes about one split in nine (`code/split_balance.gp`). Against 49,961 bases and $2^{59}$ splits each, some $2.6 \times 10^{22}$ candidates remain, and the product–balance $M \approx N \approx \sqrt{D}$ that Theorem 8.1 prefers trims this by a further factor of order 10, not of order $10^{10}$. Each surviving candidate still needs two divisibility tests on 113–digit integers. We therefore record level 61 as out of computational reach by roughly ten orders of magnitude, and note that no constraint in this paper closes that gap.

*What actually obstructs, in both problems.* It is worth being exact about where the difficulty sits, because the port dynamics is not where it sits. Along Sylvester's chain the value is $\Delta \equiv 1$ *identically*, so for #313 the value dynamics never obstructs at all; what obstructs is that the next element $B/\Delta + 1$ must be *prime*, and the chain $2, 6, 42, 1806$ stops because $1807 = 13 \cdot 139$. That single composite is the whole reason #313 is open rather than settled by Sylvester.

The comparison is then clean, and it is the honest conclusion of the transfer. #313 carries one obstruction: the primality of a constructed element. The 1–free case of #307's relaxation carries that same obstruction *and*, by Proposition 12.8, a second one that #313 does not have, no self–sustaining value to run a chain along in the first place. So the two problems are not of equal difficulty under this correspondence: #307's relaxation is strictly harder in an identifiable way, and a solution of #313 would not by itself supply one. This is a negative conclusion, but it is a proved one rather than a failed search, and it retires the transfer as a route.

$u = 2$. The first 1–free case. The condition is $D(B) = 2\beta - 4$, i.e. $\sum_{b \in B} 1/b = 2 - 4/\beta$, and then $A = \{2, \beta - 2\}$, which needs $\beta$ odd so that $\gcd(2, \beta - 2) = 1$ (0 `sorry`, `Erdos307.Coprime.two_free_case`). Reducing $D(B) = 2\beta - 4$ modulo each $b \in B$ gives the Znám–type system $\prod_{b' \neq b} b' \equiv -4 \pmod{b}$, and conversely those congruences force $\beta \mid D(B) - 2\beta + 4$, so with $1 < \sum_B 1/b < 2$ the quotient is an integer in $(-1, 0]$ and must vanish: *the local conditions imply the global one.*

**Proposition 12.12** ($u$ is the mass, so $u = 1$ is the only cheap case)**.** *In the situation of Proposition 12.6, $\sum_{b \in B} 1/b = u(\beta - u)/\beta = u - u^2/\beta$. Hence the reciprocal sum of $B$ is $u$ up to $O(u^2/\beta)$: the parameter $u$* is *the mass $B$ must carry. Consequently the cost is monotone in $u$, and $u = 1$, the case that contains the member* 1*, is the only value for which $B$ needs mass $\approx 1$ and can therefore be short.*

This settles what one might hope for from larger $u$. Since $\sum_A 1/a = 1/u + 1/(\beta - u)$ and $\sum_B 1/b$ is its reciprocal, asking for $1 \notin P \cup Q$ is asking for $u \geq 2$, which is asking for $B$ to carry mass at least $2 - 4/\beta$; $u = 3$ demands $3 - 9/\beta$, and so on. There is no cheaper $u$ hiding above 2, and no analogue

of Sylvester's invariant to be had there: the invariant $\beta - D(B) = 1$ works at $u = 1$ precisely because mass 1 is reachable in four terms. Verified exhaustively: over every $\beta \le 4 \times 10^7$ and *every* coprime factorisation of it, all set partitions of its prime powers, the discriminant $\beta^2 - 4D(B)$ is a square with admissible root in exactly 7 cases, and every one has $u = 1$. The largest $u$ attained anywhere in that range is 1 (`code/two_member_scan.rs`).

*The price of dropping the member* 1. The reduction is exact and factoring–free, but it fixes the size. Since $\sum_B 1/b = 2 - 4/\beta$ and $B$ consists of odd pairwise coprime integers, the reciprocal sum must exceed 2, and $\sum_{3 \le p \le X} 1/p$ first exceeds 2 at the 1412nd odd prime $p = 11789$ (sum $2.0000206\ldots$). Hence $|A \cup B| \ge 1414$, against 60 for the prime problem: passing from $u = 1$ to $u = 2$ costs a factor of 23 in the number of members, because the member 1 was carrying a full unit of mass that must now be found among odd reciprocals. This is the exact price of the gap on the problem page, and it is why the Sylvester descent, which produces the $u = 1$ family for free, does not reach $u = 2$: driving $e = 2\prod B - D(B)$ to a divisor of $\beta + 4$ by the recursion $e \mapsto be - \prod B$ failed over 399 seeds, the smallest $e$ reached being $2.7 \times 10^6$, while the products outrun any bound long before 1412 members (`code/two_side_descent.gp`).

*Where the neighbouring literature stands, and what it supplies.* Three recent items bear on the problem without deciding it, and it is worth saying exactly what each does and does not give.

[22] studies $\sum_i 1/x_i = 1$ with the prime factors of the $x_i$ confined to a fixed finite set $S$, motivated by the classification of integral fusion categories, and supplies algorithms for that restricted search. This is the computational problem a level of #307 presents, and the paper records the same obstruction Proposition 4.10 meets from the other side: the general bound on the solutions is doubly exponential in the number of terms. Its "low–rank" refers to the number of terms, not to the rank–one product structure that distinguishes #307 from the unrestricted problem; the two senses should not be conflated.

[23] treats an iterated arithmetic function of Erdős and Graham, $g(n) = n + \varphi(n)$, and settles the two–step case by reducing it to the single equation $\varphi(n) + \varphi(n + \varphi(n)) = n$, enumerating the solutions, and leaving a residue that requires a prime $p \ge 10^{10}$ with $\varphi((3p-1)/4) = (p+1)/2$, apparently absent. The shape is the shape of #307: a two–cycle for an iterated arithmetic function, reduced by an exact identity to one equation, with a residue that is a rarity condition. Theorem 8.1 performs the same reduction here. The difference is the scale of the residue, which the barrier places above $10^{56}$ rather than $10^{10}$, and that difference is the whole of why the same method finishes there and not here.

Its algorithms do not reach #307: they are built for one or two distinct primes with high multiplicity (the paper says "a small number of primes, say 1 or 2", and its tables are for $\{p\}$ and $\{2, p\}$), whereas a solution here needs at least 60 distinct primes each to the first power. The two regimes are disjoint, not merely separated by scale. What does transfer is one necessary condition, recorded in Theorem 4.14.

[24] bounds the distance from 1 to the best reciprocal subsum of a multiset of reciprocal mass $K$ by $\exp(-c\sqrt{K \log K})$, improving $\ll K^{-2}$. It concerns how well 1 can be approached, which is the direction Proposition 16.8 occupies, and says nothing about exact representation, which is what #307 asks.

## 13 Status and questions

Erdős #307 remains **open**. It is Conjecture 4 of Ufnarovski–Åhlander [8], so it is equivalent to the existence of a non–trivial two–cycle of the arithmetic derivative and has been open in that literature since 2003 under another name. We establish rigidity and a barrier, $|P \cup Q| \ge 60$

with $\min(\prod P, \prod Q) > 3.50 \times 10^{57}$, which voids all direct search; it appears to be the first valid unconditional bound of its kind, the only prior candidate resting on an invalid step (Section 14). The rigidity result and the barrier are Lean–verified, extremality proved rather than assumed. Level 60 is then shown empty (Proposition 5.9), both its sectors closing without factorisation, which sharpens the barrier to $|P \cup Q| \geq 61$ and $\min(\prod P, \prod Q) > 5.89 \times 10^{58}$ (Corollary 5.10).

The barrier is one instance of a single inequality. Writing $n(c) = \min\{n : T_n \geq c\}$,

$$\sum_{p \in U} 1/p \;\geq\; c \implies |U| \geq n(c), \qquad \prod_{p \in U} p \;\geq\; \Pi_{n(c)},$$

and the arithmetic of each result enters only in producing its $c$ (Remark 16.11). Read in the size it gives the double–logarithmic frontier law (Proposition 16.9); in the near–miss ratio, the cost family (Corollary 16.10), whose endpoint at $r = 1$ is the barrier and which shows $a'' \geq 2a$ needs 37,963 primes; in the period, that the barrier is uniform in $k$, so it bounds the whole of Ufnarovski–Åhlander's Conjecture 3 and not only #307 (Proposition 4.16); in the orbit length, that the derivative cannot outgrow its own support (Proposition 16.19); in the split, that $\min(\omega(a), \omega(b)) \geq 3$ whenever $|U| \leq 68$ (Proposition 16.18); and in the level, the forced core and the ceiling that pin level 59 from both sides and yield its admissible count 49,961 in closed characterisation (Propositions 16.12, 16.13, Corollary 16.14). The algebraic cores of these are Lean–verified (`Dictionary.lean`).

One analytic statement is settled here that was open through several revisions of this paper. The squarefree minus layer $\{m : m' - 2m = \square\}$ has density zero, with a rate (Theorem A.9, Proposition A.7), and likewise the plus layer, so the squarefree Pythagorean layer has density zero unconditionally. The route is not the square sieve of Appendix A, whose analytic input is foreclosed at its range by Remark A.3; it is a sieve at the far smaller $P = \log\log N/(\log\log\log N)^{1+\varepsilon}$, where two uniform estimates *are* available, Lemma A.4 for the character sums and Lemma A.6 for the local deficits. The obstruction that remark records is range–dependent rather than absolute, and that is the whole of the opening. The rate obtained, $N(\log\log\log N)^{2+\varepsilon}/\log\log N$, is very weak; the far stronger Theorem A.12 sharpens it to $N(\log\log N)^{1/2+\delta}(\log N)^{-1/4}$, a full quarter–power of $\log N$ better. What made that possible is Lemma A.5: the pretentious distance can be estimated by applying Siegel–Walfisz to the residue classes directly, never expanding in multiplicative characters, and the $\sqrt{\varphi(r)}$ coefficient mass that capped everything at polyloglog moduli is a cost of the expansion rather than of the arithmetic. Two further blockers recorded for the rate in earlier revisions are retracted after Lemma 17.4 and at the theorem itself. Nothing else in this paper depends on any of it, and the barrier is untouched.

On the constructive side we give a Thābit/Euler–type calculus and a bilinear breeder whose candidate $q$ is always integral (Proposition 9.2), a conditional reduction (H) $\Rightarrow$YES, and a quantitative ledger showing (H) is not a feasible finite search. No fixed–shape family of any growth rate can parametrise #307 (Proposition 9.4), which closes the Thābit/Euler programme rather than only its polynomial shadow. We also give a parity dichotomy, an Erdős–Wintner density theorem, and a function–field theorem localising the obstruction to the archimedean place. The heuristics favour YES, and the model behind them survives a second–moment test (Proposition 16.25); we claim no proof in either direction.

Open problems: (i) close level 61. Proposition 5.9 closes level 60 by no factorisation (the pair sector by a mass bound, the single–tail sector by an exact identity over every admissible base), and the finite verification of Proposition 4.5 is Lean–checked (`erdos307_sixty`); but 61 is not one more instance of the same computation, since past level 59 the mass condition constrains only 59 of the elements (Proposition 4.12), and the tail bound of Corollary 4.11 at level 61 already reaches $q \leq 1.1020\, D$ with $\log_{10} D = 117.84$ at the extremal base: astronomically beyond the $2^{40}$–scale

sieves that closed level 60. What method decides that regime, if any short of Proposition 4.10's tower–type bound, is open. The conditional 3/4 law is explained; it is the reciprocity switch of Remark 5.13, made exact by the class–vector functional there); (ii) decide the parity dichotomy's all–odd case (we expect it empty); (iii) sharpen the certified enclosure $0.04202 \le \delta \le 0.04223$ of Proposition 6.5, whose width is carried almost entirely by the tail window $a$; three digits are certified, and a longer envelope convolution with a smaller window certifies more; (iv) settle the density hypothesis (H) of Section 9, or exhibit a non–circular frame family escaping the $\tau(K)/G$ ceiling; (v) find a $\mathbb{Z}$–side structure detecting the archimedean obstruction of Section 7; one that must be non–local, since by Proposition 6.2 no single modulus obstructs; (vi) settle #307.

*The place of the problem: a local–global dictionary.* The pieces above assemble into a dictionary. (i) For a fixed support $U$, the subset form of Proposition 9.3, $A(\Sigma - A) = D^2$, satisfies the *Hasse principle as an equation*: an integer solution $A$ exists iff one exists over $\mathbb{R}$ and over every $\mathbb{Z}_\ell$ (the discriminant $\Sigma^2 - 4D^2$ must be a nonnegative square, and squares obey local–global). Its real–place condition is $\Sigma \ge 2D$, i.e. $\sigma(U) \ge 2$: *the barrier is precisely the real–place solvability condition of the subset form.* (ii) The barrier never needed primality: pairwise coprime integers have distinct smallest prime factors, so $\sum 1/a_i \le \sum 1/p_i$, and *any* pairwise–coprime support of mass $\ge 2$ has at least 59 elements and product at least $\Pi_{59}$. Coprimality and the ordering of $\mathbb{R}$ carry the wall; primality enters only through rigidity. (iii) What remains unsolved is the *realization* of the root $A^* = \big(\Sigma + \sqrt{\Sigma^2 - 4D^2}\big)/2$ as a sub–multiset sum of the cofactors $D/p$. Its local conditions are exactly the local anatomy of Lemma 6.1: modulo $\ell^2$ for $\ell \in U$ the subset sums occupy precisely the two cosets $\ell\mathbb{Z} \cup (D/\ell + \ell\mathbb{Z})$ ($2\ell$ residues, observed exactly (6 of 9, 10 of 25, 14 of 49, 22 of 121)) and these are the same constraints the equation itself forces: no completion separates the equation from its realization. In the language of integral points, then: #307 is everywhere locally soluble, with the barrier as its archimedean height condition, and its resolution is a *strong approximation* question for the thin, prime–structured set of subset sums a failure mode invisible to Brauer–Manin–type obstructions, consistent with every no–go above and with the existence of cycles over the neighbouring rings. We record this as a dictionary, not a method.

**Attribution.** We are careful to separate prior, concurrent, and present contributions (details in Section 14). *Prior/concurrent:* the two–cycle reformulation, and its equivalence with the arithmetic derivative, are due to Ufnarovski–Åhlander [8] (2003, Conjecture 4); the naming of it as #307 to Seva [21] (2019); an independent rederivation to Bado [28]; the bound $|P \cup Q| \ge 59$ to J. Rosen and the disjointness valuation $v_p(\sum 1/p_i) = -1$ to R. Israel [10]; the source identification to "Woett" [10]. Bado [28] independently obtained the rigidity cross–matching, the local anatomy, the parity restriction, a one–sided exact test, approximate solvability, and the union criterion of Proposition 14.1. *Present note:* the identification of #307 with Ufnarovski–Åhlander's Conjecture 4 and the resulting two–way bridge to the $n'' = n$ literature; the product barrier $2.09 \times 10^{56}$; the function–field theorem and archimedean localisation; the constructive deficit analysis and seed obstruction; the Erdős–Wintner density theorem, and the $k$–cycle barriers. Both external sources [21, 28] have been examined directly and the overlaps above reflect their actual contents.

**Acknowledgments.** This work was carried out by the author with substantial assistance from Anthropic's Claude. The AI assistant contributed to the derivations, the exact and heuristic computations, the Lean 4 formalisation, and the drafting of this note; it was also used to stress-test and attempt to refute each claim. All statements, proofs, attributions, and the final text were reviewed and are the responsibility of the author. The author thanks the contributors to the MathOverflow thread [10]; in particular R. Israel and J. Rosen, whose observations are credited above.

*Where a proof of* NO *would have to come from.* Four results of this note combine into a single statement about the shape any negative proof must take, and it is worth assembling because each of the four is easy to read as an isolated obstruction.

Proposition 5.17 shows the cycle system is soluble modulo every $m$, so it has solutions in $\widehat{\mathbb{Z}}$ and there is no congruence obstruction to find, at any modulus or any finite combination of moduli. Theorem 7.1 shows the analogue over $\mathbb{F}_q[t]$ is false, having no cycles at all, so no argument that survives the passage to function fields can decide #307 negatively: whatever a proof uses, $\mathbb{F}_q[t]$ must lack it. Proposition 15.10 shows no place carries a descent. Together these localise the obstruction at the archimedean place, which is the reading already recorded.

The fourth is the constraint. The only archimedean tool in play is the mass, and Proposition 5.18 shows it is exactly tight: raising the barrier one level requires $\sigma(a) > t_n$, the only unconditional bound available is $\sigma(a) - 1 \geq 1/a$, and $\{2, 3, 7, 41\}$ realises $\sigma = 1.00058\ldots$ on four primes, so nothing forbids $a' - a = 1$. So a proof of NO needs an archimedean argument that is not the mass, and we know of none. This is not a claim that none exists; it is a statement of what the four results jointly exclude, recorded so that the search is not repeated inside the region they cover.

**Remark 13.1** (What the single equation leaves to work with)**.** A two–cycle carries exactly one relation. If $\sigma(P)\sigma(Q) = 1$ then $a'b' = ab$, and $\gcd(a, a') = \gcd(b, b') = 1$ gives $a \mid b'$ and $b' \mid a$, so $b' = a$ and $a' = b$; the converse is immediate. Four consequences follow.

1. No congruence between $a$ and $b$ beyond the equation is available from the primes of the support: for $r \mid a$, reduction modulo $r$ and modulo $r^2$ returns $b \equiv a'$, which holds for every $a$. A counting bound in the style of Pomerance's for Lehmer's totient problem therefore has no divisibility to start from, and the fibres of $n \mapsto n'$ contain the Goldbach sets $\{pq : p + q = v\}$, so no uniform fibre bound replaces it.

2. No second identity exists: $\sum 1/p^2$ enters no consequence of the equation, and any second–moment statement is an inequality about which supports have $\sigma$ near a given value.

3. Near misses fail on the far side. For the 289 even squarefree $A$ reached from 2 by steps keeping the defect $A - A' \leq 200$ and stopped past $10^{40}$, the candidate partner $b = A'$ is never squarefree with $b' = A$: 34 are not squarefree and in the other 255, $b' < A$, by more than $0.63A$. The primary pseudoperfect numbers $6, 42, 47058, 2214502422$ give $\sigma(b) = 0.2,\ 0.024,\ 2.1 \times 10^{-5},\ 0.053$ (`code/nearmiss_cycle.gp`).

4. The populated model shows no hidden divisibility. Over the 10,292 distinct coprime norm pairs $(A, B)$ of the twisted two–cycles of $\mathbb{Z}[\sqrt{2}]$ (Proposition 7.6), and conditional on $\ell \nmid AB$, the frequency of $\ell \mid A \pm B$ equals that of random norms of the same size for $\ell = 3, 5, 7, 17, 23, 31$; every raw bias is explained by which primes the support contains (`code/sqrt2_auxmod.gp`).

The only arithmetic a proof can use is therefore the prime content of the support, and its first moment is exhausted at $|U| \geq 60$.

## 14 Prior and concurrent work

We record the overlaps with four items, each examined directly.

**Ufnarovski–Åhlander (2003) [8].** Their Theorem 5 gives $p^k \| n,\ 0 < k < p \Rightarrow p^{k-1} \| n'$, and their Theorem 4 gives $n^{(j)} \to \infty$ when $p^p \mid n$ properly; together with Corollary 2 ($n$ squarefree $\Leftrightarrow (n, n') = 1$) these force a period–two point to have squarefree members with disjoint supports.

The argument is sound: for $2 \le k < p$ two applications of Theorem 5 give $v_p(n'') = k - 2 \ne k$, while $k \ge p$ falls to Theorem 4 or to the fixed point $n = p^p$, and the edge cases $p = 2$, $k \in \{2, 3\}$ are covered by the latter because $4 = p^p$. They then state the period–two case as **Conjecture 4**: $\big(\sum_{i \le k} 1/p_i\big)\big(\sum_{j \le l} 1/q_j\big) = 1$ has no solution in distinct primes. This is #307, and priority for the two–cycle reformulation belongs here rather than to 2019: what [21] adds is the identification of that equation as #307, which [8] does not make, they cite [1] for the derivative, not [2] for the problem. The consequence is that #307 has been an open conjecture in the arithmetic–derivative literature since 2003 under another name, and that the bounds of Section 4 are bounds on Conjecture 4.

**Seva (MathOverflow 323194, 2019) [21].** Defines $s(Q) = Q \sum_{p|Q} 1/p$, derives the valuation property $v_p(s(Q)) = v_p(Q) - 1$, asks for $s(P) = Q$, $s(Q) = P$, and states this is Problem #307 "in disguise." Priority for *naming* the two–cycle question as #307 belongs here. He further asks whether every $s$–trajectory stabilises at 0; the $s$–analogue of the Ufnarovski–Åhlander conjecture, posed independently of it. (Note $s \ne n'$ globally, e.g. $s(p^\alpha) = p^{\alpha-1}$, but $s = n'$ on squarefree integers, where both two–cycle questions live.)

**Kovič (2012), Propositions 17 and 18 [9].** Proposition 18 claims $r + s \ge 34$ unconditionally ($\ge 57$ if $\min\{p_1, q_1\} \ge 3$, $\ge 110$ if $\ge 5$), together with $p_1 < r$, $q_1 < s$ and $\max\{m, n\} \ge \prod_{i \le 17} p_i \approx 1.92 \times 10^{21}$. Its proof does not establish them. Substituting $n' = m$, $m' = n$ into its two correct pairs of bounds gives

$$\text{(A)}\quad \frac{rn}{p_r} < m < \frac{rn}{p_1}, \qquad \text{(B)}\quad \frac{sm}{q_s} < n < \frac{sm}{q_1},$$

after which it asserts "Hence: $p_1 q_s < rs$ and $q_1 p_r < rs$", and deduces everything from these via $2\max\{p_r, q_s\} < N^2/4$ against $\max\{p_r, q_s\} \ge P_N$. But products of inequalities are valid only in matching directions, and (A) with (B) admits exactly two: upper $\times$ upper gives $p_1 q_1 < rs$, lower $\times$ lower gives $rs < p_r q_s$. The asserted $p_1 q_s < rs$ pairs an upper bound on $m$, carrying $p_1$, with a lower bound on $n$, carrying $q_s$. Nor is it recoverable: the cycle says precisely $\sigma(n) \in (r/p_r,\, r/p_1)$ and $\sigma(m) = 1/\sigma(n) \in (s/q_s,\, s/q_1)$, so $\sigma(n)$ lies in $(r/p_r,\, r/p_1) \cap (q_1/s,\, q_s/s)$; nonemptiness of that intersection is equivalent to those two products and to nothing further, whereas $p_1 q_s < rs$ reads $q_s/s < r/p_1$, a comparison of the two *upper* endpoints. In $n = 2 \cdot 5 \cdot 13 \cdot 37 \cdot 53 \cdot 73 \cdot 79 \cdot 89 \cdot 97 \cdot 101$, $m = 3 \cdot 7 \cdot 19 \cdot 31 \cdot 43 \cdot 59 \cdot 67 \cdot 71 \cdot 83 \cdot 103$ (squarefree, disjoint supports, $r = s = 10$) all four inequalities hold while $p_1 q_s = 206 > 100 = rs$ and $q_1 p_r = 303 > 100$ (`code/kovic_prop18_audit.gp`; the valid products and the invalidity are machine–checked in `lean/Erdos307/KovicProp18.lean`). Two arithmetic slips sit alongside: Table 2 lists $P_{11} = 27$, and the "product of the first 17 primes" is printed with 27 for 29.

What fails is the *inference*, not the *statement*: no two–cycle is known, so no counterexample to $r + s \ge 34$ exists to be given, and none is claimed. The conclusion is unproven rather than false, and is in fact true, since Theorem 4.3 gives $|U| \ge 60$ by a route touching none of this. Propositions 16 and 19 are sound, and so is 17, but 17 assumes the *smaller* member is odd and [9] notes that no estimate follows when it is even. The strongest valid unconditional bound in the prior literature is therefore that paper's computer search $x > 10^4$; Theorem 4.3 improves it by 52 orders of magnitude, and $|U| \ge 60$ is the first valid cardinality bound for arithmetic–derivative two–cycles rather than an improvement on an earlier one. None of this diminishes [9], which remains a standard reference and whose Propositions 16, 17 and 19 we use.

**Bado (preprint, Aix–Marseille, 2026) [28].** Working with $A(S) = \sum_{p \in S} M(S)/p$ and $M(S) = \prod_{p \in S} p$ (our $N_S, D_S$), Bado independently obtains: the lowest–terms lemma (his Lemma 3.1, our

Lemma 2.1); the forcing identities $A(P) = M(Q)$, $A(Q) = M(P)$ with $P \cap Q = \varnothing$ (his Thm. A, our Theorem 2.3); the two–cycle reformulation $T(T(P)) = P$ (his Thm. B); the valuation signature (his Thm. 6.1, = Israel's); the local zero–sum congruences (his Prop. 8.1, our Lemma 6.1); the parity restrictions (his Prop. 8.2, the parity part of Theorem 4.14); an exact one–sided test (his Thm. 9.1); the deviation parameter, i.e. the integer $t$ with $A(P) = n + tm$, $A(Q) = m - tn$ and solubility exactly at $t = 0$ (his Thm. 7.6, our rung $k$ of Proposition 15.1); the intrinsic union criterion (his Thm. 7.3, our Proposition 14.1 in *iff* form); uniqueness of the splitting (his Cor. 7.5); the approximation theorem (his Thm. 10.2, sharper than our greedy version in that all primes exceed a given bound), and the union discriminant / Pythagorean structure (his Thms. 7.1, 7.4). He cites only Barbeau, Bloom, and Erdős–Graham: crucially, he does *not* identify the map with the arithmetic derivative, and his note contains none of the present note's $n'' = n$ bridge, explicit barrier constant, function–field theorem, density theorem, $k$–cycle bounds, or constructive deficit analysis. We regard the overlap on the shared structural layer as strong independent validation, and import his union criterion with credit. Bado has confirmed this account in correspondence (August 2026): he agrees that the results listed above were present in his preprint and that the overlap is concurrent and independent, and he has confirmed the identification $A(S) = M(S)'$, which he had not made, and is revising his historical discussion accordingly. The concurrency is therefore recorded on both sides rather than asserted on one:

**Proposition 14.1** (Single–integer test; derivative form of Bado's Thms. 7.1, 7.4)**.** *If $(a, b)$ is a two–cycle of the arithmetic derivative, then $N = ab$ is squarefree and satisfies $N' = a^2 + b^2$, whence $N'^2 - 4N^2 = (a^2 - b^2)^2$ is a perfect square. Conversely, a squarefree $N$ with $N'^2 - 4N^2 = \square$ determines a unique unordered factorisation $N = ab$ with $N' = a^2 + b^2$, on which the cycle condition reduces to a single equation (Bado's deviation parameter $t = 0$, his Thm. 7.6). This gives a one–integer necessary test; computationally, no squarefree $N < 2 \times 10^5$ with $\omega(N) \geq 2$ passes it (*0 *of* 103,596*, consistent with Theorem 4.3, which forces $N > 4 \times 10^{112}$).*

*Proof.* $N = ab$ with $a, b$ coprime squarefree gives $N$ squarefree and, by Leibniz with $a' = b$, $b' = a$, $N' = a'b + ab' = b^2 + a^2$; then $N'^2 - 4N^2 = (a^2 + b^2)^2 - 4(ab)^2 = (a^2 - b^2)^2$. The converse is Bado's [28]; the emptiness of the test below $10^{112}$ is upgraded from a sieve to a theorem in Corollary 14.3. □

The Pythagorean equation $N' = a^2 + b^2$ deserves to be isolated from the (harder) cycle condition.

**Proposition 14.2** (Pair form; the one–equation relaxation)**.** *For coprime squarefree $a, b$, the single equation $a^2 + b^2 = (ab)'$ holds if and only if there is an integer $k$ with $a' = b + ak$ and $b' = a - bk$; a two–cycle is exactly the case $k = 0$. We call a solution of $a^2 + b^2 = (ab)'$ (any $k$) a* Pythagorean pair*.*

*Proof.* The Leibniz identity $a^2 + b^2 - (ab)' = a(a - b') - b(a' - b)$ (a polynomial identity, verified symbolically and on 2000 random pairs) makes the equation equivalent to $a(a - b') = b(a' - b)$; since $\gcd(a, b) = 1$ this forces $a \mid a' - b$ and $b \mid a - b'$ with equal quotients $k$, i.e. $a' = b + ak$, $b' = a - bk$. Setting $k = 0$ recovers $a' = b$, $b' = a$. (Equivalently, in $\mathbb{Z}[i]$: #307 asks for $z = a + bi$ with coprime coordinates and $ab$ squarefree satisfying $|z|^2 = (\operatorname{Im} z^2/2)'$ and $k = 0$.) □

**Corollary 14.3** (The single–integer test is provably empty below $10^{112}$)**.** *Every Pythagorean pair satisfies $\sum_{p|a} 1/p + \sum_{q|b} 1/q = a/b + b/a \geq 2$ (divide $a^2 + b^2 = (ab)'$ by $ab$ and apply AM–GM), so its support carries at least* 59 *primes and $N = ab \geq \prod$(first* 59 *primes) $> 7.9 \times 10^{112}$. Hence no squarefree $N < 10^{112}$ with $\omega(N) \geq 2$ satisfies $N'^2 - 4N^2 = \square$: the empirical "*0 *of* 103,596*"*

*of Proposition 14.1 is a theorem, and the barrier of Theorem 4.3 extends from two–cycles to the strictly weaker one–equation (Pythagorean) problem.*

Formalised. `Erdos307.pyth_sum_of_squares` *is* $N' = a^2 + b^2$ *from Leibniz and the cycle,* `pyth_discriminant` *the resulting square,* `split_identity` *and* `split_product` *the two–layer form of Proposition 15.12, and* `mass_ge_two_of_pythagorean` *with* `card_ge_59_of_pythagorean` *this corollary, glued by* `mass_ge_two_of_pyth_support` *into* `emptytest`*, which takes the Pythagorean equation itself rather than an assumed mass bound. The AM–GM step of Proposition 4.15 is* `sum_ge_card_of_prod_eq_one`*, valid for every* $k$ *at once and proved from* $\log x \le x - 1$ *rather than a* $k$*–term AM–GM. Bado's converse is cited, not reproved* (`lean/Erdos307/Pythagorean.lean`).

**Remark 14.4.** (a) As $N' = a^2 + b^2$ is a sum of two coprime squares, every odd prime dividing $N'$ is $\equiv 1 \pmod 4$; a cheap union–side pre–filter complementing Bado's congruence filters. (When exactly one of $a, b$ is even this is the hypotenuse of the primitive triple $(|a^2 - b^2|, 2ab, a^2 + b^2)$.) (b) Since Seva's $s$ agrees with $n'$ on squarefree integers, and his valuation property forces two–cycle members squarefree, his question is *exactly* #307, not merely an analogue. (c) Primitive Pythagorean triples form the Barning–Hall ternary tree (the orbit structure of a reflection group in $O(2, 1; \mathbb{Z})$), and a #307 solution's triple $(|a^2 - b^2|, 2ab, a^2 + b^2)$ occupies one node with generator $(a, b) = (\prod P, \prod Q)$. The tree nonetheless yields no descent on the problem: its moves $(a, b) \mapsto (2a \mp b,\, a)$, $(a + 2b,\, b)$ are linear, preserving coprimality but not squarefreeness (over random squarefree coprime generators only 37% of children keep both members squarefree) so the locus of #307–eligible nodes (squarefree generators of disjoint prime support meeting $a^2 + b^2 = (ab)'$) is not tree–invariant. The additive tree is transverse to the multiplicative constraint; the arithmetically natural recursion is instead the prime–append $\delta$–tree of Proposition 17.26, $\delta(Mp) = p\,\delta(M) + M$, which preserves squarefreeness and tracks the line parameter, which is why the classical members organise on *that* tree, not Barning–Hall.

The deviation parameter of Proposition 14.2 organises the one–equation problem into a short graded family.

# 15 The deviation ladder, the plus layer, and a second route to the barrier

The material below accreted onto the "note added" above because it grew out of the imported union criterion, and it stayed there. None of it is prior work: it is the deviation ladder, the function–field places, the mod–8 law, the plus layer, the slot anatomy, and an independent derivation of the barrier constant, and it includes two results carrying Lean formalisations (Propositions 15.10 and 15.15). It is given its own section and subsections here so that it can be found.

## 15.1 The deviation ladder

**Proposition 15.1** (The deviation ladder)**.** *Let* $(a, b)$ *be a Pythagorean pair with* $a < b$ *and parameter* $k$*. Writing* $\sigma(m) = \sum_{p|m} 1/p$*, one has* $k = \sigma(a) - b/a = a/b - \sigma(b) \in \mathbb{Z}$*, and:*

(1) $k \le 0$*, with* $k = 0$ *exactly the (open) two–cycle case. Indeed* $b' = a - bk > 0$ *and* $b > a \ge 1$ *force* $k \le 0$ *(if* $k \ge 1$ *then* $b' \le a - b < 0$*). Moreover* $|k| < \min(b/a, \sigma(b)) \le \sigma(b) \le \log\log b + O(1)$*, so the ladder has finitely many rungs; and, since* $\sigma(b)$ *barely exceeds* 2 *at the relevant scales, in practice only one or two.*

(2) (Parity.) *For odd squarefree $m$, $m' \equiv \omega(m) \pmod 2$; for even squarefree $m$, $m'$ is odd. Hence if exactly one member is even, $k \equiv \omega(\text{odd member}) \pmod 2$; if both are odd, $\omega(a) \equiv \omega(b) \pmod 2$ and $k \equiv \omega(a) + 1 \pmod 2$. The case $k = 0$ recovers the parity dichotomy of Theorem 4.14.*

(3) *Each rung is the system $a' = b + ak$, $b' = a - bk$, equivalently $(\sigma(a) - k)(\sigma(b) + k) = 1$ (a shifted copy of the cross–match $st = 1$) so every rung is exactly as rigid as $k = 0$. Since $\sigma(a) + \sigma(b) = a/b + b/a$ is independent of $k$, Corollary 14.3 holds on* every *rung: the whole ladder is empty below $10^{112}$. (We make no quantitative claim that $k = 0$ is easier than the rest.)*

(4) *On the rung $k = -1$ the equations read $a' = b - a$, $b' = a + b$ and $a'^2 + b'^2 = 2(a^2 + b^2)$: the derivative acts on $z = a + bi$ as the orientation–reversing $\sqrt{2}$–similarity $z \mapsto (-1 + i)\,\bar{z}$.*

(5) *By uniqueness of the splitting (Bado, Thm. 7.5), each squarefree $N$ with $N'^2 - 4N^2 = \square$ carries, under the convention $a < b$, a well–defined invariant $k(N) \in \mathbb{Z}_{\le 0}$; #307 holds iff $k(N) = 0$ for some $N$, and by (3) the invariant lives on a domain that is empty below $10^{112}$.*

Formalised. `Erdos307.k_nonpos` *is the bound $k \le 0$ of* (1)*; the finiteness of the ladder, $|k| < \min(b/a, \sigma(b))$, is not formalised. Further,* `rung_parity_mixed` *and* `rung_parity_odd` *are* (2)*,* `rung_cross` *and* `rung_mass_indep` *are* (3) *(the second being the invariance that carries Corollary 14.3 onto every rung), and* `rung_minus_one` *is* (4)*. Corollary 15.5 is* `k_determined`*, with the confinement to $k \in \{0, -1\}$ entering as a hypothesis rather than proved, and Bado's uniqueness of the splitting in* (5) *cited* (`lean/Erdos307/Ladder.lean`)*.*

The whole ladder is one equation in $\mathbb{Z}[i]$, and the rung index is a Gaussian multiplier.

**Proposition 15.2** (Gaussian multiplier form of the ladder)**.** *For coprime squarefree $a, b$ put $z = a+bi \in \mathbb{Z}[i]$ and let $\mathcal{D}z := a'+b'i$ be the coordinatewise arithmetic derivative. Then, unconditionally,*

$$z \cdot \mathcal{D}z \;=\; (aa' - bb') \;+\; i\,(ab)'.$$

*Consequently $(a, b)$ is a Pythagorean pair if and only if $\mathcal{D}z = \mu\bar{z}$ for a Gaussian integer $\mu$ with $\operatorname{Im}\mu = 1$; that multiplier is $\mu = k + i$ with $k$ the deviation of Proposition 15.1, and*

$$k = \frac{aa' - bb'}{a^2 + b^2}, \qquad a'^2 + b'^2 = (k^2 + 1)(a^2 + b^2) = N(\mu)\,(a^2 + b^2), \qquad \gcd(a', b') \mid k^2 + 1.$$

*Hence #307 asks for a Pythagorean pair whose multiplier is a* unit*. Since $\{k + i : k \in \mathbb{Z}\} \cap \mathbb{Z}[i]^\times = \{i\}$, the two–cycle rung $k = 0$ is the unique rung on which $\mathcal{D}$ acts as an isometry ($a'^2 + b'^2 = a^2 + b^2$); every other rung is an expanding anti–similarity of ratio $\sqrt{k^2 + 1} > 1$, and so cannot be periodic. The condition $\operatorname{Im}\mu = 1$ is the Pythagorean equation and $\operatorname{Re}\mu = 0$ is #307: one coordinate of a single Gaussian integer each.*

*Proof.* Leibniz gives $\operatorname{Im}(z\,\mathcal{D}z) = a'b + ab' = (ab)'$ and $\operatorname{Re}(z\,\mathcal{D}z) = aa' - bb'$, which is the identity (verified symbolically and on 7,501 coprime squarefree pairs). If $(a, b)$ is Pythagorean then $(ab)' = a^2 + b^2 = N(z)$, so $z\,\mathcal{D}z = (aa' - bb') + i\,N(z)$; dividing by $N(z) = z\bar{z}$ gives $\mathcal{D}z/\bar{z} = (aa' - bb')/N(z) + i$, which lies in $\mathbb{Z}[i]$ with real part the integer $k$ of Proposition 14.2. Conversely $\mathcal{D}z = \mu\bar{z}$ with $\operatorname{Im}\mu = 1$ forces $(ab)' = \operatorname{Im}(z\,\mathcal{D}z) = \operatorname{Im}(\mu)N(z) = a^2 + b^2$. Substituting $a' = b + ak$, $b' = a - bk$ gives $\mathcal{D}z = (k + i)\bar{z}$ identically, whence the norm identity $|\mathcal{D}z|^2 = |\mu|^2|z|^2$. Finally $g = \gcd(a', b')$ divides both $k(b + ak) + (a - bk) = a(k^2 + 1)$ and $(b + ak) - k(a - bk) = b(k^2 + 1)$, and $\gcd(a, b) = 1$. $\square$

The ladder of Proposition 15.1 has a twin, visible only after extending the derivative to $\mathbb{Q}$ by the quotient rule $(a/b)' = (a'b - ab')/b^2$ [8].

**Proposition 15.3** (The co–ladder and the arithmetic Riccati equation)**.** *For coprime squarefree $a \neq b$ the following are equivalent:* (R) *$u = a/b$ satisfies the arithmetic Riccati equation*

$$u' \;=\; 1 - u^2;$$

(W) *$a'b - ab' = b^2 - a^2$;* (L) *there is $k \in \mathbb{Z}$ with $a' = b + ak$* and *$b' = a + bk$; the ladder of Proposition 14.2 with the sign of one equation flipped. The deviation cancels in the quotient: on* (L)*, $a'b - ab' = (b+ak)b - a(a+bk) = b^2 - a^2$ identically, so the Riccati equation is the $k$–free form of this* second *ladder. On it:*

(i) (Positivity barrier.) *$k \geq 0$ always: with $a < b$, $k \leq -1$ would give $b' = a + bk \leq a - b < 0$. Hence $\sigma(a) + \sigma(b) = a/b + b/a + 2k > 2$, so every co–ladder pair has $\omega(ab) \geq 59$ and $ab > 7.9 \times 10^{112}$: the co–ladder is empty at every reachable scale* by positivity alone*; a second transversal relaxation of the two–cycle carrying the full barrier through a mechanism independent of the minus square. Positivity separates the two cases further. If $k \geq 1$ then $\sigma(a) = k + b/a > 2$, since $a < b$, so the* smaller *member alone carries mass exceeding 2: $\omega(a) \geq 59$ and $a \geq \Pi_{59} > 8.77 \times 10^{112}$, against $\min(a, b) \geq 2.09 \times 10^{56}$ for a two–cycle. A co–ladder point with $k \geq 1$ is therefore larger on its smaller member by a factor exceeding $4 \times 10^{56}$, and $k = 0$ is the only case the barrier permits below $\Pi_{59}$: the relaxation buys nothing in the range where anything could be found, which is the conclusion Proposition 17.18 reaches for the mass condition, here for the ladder.*

(ii) (The cycle is the intersection.) *A two–cycle is exactly a common point of the two ladders at $k_{\mathrm{sym}} = k_{\mathrm{anti}} = 0$: #307 asks for $u \in \mathbb{Q}_{>0} \setminus \{1\}$ satisfying the Riccati equation together with the Pythagorean equation of Proposition 14.2. In multiplier form the two ladders read $\mathcal{D}z = kz + i\bar{z}$ (symmetric; norm $(k^2+1)N(z) + 4kab$) and $\mathcal{D}z = (k+i)\bar{z}$ (antisymmetric; norm $(k^2+1)N(z)$), meeting at $\mathcal{D}z = i\bar{z}$.*

(iii) (Inversion symmetry.) *$(1/u)' = 1 - (1/u)^2$ is automatic from* (R)*; the Riccati layer is symmetric in $a \leftrightarrow b$, as* (L) *is.*

*Proof.* (W)⇔(L): the equation rearranges to $b(a' - b) = a(b' - a)$, and $\gcd(a, b) = 1$ forces $a \mid a' - b$, $b \mid b' - a$ with equal quotients; the converse is the displayed cancellation. (R)⇔(W) is the quotient rule. In (i), summing $\sigma(a) = b/a + k$ and $\sigma(b) = a/b + k$ gives the mass bound (strict, as $a \neq b$), and mass $> 2$ forces $\geq 59$ primes as in Theorem 4.3. In (iii), $(1/u)' = -u'/u^2 = (u^2 - 1)/u^2$. All identities were verified symbolically; exhaustively, no co–ladder pair with $ab \leq 10^7$ exists (3,918,188 candidate pairs through the $a \mid a' - b$ prefilter; the barrier's prediction), and a first small–box probe over $\mathbb{Z}[i]$ (canonical $\pi' = 1$) also finds none, though there the positivity mechanism is unavailable and the layer's status is genuinely open (`code/riccati_coladder.py`). □

The two ladders embed in a plane, and the barrier becomes a half–plane law.

**Proposition 15.4** (The deviation half–plane)**.** *For coprime squarefree $a \neq b$ call the pair* integral *if both deviations $s = (a' - b)/a$ and $t = (b' - a)/b$ are integers, and place it at $(s, t) \in \mathbb{Z}^2$; the antisymmetric ladder is the line $t = -s$, the co–ladder is $t = s$, and a two–cycle is the origin. Then*

$$\sigma(ab) \;=\; \frac{a}{b} + \frac{b}{a} + s + t \;>\; 2 + (s + t),$$

*so* every *stratum in the closed half–plane $s+t \geq 0$ carries the full barrier: $\omega(ab) \geq 59$ and $ab > 7.9 \times 10^{112}$. Both ladder barriers are the two central lines of this one inequality, and #307 sits at the origin; on the boundary of the protected region, at the crossing of its two distinguished lines. The open half–plane $s+t \leq -1$ is unprotected, and is populated from $ab = 30$ on: over $ab \leq 10^7$ exactly* 21 *strata occur, all obeying the law, the nearest being $(0,-1)$ and $(-1,0)$ at $(a,b) = (6,5)$ and $(5,6)$; off–axis strata occur ($(1,-13)$ at $(690,53)$), so integrality does not force one exact side. The $(0,-1)$ stratum with $b = a'$ is the chain equation*

$$a'' + a' = a, \qquad \textit{equivalently} \qquad \sigma(a)\big(1+\sigma(a')\big) = 1,$$

*a companion of the cross–match $\sigma(a)\sigma(b) = 1$ with the same shape and one shifted factor. Its solutions up to $10^7$ are $\{6,\ 42,\ 15590,\ 47058\}$ and they split in two. Every primary pseudoperfect number whose predecessor is* prime *solves it, trivially: then $a' = a-1$ and $a'' = 1$ ($6, 42, 47058$; 1806 fails, its predecessor $1805 = 5 \cdot 19^2$ being composite, and 2 fails as $a'' = 0$). The remaining solution $15590 = 2 \cdot 5 \cdot 1559$ is not on the pseudoperfect line at all: $15590' = 10923 = 3 \cdot 11 \cdot 331$ and $10923' = 4667$, a genuine two–step closure. It is the first solution of the chain equation whose derivative is composite; the sequence $6, 42, 15590, 47058$ does not appear in the OEIS. All census claims verified exhaustively (`code/deviation_halfplane.py`).*

*Proof.* $\sigma(a) = a'/a = b/a + s$ and $\sigma(b) = a/b + t$ sum to the display, with strict inequality by AM–GM ($a \neq b$); mass $> 2$ forces $\geq 59$ primes as in Theorem 4.3. The ladder identifications are Propositions 14.2 and 15.3; the census is a finite computation. □

**Corollary 15.5** (Below mass 5/2 the rung is not a choice but a parity)**.** *Let $(a,b)$, $a < b$, be a Pythagorean pair with $\sigma(ab) < 5/2$. Then $k \in \{0,-1\}$, and $k$ is* determined *by the parity of $\omega$:*

$$\begin{array}{ll} \textit{exactly one member even:} & k = 0 \iff \omega(\textit{odd member}) \textit{ even}; \\ \textit{both members odd:} & k = 0 \iff \omega(a) \textit{ odd}. \end{array}$$

*Hence, at these masses, the single–integer test of Proposition 14.1 together with one parity check is* equivalent *to #307, not merely necessary for it: a squarefree $N$ with $N'^2 - 4N^2 = \square$ whose splitting has the stated $\omega$–parity* is *a two–cycle. In particular the finite verifications of Proposition 4.5 and Remark 5.1 are decisive in both directions; a surviving candidate would have been a solution, not a candidate, and the whole level–59/60 regime lies in this range, its masses being confined to $(2,\ 2.00235]$, where $r = b/a \leq 1.0497$.*

*Proof.* $\sigma(ab) = r + 1/r$ with $r = b/a > 1$, and $r + 1/r < 5/2 \iff r < 2$. By Proposition 15.1(1), $k \leq 0$ and $|k| < b/a = r < 2$, so $k \in \{0,-1\}$; a set of two consecutive integers is separated by parity, and Proposition 15.1(2) supplies that parity from $\omega$. The stated equivalences are the two cases of that law at $k \equiv 0$. The mass range for $|U| \in \{59, 60\}$ is Theorem 4.3 and Proposition 4.5. □

**Corollary 15.6** (Diagonal stratification: the cost of the last lattice step)**.** *Let $m(x)$ be least with $\sum_{i \leq m(x)} 1/p_i > x$. The diagonal $s+t = -j$ carries the barrier $\sigma(ab) > 2-j$, hence $\omega(ab) \geq m(2-j)$ and $ab \geq \Pi_{m(2-j)}$:*

$$\begin{array}{lllll} j \geq 2: & \textit{mass} > 0, & \textit{no barrier}, & ab \geq 2; \\ j = 1: & \textit{mass} > 1, & \omega \geq 3, & ab \geq 2 \cdot 3 \cdot 5 = 30; \\ j = 0: & \textit{mass} > 2, & \omega \geq 59, & ab \geq \Pi_{59} > 8.77 \times 10^{112}. \end{array}$$

*The deviation plane is thus stratified by mass, one diagonal at a time, and #307 sits on the first diagonal carrying a nontrivial barrier. The $j = 1$ floor is* attained*: the census minimum on that*

*diagonal is exactly* 30*, at* $(a, b) = (5, 6)$ *and* $(6, 5)$*; so the barrier is sharp one step off the origin. The final lattice step therefore costs*

$$\Pi_{59}/30 \;\approx\; 2.9 \times 10^{111},$$

*one hundred and eleven orders of magnitude for a single unit of* $s + t$*. This is the barrier of Theorem 4.3 in its most compressed form: not a bound on solutions, but the price of the last step of an integer lattice walk, paid because the diagonal index is a mass threshold and the mass thresholds* 1 *and* 2 *sit at* 3 *and* 59 *primes.*

*The populated strata calibrate the heuristic model.* The unprotected strata are the first family of #307–type exact coincidences where the average–case model can be tested against exhaustive truth. Pricing the second exactness condition of a candidate pair at $1/b$ and re–running the census enumeration with identical filters gives expected counts 11.4, 18.3, 26.1, 34.2 at $ab \le 10^4, \dots, 10^7$ against observed $6, 12, 18, 23$: the ratio is stable near $2/3$ over four decades; the model has the right order of growth and a constant local correction, i.e. it is *calibrated* at the one–condition level. Composing two levels, however, already breaks the naive pricing: for the chain equation $a'' + a' = a$ the crude spread price predicts $E = 48.8$ against the observed 4; a twelvefold bias at reachable height, because the composition requires mass–window conditioning between the levels (the stratified–versus–naive lesson of Section 6, here measured rather than inferred). The reliability of averaging thus degrades measurably with each composed exact condition, and the two–cycle at the origin composes many. This is the quantitative face of the average–to–instance gap: the model earns trust stratum by stratum, and loses it condition by condition (`code/strata_calibration.py`).

The Pythagorean layer behaves very differently over function fields, and refines into two arithmetic sub–layers over $\mathbb{Z}$.

**Proposition 15.7** (The Pythagorean layer over function fields)**.** *Over* $\mathbb{F}_q[t]$ *with* $q$ *odd,* $a^2 + b^2 = (ab)'$ *has no coprime squarefree monic solutions:* $\deg(a^2+b^2) = 2\max(\deg a, \deg b) > \deg a + \deg b - 1 \ge \deg((ab)')$*, the leading coefficient* 1 *or* $2 \ne 0$ *preventing cancellation. Over* $\mathbb{F}_2[t]$ *the layer is nonempty: up to degree* 8 *(exhaustively,* 23,548 *coprime squarefree pairs) its only solutions are the three pairs of the Artin–Schreier tower* $x_{j+1} = x_j(x_j + 1)$*, namely* $(t, t+1)$*,* $(t^2 + t, t^2 + t + 1)$*,* $(t^4 + t, t^4 + t + 1)$*; each has* $a' = (a+1)' = 1$ *and* $(a(a+1))' = (a+1) + a = 1$*, and the climb halts at level* 4 *(there* $(x_3 + 1)' = t^2 + t \ne 1$*). These pairs lie off the cycle stratum* $k = 0$*, on the side the integer normalisation of Proposition 15.1(1) forbids. Cycles (*$k = 0$*) remain impossible over every* $\mathbb{F}_q[t]$ *(Theorem 7.1): the function–field world separates the Pythagorean layer from the cycle layer, sharpening the integer obstruction to "the archimedean order forces* $k \le 0$*."*

The function–field immunity is more fragile than the characteristic: it is destroyed by inverting a single place.

## 15.2 Function–field places

**Proposition 15.8** (Inverting one place destroys the function–field obstruction)**.** *Over the* $S$*–integers* $\mathbb{F}_q[t, t^{-1}]$*, whose unit group* $\mathbb{F}_q^\times \cdot t^{\mathbb{Z}}$ *is infinite, unit–valued derivative two–cycles exist. Over* $\mathbb{F}_3$ *the smallest has degree* 3 *and only* two *prime classes:*

$$a = t^3 + t^2 + 1 = (t-1)(t^2 - t - 1), \qquad b = t^3 - t^2 - 1 = (t+1)(t^2 + t - 1),$$

*with* $D(a) = b$ *and* $D(b) = a$ *under the assignment* $(u_1, u_2) = (t, 1)$ *on each side,* $a, b$ *coprime squarefree and both coprime to* $t$*.*

*Proof.* $D(a) = t\,(t^2 - t - 1) + 1 \cdot (t - 1) = t^3 - t^2 - 1 = b$ and $D(b) = t\,(t^2 + t - 1) + 1 \cdot (t + 1) = t^3 + t^2 + 1 = a$, both exact in $\mathbb{F}_3[t]$; the factorisations are as displayed and share no factor. The values $t$ and 1 are units of $\mathbb{F}_q[t, t^{-1}]$, so this is a unit–valued Leibniz derivative in the sense used throughout Section 7. Verified independently by exhaustive search and by direct recomputation (`code/hunt/ff_twist.rs`). □

**Remark 15.9** (What the function–field theorem is really about)**.** Proposition 15.7 is a *degree* argument: with $\pi' = 1$ one has $\deg D(a) \le \deg a - 1$, so $D(a) = b$, $D(b) = a$ forces $\deg b \le \deg b - 2$. Sign twists cannot disturb it, because $\mathbb{F}_q^\times$ consists of constants and degree is blind to them, exactly as, over $\mathbb{Z}$, twisting by $\pm 1$ retains the full barrier (Remark 7.3). But allowing the values $t^k$, units once $t$ is inverted, makes $\deg D(a) \le \deg a - 1 + K$, and the two–cycle inequalities collapse to $0 \le 2K - 2$: the obstruction is vacuous from $K = 1$, and cycles duly appear at once. A census over $\mathbb{F}_3$ confirms both halves: at $K = 0$ the count is 0 at every degree tested, and at $K = 1, 2$ it is $4, 16$ / $4, 48$ / $12, 122$ for degrees $\le 4, 5, 6$. The moral sharpens Section 7's thesis rather than softening it. What immunises $\mathbb{F}_q[t]$ is not the characteristic, nor the finiteness of its unit group (cycles exist over $\mathbb{Z}[\sqrt{-2}]$, whose unit group has order 2) but the presence of a *single* distinguished place carrying a degree function that the derivative strictly lowers. Adjoining a second place destroys it. This is the exact structural analogue of the claim made for $\mathbb{Z}$: one archimedean place, one mass function the derivative must respect, and a barrier that survives every twist available in the ring.

**Proposition 15.10** (No place of $\mathbb{Q}$ carries the function–field descent)**.** *The mechanism isolated in Remark 15.9 (a place whose valuation the derivative strictly lowers) is unavailable over $\mathbb{Q}$, and this is a classification statement rather than a failure of ingenuity.*

(i) Archimedean place. $|D(n)| = n\,\sigma(n)$, *which exceeds $n$ for every $n$ of mass $> 1$; the derivative* expands, *first at $n = 30$, and on $1.79\%$ of squarefree $n \le 10^6$. The absolute value is in any case not ultrametric, so the second half of the degree argument (*$\deg(x + y) \le \max$*) is unavailable too.*

(ii) $p$–adic places. *For squarefree $n$ with $p \mid n$, cofactor survival gives $v_p(D(n)) = 0 < 1 = v_p(n)$: the derivative lowers $v_p$ there. But it* raises *it elsewhere, there are squarefree $n$ with $p \nmid n$ and $p \mid D(n)$ (e.g. $p = 2$, $n = 15$; $p = 3$, $n = 14$; $p = 5$, $n = 6$), in comparable abundance to the lowering cases at every $p$ tested. So no $p$–adic place contracts uniformly.*

(iii) *By Ostrowski's theorem* (i) *and* (ii) *exhaust the places of $\mathbb{Q}$.*

*Hence no valuation–theoretic descent of the type that proves Proposition 15.7 can exist over $\mathbb{Q}$: the function–field argument fails to transfer for a structural reason internal to the classification of absolute values, not for want of a cleverer place. (Counts and witnesses: `code/no_place.rs`.)*

Formalised. *`Erdos307.not_descent_of_strictMono` gives* (i) *in a form stronger than stated, killing* every *strictly monotone valuation at once and not only the absolute value, on the witness* $30 \mapsto 31$ (*`der_thirty`*)*; `padic_lowers_on_support` is the lowering half of* (ii) *and is Rigidity, while `raises_two`, `raises_three`, `raises_five` check the three raising witnesses outright. Ostrowski enters `no_place_descent` as an explicit disjunction hypothesis rather than silently.* 0 *`sorry`* (`lean/Erdos307/NoPlace.lean`).

## 15.3 The mod–8 law and the plus layer

**Proposition 15.11** (Deviation calculus for $k$–cycles)**.** *For pairwise coprime squarefree $a_1, \dots, a_k$ with product $N$ and deviations $\kappa_i = (a_i' - a_{i+1})/a_i$ (indices mod $k$): the ratios $a_{i+1}/a_i$ have product*

1 *(telescoping), and* $\sigma(N) = \sum_i a_{i+1}/a_i + \sum_i \kappa_i$. *On the relaxed layer* $\sum_i \kappa_i = 0$ *(which contains the cycles* $\kappa_1 = \cdots = \kappa_k = 0$*) AM–GM gives* $\sigma(N) \geq k$, *so the distinct–prime barrier* $m_k$ *of Proposition 4.15 extends to the relaxed problem for every* $k$, *and to genuine* $k$*–cycles for* $k \leq 3$*; for* $k \geq 4$ *it inherits the pairwise–coprimality caveat of Proposition 4.15 (the bound then holds for the multiset reciprocal sum). At* $k = 2$, $\kappa_1 = -\kappa_2$ *is the deviation of Proposition 15.1 and* $\sum \kappa_i = 0$ *is* $a^2 + b^2 = (ab)'$.

**Proposition 15.12** (Split discriminant)**.** *For coprime squarefree* $a, b$ *with* $N = ab$,

$$N' + 2N - (a+b)^2 = N' - 2N - (a-b)^2 = (ab)' - a^2 - b^2.$$

*Hence* $N$ *is the product of a Pythagorean pair iff* $N' - 2N$ *and* $N' + 2N$ *are* both *perfect squares* $d^2, s^2$ *(their product is Bado's discriminant* $N'^2 - 4N^2$*), with* $a, b = (s \pm d)/2$. *The minus factor carries the obstruction:* $N' - 2N = (a-b)^2 \geq 0$ *forces* $\sigma(N) \geq 2$, *hence the* 59*–prime barrier, indeed* 0 *of* 103,596 *squarefree* $N < 2 \times 10^5$ *satisfy* $N' - 2N = \square$ *(Corollary 14.3 in one line). The plus factor is satisfiable at small scale (*114 *such* $N < 2 \times 10^5$*). In characteristic* 2 *the obstructing factor vanishes* ($a - b = a + b$) *and the criterion degenerates to* $(a+b)^2 = N'$*: the permissiveness of Proposition 15.7 and the barrier of Corollary 14.3 are the absence and presence of one and the same algebraic factor.*

Both squares of the split criterion are pinned modulo 8 by an elementary law that seems worth recording on its own.

**Proposition 15.13** (The mod–8 law of the derivative)**.** *For odd squarefree* $m$ *write* $S(m) = \sum_{p \mid m} p$. *Then*

$$m\, m' \;\equiv\; S(m) \pmod 8.$$

*Consequently:* (a) *modulo* 2 *this is the parity law* $m' \equiv \omega(m)$ *used in Theorem 4.14;* (b) *an all–odd two–cycle* $a' = b$, $b' = a$ *forces*

$$S(a) \;\equiv\; S(b) \;\equiv\; ab \pmod 8$$

*the two prime sums are congruent to each other and to the product; refining Theorem 4.14(ii);* (c) *in the mixed case* $a = 2k$ *(*$k$ *odd), the odd member obeys* $b \equiv k\big(1 + 2S(k)\big) \pmod{16}$ *and* $S(b) \equiv 2kb \pmod 8$*, and the even member the companion law* $(2k)\,(2k)' \equiv 2 + 4S(k) \pmod{16}$*;* (d) *for odd squarefree* $N$ *the plus quantity satisfies* $N' + 2N \equiv N\big(S(N) + 2\big) \pmod 8$*, so an odd plus–hit must have* $N(S(N)+2) \equiv 1 \pmod 8$ *when* $\omega(N)$ *is odd and* $\equiv 0, 4 \pmod 8$ *when* $\omega(N)$ *is even. None of these congruences is an obstruction, consistently with Proposition 6.2, but they thin every hunt over the odd sector by a fixed factor.*

Formalised. *`Erdos307.mod8_law` is the law itself, resting only on* `odd_sq_mod8`*;* `prime_sums_congruent` *is* (b)*;* `parity_law` *is* (a) *and is stated over* $\mathbb{N}$ *with* mod *, so it needs no* `ZMod` *API, and* `omega_even_of_even_deriva` *is the step Theorem 4.14*(i) *uses. Part* (b)*'s second half,* $S(a) \equiv ab$, *is* `prime_sum_eq_product`*; part* (d) *is* `plus_quantity_mod8` *with the square–residue filter* `plus_hit_residue` *and the* $\omega$*–parity split* `plus_hit_residue_split`*, which supplies the* $\equiv 1$ *versus* $\equiv 0, 4$ *dichotomy; part* (c) *is* `mixed_member_mod16`*,* `even_member_mod16` *and* `mixed_prime_sum_mod8`*, the mod–*16 *statements chaining onto the law through its integer form* `mod8_law_int`*. All four parts are therefore formal* (`lean/Erdos307/Mod8.lean`).

*Proof.* $q^2 \equiv 1 \pmod 8$ for odd $q$, so $m/q \equiv mq \pmod 8$; summing over $q \mid m$ gives $m' \equiv m\, S(m) \pmod 8$, and multiplying by $m$ (with $m^2 \equiv 1$) gives the law. (a) is the law modulo 2, since $S(m) \equiv \omega(m)$ for odd primes. (b) is the law at $a$ and at $b$ with $a' = b$, $b' = a$. In (c), Leibniz gives $b = (2k)' = k + 2k'$, and the law at $k$ gives $k' \equiv kS(k) \pmod 8$, whence $b \equiv k + 2kS(k) \pmod{16}$;

the second congruence is the law at $b$ with $b' = 2k$, and the companion follows from $2k(k+2k') \equiv 2k^2 + 4kk' \pmod{16}$ with $k^2 \equiv 1 \pmod 8$ and $kk' \equiv S(k) \pmod 8$. In (d), $s^2 \bmod 8 \in \{0,1,4\}$ and $s \equiv N' \equiv \omega(N) \pmod 2$. All parts were verified exhaustively over the 405,285 odd squarefree $m \le 10^6$ and all odd plus–hits below $10^6$: zero exceptions (`code/mod8_plusthin.py`). □

**Proposition 15.14** (Structure of the plus–layer)**.** *Call squarefree $N$ with $\omega(N) \ge 2$ a* plus–hit *if $N' + 2N$ is a perfect square; the first are $130, 145, 305, 689, 1745, 1870, 1970, 2353, \ldots$ (114 below $2 \times 10^5$; OEIS [39]).* (i) *For a semiprime $N = pq$, $2(N' + 2N) + 1 = (2p+1)(2q+1)$, so a plus–hit forces $(2s)^2 \equiv -2 \pmod \ell$ for every prime $\ell \mid (2p+1)(2q+1)$, whence $\ell \equiv 1, 3 \pmod 8$ and $p \equiv q \equiv 1 \pmod 4$ (this explains the dominance of 5 and 13). The identity inverts the search: a semiprime plus–hit is exactly a factorisation $2r^2 + 1 = (2p+1)(2q+1)$, so one scans $r \le \sqrt{7N/3}$ and sieves $2r^2+1$ in $r$, at cost $O(\sqrt{N})$ in place of $O(N)$. This gives 243,129 semiprime plus–hits with $v \le 10^{13}$ in eighty seconds, and $p \equiv q \equiv 1 \pmod 4$ holds for every one of them* (`code/plushit_inverted.rs`, which reproduces the 1,787 below $3 \times 10^8$ exactly, with no miss and no extra). (ii) [*empirical*] *$\#\{\text{plus–hits} \le X\} \approx 0.25\sqrt{X}$ (ratios $0.250, 0.253, 0.255$ at $X = 5\times10^4, 10^5, 2\times10^5$): the plus factor behaves like a random integer of its size, so (with Proposition 15.12) the whole integer obstruction sits in the minus factor.* (iii) *Fixing $p = 5$, $q = (s^2 - 5)/11$ with $s$ even and $s \equiv \pm 4 \pmod{11}$ gives $q \equiv 1 \pmod 4$, and each prime value yields the plus–hit $N = 5q$ (68 with $s < 2000$, up to $N = 1{,}774{,}805$); Bunyakovsky's conjecture for this quadratic thus implies infinitely many plus–hits. The two discriminant factors separate cleanly: the plus–layer is Bunyakovsky–class, the minus–layer is the barrier.* (iv) *[Remark] The full cycle condition states that $(d^2, N', s^2)$ is a three–term arithmetic progression with common difference $2N$ whose outer terms are squares; a congruent–number–shaped configuration; the elliptic–curve machinery of that theory does not transfer because $N'$ is not an algebraic function of $N$. A 251–term b–file to $N < 10^6$ (counting ratio 0.251) and an OEIS submission draft accompany this note; the plus–hit sequence was not in OEIS when we searched; it is now A396915, submitted by us and cited above, so the entry is not independent corroboration.* (v) *[P, C]* General local law (QR tournament). *For every odd prime $\ell \mid N$ of a plus–hit, $N' + 2N \equiv N/\ell \pmod \ell$ (all cofactor terms $N/p$ with $p \ne \ell$ carry a factor of $\ell$; only $N/\ell$ survives), so $N/\ell$ is a nonzero quadratic residue mod $\ell$. Verified with zero violations on all 114 plus–hits below $2 \times 10^5$ and all 251 terms to $10^6$; the semiprime $p \equiv q \equiv 1 \pmod 4$ theorem of* (i) *is its $\omega = 2$ shadow via quadratic reciprocity.*

(vi) *Parity of $\omega$ on odd plus–hits, and its single failure.* Among the odd plus–hits the number of prime factors is almost always even: to $2 \times 10^7$ all 598 have $\omega$ even, and to $10^8$ all but one of 1,252 do. The exception is

$$v = 30735705 = 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 23 \cdot 89, \qquad \omega(v) = 7, \qquad v' + 2v = 89094721 = 9439^2,$$

the least odd plus–hit with $\omega$ odd. It is not isolated: to $3\times10^8$ there are three, 30735705, 165064305 and 230839455, all with $\omega = 7$ and with $t = 4, 4, 5$, against 2,143 with $\omega$ even, a rate of 0.14% (`code/plushit_omega.rs`). So the parity is a bias of about 99.9% and not a law, which is worth recording because the bias is strong enough to look like one over any range short of $3 \times 10^7$. What does hold without exception is the genus constraint: writing $t = \#\{p \mid v : p \equiv 3 \bmod 4\}$, every one of the 2,146 to $3 \times 10^8$ has $t \equiv 0$ or $1 \pmod 4$, as the identity $\prod_{p|v} \left(\frac{v/p}{p}\right) = (-1)^{\binom{t}{2}}$ requires, the counterexample included with $t = 4$. The observed pairs $(\omega, t)$ are $(2,0)$ with 1,787 instances, $(4,1)$ with 339, $(6,4)$ with 17, $(7,4)$ with two and $(7,5)$ with one.

The empirical $\frac{1}{4}\sqrt{X}$ law of part (ii) can be proved, up to logarithms, on the stratum that dominates the small–scale census.

**Proposition 15.15** (The semiprime plus–layer is provably thin)**.** *The number of semiprime plus–hits $N = pq \le X$ is $O\big(\sqrt{X}\,(\log X)^2\big)$, by a fully elementary argument; Erdős's bound $\sum_{s\le Y}\tau(2s^2+1) \ll Y\log Y$ [48] sharpens this to $O(\sqrt{X}\log X)$. Under Bunyakovsky's conjecture the $p=5$ family of Proposition 15.14(iii) alone contributes $\gg \sqrt{X}/\log X$, so conditionally*

$$\frac{\sqrt{X}}{\log X} \;\ll\; \#\{\textit{semiprime plus–hits} \le X\} \;\ll\; \sqrt{X}\,\log X :$$

*the $\sqrt{X}$ order of the empirical law is genuine on the semiprime stratum, not an artefact of the tested range.*

*Proof.* For a semiprime plus–hit $N = pq \le X$ one has $s^2 = N' + 2N = p + q + 2pq \le 3X + 1$ (as $p + q \le pq + 1$), and Proposition 15.14(i) gives $2s^2 + 1 = (2p + 1)(2q + 1)$: the map $N \mapsto \big(s, \{2p+1, 2q+1\}\big)$ into pairs (root, unordered nontrivial factorisation) is injective, so the count is at most $\sum_{s\le Y}\tau(2s^2+1)$ with $Y = \sqrt{3X+1}$. The polynomial $2s^2+1$ has discriminant $-8$, prime to every odd modulus, so its roots modulo an odd prime are simple and lift uniquely by Hensel: $2s^2 + 1 \equiv 0 \pmod u$ has at most $2^{\omega(u)} \le \tau(u)$ solutions modulo $u$ (none for even $u$). Pairing each divisor of $2s^2+1$ with its cofactor and keeping the smaller member $u \le \sqrt{2s^2+1} \le \sqrt{2}\,Y$,

$$\sum_{s\le Y}\tau(2s^2+1) \;\le\; 2\sum_{u\le\sqrt{2}Y}\#\{s\le Y : u \mid 2s^2+1\} + Y \;\le\; 2\sum_{u\le\sqrt{2}Y}\tau(u)\Big(\frac{Y}{u}+1\Big) + Y \;\ll\; Y(\log Y)^2,$$

by Dirichlet's $\sum_{u\le Z}\tau(u) \sim Z\log Z$ and $\sum_{u\le Z}\tau(u)/u \sim \frac{1}{2}(\log Z)^2$. For the conditional lower bound, the family of Proposition 15.14(iii) has $\asymp \sqrt{X}$ eligible $s \le \sqrt{2.2X+5}$ (two residue classes modulo 22), of which Bunyakovsky for $(s^2-5)/11$ makes $\gg \sqrt{X}/\log X$ prime. The injection, the Hensel root count, and the censuses were verified numerically (59 semiprime hits below $2\times10^5$, all reproduced; $\sum_{s\le Y}\tau(2s^2+1)/(Y\log Y)$ stable near 1.05–1.09 for $Y = 500$–$2000$; `code/mod8_plusthin.py`). The $\omega \ge 3$ stratum remains empirical. □

*What of the last argument is formal.* The proof splits into an elementary skeleton and an analytic tail, and the split is worth making explicit because the two have different status. The skeleton is formalised: `Erdos307.plus_semiprime_identity` is the factorisation $2s^2 + 1 = (2p + 1)(2q + 1)$, `plus_range` the bound $s^2 \le 3N + 1$ that confines $s$ to $Y$, `plus_recover` the injectivity, and `count_le_divisor_sum` the reduction of a count to $\sum_{s\le Y}\tau(2s^2+1)$ by injection into the disjoint union of the divisor sets. That reduction is stated abstractly, over any finite set carrying a root and a divisor: the notion "plus–hit" is not itself defined in Lean, so the instantiation to this layer is done in the prose above and not machine–checked. `card_large_le_card_small` and `tau_le_two_small` are the divisor–pairing step that reorganises that sum over the small member alone, and `two_roots_quadratic` is the Hensel root count at the prime level, in a form needing only $2 \ne 0$ rather than the discriminant. 0 `sorry` (`lean/Erdos307/PlusThin.lean`).

The rest of the elementary chain is formal as well. The count of $s \le Y$ in the admissible residue classes is `count_in_residue_classes`, and the divisor sum is bounded in the printed shape by `sum_tau_div_le_log_sq`, $\sum_{u\le Z}\tau(u)/u \le (1 + \log Z)^2$, from Mathlib's `harmonic_le_one_add_log`. Only Dirichlet's *asymptotics* are absent from the library, and the upper bound does not use them.

The step that was missing is now supplied. The chain needs $|R| \le \tau(u)$ for $R$ the roots of $2s^2 + 1 \equiv 0$ modulo $u$, and the half $2^{\omega(u)} \le \tau(u)$ was already `two_pow_omega_le_tau`. The other half, $|R| \le 2^{\omega(u)}$, is the multiplicativity of the root count, and is now `Erdos307.rootSet_card_le_two_pow_omega`, with the composition `rootSet_card_le_tau`. Two steps, both elementary in $\mathbb{Z}/u$: the Chinese remainder ring equivalence carries a root modulo $ab$ to a pair of roots injectively, so the count is

submultiplicative on coprime factors (`rootSet_card_mul_le`); and modulo an odd prime power there are at most two roots, because $(x-y)(x+y)=0$ while $p$ cannot divide the representatives of both factors, their sum representing the unit $2x$ (`rootSet_card_prime_pow_le_two`). Modulo a power of 2 there are none. 0 `sorry` (`lean/Erdos307/RootCount.lean`).

What remains outside Lean is the *instantiation*, not a lemma: "plus–hit" is not defined in Lean, so tying the abstract reduction to this layer is done in the prose above. Erdős's sharpening and Bunyakovsky's lower bound stay cited, and neither is used by the upper bound.

The semiprime argument generalises: the slot calculus of Proposition 17.13 counts.

**Proposition 15.16** (Slot counting: every fixed–$\omega$ stratum of the plus–layer is sparse)**.** *Fix $r \geq 2$. The number of plus–hits $N \leq X$ with $\omega(N) = r$ is $O\big(X^{1-1/r+o(1)}\big)$.*

*Proof.* Write $p = P^+(N)$ for the largest prime of $N$ and $M = N/p$, so $\omega(M) = r-1$ and, since $N < p^r$, $M \leq N^{1-1/r} \leq X^{1-1/r}$. The slot identity of Proposition 17.13 (base $M$, slot $p$) gives $N' + 2N = p\,E_0(M) + M$ with $E_0(M) = M' + 2M$, so a plus–hit $s^2 = N' + 2N$ forces

$$s^2 \equiv M \pmod{E_0(M)}, \qquad p = \frac{s^2 - M}{E_0(M)},$$

and the map $N \mapsto (M, s)$ is injective. For squarefree $M$ one has $\gcd(M, M') = 1$ (modulo $q \mid M$ only the cofactor $M/q$ survives in $M'$), hence $\gcd(M, E_0(M)) = 1$: at every odd prime $\ell \mid E_0(M)$ a root of $s^2 \equiv M$ has $\ell \nmid s$, is nonsingular, and lifts uniquely (Hensel), so the congruence has at most $4 \cdot 2^{\omega(E_0)} = X^{o(1)}$ roots modulo $E_0(M)$. Since $s^2 = N\big(\sigma(N)+2\big)$ and $\sigma(N) \ll \log\log X$, each base carries at most $X^{o(1)}\big(X^{1/2}/E_0(M) + 1\big)$ admissible $s$, and with $E_0(M) \geq 2M$,

$$\#\{N\} \;\leq \sum_{\substack{M \leq X^{1-1/r} \\ \omega(M) = r-1}} X^{o(1)}\Big(\frac{X^{1/2}}{M} + 1\Big) \;\ll\; X^{1/2+o(1)} + X^{1-1/r+o(1)} \;=\; X^{1-1/r+o(1)}.$$

(For $r = 2$ this recovers the order of Proposition 15.15, which is sharper by logarithms.) The injection, the identity, $\gcd(M, M') = 1$ and $P^+(N) \geq N^{1/r}$ were verified on all 251 plus–hits below $10^6$: zero exceptions (`code/strata_count.py`). □

*What separates the strata from the full layer.* The bound is honest but far from the truth: at $X = 10^6$ the strata $\omega = 2, \ldots, 5$ hold 124, 69, 50, 8 hits (every one below $\sqrt{X}$) because the congruence $s^2 \equiv M \pmod{E_0(M)}$ is usually *insoluble*: over random odd semiprime bases the average number of roots is 0.34, the no–residue branch of the trichotomy of Proposition 17.13 dominating. Summing the proposition over $r$ does *not* give the full layer density zero, because the $X^{o(1)}$ root–count factors are not uniform in $r$; an average bound for $\tau(M' + 2M)$ (a Titchmarsh–divisor–type sub–problem) remains the route to the true ($\sqrt{X}$–order) count. Density zero itself, however, needs less: a smooth/rough dichotomy bypasses the average entirely, because roughness buys a factor $X^{-1/\log\log X}$ against which the divisor bound costs only $X^{\log 2/\log\log X}$, and $\log 2 < 1$.

**Proposition 15.17** (The plus–layer has density zero)**.** *Unconditionally, with $\log_3 = \log\log\log$,*

$$\#\{\textit{plus–hits } N \leq X\} \;\ll\; \frac{X}{(\log X)^{(1+o(1))\log_3 X}}.$$

*In particular the plus–layer has density zero. The proof applies verbatim to $\{N \text{ squarefree}: N' + cN = \square\}$ for any fixed $c \geq 1$.*

*Proof.* Split at $y = X^{1/u}$ with $u = \log\log X$. *Smooth part.* The number of $N \le X$ with $P^+(N) \le y$ is $\Psi(X,y) = X\,u^{-u(1+o(1))} = X(\log X)^{-(1+o(1))\log_3 X}$ (Rankin's bound; [19], III.5). *Rough part.* If $P^+(N) = p > y$, the injection of Proposition 15.16 applies with $M = N/p \le X/y = X^{1-1/u}$: the hit determines $(M,s)$ with $s^2 \equiv M \pmod{E_0(M)}$, a congruence with at most $\rho(E_0) \le 4\cdot 2^{\omega(E_0)} \le \exp\big((\log 2{+}o(1))\log X/\log\log X\big)$ roots by the maximal order of the divisor function [18]; each base therefore carries at most $\rho(E_0)\big(X^{1/2+o(1)}/E_0(M)+1\big)$ values of $s$. Summing, with $E_0(M) \ge 2M$:

$$\#\{\text{rough hits}\} \;\ll\; X^{1/2+o(1)} + X^{1-1/u}\exp\Big((\log 2{+}o(1))\frac{\log X}{\log\log X}\Big) \;=\; X^{1/2+o(1)} + X\exp\Big(-(1{-}\log 2{-}o(1))\frac{\log X}{\log\log}$$

and $1 - \log 2 = 0.3068\ldots > 0$: the rough part is smaller than every $X/(\log X)^A$. The smooth part dominates and gives the stated rate. (At $X = 10^6$ the split reads $251 = 225$ rough $+\,26$ smooth, the injection collision–free; `code/pluszero.py`.) □

On a derivative *line* the slot is not merely constrained; it is pinned, and the counting sharpens to every line at once.

### 15.4 Slot and line anatomy

**Proposition 15.18** (The determined slot: all derivative lines are thin, uniformly)**.** *For integers $e \ge 1$ and $c$,*

$$\#\{N \le X \text{ squarefree}:\; N' = eN + c\} \;\ll\; \frac{X}{(\log X)^{(1+o(1))\log_3 X}},$$

*uniformly in $e$ and $c$. In particular the primary pseudoperfect line $n' = n-1$, the Giuga quotient–one line $n' = n+1$, every line $n' = n \pm d$ of [35], and every slope–two line $n' = 2n + c$ obey this bound. (Continuity of the limit law of Proposition 6.5 already forces each line to have density zero; the content is the rate, far beyond what concentration inequalities give, and the uniformity. Classical lines with extra divisibility structure may well admit sharper bounds; we claim none.)*

*Proof.* Split at $y = X^{1/u}$, $u = \log\log X$; smooth members number $\Psi(X,y) = X(\log X)^{-(1+o(1))\log_3 X}$ (Rankin). For a rough member $N = Mp$, $p = P^+(N) > y$, Leibniz gives $N' - eN = p\,(M' - eM) + M$, and $M' - eM \ne 0$: $\sigma(M) = e$ would make the auto–reduced fraction of Theorem 2.1 an integer, forcing $D_M = 1$, $M = 1$. Hence

$$p \;=\; \frac{c - M}{M' - eM}$$

is *determined* by $M$ (the line has no free root at all) so $N \mapsto N/P^+(N)$ is injective on rough members, whose number is at most $X^{1-1/u}$. The smooth part dominates. Two classical objects fall out of the formula: on the stratum $M' - M = -1$ of the two slope–one lines it reads $p = M + 1$ ($c = -1$) and $p = M - 1$ ($c = +1$) (the oblong chains $6 \mapsto 42 \mapsto 1806$ and $30 = 6\cdot 5$, $1722 = 42\cdot 41$) while off–stratum members stay slot–determined ($858 = 66\cdot 13$ with $M' - M = -5$; $47058 = 1518\cdot 31$ with $M' - M = -49$). Verified: the formula reproduces $P^+(N)$ on all listed members and identically on the 51,201 squarefree $N \le 10^5$ with $\omega \ge 2$ (`code/lines_slot.py`). □

The members of a line carry far more structure than thinness: every prime factor is pinned modulo its cofactor, and the top of the factorisation is capped by the bottom.

**Proposition 15.19** (Line anatomy: pinning, coupling, caps)**.** *Let $M$ be squarefree on the line $M' = eM + c$, let $p > q$ be its two largest primes and $R = M/(pq)$.*

(a) (Pinning.) $M/t \equiv c \pmod t$ *for* every *prime* $t \mid M$*; the Giuga congruence of the line, independent of* $e$*. Equivalently, each prime factor lies in one determined residue class modulo its cofactor:* $t \equiv c\,(M/(tr))^{-1} \pmod r$ *for all primes* $r \mid M/t$*.*

(b) (Top cap.) *If* $M/p \neq c$ *then* $p \le \sqrt{M} + |c|$*: line members are* $(\sqrt{M} + |c|)$*–smooth.*

(c) (Coupling.) $pq \mid R(p+q) - c$*. Here* $\gcd(pq, c) = 1$ *is automatic:* $p \mid c$ *together with* $p \mid qR - c$ *would give* $p \mid qR$*, impossible as* $p$ *exceeds every prime of* $qR$*.*

(d) (Second cap.) *If* $R(p+q) \neq c$ *then* $q < R + \sqrt{R^2 + |c|}$*; the degenerate stratum* $R(p+q) = c$ *forces* $c \ge 2\sqrt{RM}$*, so it is empty on every line with* $c < 2\sqrt{M}$*; in particular on all the near–critical minus lines.*

*Proof.* (a) Cofactor survival gives $M' \equiv M/t \pmod t$ while $eM \equiv 0$; the second form is CRT. (b) $M/p - c$ is a multiple of $p$; if nonzero, $M/p \ge p - |c|$. (c) $p \mid qR - c$ and $q \mid pR - c$ give $pq \mid (qR-c)(pR-c) = pqR^2 - cR(p+q) + c^2$, so $pq \mid c\,(R(p+q) - c)$, and the displayed coprimality removes $c$. (d) If $R(p+q) \neq c$ then $pq \le |R(p+q) - c| \le R(p+q) + |c| < 2Rp + |c|$, whence $q^2 < 2Rq + |c|$; if $R(p+q) = c$ then $c \ge 2R\sqrt{pq} = 2\sqrt{RM}$ by AM–GM. All parts were verified on the 7,184,234 squarefree $n \le 2 \times 10^7$ with $\omega \ge 3$, for both $e = 1, 2$ with $c := n' - en$ (zero exceptions, the second cap attained with slack 1 at $n = 70$) and on the classical members, where the coupling quotient $\big(R(p+q) - c\big)/pq$ equals 1 on the oblong–chain members and 5 on both off–stratum members 47058 and 66198 (`code/line_anatomy.py`). □

The lines themselves close into a family under peeling, and the closure explains why no reorganization of the counting has broken through.

**Proposition 15.20** (The peeling groupoid)**.** *For integers* $a \ge 1$*,* $b$*,* $c$ *let* $L(a, b, c) = \{n \text{ squarefree} : an' = bn + c\}$*.*

(i) (Closure; the slope is a mass.) *If* $n = qm \in L(a, b, c)$ *with* $q = P^+(n)$*, then* $m \in L(aq,\ bq-a,\ c)$*, and the slope moves by* $b/a \mapsto b/a - 1/q$*: the full peel chain of* $n$ *descends the partial sums of* $\sigma(n)$*; the classical lines, their* $\mu$*–Sondow shifts, and the slope–two lines of the minus square are one orbit family.*

(ii) (Rigidity at every level.) *Along the chain of* $n$ *on an integer line* $(a, b) = (1, e)$*, the level–*$j$ *denominator vanishes iff* $\sigma(m_j) = b_j/a_j$*, which telescopes to the single* $j$*–independent condition* $\sigma(n) = e$*; impossible, since an integer mass forces* $n = 1$ *by Theorem 2.1. So every denominator on every genuine chain is nonzero.*

(iii) (Determined slot at every level.) *Consequently* $q = (c - am)/(am' - bm)$ *at each step: a line member is a tower of single determined candidates over its own core.*

*Every partial–peel reorganization of the line counts (peeling one prime, several, in any order, from either layer of the split criterion) stays inside this groupoid, where part* (iii) *reproduces the one–candidate–per–cofactor obstruction verbatim. The obstruction is therefore* invariant *under the natural changes of viewpoint: a proof must inject input from outside the groupoid; primality along determined thin sequences, or a decomposition of integers not indexed by their primes.*

*Proof.* (i) is Leibniz: $a(qm'+m) = bqm+c$ gives $aq\,m' = (bq-a)m+c$, and $(bq-a)/(aq) = b/a-1/q$. In (ii), $\sigma(m_j) = b_j/a_j = e - \sum_{i<j} 1/q_i$ says $\sigma\big(m_j \prod_{i<j} q_i\big) = \sigma(n) = e$; by the auto–reduced fraction of Theorem 2.1 an integer mass forces $D = 1$. (iii) is the display above (i). Verified: closure and slot on $10^5$ random $(n, a, b)$ (the only slot exceptions being exactly the $\sigma(m) = b/a$ locus, excluded

on genuine chains, 73 of $10^5$ random triples, 0 of 310,788 genuine chain steps at $e = 1, 2, 3$), and the slope telescope exact on every chain (`code/peeling_groupoid.py`). □

The determined slot bounds more than itself: unwound one level further it constrains the *second* prime by a mass.

### 15.5 A second derivation of the barrier constant

**Proposition 15.21** (Mass–defect bound on the second prime)**.** *Let $M$ be squarefree with $\omega(M) \geq 3$ on the line $M' = eM + c$, and write $p = P^+(M)$, $m = M/p$, $P = P^+(m)$, $n = m/P$. If $\sigma(m) < e$ then*

$$P\big(e - \sigma(n)\big) \;<\; 2 - \frac{c}{m}, \qquad \textit{equivalently} \qquad P^2\,(en - n') \;<\; 2Pn - c.$$

*For $e = 2$ the hypothesis is automatic below $\Pi_{59}$; that is, throughout the range in which the minus layer is counted. On the near–critical slope–two lines ($|c|$ small) the bound reads $P < 2/(2-\sigma(n)) + O(|c|/n)$:* the second–largest prime of $M$ is bounded by twice the reciprocal of the mass defect of the remaining cofactor.

*Proof.* $M$ squarefree gives $p > P$. The determined slot of Proposition 15.20 is $p = (c-m)/(m'-em)$, and $\sigma(m) < e$ makes the denominator negative, so $p = (m-c)/\big(m(e-\sigma(m))\big)$ and $p > P$ reads $P\,m(e-\sigma(m)) < m-c$, i.e. $P(e-\sigma(m)) < 1-c/m$. Since $m = Pn$ and $\sigma(m) = \sigma(n)+1/P$, the left side is $P(e-\sigma(n))-1$, which gives the first display; multiplying by $Pn$ clears denominators. Verified with 0 violations on every squarefree $M \leq 2 \times 10^6$ with $\omega \geq 3$, for $e = 1, 2, 3$; at $e = 1$ the 21,146 exceptions are exactly the $\sigma(m) \geq 1$ cases excluded by the hypothesis, and the bound is within a factor 2 of equality for 11% of members and within 100 for 59% (`code/slot_massbound.rs`). □

**Remark 15.22** (The exact form, and a window that is real but inert)**.** Proposition 15.21 has an exact ancestor. With $N = pm$, $p = P^+(N)$, $k = 2m - m'$, Leibniz gives $N' = pm' + m$, so on the minus layer

$$d^2 \;=\; N' - 2N \;=\; m - pk, \qquad \text{whence} \qquad \sigma(N) \;=\; 2 + \frac{1}{P^+(N)} - \frac{k}{m}, \qquad k \geq 1.$$

Since $k \geq 1$ this is a *two–sided* window; the barrier supplies only the lower half:

$$2 \;<\; \sigma(U) \;<\; 2 + \frac{1}{\max U}, \qquad \text{equivalently} \qquad \sigma\big(U \setminus \{\max U\}\big) < 2,$$

so a minus–layer support is *mass–minimal*: deleting any one of its primes drops the mass below 2 (the largest is the binding case). The upper half is nonetheless *inert* where it can be tested: all 49,961 admissible level–59 supports already satisfy it, the tightest margin being $1.26 \times 10^{-3}$, so it prunes nothing there. We record it because it is exact, because it is the identity behind Proposition 15.21, and because it may bind at higher levels, where the largest admissible prime grows. (The enumeration also reproduces the count 49,961 of Proposition 4.5 independently, from a different traversal; `code/masswindow.rs`.)

**Remark 15.23** (The minus layer is the anatomy of a quadratic polynomial)**.** The identity $d^2 = m - pk$ of Remark 15.22 can be read as a parametrisation rather than a constraint. On the minus layer $d^2 = N' - 2N \geq 0$ is a square, so

$$m \;=\; pk + d^2,$$

and for *fixed* $(p,k)$ the cofactor $m$ runs over the values of the quadratic polynomial $f_{p,k}(d) = d^2 + pk$. The minus layer is therefore not an arbitrary exact coincidence but a question about the *anatomy of quadratic–polynomial values*: for which $d$ does $f_{p,k}(d)$ carry mass $\sigma = 2 - k/f_{p,k}(d)$, i.e. $\omega \approx 58$ prime factors concentrated on the small primes, with $P^+(f_{p,k}(d)) < p$?

This is the first place in this circle where a *named* toolkit applies. The anatomy of polynomial values is a developed subject; Erdős–Kac for $\omega(f(n))$, Halberstam–Richert sieve bounds for almost–primes in polynomial sequences, and the divisor sums of [48] already used in Proposition 15.15; whereas "primality of a determined value" (Remark 17.11) had none. What the toolkit must supply is unusual: not the typical behaviour of $\omega(f(d))$, which is $\log\log$ by Erdős–Kac, but a *large deviation* to $\omega \approx 58$ with the prime factors forced onto the bottom of the spectrum, since only the small primes carry mass. Existing polynomial–anatomy results control typical and almost–prime behaviour, not deviations of this shape; so the reformulation names the tool without yet applying it. We record it as the most promising translation of the residual question, and note that the $p = 5$ family of Proposition 15.14(iii) is the plus–layer shadow of the same quadratic mechanism. (Identity verified on all 1,606,956 squarefree $M \le 3 \times 10^6$ with $\omega \ge 2$: 0 violations.)

**Remark 15.24** (The residue amplifier is real, and exactly paid for)**.** Sieving $f_c(d) = d^2 + c$ by its roots exposes a mechanism the union–side reading of Corollary 17.12 misses. A prime $\ell$ divides some value iff $-c$ is a quadratic residue modulo $\ell$, which removes half the primes; the $2^{-\omega}$ penalty the corollary records. But an *admissible* odd $\ell$ has *two* roots, so it divides values with density $2/\ell$ rather than $1/\ell$: on the primes that survive, the weight is doubled. Only the penalty has ever been counted, and one might expect the doubling to tilt the heuristic. It does not: the two effects cancel exactly.

Per fixed $c$ the effect is genuine. Writing $\sigma_B$ for the mass on primes $\le B$, measured over $d \le 2 \times 10^6$ with $B = 2 \times 10^4$: a generic $c = 1009$ gives mean $\sigma_B = 0.368$, a friendlier $c = 4966$ (admitting 35 of the 46 primes below 200) gives 0.400, against 0.452 for random integers; a *perfectly* friendly $c$, admitting every odd prime, would give $\frac{1}{4} + 2\sum_{\ell>2} \ell^{-2} = 0.654$, a factor 1.45 above random. So friendliness helps, and in the limit beats a random integer outright.

But friendliness is priced. Each odd $\ell$ is admissible for only half of the $c$, so a $c$ admitting the first 58 odd primes has density $2^{-58}$, exactly cancelling the $2^{58}$ gained from the doubled densities:

$$2^{-58} \cdot \tfrac{1}{2} \prod_{i=2}^{59} \frac{2}{\ell_i} \;=\; \prod_{i=1}^{59} \frac{1}{\ell_i},$$

the random–integer probability. Averaging over $c = 1, \dots, 400$ confirms it numerically: the mean mass ratio is 0.9990, and the tail ratios $\Pr[\sigma_B > T]$ are 0.998, 0.996, 0.978 at $T = 0.8, 1.0, 1.2$; unity to within sampling, while individual $c$ range from 0.257 to 0.635 in mean. *The quadratic reformulation is heuristically neutral*: it neither strengthens nor weakens the YES lean of Remark 5.16, and the $2^{-59}$ filter of Corollary 17.12 is correctly priced after all. What Remark 15.23 buys is a toolkit, not a bias (`code/quad_anatomy.rs`, `code/amplifier_cost.rs`).

The bound has enough force to regenerate the barrier on its own.

**Corollary 15.25** (A second, independent derivation of the barrier)**.** *Let $N$ be a minus–layer hit with $\omega(N) \ge 3$; more generally a member of any line $N' = 2N + c$ with $c \ge 0$. Write $p = P^+(N)$, $m = N/p$, $P = P^+(m)$, $n = m/P$. Then $N > 10^{112.94}$, without using the arithmetic–geometric mean inequality.*

*Proof.* If $\sigma(m) \ge 2$ then $m$ carries at least 59 primes and $N > m \ge \Pi_{59}$ already. Otherwise Proposition 15.21 applies, and $c \ge 0$ makes its right side at most 2: $P\big(2 - \sigma(n)\big) < 2$. Since

$P > P^+(n)$ this gives

$$2 - \sigma(n) \;<\; \frac{2}{P^+(n)}, \qquad \text{i.e.} \qquad \sigma(n) > 2 - \frac{2}{P^+(n)}.$$

But $\sigma(n) \le \sum_{q \le P^+(n)} 1/q$, so $P^+(n)$ must satisfy $\sum_{q \le p} 1/q > 2 - 2/p$. The least such prime is $p = 269$, the 57th: at $p = 269$ the sum is $1.99505 > 1.99257$, while at $p = 263$ it is $1.99133 < 1.99240$. Hence $\omega(n) \ge 57$, $P^+(n) \ge 269$, and $n \ge \Pi_{57} > 10^{108.07}$; with $p > P > P^+(n)$ the two further primes give $N > \Pi_{57} \cdot 271 \cdot 277 > 10^{112.94}$. □

*Two mechanisms, one constant.* Corollary 15.25 reaches $10^{112.94}$, and $\Pi_{59} > 8.77 \times 10^{112}$ is $10^{112.94}$: the two derivations agree to two decimal places. They share no step. Theorem 4.3 divides the defining equation by $ab$ and applies AM–GM to $a/b + b/a$; the present route never forms that quotient, using instead the determined slot of Proposition 15.20, the trivial inequality $p > P^+(N/p)$, and the additivity $\sigma(m) = \sigma(n) + 1/P$. What they share is the arithmetic input underneath (how many primes Mertens needs to reach mass 2) which is why the constants coincide rather than merely being comparable. The barrier is thus not an artefact of the mass inequality one happens to use: it is the same Mertens threshold seen through two independent structures, and the critical set $S_2 = \{n : 2 - \sigma(n) < 2/P^+(n)\}$ of the second structure is empty below $10^{108}$ (verified empty for $n \le 5 \times 10^6$, `code/critical_set.rs`).

**Remark 15.26** (The minus side: coercive versus degenerate moduli)**.** Proposition 15.17 rests on the plus modulus being *coercive*: $E_0^+(M) = M' + 2M \ge 2M$. The minus–square obeys the same slot identity with $E_0^-(M) = M' - 2M = M(\sigma(M) - 2)$, a modulus that *vanishes exactly on the mass–2 wall*: the degeneration locus of the counting method is the barrier stratum itself; the counting incarnation of the form degeneration $N_0(2 - \sigma) \to 0$ of Proposition 17.14. The minus–layer is the union of the slope–two lines $N' = 2N + d^2$ over square heights, and three facts are unconditional. (i) A rough minus–hit ($P^+(N) = p > y$) forces its base upward: if $\sigma(M) < 2 - 1/y$ then $d^2 = p\,E_0^-(M) + M < M(1 - p/y) \le 0$, impossible; so $\sigma(M) \ge 2 - 1/y$, and once $y > (2 - T_{58})^{-1} = 793.7\ldots$ this yields $\omega(M) \ge 59$ and $M \ge \Pi_{59}$: *the* 59*–prime wall propagates from the hit to its base*, and $N \ge \Pi_{59}\, y$. (ii) Bases with $\sigma(M) \ge 2 + 1/y$ have moduli $E_0^- \ge M/y$, and contribute as in Proposition 15.17. (iii) The remaining bases are near–critical, $\sigma(M) \in [2 - 1/y, 2 + 1/y)$; exactly the low slope–two lines $M' - 2M = \pm k$, $k \le M/y$. Each is thin by Proposition 15.18, but at a degenerate modulus the $d$–freedom is $\sqrt{X}$–sized, and summing the uniform bound does not close the layer. What would: *square–root bounds on slope–two lines*, $\#\{M \le Z : M' - 2M = \pm k\} \ll Z^{1/2+o(1)}$ uniformly in $k$. There is a second, sharper route to that bound, obtained by abandoning the prime–indexed decomposition altogether. Split $M = uv$ by *size* rather than by largest prime, $u$ the coprime divisor nearest $\sqrt{M}$; Leibniz gives the exact bilinearisation

$$M' - 2M \;=\; v\,\delta(u) + u\,\delta(v), \qquad \delta(n) = n' - n,$$

so a slope–two line becomes a bilinear equation in $(u, v)$ with both variables of comparable size and, for fixed $u$, the exact line $u\,v' - (2u - u')\,v = c$, i.e. $v \in L\big(u,\, 2u - u',\, c\big)$ in the notation of Proposition 15.20. Writing $N(u)$ for the number of $v$ on that line in the balanced window $v \asymp M/u$,

$$\#\{M \le Z \text{ on the line}\} \;\le\; \sum_{u \le Z^{1/2}} N(u),$$

so the missing square–root bound follows from $N(u) \ll Z^{o(1)}$ on average; a *short–window line–multiplicity* hypothesis. This is a strictly sharper target than the one–class–per–cofactor count: the

summation length drops from $Z$ cofactors to $Z^{1/2}$ balanced divisors (empirically $Z^{0.43}$, the median of $\log u/\log M$ over nearest–to–$\sqrt{M}$ coprime splits being 0.432), and the surviving object is bilinear in a short window; the shape on which dispersion and Fouvry–Iwaniec methods operate, rather than a worst–case single–class count. The measured multiplicity is exactly 1 (mean 1.00, no sampled line carrying two solutions), so the hypothesis is not merely plausible but numerically flat; proving it, however, runs back into the determined slot, since $L(u, 2u-u', c)$ is itself a groupoid object. All identities verified exhaustively on the 591,423 squarefree $M \le 10^6$, in exact rationals for the line form (`code/bilinear_balanced.py`). By Proposition 15.19(a) that missing bound is a *one–class–per–cofactor* prime count: a line member is $M = pm$ with $p$ prime in a single determined class modulo its own cofactor $m$, so the count is $\sum_m \#\{p \le Z/m : p \equiv \alpha_c(m) \pmod m\}$, and the trivial estimate $Z/m^2 + 1$ per cofactor fails exactly on its $+1$: one candidate integer per $m$, thinned only by the primality of that single determined value. Bounding worst–case single–class counts summed over all moduli lies beyond Bombieri–Vinogradov and beyond Linnik uniformly; the same species of count guards the true $\sqrt{X}$ order of the plus layer. The two layers' remaining questions are one wall. The ledger: plus layer closed by coercivity; minus layer reduced to the near–critical spectrum. (A small unconditional companion, from Proposition 15.13: an odd member of $N' = 2N + c$ has $S(N) \equiv cN + 2 \pmod 8$.)

*The primality can be dropped: the missing bound is a divisibility count.* The formulation above asks for a bound on how often a determined value is *prime*, which is what places it beyond Bombieri–Vinogradov. That requirement can be removed. The slot $(c-m)/(m'-em)$ is an integer at all only when

$$(m' - em) \mid (c - m),$$

and line members inject into the set of $m$ satisfying that divisibility, so

$$\#\{m \le Z : \ (m' - em) \mid (c - m)\} \ \ll \ Z^{1/2+o(1)} \quad \text{uniformly in } (e, c)$$

already implies the square–root bound, with no prime–distribution input whatever.

The substitution costs almost nothing, because integrality is where the thinness lives. Counting both conditions separately on the two lines whose members are known, over $m \le 2 \times 10^7$ (so $\log Z = 16.8$): the line $n' = n - 1$ has 21 integral slots of which 16 are prime, and $n' = n + 1$ has 29 of which 19 are prime. Integrality alone is already $O(\log Z)$; primality removes a further factor of about 1.3 and no more. The prime–counting requirement was carrying a constant, while the divisibility was carrying the whole saving.

What remains is therefore a purely multiplicative question about the derivative, of the same shape as Proposition 15.20 but with the primality stripped out, and with roughly four orders of slack against the target at the ranges where the count can be observed. We record the substitution because it changes which literature the problem faces: divisibility counts for $m' - em$ rather than primes in determined classes.

## 16 Two closures on the existence route

Atom A10 asks for a prime term of the ternary recurrence of Remark 5.28. Before attacking it, two families of approach can be removed: the relaxations that would give the construction more freedom, and the algebraic methods that would avoid the archimedean place. Both are closed outright.

**Proposition 16.1** (No free slot)**.** *Let $P, Q$ be finite sets of primes with $\sigma(P)\sigma(Q) = 1$, and put $a = \prod P$, $b = \prod Q$. Then $b = a'$ and $a = b'$ exactly. In particular $Q$ is* determined *by $P$: it is the set of prime factors of $a'$, and the existence route carries exactly one parameter, the set $P$.*

Formalised. *`Erdos307.Q_determined`: given $a' = b$, the set $Q$ is exactly the set of prime factors of $a'$. The one–prime normal form of Proposition 8.4 is `oneprime_forced`, `oneprime_cycle`, `oneprime_converse` (which closes the equivalence), `oneprime_mass_identity` and `oneprime_mass_bound`, the last two giving part* (b)*: the mass identity holds for* every *witness, prime or composite, so the exactness layer alone carries the barrier. The slot calculus of Proposition 17.13 is `slot_layers` and `slot_recovery`, the filter of Corollary 17.12 is `qr_filter`, resting on `cofactor_survival`, the congruence $N' + 2N \equiv N/\ell \pmod{\ell}$, which is proved rather than assumed (stating `qr_filter` with that arithmetic as a hypothesis on abstract residues would leave it contentless); Lemma 17.1 is `Frame.csum_eq_mul_recipSum` for its first half and `symbolfact` for the Jacobi factorisation; and the reduction mechanism behind Proposition 7.9, Rigidity and Lemma 6.1 alike is isolated once as `dvd_sum_erase_iff`* (`lean/Erdos307/NormalForm.lean`)*.*

*Proof.* $\sigma(a) = a'/a$ and $\sigma(b) = b'/b$, both in lowest terms since $\gcd(m, m') = 1$ for squarefree $m$. Then $\sigma(a)\sigma(b) = 1$ reads $a'b' = ab$. From $\gcd(a', a) = 1$ and $a' \mid ab$ we get $a' \mid b$, say $b = a'k$; symmetrically $b' \mid a$, say $a = b'j$. Substituting, $a'b' = ab = b'j \cdot a'k$, so $jk = 1$ and $j = k = 1$. □

**Corollary 16.2** (The multi-slot relaxations are vacuous)**.** *There is no version of the construction in which two or more primes of $Q$ may be chosen freely. In particular one cannot fix a base $B \subset Q$ and solve $1/q_1 + 1/q_2 = r/s$ for the remaining two primes by ranging over the $\tau(s^2)$ factorisations $(rq_1 - s)(rq_2 - s) = s^2$: by Proposition 16.1 the pair $(q_1, q_2)$ is already forced to be the corresponding part of the factorisation of $a'$. The exponential parametrisation of Remark* 5.28 *is therefore intrinsic, not an artefact of the one-free-prime ansatz, and no amount of extra slots converts it into a Bunyakovsky one. What does dispose of the exponential growth is not extra slots but Proposition 4.7, which bounds the tail prime outright and so replaces the orbit by a finite divisor search.*

The second closure is not new here, but it belongs in this list because it is what forces the whole existence route to be analytic. Theorem 7.1 of Section 7 shows that over $\mathbb{F}_q[t]$ the arithmetic derivative satisfies $\deg f' \le \deg f - 1$, hence has no two–cycles and no nonzero fixed points, and Corollary 7.2 concludes that the function–field Erdős–Barbeau equation is insoluble outright. (An independent brute-force recheck, 8192 squarefree monics over $\mathbb{F}_2$ to degree 13 and 6561 over $\mathbb{F}_3$ to degree 8, 0 violations and 0 two–cycles: `code/ff_nocycle.rs`.) What is added here is not that statement but its consequence for *method*, recorded next.

*Why the existence route cannot be algebraic.* The mechanism of Theorem 7.1 is the ultrametric inequality. The cycle equation is $\sigma(a)\sigma(b) = 1$, so one of the two masses must reach 1; over $K[t]$ the mass satisfies $|\sigma(f)| = \max_i |1/\pi_i| < 1$ by ultrametricity and can never do so, whereas over $\mathbb{Z}$ the archimedean absolute value permits $\sum_{p|n} 1/p > 1$. So #307 is possible at all only because $|\cdot|_\infty$ is not ultrametric.

This has a consequence for method. Any argument for *existence* that is purely algebraic, in the sense that it applies verbatim with $\mathbb{Z}$ replaced by $K[t]$, would prove a statement that Theorem 7.1 refutes; hence no such argument exists. Together with Proposition 8.6 (no polynomial parametrisation) and Proposition 15.10 (no place of $\mathbb{Q}$ carries the descent) this closes the algebraic and geometric toolkit on *both* routes: the negative route has no place to descend at, and the positive route has no place to construct at. What remains for A10 is archimedean and analytic, which is precisely the regime in which lower bounds for primes in sparse sequences are unavailable. That

is a sharper statement of the obstruction than Remark 5.28 gives, and it is the reason the present work does not resolve the existence direction.

A third closure was claimed here in place of what follows, and it was wrong. The 34 immune families of Remark 5.4 are precisely the tail families the congruence campaign cannot kill, so each is a live candidate and the obvious move is to test them one at a time. That move *is* available, and the argument that said otherwise is worth setting out, because its error is a natural one.

**Remark 16.3** (Why the immune families looked untestable, and why they are not)**.** Deciding whether the tail family $S \cup \{q\}$ contains a Pythagorean pair is, by Remark 5.28, deciding whether $B_S x^2 - A_S y^2 = -4D_S^2$ has an integral solution with $q = (x^2 - D_S)/A_S$ prime, and two routes to that decision present themselves.

(i) *Global route.* Enumerating *all* integral points on the conic requires the fundamental unit, equivalently the class group, of the real quadratic order of discriminant $4A_S B_S$. On the first immune base that discriminant has 225 digits and is a nonsquare, so the subexponential index-calculus cost $L_\Delta(1/2) = \exp\bigl(\sqrt{\log \Delta \log\log \Delta}\bigr)$ evaluates to $10^{24.7}$ operations. This remains true, and it is what blocks the squarefree–density argument discussed above, which genuinely needs the Pell orbit modulo $p^2$.

(ii) *Parametric route.* Since $A_S$ is prime and $(D_S \mid A_S) = +1$, a square root $x_0^2 \equiv D_S \pmod{A_S}$ exists and the solutions are parametrised by $x = x_0 + tA_S$ with $q(t) = \bigl((x_0 + tA_S)^2 - D_S\bigr)/A_S$. A solution additionally needs $B_S q(t) + D_S$ to be a square, a quantity of 223 digits already at $t = 0$, so the heuristic hit rate is $10^{-112}$ per trial and the parameter $t$ appears to range without bound.

It does not. The earlier reading of (ii) overlooked the bound this paper had already proved: Proposition 4.7 forces $q \le tD_S$ and hence $x < A_S$ (Proposition 5.8), so $t = 0$ and the two square roots $x_0$ and $A_S - x_0$ exhaust the parametrisation. The decision is therefore a pair of primality tests, not an unbounded sweep, and Proposition 5.5 carries it out: all 34 immune families are empty. The global route (i) is not needed because the decision never asks for all integral points on the conic, only for those in a box the tail bound has already made small. The error was to treat a parametrisation as a search space without asking what the accompanying inequality did to it, and it cost the paper a false impossibility claim.

One hope survives all of this on its face: a lower bound for *some term of some sequence in a family* is formally weaker than one for a single sequence, and Proposition 4.5 supplies 49,961 of them. It fails, and it fails for a reason no enlargement of the family can repair.

**Proposition 16.4** (Multiplicity cannot help)**.** *Every known lower-bound method for primes in a set requires that set to have counting function at least $X^\theta$ for some fixed $\theta > 0$. The union over all* 49,961 *bases of the terms of their sequences below $X$ has counting function $C \log X$ with $C = 49{,}961/\log \varepsilon$, since each sequence grows like $\varepsilon^n$ and so contributes only $O(\log X)$ terms. Hence*

$$\frac{\log(\textit{union below } X)}{\log X} = \frac{\log\bigl(C \log X\bigr)}{\log X} \longrightarrow 0,$$

*and the family size $C$ is powerless because it sits inside a logarithm.*

*Measurement.* On an immune base $\varepsilon$ has 224 digits, so at $X = 10^{500}$ each sequence contributes 2.2 terms and the whole family contributes $10^{5.05}$, a density exponent of 0.0101. Reaching $\theta = \frac{1}{2}$ at that $X$ would need $10^{249.7}$ bases against the $10^{4.7}$ available, a shortfall of $10^{245}$. Even the absolute ceiling on the family at *every* level combined, all $2^{139} = 10^{41.8}$ subsets of the primes below 800, falls short by $10^{207.8}$ (`code/a10_multiplicity.gp`). □

Each of the four blocks was then attacked on its own terms. None fell; each is now located more precisely, and two further routes are closed.

**Proposition 16.5** (The number–field barrier transfer is FALSE)**.** *Theorem 7.1 kills the algebraic route by ultrametricity, which suggests replacing $K[t]$ by a number ring $\mathcal{O}_K$, where the absolute value is archimedean and the obstruction disappears. It does disappear, so it is not the case that for every number field $K$ the least $\prod N(\pi)$ over prime ideals with $\sum 1/N(\pi) \geq 2$ is at least that of $\mathbb{Q}$. That assertion is false. What holds is the following.*

(i) The true infimum is 16. *In the biquadratic field $K = \mathbb{Q}(\sqrt{17}, \sqrt{33})$, of discriminant* 314,721*, the prime* 2 *splits completely: there are four prime ideals of norm* 2*. They carry mass $4 \cdot \frac{1}{2} = 2$ exactly, and their product of norms is $2^4 = 16$. Conversely* 2 *is the least possible ideal norm, so mass* 2 *needs at least four ideals of norm* 2 *and any admissible family has product at least* 16*; a field can carry at most $[K : \mathbb{Q}]$ ideals of norm* 2*, so degree* 4 *is also minimal. Hence* 16 *is the exact infimum over all number fields, against $\prod_{i \leq 59} p_i > 8.77 \times 10^{112}$, a factor $10^{112}$ apart.*

(ii) Where the argument broke. *The Chebotarev computation is correct as a statement about* density*: in a degree-$d$ Galois field the completely split primes have density $1/d$, so they carry mass $d \cdot (1/d) \log \log Y = \log \log Y$, exactly as over $\mathbb{Q}$, while each costs $d \log p$, and inert primes contribute $1/p^d$, that is, nothing. The error was to read a statement about the tail as a statement about the minimum. The barrier consumes only* finitely many *primes, and for any finite set of small primes one may choose $K$ in which all of them split; Chebotarev constrains how often splitting happens, not which particular primes split.*

(iii) What survives, and what closes the route instead. *For a* fixed *field the measurement stands: $\mathbb{Q}$ needs* 59 *primes and $10^{112.9}$, whereas $\mathbb{Q}(i)$ needs* 226 *prime ideals and $10^{735.1}$* (`code/a10_fieldbarrier.gp`)*. What does not survive is the use of this proposition as a* block*. The route is nevertheless closed, and by a sounder argument, which is Proposition 16.6 below: the ideal–theoretic #307 is* trivially soluble*, so it carries no information about the rational problem.*

**Proposition 16.6** (The difficulty of #307 is distinctness, not mass)**.** *Allow $P$ and $Q$ to be finite* multisets *of primes rather than sets. Then the #307 equation is soluble with four primes in total: $\left(\frac{1}{2} + \frac{1}{2}\right)\left(\frac{1}{2} + \frac{1}{2}\right) = 1$. With $P, Q$ genuine sets the same equation forces $|P \cup Q| \geq 59$ and $\prod > 10^{112.9}$ by Theorem 4.3. The barrier is therefore a statement about the injectivity of $p \mapsto 1/p$ on the primes, not about the arithmetic of the mass condition.*

*Two consequences, and the second is the operative one.*

(i) The ideal–theoretic problem is trivial. *In $K = \mathbb{Q}(\sqrt{17}, \sqrt{33})$ the prime* 2 *splits completely into four pairwise coprime prime ideals of norm* 2*; taking $P = \{\pi_1, \pi_2\}$ and $Q = \{\pi_3, \pi_4\}$ gives $\sigma(P) = \sigma(Q) = 1$ and $|P \cup Q| = 4$* (`code/ideal307.gp`)*. Nor does the derivative transfer: $\pi_1 + \pi_2$ is the unit ideal, so $a'$ collapses. This is the correct reason the number–field route is closed, and it explains why refuting the previous proposition reopened nothing: the transferred problem was never the same problem.*

(ii) A no–go for transfers generally. *Any setting admitting two distinct objects of the same norm $n \geq 2$ on each side solves the mass equation at size four, whatever the setting is. So a dictionary permitting repeated denominators* dissolves *#307 rather than relocating its difficulty, and can carry no information about it. This prunes a class of attacks in one line, and it is the test any future transfer should face first. Corollary 4.6 passes it only because pairwise–coprime integers still have distinct least prime factors, which is distinctness re-entering through the back door.*

Formalised. *`Erdos307.multiset_solution` is the four–element witness,* *`distinctness_carries_the_barrier`* *the contrast against* *`card_ge_59_of_recipSum_ge_two`, stated as a conjunction so the two halves cannot drift apart, and* *`transfer_dissolves`* *the general no–go* (`lean/Erdos307/Distinctness.lean`).

*The three other blocks, attacked.* (i) Against Proposition 16.1. The rigidity $\prod Q = a'$ would dissolve if disjointness were an assumption one could relax. It is not: $P \cap Q = \varnothing$ is *derived* in Theorem 2.3 by the Israel mechanism, $r \in P \cap Q$ forcing $v_r(st) = -2 \neq 0$. There is nothing to relax.

(ii) The block that fell. A block was claimed here on the ground that deciding an immune family needs integral points on the conic $B_S x^2 - A_S y^2 = -4D_S^2$, hence the fundamental unit of a 225–digit discriminant. The objection to it was that conics obey Hasse–Minkowski, so with the local solubility of Remark 5.1 rational solubility is unconditional; the reply was that what is missing is integrality, not solubility. Both sides of that exchange were beside the point. Integral points are not needed either, because Proposition 4.7 confines the solution to a box containing at most two candidates, and Proposition 5.5 simply tests them. Remark 16.3 records the error. The lesson is worth more than the block was: an obstruction stated in terms of a general algorithmic problem is only as good as the argument that the problem instance is general, and here it was not.

(iii) Against Proposition 16.4. The template that exploits a family without density is Heath-Brown's: Artin's conjecture holds for all but at most two primes, with no means of saying which. It does not transfer. That argument needs each failure to impose a *detectable* structure, which several failures then cannot share. Here a failing family would be one all of whose $q_n$ are composite, and the only known mechanism for that is a covering system, which Proposition 5.17 proves does not exist. A failure would therefore be structureless, and structureless failures cannot be played against one another.

All of the above bounds the negative direction. The positive direction admits a quantitative shadow that had not been measured: a solution is exactly $r(a) := \sigma(a)\sigma(a') = 1$, so one may ask how close $r$ actually comes to 1. The answer is instructive, and it disposes of the usual way such heuristics are calibrated.

**Proposition 16.7** (The near-miss spectrum, and why the heuristic cannot be calibrated)**.** *Over all* 4,385,727 *integers* $a \leq 10^7$ *with* $a$ *and* $a'$ *both squarefree, the maximum of* $r(a)$ *is* 0.5535*, attained at* $a = 3{,}972{,}254$*; the frontier advances by decade as* 0.333, 0.345, 0.502, 0.502, 0.536, 0.554*, and at* $10^8$ *it reaches* 0.586076 *at* $a = 23{,}667{,}105$ (`code/sweep1e8.rs`, which reproduces every earlier decade independently)*. This last decade was run as a test of Proposition 16.9: the bound* $R(10^8) \leq 0.67270$ *and the range* $[0.55, 0.62]$ *suggested by the earlier decades were both recorded before the computation, and both hold. In particular* $r \geq 1$ *never occurs, the defect* $|a'' - a|$ *is of size* $a^{0.9}$ *or larger in all but three cases, and the smallest relative defect is* 0.4465*. Consequently the constant in the heuristic count of two-cycles,* $f(1)\log X$ *with* $f$ *the density of* $r$ *at* 1*, is* not measurable*, and the obstruction is two–sided.*

*By Theorem* 4.3 *the region* $r \geq 1$ *lies above* $10^{112.9}$*. The window immediately to the* left *of* 1 *is barred as well, by AM–GM rather than by the barrier hypothesis:* $\sigma(a)+\sigma(a') \geq 2\sqrt{\sigma(a)\sigma(a')} = 2\sqrt{r}$*, and* $2\sqrt{r} > T_{58} = 1.99874\ldots$ *as soon as* $r > (T_{58}/2)^2 = 0.998740\ldots$*, whence* $|P \cup Q| \geq 59$ *and the product again exceeds* $\Pi_{59} > 8.77 \times 10^{112}$*. So the whole neighbourhood* $r \in (0.998740, \infty)$ *of the value* 1 *lies above the barrier, not merely the half–line* $r \geq 1$*, and no sweep reaches any of it* (`code/a10_nearmiss.rs`, `code/a10_density1.rs`)*. Barring only* $r \geq 1$ *would leave the left window formally open and the unmeasurability argument one–sided.*

**Proposition 16.8** (An effective approximation theorem, with a certified witness)**.** *Bado's Theorem 10.2 [28] asserts that for every* $Y$ *and every* $\varepsilon > 0$ *there are disjoint finite sets of primes*

*$P, Q$, all greater than $Y$, with $0 \le \sigma(P)\sigma(Q) - 1 < \varepsilon$. The proof is a greedy accumulation and is ineffective. Made effective it is expensive: to land in $[T, T+\eta)$ the accumulation must start above $\approx 1/\eta$, and Mertens gives $\sum_{Y_1 < p \le W} 1/p \approx \log\log W - \log\log Y_1 = T$, so $W = Y_1^{e^T}$; two stages with $T \approx 1$ put the largest prime and the cardinality both at $\approx \varepsilon^{-e^2} = \varepsilon^{-7.389\ldots}$.*

*A Newton ladder does the same work with a handful of primes. Take $Q = \{2, 3, 5\}$, so $\sigma(Q) = 31/30$ and $P$ must approach $30/31$; from a pool, repeat $p \leftarrow$ the least prime $\ge 1/d$ and $d \leftarrow d - 1/p$, where $d$ is the running deficit. Each rung squares it, $d_{\text{new}} = d - 1/p \approx t\, d^2$ with $t$ the local prime gap, so the largest prime is $\approx \varepsilon^{-1/2}$ and the number of rungs is $O(\log\log(1/\varepsilon))$. Taking the pool to be the primes $7 \le p \le 271$ and eleven rungs gives $|P| = 66$, $|Q| = 3$, largest prime of 2,010 digits, and, in exact rational arithmetic,*

$$\left|\sigma(P)\sigma(Q) - 1\right| \;=\; 4.98523\ldots \times 10^{-4016},$$

*a rational whose numerator has 129 digits and whose denominator has 4,145. Every rung prime is* proved *prime by* APR–CL*, not merely a pseudoprime* (`code/nearmiss_effective.gp`)*. The certificate is tiered, and a reader may stop where they like:*

| *rungs* | 9 | 10 | 11 |
|---|---|---|---|
| *largest prime (digits)* | 505 | 1,007 | 2,010 |
| $\left\lvert\sigma(P)\sigma(Q) - 1\right\rvert$ | $9.57 \times 10^{-1007}$ | $2 \times 10^{-2010}$ | $4.99 \times 10^{-4016}$ |
| *cumulative certification* | $< 10$ *s* | $\approx 110$ *s* | $\approx 27$ *min* |

*The statement is machine–checked at the three–rung tier, where the largest prime is 2,348,039,453 and the defect is $2.1111\ldots \times 10^{-18}$: the primality of all 58 primes, the disjointness of the two sides, the exact value of $\sigma(P)$ as a ratio of a 126– and a 129–digit integer, and the bound $|\sigma(P)\sigma(Q) - 1| < 10^{-17}$ are all verified in `Erdos307.Witness.effapprox_witness`, on the standard axioms and with no appeal to kernel–external evaluation. Beyond that tier the primes are certified by APR–CL, which Mathlib does not carry; the higher rungs are therefore cited to the* PARI/GP *certificate rather than formalised.* 0 `sorry` (`lean/Erdos307/Witness.lean`).

The gap is not marginal. At $\varepsilon = 10^{-13}$ the greedy needs a largest prime near $10^{94}$ and some $10^{92}$ primes; the ladder needs a largest prime near $10^7$ and three rungs. The ladder is exponentially better in the size of the largest prime and doubly exponentially better in the count, and the reason is visible: greedy convergence is linear in the number of primes, Newton convergence is quadratic. Truncating gives certificates of any size, five rungs and a 34–digit largest prime already give $6.07 \times 10^{-66}$, and seven rungs give $3.22 \times 10^{-254}$. The witness above approaches 1 from below; taking the last rung's prime just under $1/d$ instead overshoots, so either side is available and the statement of Theorem 10.2 as printed is met.

What this is not: progress towards a solution. A free pair of disjoint prime sets carries none of the rigidity an exact hit requires ($\sigma(P)\sigma(Q) = 1$ forces $\prod P = (\prod Q)'$ and $\prod Q = (\prod P)'$, and nothing in the ladder produces that) and by Proposition 16.7 the size of $|\sigma(P)\sigma(Q) - 1|$ does not measure progress towards the integer identity at all. The value of the construction is that it makes an existence theorem effective and exhibits a witness anyone can check.

**Proposition 16.9** (The near–miss frontier obeys a double–logarithmic law)**.** *For $X \ge 2$ put $R(X) = \max\{r(a) : a \le X$ squarefree$\}$, where $r(a) = \sigma(a)\sigma(a')$. Let $\omega_{\max}(X)$ be the largest $k$ with $\Pi_k \le X$, let $\sigma_{\max}(X) = T_{\omega_{\max}(X)}$, and let $n(X)$ be the largest $n$ with $\Pi_n \le X^2 \sigma_{\max}(X)$. Then*

$$R(X) \;\le\; \left(\tfrac{1}{2}\, T_{n(X)}\right)^2.$$

*Proof.* Rigidity gives $\gcd(a,a')=1$, so $U=\operatorname{supp}(a)\sqcup\operatorname{supp}(a')$ has $|U|=\omega(a)+\omega(a')$ distinct primes and $\prod_{p\in U}p\mid aa'$. By AM–GM, $\sum_{p\in U}1/p=\sigma(a)+\sigma(a')\ge 2\sqrt{\sigma(a)\sigma(a')}=2\sqrt{r}$, while the same sum is at most $T_{|U|}$ by the extremality of the smallest primes; hence $r\le(T_{|U|}/2)^2$. For the size, $\Pi_{|U|}\le\prod_{p\in U}p\le aa'=a^2\sigma(a)\le X^2\sigma_{\max}(X)$, so $|U|\le n(X)$, and $T$ is increasing. □

The bound is unconditional, needs no squarefreeness of $a'$, and is close: against the measured frontier of Proposition 16.7 it reads

| $X$ | $10^2$ | $10^3$ | $10^4$ | $10^5$ | $10^6$ | $10^7$ |
|---|---|---|---|---|---|---|
| bound | 0.4014 | 0.4920 | 0.5296 | 0.5879 | 0.6129 | 0.6342 |
| $\max r$ observed | 0.333 | 0.345 | 0.502 | 0.502 | 0.536 | 0.554 |

(`code/frontier_bound.gp`). Two consequences. First, since $T_{n(X)}=\log\log\bigl(X^2\sigma_{\max}\bigr)+M+o(1)$ by Mertens, the frontier can climb only like

$$R(X)\ \le\ \tfrac14\bigl(\log\log X^2+M+o(1)\bigr)^2,$$

a *doubly logarithmic* law: this is why the measured frontier crawls, and why an extrapolation of its per–decade advance is worthless. Second, the bound reaches 1 exactly when $T_n\ge 2$, that is $n\ge 59$, that is $X^2\sigma_{\max}\ge\Pi_{59}$, giving $X\ge 2.0942\times10^{56}$, the constant of Theorem 4.3, recovered here from the frontier side rather than from the cycle equations. The two derivations are independent and agree to all printed digits.

**Corollary 16.10** (The cost of a near–miss; the barrier as a function of $r$)**.** *For $r>0$ put $n(r)=\min\{n: T_n\ge 2\sqrt{r}\}$. Every squarefree $a$ with $r(a)\ge r$ has $|U|\ge n(r)$, hence $aa'\ge\Pi_{n(r)}$ and*

$$\min(a,a')\ \ge\ \sqrt{\Pi_{n(r)}/T_{n(r)}}.$$

*Proof.* The first display of the proof of Proposition 16.9 gives $2\sqrt{r}\le\sum_{p\in U}1/p\le T_{|U|}$, so $|U|\ge n(r)$; the rest is Theorem 4.3 verbatim with 59 replaced by $n(r)$. □

At $r=1$ this returns $n=59$ and $2.0929\times10^{56}$, so Theorem 4.3 is the endpoint of a one–parameter family rather than an isolated statement. The family is what makes the crawl quantitative (`code/nearmiss_cost.gp`):

| $r$ | 0.5 | 0.554 | 0.6 | 0.7 | 0.8 | 0.9 | 0.99 | 1 |
|---|---|---|---|---|---|---|---|---|
| $n(r)$ | 8 | 9 | 11 | 16 | 24 | 37 | 56 | 59 |
| $\min(a,a')\ge$ | $2.6{\cdot}10^3$ | $1.2{\cdot}10^4$ | $3.6{\cdot}10^5$ | $4.4{\cdot}10^9$ | $1.2{\cdot}10^{17}$ | $4.3{\cdot}10^{30}$ | $4.7{\cdot}10^{52}$ | $2.1{\cdot}10^{56}$ |

Each increment in $r$ is paid for exponentially in the product, which is the same double–logarithmic law seen from the other side: $r$ enters the bound only through $2\sqrt{r}$, and $T_n$ grows like $\log\log p_n$. It also explains the shape of the swept data. The record $r=0.5535$ needs $\min(a,a')\ge1.2\times10^4$ and was found at $a=3{,}972{,}254$; the next decade of $r$ costs four decades of $a$, and by $r=0.7$ the requirement has passed $10^9$, beyond any sweep that has been run.

The restriction $r\le1$ is not used in the proof, and above 1 the corollary says how expensive it is for the second derivative to *exceed* its argument:

| $r$ | 1 | 1.03944 | 1.25 | 1.5 | 2 | 9/4 |
|---|---|---|---|---|---|---|
| $n(r)$ | 59 | 71 | 212 | 940 | 37,963 | 361,139 |
| $\min(a,a')\ge$ | $10^{56.3}$ | $10^{71.3}$ | $10^{274.8}$ | $10^{1579.7}$ | $10^{98275.7}$ | $10^{1127769.3}$ |

So $a''\ge 2a$ forces a union support of 37,963 primes and $\min(a,a')>10^{98275}$: the second derivative cannot double its argument except at wholly inaccessible size. The witness of Theorem 16.33, with $r=1.03944$, needs $|U|\ge71$ and $\min\ge10^{71.3}$, and its 142 digits supply both.

**Remark 16.11** (One inequality, six barriers)**.** Corollary 16.10 and Propositions 4.16 and 4.18 are the same statement read at different constants. Every barrier in this note is an instance of

$$\sum_{p\in U}\frac{1}{p}\ \geq\ c\quad\Longrightarrow\quad |U|\ \geq\ n(c),\qquad \prod_{p\in U} p\ \geq\ \Pi_{n(c)},$$

with $n(c)=\min\{n : T_n \geq c\}$, and the arithmetic enters only in producing $c$:

| configuration | $c$ | $n(c)$ | recorded as |
|---|---|---|---|
| near–miss ratio $r$ | $2\sqrt{r}$ | varies | Corollary 16.10 |
| $r=1$, i.e. a two–cycle | 2 | 59 | Theorem 4.3 |
| $k$–cycle | $k/\lfloor k/2\rfloor$ | 59 ($k$ even) | Proposition 4.16 |
| three–cycle | 3 | 361,139 | Proposition 4.15 |
| all–odd, $\min U\geq 3$ | 2 on $p\geq 3$ | 1,412 | Theorem 4.14 |
| $a''\geq 2a$ | $2\sqrt{2}$ | 37,963 | above |
| orbit of growth $G$ in $L$ steps | $\frac{L}{\lceil L/2\rceil}G^{1/L}$ | varies | Proposition 16.19 |

The coincidence $2\sqrt{9/4}=3=c(3)$ is why $r=9/4$ and the three–cycle carry the identical bound 361,139: they are the same value of $c$. What looked like six separate barriers is one mass–to–size dictionary, and the only mathematics specific to each row is the derivation of its $c$.

**Proposition 16.12** (The mass window, and the forced core)**.** *Let $(a,b)$ be a two–cycle with $|U|=n$ and let $t_n>1$ solve $t+1/t=T_n$. Then*

$$\frac{1}{t_n}\ \leq\ \sigma(a),\,\sigma(b)\ \leq\ t_n,$$

*and every prime $p$ with $1/p\geq T_{n+1}-2$ lies in $U$.*

*Proof.* The masses are positive with product 1 and sum at most $T_n$, so each is between the roots of $z^2-T_nz+1$. If $p\notin U$ then the $n$ primes of $U$ carry at most $T_{n+1}-1/p$, and $\sigma(U)>2$ forces $1/p<T_{n+1}-2$. □

Both halves are sharp and both are informative at $n=59$. The window is [0.952682, 1.049668]: *each member's mass is pinned within five per cent of* 1. And the forcing puts every prime up to 167 in $U$, 39 of the 59, so two thirds of the support is determined before any search begins. The window also recovers Proposition 16.18: the smaller member needs $T_k\geq 1/t_n$, which gives $k\geq 3$ for $n\leq 68$, $k\geq 2$ from $n=69$ and $k\geq 1$ at $n=1413$, reproducing that proposition's three thresholds exactly. The forcing decays with the level:

| $n$ | 59 | 60 | 61 | 62 | 63 | 65 | 68 | 72 |
|---|---|---|---|---|---|---|---|---|
| all $p\leq$ | 167 | 103 | 73 | 61 | 47 | 37 | 23 | 19 |
| forced | 39 | 27 | 21 | 18 | 15 | 12 | 9 | 8 |

(`code/forced_core.gp`). This accounts for the size of the level–59 enumeration. The 39 forced primes carry mass 1.91162…, so the remaining 20 must supply more than 0.08838 from primes above 167; the twenty smallest such primes reach exactly that, at 277. The free part is therefore a choice of 20 primes from a narrow band, which is why the admissible count is 49,961 and not astronomical. At level 60 the forced core has fallen to 27 and the free part to 33, and by level 68, the top of the window Proposition 16.18 says a level search must cross, only 9 primes remain forced.

**Proposition 16.13** (The free band; why exactly one level is finite)**.** *Let $U$ be an admissible support with $|U| = n$ and let $q = \max U$. If $T_{n-1} < 2$ then*

$$q \;<\; \frac{1}{2 - T_{n-1}}.$$

*For $n \le 58$ no admissible support exists, since $T_n < 2$. For $n = 59$ the hypothesis holds, $T_{58} = 1.99874\ldots$, and $q < 793.68$, so $q \le 787$. For $n \ge 60$ it fails, since $T_{59} = 2.00235\ldots > 2$, and no bound on $q$ is available.*

*Proof.* The $n-1$ primes of $U$ other than $q$ carry at most $T_{n-1}$, so $\sigma(U) > 2$ forces $1/q > 2 - T_{n-1}$. □

This is the mechanism of Proposition 4.8 in its sharpest form. Level–by–level enumeration is finite at 59 and infinite from 60, and the switch is the single fact

$$T_{58} \;<\; 2 \;<\; T_{59}.$$

Below 59 there is not enough mass to reach 2 at all; at 59 every prime is capped at 787; at 60 the first 59 primes already carry mass $2.00235 > 2$, so appending *any* prime leaves an admissible support and the count is infinite at once. There is nothing gradual about the transition, and it happens at one level.

Together with Proposition 16.12 this closes level 59 from both sides. The support contains every one of the 39 primes $\le 167$ and is contained in the 138 primes $\le 787$, so the free part is a choice of 20 primes from the 99 candidates in $(167, 787]$. Unconstrained that is $\binom{99}{20} = 4.288 \times 10^{20}$ choices; the mass condition $\sigma(U) > 2$ cuts it to 49,961, a reduction by a factor $8.6 \times 10^{15}$ (`code/free_band.gp`). The level–59 computation of Proposition 4.5 is therefore not a brute force over an unbounded space but an exact traversal of a set whose two ends are both pinned by the same inequality.

**Corollary 16.14** (The level–59 count, from the two pins)**.** *The admissible supports at level 59 are exactly the sets*

$$\{p : p \le 167\} \cup S, \qquad S \subset \{p : 167 < p \le 787\}, \quad |S| = 20, \quad \sum_{p \in S} \frac{1}{p} \;>\; 2 - T_{39} \;=\; 0.0883759429\ldots,$$

*and there are 49,961 of them.*

*Proof.* Membership of the core is Proposition 16.12, the ceiling is Proposition 16.13, and the displayed inequality is $\sigma(U) > 2$ after subtracting the forced mass $T_{39} = 1.9116240570\ldots$. The count is a 20–from–99 knapsack, evaluated in `code/level59_count.rs`. □

This is not a closed formula, subset–sum counts do not have one, but it is a closed *characterisation*, and it reduces the level to a search of 119,509 nodes in place of the original enumeration. That the two independent routes agree on 49,961 is the point: the pins are not merely necessary conditions that happen to hold, they cut out the level exactly, and nothing outside them is doing any work. The evaluation also reports no subset sum within $10^{-12}$ of the threshold, so the floating comparison is measured to be safe rather than assumed. For orientation, the baseline 20 smallest candidates carry $0.0907260944\ldots$, exceeding the threshold by $T_{59} - 2 = 0.0023501514\ldots$, and every admissible $S$ is a perturbation of that baseline within this budget.

**Proposition 16.15** (How many rungs the ladder has)**.** *Let $(a, b)$ be a Pythagorean pair with $a < b$, $t = b/a$ and $|U| = n$. Positivity of the two masses gives $-t < k \le 0$, so $k$ takes at most $\lceil t \rceil$ values,*

*and* $t \le t_n$ *where* $t_n + 1/t_n = T_n$*. Hence the number of rungs available at level* $n$ *is at most* $\lceil t_n \rceil$*, and*

$$t_n > m \iff T_n > m + \tfrac{1}{m}.$$

*In particular the ladder has exactly the two rungs* $k \in \{0, -1\}$ *at every level* $n \le 1412$*, and the third rung* $k = -2$ *opens only at* $n = 1413$*.*

Two readings. First, the hypothesis $\sigma(ab) < 5/2$ under which Proposition 15.5 confines $k$ to $\{0, -1\}$ is not a restriction anywhere a search operates: it is equivalent to $t < 2$, hence to $n \le 1412$, and at level 59 one has $t = 1.049668$ with $\lceil t \rceil = 2$. The confinement is unconditional throughout the range the rest of this note inhabits, and the hypothesis only becomes real past level 1412.

Second, 1413 is not a new constant. It is the level at which $T_n$ reaches $5/2$, and that is the same threshold as the all–odd case of Theorem 4.14: an all–odd cycle needs mass 2 on primes $\ge 3$, hence $T_n \ge 2 + \frac{1}{2}$. The level at which the ladder acquires a third rung and the level at which an all–odd cycle becomes possible are the same level, for the same reason (`code/level_structure.rs`). The fourth rung would need $T_n > 10/3$, beyond $\pi(4 \times 10^7)$.

**Proposition 16.16** (The admissible supports are not a tree)**.** *There are admissible supports at level* 60 *with no admissible subset at level* 59*. Explicitly,*

$$U \;=\; \{p : p \le 271\} \;\cup\; \{809,\, 811\}$$

*has* $|U| = 60$ *and* $\sigma(U) = 2.0012091\ldots > 2$*, while every* 59*–element subset has mass below* 2*.*

*Proof.* Deleting the largest prime loses the least mass, so if that deletion is inadmissible every deletion is. Here $\sigma(U) - 1/811 = 1.9999761\ldots < 2$. □

Such supports are exactly those of the form $V \cup \{q\}$ with $|V| = 59$, $\sigma(V) \le 2$, $q > \max V$ and $q < 1/(2 - \sigma(V))$, and they are not exceptional. Taking $V = \{p \le 271\} \cup \{r\}$ with $r > 794$, so that $\sigma(V) < 2$, each $r$ admits every prime $q$ in $\big(r,\ 1/(2 - \sigma(V))\big)$: at $r = 797$ that interval reaches 190,414 and contains 17,061 primes, and summing over $r < 4000$ gives 60,831 such supports from this one shape alone (`code/level_structure.rs`).

The consequence is for the organisation of a level search, and it is independent of Proposition 4.8. That proposition says level 60 has infinitely many admissible supports; this one says that even the part reachable with bounded primes is not exhausted by the base–plus–tail organisation, because the admissible supports do not form a tree under adjunction. Closing every one–new–prime family at level 60, which is question (i) of Section 13, would therefore still not close level 60. The searches of Propositions 5.2 and 4.5 are stated over one–new–prime families throughout and are unaffected; what changes is the reading of what completing them would buy.

**Proposition 16.17** (The level–minimal structure at any $c$)**.** *Let* $U$ *satisfy* $\sum_{p \in U} 1/p \ge c$ *with* $|U| = n(c) = \min\{n : T_n \ge c\}$*. Then every prime* $p$ *with* $1/p \ge T_{n(c)+1} - c$ *lies in* $U$*, and* $\max U < 1/\big(c - T_{n(c)-1}\big)$*, the denominator being positive by minimality of* $n(c)$*.*

*Proof.* Both are Propositions 16.12 and 16.13 with 2 replaced by $c$; neither argument uses the value. □

The slack $T_{n(c)} - c$ is the total mass a support may waste, and it collapses as $c$ grows, so the

support becomes more forced, not less (`code/level_structure.rs`):

| $c$ | 2 | 9/4 | 7/3 | 5/2 | 3 |
|---|---|---|---|---|---|
| $n(c)$ | 59 | 231 | 400 | 1,413 | 361,139 |
| $T_{n(c)} - c$ | $2.35{\cdot}10^{-3}$ | $2.33{\cdot}10^{-4}$ | $2.44{\cdot}10^{-4}$ | $2.07{\cdot}10^{-5}$ | $3.09{\cdot}10^{-8}$ |
| forced primes | 39 | 181 | 259 | 1,175 | 314,407 |
| forced fraction | 66% | 78% | 65% | 83% | 87% |
| $\max U <$ | 793.7 | 2,194 | 8,284 | 15,592 | $6.19{\cdot}10^{6}$ |

The row $c = 2$ is Propositions 16.12 and 16.13, recovered. The row $c = 3$ is the three–cycle of Proposition 4.15: at its minimal level a three–cycle contains *every* prime up to 4,476,608, that is 314,407 of its 361,139, and no prime beyond $6.19 \times 10^6$. Its support is almost completely determined, with a slack of $3.09 \times 10^{-8}$ to distribute among the remaining 46,732 places.

The same reading applies with the mass restricted to primes $\geq P$, which is the setting of Proposition 4.18. At $P = 3$, the all–odd case of Theorem 4.14, the minimal level is $n = 1412$ with slack $2.069 \times 10^{-5}$; every odd prime up to 9,484 is forced, 1,174 of the 1,412 or 83%, no prime exceeds 15,592, and the product is $10^{5040.1}$. At $P = 5$ the figures are $n = 40{,}259$, 24,770 forced, $\max U < 1.61 \times 10^6$ and $10^{209582}$, reproducing Proposition 4.18 from the structural side.

That the all–odd case is 83% forced does not make it a feasible search, and it is worth recording why, since the shape invites the comparison with level 59. There the free part was 20 primes from 99 candidates and the count was 49,961. Here it is 238 from 643: the forced mass is $1.97751561\ldots$, so the free primes must carry more than $0.02248438\ldots$ against a baseline of $0.02250507\ldots$, an excess of $2.069 \times 10^{-5}$. A traversal capped at $4 \times 10^8$ nodes reaches $1.977 \times 10^8$ admissible supports without exhausting the space (`code/level_structure.rs`). The slack per free place, not the forced fraction, is what decides enumerability, and here it is 87 times larger per place than at level 59. Question (ii) of this section does not close this way.

**Proposition 16.18** (The barrier stratified by the smaller support)**.** *Let $(a, b)$ be a two–cycle and put $k = \min(\omega(a), \omega(b))$. Then*

$$k = 1 \;\Rightarrow\; |U| \geq 1413, \qquad k = 2 \;\Rightarrow\; |U| \geq 69, \qquad k \geq 3 \;\Rightarrow\; |U| \geq 59,$$

*and the last is attained. Consequently every two–cycle with $|U| \leq 68$ has $\min(\omega(a), \omega(b)) \geq 3$.*

*Proof.* Write $T = T_{|U|}$. The supports are disjoint, so $\sigma(a) + \sigma(b) \leq T$, and $\sigma(a) \leq T_k$ for the member with $k$ primes. The product $x(T - x)$ is increasing for $x \leq T/2$, so $\sigma(a)\sigma(b) \leq m(T - m)$ with $m = \min(T_k, T/2)$, and $\sigma(a)\sigma(b) = 1$ forces $m(T - m) \geq 1$. For $k \geq 3$ one has $T_k \geq T_3 = 31/30 > T/2$ in the relevant range, so $m = T/2$ and the condition is $T \geq 2$, i.e. $|U| \geq 59$. For $k \leq 2$ one has $T_k < T/2$, so $m = T_k$ and the condition is $T \geq T_k + 1/T_k$: at $k = 1$, $T \geq 2.5$ and $|U| \geq 1413$; at $k = 2$, $T \geq 2.0333\ldots$ and $|U| \geq 69$. □

The stratification is sharp at each end. At $k = 1$ the member is a prime $p$, so $\sigma(b) = p \geq 2$ and $b$ must carry mass 2 on primes other than $p$, which needs 1412 of them; at $k = 2$ the best case is $a = 6$, whence $\sigma(b) \geq 6/5$ on primes $\geq 5$, which needs 67. This contains and extends Proposition 16 of [9], that no two–cycle has both members a product of two primes: the present statement rules out *either* member having two prime factors unless $|U| \geq 69$, and either having one unless $|U| \geq 1413$. Since Proposition 4.5 gives $|U| \geq 60$, the window $60 \leq |U| \leq 68$ that any level–by–level search must cross admits only splits with both $\omega(a) \geq 3$ and $\omega(b) \geq 3$ (`code/split_bound.gp`).

**Proposition 16.19** (The derivative cannot outgrow its own support)**.** *Let $a_0 \to a_1 \to \cdots \to a_L$ be an orbit segment of the arithmetic derivative with every $a_i$ squarefree, put $G = a_L/a_0$ and let $U = \bigcup_{i<L} \operatorname{supp}(a_i)$. Then*

$$\sum_{p\in U} \frac{1}{p} \;\geq\; \frac{L}{\lceil L/2\rceil}\, G^{1/L}, \qquad \textit{equivalently} \qquad G \;\leq\; \Big(\frac{\lceil L/2\rceil}{L}\, T_{|U|}\Big)^{L},$$

*and for even $L$ the geometric growth rate satisfies $G^{1/L} \leq T_{|U|}/2$.*

*Proof.* Squarefreeness gives $\gcd(a_i, a_i') = 1$, so consecutive members are coprime and $S_p = \{i < L : p \mid a_i\}$ has no two consecutive elements: it is independent in the path on $L$ vertices, whence $|S_p| \leq \lceil L/2\rceil$. Then $\sum_{i<L} \sigma(a_i) = \sum_{p\in U} |S_p|/p \leq \lceil L/2\rceil \sum_{p\in U} 1/p$, while $\prod_{i<L} \sigma(a_i) = a_L/a_0 = G$ gives $\sum_{i<L} \sigma(a_i) \geq LG^{1/L}$ by AM–GM. The second form inverts the first using $\sum_{p\in U} 1/p \leq T_{|U|}$. □

This is the general shape of which the rest of the dictionary is a specialisation: at $L = 2$ it reads $\sum_{p\in U} 1/p \geq 2\sqrt{G}$, which is Corollary 16.10 with $G = r$; at $G = 1$ it returns $c = L/\lceil L/2\rceil \to 2$, which is Proposition 4.16 on the path in place of the cycle. The statement it adds is dynamical rather than Diophantine, and it bounds the growth branch of Ufnarovski–Åhlander's Conjecture 2: an orbit may tend to infinity, but only slowly unless it accumulates an enormous support (`code/growth_bound.gp`):

| rate $g$ | 1.05 | 1.10 | 1.25 | 1.50 | 1.75 | 2 |
|---|---|---|---|---|---|---|
| $\lvert U\rvert \geq$ | 97 | 170 | 1,413 | 361,139 | $4.6\times10^9$ | $4.3\times10^{16}$ |
| largest prime $\geq$ | $5.4{\cdot}10^2$ | $1.0{\cdot}10^3$ | $1.2{\cdot}10^4$ | $5.2{\cdot}10^6$ | $1.2{\cdot}10^{11}$ | $1.8{\cdot}10^{18}$ |

A squarefree orbit that doubles at every step needs some $4.3 \times 10^{16}$ distinct primes in its supports and a prime past $1.8 \times 10^{18}$: sustained rapid growth is not merely unobserved but impossible on a small support. The hypothesis that every $a_i$ is squarefree is essential and is not generic along orbits; it is automatic on cycles, by Ufnarovski–Åhlander, which is why the cycle rows of the dictionary carry no such caveat.

**Corollary 16.20** (Sustained growth needs scale, and the scale is enormous)**.** *Let $a_0 \to \cdots \to a_L$ be a squarefree orbit segment with every member at most $X$ and geometric growth rate at least $g > 1$. Then*

$$\log_2 X \;\geq\; \sqrt{n(2g)\,\log_2 g}\,, \qquad n(c) = \min\{n : T_n \geq c\}.$$

*Proof.* Proposition 16.19 gives $\sum_{p\in U} 1/p \geq 2g$, so $|U| \geq n(2g)$. Every member is at most $X$, so $\omega(a_i) \leq \log_2 X$ and $|U| \leq \sum_{i<L} \omega(a_i) \leq L \log_2 X$; growth at rate $g$ from $a_0 \geq 2$ gives $a_L \geq g^L$, so $L \leq \log_g X = \log_2 X/\log_2 g$. Combining, $(\log_2 X)^2/\log_2 g \geq n(2g)$. □

The threshold is explosive (`code/growth_bound.gp`):

| $g$ | 1.10 | 1.25 | 1.50 | 1.75 | 2 |
|---|---|---|---|---|---|
| members must exceed | $10^{1.5}$ | $10^{6.4}$ | $10^{138}$ | $10^{18420}$ | $10^{6.23\times10^7}$ |

The arithmetic derivative can therefore grow slowly at any scale and quickly only at scales no computation reaches. Rate 1.25 needs members past $2.6 \times 10^6$, so it is visible inside the sweep of Proposition 16.7; rate 1.5 needs $10^{138}$; and a squarefree orbit cannot double at every step unless its members exceed $10^{6.2\times10^7}$.

Inverting the corollary bounds the rate itself. The largest $g$ a squarefree orbit can sustain up to $X$ solves $n(2g)\log_2 g = (\log_2 X)^2$, and since $n(c)$ grows like $\mathrm{Li}\big(e^{e^{c-M}}\big)$ this gives $e^{e^{2g}} \asymp (\log_2 X)^2$, that is

$$g \;\sim\; \tfrac{1}{2}\log\log\log X.$$

| $X$ | $10^8$ | $10^{20}$ | $10^{100}$ | $10^{1000}$ | $10^{10^5}$ | $10^{10^7}$ |
|---|---|---|---|---|---|---|
| max sustainable $g$ | 1.2768 | 1.3590 | 1.4808 | 1.6174 | 1.8134 | 1.9537 |

The ceiling is a triple logarithm, so it is effectively a constant: even at $X = 10^{10^7}$ the rate stays below 2, and for any range a computation can address it is below 1.5. The reading for the escape branch of Ufnarovski–Åhlander's Conjecture 2 is that an orbit tending to infinity while remaining squarefree must do so sub–geometrically in every bounded window, its rate decaying like the triple logarithm of that window. Escape is possible; escape at a fixed geometric rate is not.

The mass window transfers to the normal form as well. In Proposition 8.4 a solution is $a = Mp$, so $\sigma(a) = \sigma(M) + 1/p$, and Proposition 16.12 places $\sigma(a)$ in $[1/t_n, t_n]$. Hence

$$\sigma(M) \;\in\; \big[\,1/t_n - 1/p,\; t_n\,\big],$$

which at level 59 is $[0.9527 - 1/p,\ 1.0497]$: the base of the one–prime normal form carries mass within five per cent of 1, just as each member does, and $\omega(M) \geq 2$ by Proposition 16.18.

The region is not unreachable, only unsweepable. The witness $n_0$ of Theorem 16.33 has $n_0$ and $n_0'$ both squarefree and $r(n_0) = \sigma(n_0)\sigma(n_0') = 1.03944\ldots$, so $n_0'' > n_0$: the value 1 that a solution must hit is interior to the range of $r$, not extremal, and no inequality on $r$ alone can settle #307. The same construction approaches 1 from below. Tuning $\sigma(a)$ against the forced $\sigma(a') = \frac{31}{30} + \varepsilon$ gives

$$a = 3359 \prod_{\substack{7\leq p\leq 269\\ p\notin\{229,271\}}} p, \qquad a' = 2\cdot 3\cdot 5\cdot 383\cdot 113717\cdot P_{99},$$

with $P_{99}$ prime, both $a$ and $a'$ squarefree, 108 digits, and

$$r(a) = 0.99207796\ldots, \qquad |r(a) - 1| = 7.9220\times 10^{-3},$$

against the smallest relative defect 0.4465 over the swept range (`code/nearmiss_tune.gp`).

The same tuning with its acceptance window mirrored, $\sigma(a)$ just *above* 30/31 rather than just below, reaches 1 from the other side and closer. With

$$a = 3301 \prod_{\substack{7\leq p\leq 293\\ p\notin\{137,263\}}} p, \qquad a' = 2\cdot 3\cdot 5\cdot P_{117},$$

where $P_{117}$ is a 117–digit prime, one has $a$ of 118 digits and, exactly,

$$r(a) = 1.001014244\ldots, \qquad |r(a) - 1| = 1.0142\times 10^{-3}, \qquad a'' > a.$$

This is exact rather than a lower bound, since $a'$ factors completely: the cofactor after trial division is prime, so $\sigma(a')$ is known and not merely bounded below. It improves the record on both sides at once, against $7.9220\times 10^{-3}$ from below and $3.944\times 10^{-2}$ from above, the latter being $r(n_0)$ of Theorem 16.33 (`code/nearmiss_above.gp`, re–checked independently in `code/nearmiss_above_verify.gp`).

That record is not the natural limit, and seeing why identifies what the near–miss problem actually is. In every witness of this shape $a' = 2\cdot 3\cdot 5\cdot P$ with $P$ prime, so $\sigma(a') = \frac{31}{30} + P^{-1}$ with $P^{-1}$ of size $10^{-118}$, and therefore

$$|r(a) - 1| = \tfrac{31}{30}\,\big|\sigma(a) - \tfrac{30}{31}\big|$$

to any precision that matters. The quantity $a'$ has dropped out. What remains is the Diophantine question of how closely a sum of *distinct prime reciprocals* approaches 30/31 from above, and the $10^{-3}$ above is only the granularity of adjusting by a single prime below 4000.

Closing the deficit in stages removes that ceiling. With $d = \frac{30}{31} - \sigma(S)$ for $S$ the primes in $[7, 271]$, take $p_1$ the least prime exceeding $1/d$, then $p_2$ the least exceeding $1/(d - p_1^{-1})$, and finally $p_3$ the greatest prime below the reciprocal of what remains; the overshoot is then of order $\mathrm{gap}(p_3)/p_3^2$. Each rung squares the precision. Sweeping the choice at the first two rungs and retaining those that satisfy the forcing congruence and have $a' = 2 \cdot 3 \cdot 5 \cdot P$ gives, with $(p_1, p_2, p_3) = (569, 1949, 15461)$,

$$a = 569 \cdot 1949 \cdot 15461 \prod_{7 \le p \le 271} p \quad (58 \text{ primes, } 120 \text{ digits}), \qquad a' = 2 \cdot 3 \cdot 5 \cdot P_{118},$$

and, exactly,

$$r(a) = 1 + 6.278713 \times 10^{-9}, \qquad a'' > a.$$

This is $1.6 \times 10^5$ times closer to 1 than the preceding record.

That record is on the wrong side of 1, and identifying the family says why. Every witness here has $a' = 2 \cdot 3 \cdot 5 \cdot P$ with $P$ prime, which is exactly the stratum $M = 30$ of Proposition 8.4: with $M' = 31$ the equation $MN - M'N' = M^2$ reads $30N - 31N' = 900$, and the proposition's $N = M + M'p$, $N' = Mp$ become $a = 31P + 30$, $a' = 30P$. Two identities follow, both exact: $a'' = (30P)' = 31P + 30$, so $r = a''/a = (31P + 30)/a$, and a solution of the stratum is precisely $30a - 31a' = 900$. Hence

$$\sigma(a) = \frac{30P}{31P + 30} = \frac{30}{31} - \frac{900}{31a} < \frac{30}{31}$$

for every solution of this stratum, strictly. Approaches to 1 from *above*, which is what the $6.28 \times 10^{-9}$ above and the $3.76 \times 10^{-9}$ of a wider sweep are, therefore lie on a side the stratum cannot populate. They stand as records for the spectrum of $r$ and cannot converge on a solution.

Aimed below instead, where the stratum's solutions must lie, the same ladder gives

$$a = 431 \cdot 68111 \cdot 2792213 \prod_{7 \le p \le 271} p \quad (58 \text{ primes, } 123 \text{ digits}), \qquad a' = 2 \cdot 3 \cdot 5 \cdot P_{122},$$

and, exactly,

$$r(a) = 1 - 1.654663 \times 10^{-13}, \qquad \sigma(a) < \tfrac{30}{31}, \qquad a'' < a,$$

which is $4.8 \times 10^{10}$ times closer to 1 than the previous record from below. For this witness $30a - 31a'$ is a 112–digit number where a solution needs 900, which is the honest measure of the distance remaining: the ratio is close, the integer is not.

Neither construction is exhausted. Over the same sweeps the smallest *available* deficit was $2.043 \times 10^{-18}$ from below and $1.209 \times 10^{-14}$ from above; what limits the reported values is the rate at which $a'$ takes the form $2 \cdot 3 \cdot 5 \cdot P$, not the tuning (`code/nearmiss_ladder.gp`, `code/nearmiss_below.gp`, re–checked in `code/nearmiss_ladder_verify.gp` and `code/nearmiss_below_verify.g`

We stop here, and the reason is a measurement rather than a budget. Between the two from–below witnesses the ratio improved by three orders of magnitude while $30a - 31a'$ fell from 114 digits to 112, against the 3 a solution requires. Exhausting the available deficit would gain perhaps five more digits of the 112. The near–miss frontier measures $|r - 1|$, a ratio; membership of the stratum is the integer identity $30a - 31a' = 900$, and the two are not commensurable at this scale. Driving the ratio further is therefore not evidence about the integer, and the line is closed as a line of attack. What survives it is structural and is stated above: the family is the $M = 30$ stratum

of Proposition 8.4, its solutions satisfy $\sigma(a) < 30/31$ strictly, and approaches from above cannot reach one. Those three facts are machine–checked in `lean/Erdos307/StratumM30.lean`.

The constant 30/31 is not special. In the normal form of Proposition 8.4 a solution of the $M$–stratum is $a = Mp$, $b = N = M + M'p$, so $\sigma(a) = \sigma(M) + 1/p$ and, by the identity $(\sigma(M) + 1/p)\sigma(N) = 1$ of part (b), $\sigma(b) = 1/\sigma(a) < 1/\sigma(M) = M/M'$, with defect exactly $M^2/(M'(M + M'p))$. Hence *every* stratum is one–sided: for no squarefree $M$ does the half–space $\sigma(b) \geq M/M'$ contain a solution, and $M = 30$ recovers 30/31. This is a corollary of Proposition 8.4(b) rather than a new result, but it is the constraint that governs search direction, so it is machine–checked in `lean/Erdos307/StratumGeneral.lean`. The consequence for the near–miss programme is that the mirrored tuning of the previous paragraph is not merely unlucky at $M = 30$; the corresponding tuning is empty in every stratum. Consistently with the two–sided statement above, $a$ lies past the barrier: $118 > 113$ digits, and $r$ sits inside the barred neighbourhood $(0.998740, \infty)$.

What remains unmeasurable is the density of $r$ at 1, since constructed points do not estimate a density, and that is what the heuristic constant needs.

The situation is not the usual one. Heuristics of Bateman–Horn or Hardy–Littlewood type are normally checked against data, and the agreement is what justifies confidence in them. Here the prediction that solutions exist rests on a constant that cannot be estimated by sampling, because the barrier and the sampleable range are disjoint. The expected-count heuristic for #307 is therefore unverifiable in the usual way.

It is not, however, beyond computation, and the reason is that the constant is the barrier.

**Proposition 16.21** (The heuristic constant is the barrier's own probability)**.** *Model a random squarefree $a$ by $\mathbb{P}(p \mid a) = 1/(p+1)$, and $a'$ by the coprimality model that Proposition* 16.25 *validates, so that $\mathbb{P}(q \mid a') = 1/q$ for $q \nmid a$. Then each prime independently lies in $a$, in $a'$, or in neither, with probabilities $\frac{1}{p+1}, \frac{1}{p+1}, 1 - \frac{2}{p+1}$, and*

$$\sigma(a) + \sigma(a') = \sum_{p \mid aa'} \frac{1}{p}, \qquad \mathbb{P}(p \mid aa') = \frac{2}{p+1}.$$

*Since a two-cycle forces $\sigma(a)\sigma(a') = 1$ and hence $\sigma(a) + \sigma(a') \geq 2$ by* AM–GM*, the density $f$ of $r = \sigma(a)\sigma(a')$ is supported in the event $B : \sigma(a) + \sigma(a') \geq 2$, so $f(1) \leq C\,\mathbb{P}(B)$ with $C$ the conditional density of $r$ at 1 given $B$. Numerically*

$$\mathbb{P}(B) = 2.74 \times 10^{-90}.$$

The point is the order and not the digits. $\mathbb{P}(B)$ is a *tail* probability of a one-dimensional sum, not a density, which is what makes it stable: it moves from 4.09 to 3.01 to $2.74 \times 10^{-90}$ as the grid is refined from $2 \times 10^4$ to $8 \times 10^4$ bins and the prime range from $10^4$ to $10^6$, while a direct attempt at $f(1)$ on a two-dimensional grid moves by four orders of magnitude per refinement and converges to nothing. Two checks support it. The same computation with $\mathbb{P}(p \mid n) = 1/p$ returns $\mathbb{P}(\sigma(n) \geq 1) = 0.0420$, reproducing the measured density of prime-abundant integers to four places; and the single configuration in which $aa'$ absorbs exactly the first 59 primes has probability $\prod_{p \leq 277} 2/(p+1) = 10^{-96.0}$, below $\mathbb{P}(B)$ by the factor one expects from the many other prime sets of mass 2 (`code/barrier_probability.rs`).

One more thing follows, and it disarms the near-miss spectrum entirely.

**Proposition 16.22** (The near-miss infimum is zero, so closeness is not evidence)**.**

$$\inf\Big\{\, \big|\sigma(P)\sigma(Q) - 1\big| \;:\; P, Q \text{ finite disjoint prime sets} \,\Big\} = 0,$$

*and #307 asks exactly whether the infimum is attained.*

*Proof.* Split the primes into two infinite classes, say by parity of index. Each class has terms tending to 0 and divergent reciprocal sum, so by the classical subseries theorem each carries a subseries summing to exactly 1. This produces disjoint *infinite* $P_\infty, Q_\infty$ with $\sigma(P_\infty) = \sigma(Q_\infty) = 1$. Truncating both at height $y$ gives finite disjoint sets with $\sigma(P_y)\sigma(Q_y) \to 1$. □

The analytic content is formalised, and in a slightly stronger form than the proof above uses. For the infimum one does not need the subseries theorem, only that each mass can be placed in $(1-\delta, 1)$: `Erdos307.exists_block_sum_near` takes the first consecutive block whose sum exceeds $T - \varepsilon$ and observes that it overshoots by less than one term, and `infimum_zero` assembles two such blocks into a product within $\varepsilon$ of 1. Both carry 0 `sorry` and the three standard axioms (`lean/Erdos307/InfZero.lean`). The arithmetic input, that the primes split into two infinite classes each with reciprocal terms tending to 0 and unbounded partial sums, is cited rather than reproved, as in Theorem 16.29.

The greedy form of that argument reaches $|\sigma(P)\sigma(Q)-1| = 6.50\times10^{-9}$ with 144 primes, against 0.4465 for the swept frontier of Proposition 16.7 (`code/infimum_zero.gp`). So the analytic half of #307 is empty: the mass equation can be approximated as closely as one likes, and no near-miss, however good, is evidence that a solution exists.

What the near-misses do measure is a trade-off, and it is the honest picture of the obstruction. A solution must kill two defects at once, the mass defect $|\sigma(P)\sigma(Q) - 1|$ and the size defect $\log_{10}\big(\prod Q/\sigma(P)\prod P\big)$, the latter vanishing exactly when $\prod Q = N_P$, which is Theorem 2.3. Each is reachable alone and neither construction moves the other:

| construction | mass defect | size defect |
|---|---|---|
| greedy, $Q$ free | $6.50 \times 10^{-9}$ | $10^{341.3}$ |
| $Q = \text{primes}(a')$, `nearmiss_tune.gp` | $7.92 \times 10^{-3}$ | 0 |

Fixing $Q$ as the prime set of $a'$ makes the size defect vanish identically and leaves the mass defect at $10^{-3}$; choosing $Q$ freely drives the mass defect to $10^{-9}$ and costs 341 orders of magnitude in size. #307 is the demand that both columns read zero simultaneously, and nothing in this paper moves them together.

*The conditional shape of the near–miss law.* Proposition 16.7 records the maximum of $r(a) = \sigma(a)\sigma(a')$ by decade. The conditional maximum, given the value of $\sigma(a)$, is more informative, because it is where the two–sided obstruction becomes visible. Over the squarefree $a \le 8 \times 10^6$ with $a'$ squarefree and $\sigma(a) \in [0.90, 1.12]$, the largest $\sigma(a')$ in each bucket of width 0.01 decreases through the critical region, from 0.5481 at $\sigma(a) = 0.90$ to 0.3110 at $\sigma(a) = 1.12$, and equals 0.4436 at $\sigma(a) = 1.00$, where a solution requires 1. Quadrupling the range from $2 \times 10^6$ leaves the bucket maxima at $\sigma(a) = 1.00$ and 1.05 unchanged, at 0.4436 and 0.3786, and moves the window maximum of $r$ only from 0.5016 to 0.4955, attained at $a = 5{,}767{,}190$ (`code/twoside_conditional.rs`).

The mechanism is the barrier seen from the other side. A mass $\sigma(a)$ near 1 requires $a$ to carry the small primes; $\gcd(a, a') = 1$ then denies them to $a'$, whose primes are larger and whose mass is correspondingly smaller. This is why the primary pseudoperfect numbers, which satisfy the one–sided equation with $|\sigma(n) - 1| = 1/n$ and so refute any one–sided route to an absolute constant, never come near a cycle: for them $a' = a - 1$ is prime or nearly prime and $\sigma(a')$ collapses, at $a = 47{,}058$ to about $2 \times 10^{-5}$. The suppression is not special to that family but holds across the window. The label is HEURISTIC: by Proposition 16.7 the region $r \in (0.998740, \infty)$ lies above the barrier, so no sweep tests the claim where it would matter, and the measurement bounds nothing.

**Remark 16.23** (What the calibrated heuristic says, and it is not encouraging)**.** With $\mathbb{E}(X) \sim f(1)\,\delta \log X$ and $\delta = 0.3721$, Proposition 16.21 gives an expected number of two-cycles below the

proved barrier of about $10^{-88}$, and $\mathbb{E}(X)$ does not reach 1 until $X$ is of order $10^{4\times10^{89}}$. So the heuristic favours YES only in the limiting sense that $\log X$ diverges; at every scale that any search or any barrier argument will reach, it predicts nothing. Two consequences, both negative and both worth stating. A proof of the negative direction would have to contradict a divergence, and a search for a solution is hopeless not merely at present but at any scale expressible in the notation of this paper. The constant is small for a reason that is not incidental: it *is* the barrier, seen as a probability rather than as a counting bound, which is the third independent appearance of the same Mertens threshold in this work.

The label is HEURISTIC and the model is doing the work. What is claimed is only that the constant, said above to be unavailable to sampling, is available to computation once the coprimality model of Proposition 16.25 is granted.

**Remark 16.24** (What is left, on both sides)**.** The two directions can now be specified rather than merely called open.

*Negative.* Theorem 16.31 closes the certificate route on both branches, so a proof must count rather than certify; and by Remark 16.32 the count it must defeat diverges. A proof of the negative direction would therefore have to contradict a divergent expected count, which is to say that the direction is not merely unproved but probably false.

*Positive.* Three routes are available in principle, and two are closed quantitatively.

(a) *Exhibit a witness.* Excluded by Remark 16.23: the first is expected near $10^{4\times10^{89}}$.

(b) *Bound the count below by analytic means.* Excluded by size. At the proved barrier the main term $f(1)\delta\log X$ lies $10^{200.6}$ below the trivial count $X$, and the saving a method would need in order to keep the main term above its own error grows like $X$ itself. Sieve and circle-method errors save a power of $X$ or a power of $\log X$. So an analytic proof here would have to be *exact*, with error identically zero, rather than sharp.

(c) *Construct exactly.* This is the only branch alive, and it exists: the Pell orbit of Remark 5.28. It is alive but narrower than it was, and the narrowing should not be mistaken for proximity. Two cautions, both quantitative. The searches of Remark 5.19 came back empty over $3.99\times10^{11}$ candidates, but empty was the *predicted* outcome: the expected count was $2.4\times10^{-56}$, so the computation confirms the heuristic model rather than deciding anything, and a confirmed model is not a decided problem. And by Proposition 4.8 those searches cannot be continued upward at all, since the admissible supports are infinite at every level past 59. The unconditional exclusions of this note stop where enumeration stops, which is now. Proposition 4.7 bounds the tail prime by $4N$, so the orbit contributes nothing past a bounded initial segment and the construction is a finite search over the divisors of $4D$ rather than a Lehmer primality question; Remark 5.19 then makes that finite search heuristically empty on all 34 immune families, by two independent estimates. What survives is the branch itself, not its expectation: the construction must now either come from a base outside the immune list or exploit the parametrisation $q=2(N\pm k)/m^2$ directly.

The two exclusions interlock, and the interlock is the real statement. Exactness is what algebra supplies, and Theorem 7.1 excludes every argument that survives the passage to $K[t]$, where the conclusion is false. The Pell construction escapes that exclusion for one reason only: its final input is the primality of a determined term, and primality does not survive to $K[t]$. So a successful method must be *archimedean and exact at the same time*, a combination which is unusual but not unprecedented, continued fractions solving Pell being the standard instance, and Remark 5.28 is exactly the analogue here.

The labels differ across the three branches and the difference matters. The exclusions of (a) and (b) are HEURISTIC: both rest on the value of $f(1)$ from Proposition 16.21, hence on the coprimality model, and a reader who rejects that model recovers both branches. The interlock itself is not heuristic: Theorem 7.1 and Proposition 8.6 are PROVED, and so is the reduction of branch (c) to a primality question. So what follows is stated at the strength of its weakest link, which is the model.

What remains of #307 is therefore one question and not a programme: whether some base admits a Pythagorean tail prime, together with the sign selection $k = 0$, which Remark 5.28 shows is a choice between $\{0, -1\}$ rather than a coincidence of measure zero. By Proposition 8.6 the orbit parametrising the Pythagorean locus is forced to be exponential rather than polynomial. That was previously read as placing the question in the Lehmer class rather than the Schinzel class, hence open for every non-degenerate sequence; but by Proposition 4.7 a prime tail satisfies $q \le 4N$, so the exponential orbit supplies no candidate past a bounded initial segment and the question is neither Lehmer nor Schinzel. It is a finite search over the divisors $m \mid 4D$ with $AB + Dm^2$ a square, which Remark 5.19 makes heuristically empty on all 34 immune families. The residue is thus smaller and better located than it was, and still undecided: the bound is proved, the emptiness is not, and no base outside the immune list has been examined at all. That is why nothing in this paper decides the problem.

The measurement does expose a sharp structural fact, which turns out to be exactly the barrier seen statistically.

**Proposition 16.25** (The barrier is statistically complete)**.** $\mathbb{E}[\sigma(a')]$ *falls by an order of magnitude as* $\sigma(a)$ *grows, from* 0.2035 *unconditionally to* 0.0207 *on* $\sigma(a) \ge 1.3$*. The whole of this suppression is coprimality. Writing the prediction for an integer chosen at random subject to* $\gcd(n, a) = 1$*, namely* $\sum_{p \nmid a} p^{-2}$*, the ratio of measured to predicted is*

$$0.65, \quad 0.82, \quad 0.91, \quad 0.92, \quad 0.95, \quad 1.01$$

*at* $\sigma(a) \ge 0$, 0.5, 0.9, 1.0, 1.2, 1.3*: it converges to* 1 *precisely in the regime a solution would inhabit* (`code/a10_anticorr.rs`).

The mean is not the moment that $f(1)$ depends on. Since $f(1)$ is an integral of the joint density of $(\sigma(a), \sigma(a'))$ along $\sigma(a)\sigma(a') = 1$, a model could match the conditional mean and still misstate the density; the next moment is therefore the sharper test. The model makes a prediction for it too, and a per–$a$ one: given $\mathrm{supp}(a)$, each $q \nmid a$ lands in $a'$ independently with probability $1/q$, so

$$\mathbb{E}[\sigma(a')] = \sum_{q \nmid a} \frac{1}{q^2}, \qquad \mathrm{Var}[\sigma(a')] = \sum_{q \nmid a} \frac{1}{q^3}\Big(1 - \frac{1}{q}\Big).$$

Measured against these over all $a \le 10^7$ with $a, a'$ squarefree (`code/f1_variance.rs`), the ratio of measured to predicted variance is

$$1.111,\ 0.968,\ 0.894,\ 0.956,\ 0.938,\ 0.860,\ 0.948,\ 1.030$$

at $\sigma(a) \ge 0$, 0.5, 0.7, 0.9, 1.0, 1.1, 1.2, 1.3, against $0.648, \ldots, 1.010$ for the mean. The variance is predicted as well as the mean, it shows no systematic drift, and the two converge together. The top band rests on 438 values and is correspondingly noisy, so we claim only the absence of a trend, not the precise value. What the test was designed to detect (a model matching the first moment while its tail degrades, which would inflate $f(1)$ and could in principle send the true value to 0) does not occur.

So on the sampled range $a'$ is mass-poor for one reason only, that it is coprime to $a$ and $a$ has absorbed the small primes, and there is no further effect to exploit. This is an independent confirmation of Proposition 17.18: the barrier is not merely near-optimal as a bound, it already contains the entire statistical structure of the obstruction, and no refinement by correlation is available.

The range matters, and the statement is about averages. Every $a \le 4 \times 10^8$ with $\sigma(a) > 1$ is even, so the measurement never sees an $a$ that omits the small primes. Such $a$ exist, since the prime reciprocals diverge, and for them $a'$ is free to absorb 2, 3 and 5 and is not mass-poor at all: Theorem 16.33 exhibits one with $\sigma(a') = \frac{31}{30}$. Proposition 16.25 bounds an expectation on a range, and nothing here bounds a supremum off it.

## A Lyapunov candidate for the negative direction

Theorem 7.1 rules out cycles over $\mathbb{F}_q[t]$ by exhibiting a function that provably decreases, namely the degree. Proposition 15.10 shows no *absolute value* on $\mathbb{Q}$ plays that role. It does not exclude functions that are not absolute values, and the following is the outcome of a search among such functions. It is a conjecture, not a theorem, and it is recorded because it is falsifiable and because the mechanism protecting it is one of the paper's own results.

**Definition 16.26.** For $\lambda > 0$ and squarefree $n$ set $\delta_\lambda(n) = \log n + \lambda\,\sigma(n)$.

**Proposition 16.27** (A decreasing $\delta_\lambda$ would settle #307 negatively)**.** *If for some $\lambda > 0$ one has $\delta_\lambda(n') < \delta_\lambda(n)$ for every squarefree $n$ with $n'$ squarefree, then the arithmetic derivative has no cycle of any length among such $n$, and in particular #307 has answer* NO.

*Proof.* A cycle would give $\delta_\lambda(n) < \delta_\lambda(n)$. □

Since $n' = n\sigma(n)$, the requirement is the pointwise inequality

$$\log \sigma(n) \;<\; \lambda\big(\sigma(n) - \sigma(n')\big). \tag{$*$}$$

Around a two-cycle the two instances of $(*)$ have increments summing to $\log(\sigma(a)\sigma(b)) = 0$, so they cannot both hold: the admissible range of $\lambda$ closes exactly at a cycle. The content is therefore whether that range is non-empty.

Below the barrier the criterion is not conjectural: it holds with the explicit value $\lambda = \frac{1}{2}$, by an elementary inequality.

**Proposition 16.28** (The obstruction is to the shape of $\delta_\lambda$, not to $\lambda$)**.** *The closing of the admissible range of $\lambda$ is not special to a linear dependence on $\sigma$. Let $F(n) = \log n + G(\sigma(n))$ for an arbitrary $G$. Since $n' = n\,\sigma(n)$, one step gives*

$$F(n') - F(n) \;=\; \log \sigma(n) + G\big(\sigma(n')\big) - G\big(\sigma(n)\big),$$

*so around a two–cycle $(a, b)$ the two $G$–terms telescope and the increments sum to $\log\big(\sigma(a)\sigma(b)\big) = 0$ for* every *$G$. A strictly decreasing $F$ of this shape would settle #307, and none can be exhibited without already excluding cycles: the $G$–part cancels identically and only the mass identity survives. The relevant identity is $\sigma(n)\sigma(n') = n''/n$, so $\sigma(n)\sigma(n') = 1$ holds exactly when $n'' = n$: the configuration that defeats the class occurs at a cycle and nowhere else. Any Lyapunov argument must therefore use a potential that sees more of $n$ than $\sigma(n)$ alone.*

**Theorem 16.29** (The half-mass Lyapunov function)**.** *Put $\delta(n) = \log n + \frac{1}{2}\sigma(n)$. Let $n$ be squarefree with $n'$ squarefree. If $\sigma(n\,n') < 2$ then $\delta(n') < \delta(n)$.*

*Proof.* By Theorem 2.3 $n$ and $n'$ are coprime, and both are squarefree, so $\sigma(n\,n') = \sigma(n) + \sigma(n')$. The hypothesis therefore gives $\sigma(n') < 2 - \sigma(n)$, hence

$$\tfrac{1}{2}\big(\sigma(n) - \sigma(n')\big) \;>\; \sigma(n) - 1 \;\geq\; \log \sigma(n),$$

the last step by $\log x \leq x - 1$. If $\sigma(n) = 1$ the two ends read 0 and the first inequality is still strict, since $\sigma(n) + \sigma(n') < 2$ forces $\sigma(n') < 1$. Now $n' = n\sigma(n)$ gives $\delta(n') - \delta(n) = \log \sigma(n) - \frac{1}{2}(\sigma(n) - \sigma(n')) < 0$. □

Verified over all 7,480,605 squarefree $n \leq 2 \times 10^7$ with $n'$ squarefree lying in the region: 0 violations, least slack 0.242 (`code/invent_lyapunov.rs`). The analytic content is formalised: `Erdos307.log_lt_half_sub` proves $\log s < (s - t)/2$ for $s > 0$ and $s + t < 2$, `delta_decreasing` gives the increment form, and `two_le_add_of_mul_eq_one` is the AM–GM step of Corollary 16.30; all three carry 0 `sorry` and only the three standard axioms (`lean/Erdos307/HalfLyap.lean`). The arithmetic inputs, that $n$ and $n'$ are coprime and that $n' = n\sigma(n)$, are Theorem 2.3 and the definition, so the statement is VERIFIED modulo those.

**Corollary 16.30** (A Lyapunov proof of the barrier)**.** *The arithmetic derivative has no cycle, of any length, all of whose consecutive pairs satisfy $\sigma(n\,n') < 2$. In particular a two-cycle $a' = b$, $b' = a$ with $a \neq b$ has $\sigma(ab) = \sigma(a) + \sigma(b) \geq 2\sqrt{\sigma(a)\sigma(b)} = 2$, with equality only if $\sigma(a) = \sigma(b) = 1$ and hence $a = b$; so $\sigma(ab) > 2$ and therefore $ab \geq \Pi_{59}$.*

*Proof.* A cycle would give $\delta(n) < \delta(n)$ by Theorem 16.29. The second statement is AM–GM together with the mass estimate of Theorem 4.3. □

There are exactly two ways to exploit the fact that increments cancel around a cycle, and the second is foreclosed. Around a two-cycle the two increments of any function sum to zero, so an obstruction must pin down either their *sign*, which is the Lyapunov route above, or their *value in a finite group*: if some $\varphi$ satisfied $\varphi(n') - \varphi(n) \equiv c \pmod m$ for all $n$ with $2c \not\equiv 0$, a two-cycle would give $0 \equiv 2c$. No such $\varphi$ exists, and not for want of searching: Proposition 5.17 proves the cycle system soluble modulo every $m$, so any obstruction expressible as a congruence is empty a priori. Measured, the increments of $\omega(n)$, of $\#\{p \mid n : p \equiv 3\ (4)\}$, of $n$, and of $n+\omega(n)$ each hit every residue class modulo 2, 3 and 4 (`code/invent_lyapunov.rs`). The sign route is therefore the only one of the two available, which is another form of the thesis that the obstruction is ordered–archimedean rather than congruential (Remark 7.3).

The sign route is available but not weaker than the problem. Both branches of the dichotomy can now be closed at once.

**Theorem 16.31** (Both branches are closed)**.** *Let $S$ be the set of squarefree $n \geq 2$ with $n'$ squarefree.*

(1) (Finite-group branch.) CONJECTURE*, downgraded. The claim that there is no $\varphi$ on $S$ and modulus $m$ with $\varphi(n') - \varphi(n) \equiv c \pmod m$ for all $n$ and $2c \not\equiv 0 \pmod m$ is* not *established. Proposition 5.17 settles only the moduli it covers, $m \leq 210$, and the general statement is equivalent to the target rather than independent of it: on a functional graph with no periodic orbit one may build a constant-increment $\varphi$ for any $(m, c)$ by assigning $\varphi$ along depth, so nonexistence for all $m$ is equivalent to the existence of a periodic orbit. Branch (1) therefore has exactly the status of branch (2), which Proposition 16.37 already proves for the sign side. What is proved is the mechanism recorded after the proof.*

(2) (Sign branch, not weaker than the target.) *A $\delta$ on $S$ with $\delta(n') < \delta(n)$ throughout exists if and only if $n \mapsto n'$ has no periodic orbit in $S$; that condition implies a negative answer to #307 and is not implied by it.*

*Proof.* (1) is not proved; see the statement. Proposition 5.17 makes the cycle system soluble modulo every $m \le 210$, which does not reach the quantifier the branch needs. (2) is Proposition 16.37; a two-cycle is a periodic orbit, and periodic orbits of length other than two are not excluded by a negative answer to #307. □

The mechanism of branch (1) is formalised. `Erdos307.two_nsmul_eq_zero_of_period_two` proves that a constant increment $c$ in any abelian group forces $2c = 0$ at a point of period two, `const_increment_iterate` gives the general $\varphi(D^{[k]}n) = \varphi(n)+kc$, `no_period_two_of_two_nsmul_ne_zero` is the certificate such a $\varphi$ would supply, and `zmod_two_mul_eq_zero_of_period_two` states it over $\mathbb{Z}/m$; all with 0 `sorry` and on `propext` alone (`lean/Erdos307/Congruential.lean`). What is cited rather than formalised is the *emptiness* of the class, that no such $\varphi$ with $2c \neq 0$ exists for the arithmetic derivative, which is Proposition 5.17 and is arithmetic. Branch (1) is therefore VERIFIED in its mechanism and PROVED in its emptiness, and the two should not be conflated.

The taxonomy that makes this a closure rather than two isolated facts is the displayed thesis above, that an invariant whose increments cancel around a cycle can be exploited only through their sign or their value in a finite group. That thesis is not a theorem, and an argument outside it is not excluded by anything here.

This is a statement about methods, not about #307, and it is the natural endpoint of the structural programme: Proposition 15.10 removes the absolute values, Theorem 7.1 removes the arguments that survive to $K[t]$, Proposition 5.17 removes the congruences, and Theorem 16.31 removes what is left of the pointwise route. A proof of the negative direction, if there is one, must be global: it must count, and not certify.

**Remark 16.32** (What a counting proof would have to defeat)**.** The count it would have to defeat diverges. A two-cycle is exactly $r(a) = 1$ for $r(a) = \sigma(a)\sigma(a') = a''/a$, so if $r$ has a limiting density $f$ on the class of squarefree $a$ with $a'$ squarefree, then $\mathbb{P}(a'' = a) = \mathbb{P}\big(r \in [1, 1 + \frac{1}{a})\big) \sim f(1)/a$ and

$$\mathbb{E}\big(\#\{a \le X : a'' = a\}\big) \;\sim\; f(1)\,\delta \log X, \qquad \delta = 0.3721\ldots,$$

$\delta$ being the measured density of the admissible class (`code/region_shape.rs`, 3,721,148 admissible $a \le 10^7$). This diverges, so the heuristic predicts infinitely many two-cycles, and any proof of the negative direction must account for a divergent expected count producing no solutions at all.

The same formula locates them, and shows why the location is not knowable. Setting the expectation to 1 gives a first solution near $\exp\big(1/(\delta f(1))\big)$, which is exponential in $1/f(1)$: at $f(1) = 10^{-2}$ it is $10^{116.7}$, at $f(1) = 5 \times 10^{-3}$ it is $10^{233.4}$, at $f(1) = 10^{-3}$ it is $10^{1167}$. Halving the constant squares the answer. Consistency with Theorem 4.3 needs $f(1) < 0.01034$, which is evidence of the softest kind, since an expectation near 1 with no solution observed is an ordinary event and not a contradiction. What Proposition 16.7 shows is that $f(1)$ is the one quantity here that cannot be sampled (`code/heuristic_location.gp`). That $f(1) > 0$ is not established either; the constructed witnesses of Theorem 16.33 and of the near-miss frontier show only that $r$ takes values on both sides of 1 and within $8 \times 10^{-3}$ of it.

Corollary 16.30 recovers the barrier by a route independent of the minimisation used in Theorem 4.3: no extremality argument, no enumeration, only $\log x \le x - 1$ and the coprimality of Theorem 2.3. It is also stronger in one direction, since it excludes cycles of every length rather than two-cycles alone. What it does not do is decide #307: the region it covers is exactly the region where the barrier already says nothing can happen, and above $\Pi_{59}$ the hypothesis $\sigma(n\,n') < 2$ fails, which is where any solution must live. The general conjecture is what remains.

Over all 153,397,754 such $n \le 4 \times 10^8$, $(*)$ holds for every $\lambda \in (0.275438,\ 1.250828)$ and fails for no step (`code/lyap_battery.rs`). The interval narrows with the range, of width 1.654, 1.267,

1.237, 1.116, 1.092, 1.004, 0.975 at cutoffs $10^4$ through $4 \times 10^8$; the binding lower constraint is at $n = \Pi_9 = 223{,}092{,}870$. Six candidate functions were tested and five died, among them the control $\delta = \log n$, which fails first at $n = 30$ where $\sigma = 1.0333$; a battery that did not kill it would not be testing anything (`code/invent_lyapunov.rs`).

That interval is an artefact of the range swept. No $\lambda$ works.

**Theorem 16.33** (The criterion fails, for every $\lambda$)**.** *There is no real $\lambda$ for which $(*)$ holds at every squarefree $n$ with $n'$ squarefree. Two witnesses suffice. Let*

$$n_0 = \prod_{\substack{7 \le p \le 373 \\ p \notin \{307,\,317,\,359\}}} p,$$

*a squarefree integer with 68 prime factors and 142 decimal digits. Then $n_0' = 2 \cdot 3 \cdot 5 \cdot C$ with $C$ a prime of 141 digits, so $n_0'$ is squarefree and $\sigma(n_0') = \frac{31}{30}$ exactly, while $\sigma(n_0) = 1.0059067\ldots$. Since $\sigma(n_0) > 1$ and $\sigma(n_0') > \sigma(n_0)$,*

$$\log \sigma(n_0) \;>\; 0 \;>\; \lambda\big(\sigma(n_0) - \sigma(n_0')\big) \qquad (\lambda > 0),$$

*so $(*)$ fails at $n_0$ for every positive $\lambda$. At $n = 30$, with $30' = 31$ prime, $\sigma(30) = \frac{31}{30} > 1$ and $\sigma(30) - \sigma(31) = \frac{931}{930} > 0$, so $\log \sigma(30) > 0 \ge \lambda(\sigma(30) - \sigma(31))$ for every $\lambda \le 0$. Together the two exclude every real $\lambda$.*

*Proof.* Both statements are finite verifications in exact rational arithmetic. For $n_0$: $\sigma(n_0)$ and $\sigma(n_0')$ are computed as rationals, $\gcd(n_0, n_0') = 1$ and $\sigma(n_0) = n_0'/n_0$ confirm Theorem 2.3, the factorisation $n_0' = 2 \cdot 3 \cdot 5 \cdot C$ multiplies back to $n_0'$ and has all exponents 1, and $C$ carries an Atkin–Morain certificate, generated and validated (`data/certs/lyap_refute_cofactor_ecpp.txt`). The comparisons $\sigma(n_0) > 1$ and $\sigma(n_0') > \sigma(n_0)$ are exact, and $\sigma(n_0') = \sigma(n_0)$ is impossible in any case by Lemma 17.2. The case $n = 30$ is arithmetic (`code/lyap_refute_verify.gp`). □

The argument is formalised. `Erdos307.criterion_fails_of_pos` and `criterion_fails_of_nonpos` give the two sign clashes, `no_lambda_works` combines them, and `lyapunov_criterion_false` instantiates the combination at $\sigma(n_0)$, carried as the exact quotient $n_0'/n_0$, and at the pair $(\frac{31}{30}, \frac{1}{31})$ of $n = 30$; the two numeric comparisons $1 < \sigma(n_0)$ and $\sigma(n_0) < \frac{31}{30}$ are checked on the 142-digit numerals themselves. All six declarations carry 0 `sorry` and only `propext`, `Classical.choice`, `Quot.sound`, with no `native_decide` (`lean/Erdos307/LyapFalse.lean`). What Lean does not see is the factorisation $n_0' = 2 \cdot 3 \cdot 5 \cdot C$ and the primality of $C$, which is the Atkin–Morain certificate; the status is therefore VERIFIED modulo that certificate, in the same sense that Theorem 16.29 is VERIFIED modulo Theorem 2.3.

The witness is built, not found. For squarefree $n$ and a prime $w \nmid n$ one has $n' \equiv n \sum_{p \mid n} p^{-1} \pmod{w}$, hence

$$w \mid n' \quad \Longleftrightarrow \quad \sum_{p \mid n} p^{-1} \equiv 0 \pmod{w}. \tag{F}$$

Imposing (F) for $w \in \{2, 3, 5\}$ forces $\sigma(n') \ge \frac{31}{30}$, and building $n$ from primes $\ge 7$ with $\sigma(n)$ just above 1 gives $\sigma(n') > \sigma(n)$. Reaching mass 1 without $2, 3, 5$ needs primes only to about 360, which is what keeps $n_0'$ small enough to factor completely; the squarefreeness of $n_0'$ is therefore verified rather than assumed. Three of the larger pool primes are dropped to satisfy (F), at negligible cost in mass (`code/lyap_refute.gp`).

The sweep could not have seen this. Every $n \le 4 \times 10^8$ with $\sigma(n) > 1$ is even, so the battery measured only $n$ containing 2, whereas the failure needs $n$ to omit 2, 3 and 5 so that $n'$ may absorb

them. By Theorem 16.34 below, any failure must have $n\,n' \geq \Pi_{59}$, and $n_0 n_0'$ has 284 digits: the region was out of reach of any sweep, and only a construction enters it.

One route to the criterion had looked provable rather than measured, and it fails for the same reason. Write $\rho = \sup\{\sigma(n')/\sigma(n) : \sigma(n) > 1\}$, the supremum being over the same class of $n$. If $\rho \leq c < 1$ then $\sigma(n) - \sigma(n') \geq (1-c)\sigma(n)$, and since $\log x/x \leq 1/e$,

$$L := \sup_{\sigma(n)>1} \frac{\log \sigma(n)}{\sigma(n) - \sigma(n')} \;\leq\; \frac{1}{e(1-c)}.$$

Measured over $n \leq 4 \times 10^8$, $\rho = 0.519746$, which would give $L \leq 0.766$ against a measured upper demand of 1.274; a proof of $\rho < 1 - 1/(eU)$, for $U$ a lower bound on the upper demand, would then have given $(*)$. The measurement is not the truth. Already

$$a_0 = \prod_{5 \leq p \leq 163} p, \qquad a_0' = 2 \cdot 3 \cdot 173 \cdot 227 \cdot 1973 \cdot 82890547 \cdot 68126695363 \cdot P_{36},$$

with $P_{36}$ prime, is squarefree with squarefree derivative, $\sigma(a_0) = 1.0723027\ldots > 1$ and $\sigma(a_0') = 0.8440258\ldots$, so $\rho \geq 0.7871152\ldots$; the bound the route would yield is $1/(e(1-\rho)) \geq 1.7280$, already above the measured upper demand, so it constrains nothing. The witness $n_0$ of Theorem 16.33 gives $\rho \geq 1.02726$, so $\rho < 1$ is false outright (`code/rho_verify.gp`).

The mechanism advanced for $\rho < 1$ was that $\sigma(n) > 1$ forces $n$ to absorb the small primes while $n'$ stays coprime to $n$. It does not: the reciprocals of the primes diverge, so $\sigma(n) > 1$ is reachable from primes $\geq 7$ alone, and then $n'$ is free to take 2, 3 and 5. Both $a_0$ and $n_0$ are of that shape. The earlier probe of this regime evaluated $\sigma(n')/\sigma(n)$ at the least odd $n$ with $\sigma(n) > 1$, obtaining 0.0312, and read the regime off that single point; the quantity to be bounded is a supremum, and the least member of a family does not report it. This is the error recorded in Remark 17.19, in a second instance.

What survives is the location of the failure, and it is exactly where the barrier puts it.

**Theorem 16.34** ($\rho < 1$ below the barrier)**.** *Let $n$ be squarefree with $n'$ squarefree and $\sigma(n) > 1$. If $\sigma(n') \geq \sigma(n)$ then $n\,n' \geq \Pi_{59} > 8.77 \times 10^{112}$. Consequently $\rho < 1$ for every $n$ with $n\,n' < \Pi_{59}$, and since $\sigma(n) > 1$ gives $n' > n$, for every $n < \sqrt{\Pi_{59}}$, that is for every $n < 10^{56.5}$.*

*Proof.* $\sigma(n') \geq \sigma(n) > 1$ gives $\sigma(n) + \sigma(n') > 2$. By Theorem 2.3 $n$ and $n'$ are coprime, and both are squarefree, so $\sigma(n\,n') = \sigma(n) + \sigma(n') > 2$. A squarefree integer of mass at least 2 has at least 59 prime factors and is therefore at least $\Pi_{59}$, since $\sigma(\Pi_{58}) = 1.998740 < 2 \leq 2.002350 = \sigma(\Pi_{59})$; this is the mass estimate underlying Theorem 4.3. Finally $\sigma(n) > 1$ gives $n' = n\sigma(n) > n$, so $n^2 < n\,n'$. □

The counting step is formalised. `Erdos307.card_ge_59_of_recipSum_ge_two` isolates the mass estimate, that a set of primes of reciprocal sum at least 2 has at least 59 elements, and `Erdos307.card_union_ge_59_` applies it to two disjoint sets each of mass exceeding 1; both carry 0 `sorry` (`lean/Erdos307/RhoBarrier.lean`). Their axiom lists are those of Theorem 4.3 itself, the three standard axioms together with the `native_decide` evaluation of the first 58 primes. The same core proves `card_ge_59` of the barrier, which assumes $\sigma(P)\sigma(Q) = 1$ instead; the two hypotheses differ and the shared step has been extracted rather than duplicated.

Theorem 16.34 therefore locates the failure before exhibiting it: $\rho \geq 1$ forces $n\,n' \geq \Pi_{59}$, so no sweep can meet a counterexample, and any counterexample must have $n$ of at least 57 digits. The witness $n_0$ has 142, and $n_0 n_0'$ has 284. The regime in which $\rho \geq 1$ is possible is the regime in which a two-cycle is possible, since both need two coprime squarefree integers of total mass at least

2; what Theorem 16.33 shows is that the first of those is genuinely non-empty, which says nothing about the second.

**Corollary 16.35** (The half-mass hypothesis cannot be relaxed)**.** *For every $\lambda > 0$ and every $c > 2.03925$, the implication "$\sigma(n\,n') < c$ implies $\delta_\lambda(n') < \delta_\lambda(n)$" is false. Indeed $\sigma(n_0\,n_0') = 2.039240\ldots$ while $\delta_\lambda(n_0') - \delta_\lambda(n_0) = \log\sigma(n_0) + \lambda(\sigma(n_0') - \sigma(n_0)) > 0$ for every $\lambda > 0$.*

Two is also the exact threshold for the argument of Theorem 16.29, in the following sense, which concerns the inequality alone and not the arithmetic of $\sigma$.

**Proposition 16.36** (The mass-sum relaxation stops at 2)**.** *Let $c > 0$. There is $f : (0,\infty) \to \mathbb{R}$ with $f(x) - f(y) > \log x$ for all $x, y > 0$ satisfying $x + y < c$ if and only if $c \le 2$. For $c = 2$ the linear solutions are exactly $f(x) = \frac{1}{2}x + const.$*

*Proof.* The region $x + y < c$ is symmetric, so a solution gives both $f(x) - f(y) > \log x$ and $f(y) - f(x) > \log y$; adding, $0 > \log(xy)$. Taking $x = y \uparrow c/2$ forces $c^2/4 \le 1$, that is $c \le 2$. For $c \le 2$ take $f(x) = \frac{1}{2}x$: if $x + y < 2$ then $\frac{1}{2}(x - y) > x - 1 \ge \log x$. If $f = \lambda x$ then letting $y \uparrow 2 - x$ gives $2\lambda(x - 1) \ge \log x$ for all $x \in (0, 2)$, and dividing by $x - 1$ and letting $x \to 1$ from each side gives $2\lambda \ge 1$ and $2\lambda \le 1$. □

Both halves are formalised: `Erdos307.half_works` for $c \le 2$ and `no_f_above_two` for $c > 2$, the latter by the diagonal point $x = y$ with $1 < x < c/2$, where the right side vanishes and the left is positive; `mass_sum_threshold` states the two together. The uniqueness of the constant is formalised as well: `lambda_eq_half` shows that a linear solution at $c = 2$ forces $\lambda = \frac{1}{2}$, by the local-minimum route rather than a two-sided limit, since $g(x) = 2\lambda(x - 1) - \log x$ is nonnegative on $(0, 2)$ and vanishes at 1, so its derivative $2\lambda - 1$ vanishes there (`lean/Erdos307/NoInvariant.lean`).

Any strengthening must therefore use an input about $n$ beyond the pair $(\sigma(n), \sigma(n'))$, and by the next proposition there is no room in the choice of function either.

**Proposition 16.37** (The ansatz is universal, so the search was the problem)**.** *Let $S$ be the set of squarefree $n \ge 2$ with $n'$ squarefree. The following are equivalent:*

(1) *$n \mapsto n'$ has no periodic orbit inside $S$;*

(2) *there is $\delta : S \to \mathbb{R}$ with $\delta(n') < \delta(n)$ whenever $n, n' \in S$;*

(3) *there is $f$ on $\sigma(S)$ with $f(\sigma(n)) - f(\sigma(n')) > \log\sigma(n)$ for all such $n$.*

*Proof.* (2) implies (1), since a periodic orbit would give $\delta(n) < \delta(n)$. For (1) implies (2), the transitive closure of $n \mapsto n'$ is irreflexive by (1) and transitive by construction, hence a strict partial order on the countable set $S$; it has a linear extension, and a countable linear order embeds order-preservingly in $\mathbb{Q}$, which supplies $\delta$. For (2) and (3), $\sigma$ is injective on squarefree integers by Lemma 17.2, so every $\delta$ on $S$ is $\log n + f(\sigma(n))$ for exactly one $f$, namely $f(\sigma(n)) = \delta(n) - \log n$; and $n' = n\sigma(n)$ turns $\delta(n') < \delta(n)$ into the inequality of (3). □

The collapse of the ansatz is formalised. `Erdos307.exists_f_of_injective` proves that injectivity of $\sigma$ forces *every* $\delta$ into the shape $\log n + f(\sigma(n))$, `increment_iff` converts $\delta(n') < \delta(n)$ into $(*)$ verbatim, and `no_two_cycle_of_decreasing` is the implication of Proposition 16.27; all with 0 `sorry` and the three standard axioms (`lean/Erdos307/NoInvariant.lean`).

So Definition 16.26 restricts nothing. The tournament that produced $\delta_\lambda$ was searching the space of all functions on $S$, and by Proposition 16.37 the existence of any member of that space is equivalent to the absence of cycles, which is strictly stronger than a negative answer to #307.

The criterion was never a weakening of the problem, and Theorem 16.33 settles the one form of it that had been left open. What remains true, and is the residue of the whole construction, is Theorem 16.29 together with Corollary 16.30: an elementary Lyapunov proof of the barrier, valid exactly up to mass 2 and, by Corollary 16.35, not one step further.

*Three independent blocks on* A10. The existence route is therefore obstructed at three separate levels, and no two of them are the same obstruction. Proposition 16.1 removes the freedom: the construction carries one parameter and cannot be relaxed into a denser family. Theorem 7.1 removes the method: no argument that survives the passage to $K[t]$ can prove existence, because there the statement is false. Proposition 16.4 removes the last refuge, the hope that a large family of sequences is easier than one: the union stays logarithmically thin however many sequences it has. A fourth block was claimed here and does not stand: it held that the 34 immune families could not be examined, and Proposition 5.5 examines them and finds them empty (Remark 16.3). Removing it strengthens nothing and is recorded because the count of independent obstructions is exactly the kind of claim that should not drift. What is left is a lower bound for primes in a sparse exponential sequence, which exists for no non-degenerate linear recurrence at all. We state plainly that the present work does not decide the existence direction and does not contain a route to deciding it.

# 17 The square sieve at the mass-two wall

Remark 17.9 left the residue inside $\{\sigma(m) \geq 2 - \varepsilon\}$. We now attack that set directly, by a change of question. The minus layer is $m' - 2m = d^2$, so $\sigma(m) = 2 + d^2/m > 2$: it lives *above* the wall, and the layer is exactly the set on which the arithmetic function $m \mapsto m' - 2m$ takes square values. Atom A9 is therefore an instance of *how often is an arithmetic function a perfect square*, and for that question there is a tool: Heath-Brown's square sieve. It has not previously been applied to the arithmetic derivative, because $m'$ is not a polynomial and no Weil bound is available. The obstruction dissolves because the relevant symbol factorises.

**Lemma 17.1** (Factorisation of the symbol)**.** *Let $r$ be odd and let $m$ be squarefree with* $\gcd(m, r) = 1$ *(equivalently, no prime factor of $m$ divides $r$). Put $\sigma_r(m) = \sum_{p \mid m} p^{-1} \in \mathbb{Z}/r$. Then $m' \equiv m\,\sigma_r(m) \pmod r$ and*

$$\Big(\frac{m' - 2m}{r}\Big) = \Big(\frac{m}{r}\Big)\Big(\frac{\sigma_r(m) - 2}{r}\Big).$$

*Proof.* $m' = \sum_{p \mid m} m/p$ and $m/p \equiv m\,p^{-1} \pmod r$ for each $p \mid m$, since $p$ is invertible modulo $r$. Summing gives $m' \equiv m\sigma_r(m)$, hence $m' - 2m \equiv m(\sigma_r(m) - 2)$, and the Jacobi symbol is completely multiplicative in its numerator. □

Verified exhaustively for $r = 101, 1009, 10007$ over all admissible squarefree $m \leq 3 \times 10^7$: 54,514,790 checks, 0 violations (`code/sqsieve_lemmas.rs`).

The point of Lemma 17.1 is that $\sigma_r$ is *additive* over the distinct prime divisors, so after a Gauss-sum expansion the second factor becomes a product over $p \mid m$ and the whole sum turns multiplicative. That is what puts it inside Halász's theorem.

## 17.1 The square sieve and Theorem A.9

The argument recorded in this section is a square sieve at $P \asymp (\log N)^{1/4}$ whose analytic input does not close in the form attempted there, and why Remark 16.24 keeps this material off the critical path; the rate is nonetheless proved, by a different route, in Theorem A.12. The density statement itself, Theorem A.9, is proved, by a different sieve at a far smaller range (Proposition A.7). The

section occupied eight pages in the main line. It is moved verbatim to Appendix A; nothing is omitted and every label is unchanged, so the cross-references from §17.2 still resolve. What the argument establishes, what it assumes, and exactly where the gap sits are stated there.

## 17.2 Pair averages and the mass–two wall

**Lemma 17.2** ($\sigma$ is injective on squarefree integers)**.** *For squarefree $d, d'$, $\sigma(d) = \sigma(d')$ forces $d = d'$. Consequently $F(d,d') := D(d)\,d' - D(d')\,d = dd'(\sigma(d) - \sigma(d'))$ vanishes only on the diagonal.*

*Proof.* By Theorem 2.3, $\sigma(d) = D(d)/d$ is already in lowest terms, since $\gcd(D(d), d) = 1$ for squarefree $d$; so $d$ is recoverable from $\sigma(d)$ as its denominator. □

Verified over all 182,377 squarefree $d \le 3 \times 10^5$: 0 collisions (`code/sqsieve_pairavg.rs`). The lemma is formalised: in the notation of Section 11, `Erdos307.dprod_eq_of_mass_eq` and `Erdos307.csum_eq_of_mass_eq` derive it from `rigidity_coprime`, with 0 `sorry` and axioms `propext, Classical.choice, Quot.sound` (`lean/Erdos307/Injective.lean`). Its label is therefore VERIFIED.

**Theorem 17.3** (The average pair bound)**.** *Let $\mathcal{R} = \{pq : p \ne q \in \mathcal{P}\}$ and $P(D,r) = \#\{(d,d') \sim D : \sigma_r(d) \equiv \sigma_r(d')\}$. Then*

$$\sum_{r \in \mathcal{R}} P(D,r) \;\ll\; |\mathcal{R}|\,D + D^{2+o(1)}.$$

*Proof.* As $\gcd(d,r) = \gcd(d',r) = 1$, the condition $\sigma_r(d) \equiv \sigma_r(d')$ is exactly $r \mid F(d,d')$, so $\sum_{r\in\mathcal{R}} P(D,r) = \sum_{(d,d')} \#\{r \in \mathcal{R} : r \mid F(d,d')\}$. On the diagonal $F = 0$ is divisible by every $r$, contributing $|\mathcal{R}| \cdot \#\{d \le D : \mu^2(d) = 1\}$. Off it $F \ne 0$ by Lemma 17.2, and $|F| \le 2D^2\sigma_{\max}$, so $\#\{r \in \mathcal{R} : r \mid F\} \le \tau(F) \ll D^{o(1)}$, contributing $D^{2+o(1)}$. □

Measured against $D^2/r + D$ the average has ratio 1.011 at $|\mathcal{R}| = 78$ (`code/sqsieve_pairavg.rs`). Note the proof uses no equidistribution of $\sigma_r$ and no uniformity in $r$: only the divisor bound and Lemma 17.2, the latter being the same rigidity that makes #307 a two-cycle problem at all. The counting skeleton is formalised as `Erdos307.pairavg_bound`, with the divisor step `card_filter_dvd_le` and the double count `sum_filter_comm`; 0 `sorry`, standard axioms (`lean/Erdos307/PairAvg.lean`). The arithmetic input enters only through the hypothesis that $F$ vanishes exactly on the diagonal, which is Lemma 17.2, itself VERIFIED and transferred from prime sets to squarefree integers by `Erdos307.eq_of_cross`, so that the hypothesis consumed by the skeleton is exactly the statement the paper uses.

The decomposition demanded by Remark A.20 can be bypassed entirely. Rather than splitting $\sum_{m \le N}$ into bilinear pieces, one may reverse the roles of the two arguments of the symbol and appeal to the quadratic large sieve, which averages over the modulus. This is the more promising route and we record it, together with the one hypothesis it still needs.

**Lemma 17.4** (Reversal)**.** *Write $a_m = m' - 2m$ and $a_m = \pm s_m^2 k_m$ with $k_m$ squarefree. For odd squarefree $r$,*

$$T_r(N) = \sum_k b_k^+\Big(\frac{k}{r}\Big) + \Big(\frac{-1}{r}\Big)\sum_k b_k^-\Big(\frac{k}{r}\Big) \;+\; O\big(\#\{m \le N :\; p^2 \mid a_m \text{ for some } p \mid r\}\big),$$

*where $b_k^{\pm} = \#\{m \le N : \mu^2(m) = 1,\ k_m = k,\ \pm a_m > 0\}$ depend on $N$ only. For $r = pq$ with $p, q \asymp N^{1/2}$ one would want the error term to be $O(1)$, which would need Lemma A.8 in the form*

$\#\{m \le N : p^2 \mid a_m\} \ll N/p^2$. *That form is the one that lemma explicitly disowns, since its implied constant depends on $p$ while $p$ here grows with $N$; the error term is not controlled at this range.*

Regime caveat, and its retraction. *That $O(1)$ holds only at $p \asymp N^{1/2}$; at $P \asymp (\log N)^{1/4}$ the same estimate gives only $N/p^2 \asymp N/(\log N)^{1/2}$. Two earlier revisions read that as a second, independent reason for the* CONJECTURE *label then carried by Theorem A.12. It is not, for two reasons. The comparison there was made against $N$, whereas the benchmark is that statement's own target, now $N(\log\log N)^{1/2+\delta}(\log N)^{-1/4}$ (Theorem A.12), against which $N(\log N)^{-1/2}$ is* smaller*, an error term below the claimed rate cannot obstruct it. And Theorem A.12 does not travel this route at all: re–derived as in Proposition A.7 it needs no kernel reversal, and Proposition 17.6 shows the reversal is no reduction in any case. The operative blocker is recorded at the corollary itself.*

*Proof.* If $\gcd(a_m, r) = 1$ then $(a_m/r) = (\pm 1/r)(k_m/r)(s_m/r)^2 = (\pm 1/r)(k_m/r)$. The two sides can differ only when some $p \mid r$ divides $s_m$ but not $k_m$, that is when $p^2 \mid a_m$, which is the error term. The coefficients $b_k^{\pm}$ carry no dependence on $r$. □

Measured at $N = 5 \times 10^6$: 30.0% of the $a_m$ are non-squarefree, but only 0.035% have $s_m > 10^3$, so for $p \asymp N^{1/2}$ the error term is empty in practice (`code/sqsieve_holes.rs`). The kernels $k_m$ are even 22.2% of the time, so the moduli are split into the four classes mod 8 before the sieve is applied, at a cost of a factor 4.

**Hypothesis 17.5** (Kernel collisions)**.** $\sum_k \big(|b_k^+|^2 + |b_k^-|^2\big) \ll N^{1+o(1)}$, equivalently $\#\{(m, m') \text{ squarefree}, \le N : \ a_m a_{m'} \text{ is a perfect square}\} \ll N^{1+o(1)}$.

**Proposition 17.6** (The route is circular)**.** *Hypothesis* 17.5 *implies the bound it would be used to prove, so it cannot serve as a reduction. Indeed $b_1^+ = \#\{m \le N : \ \mu^2(m) = 1,\ a_m \text{ is a positive perfect square}\}$ is precisely the minus layer, and $(b_1^+)^2 \le \sum_k |b_k^+|^2$, so Hypothesis* 17.5 *forces $b_1^+ \ll N^{1/2+o(1)}$, which is the conclusion.*

*Why computation cannot detect this.* The circularity is invisible to measurement. Below the barrier $\sigma(m) > 2$ is impossible, so $a_m > 0$ never occurs and $b_1^+ = 0$ identically: at $N = 5 \times 10^6$ one finds $b_1^+ = 0$ while $b_1^- = 1213$, the latter contributing 12.5% of $\sum_k |b_k|^2$ (`code/sqsieve_kernel.rs`). Every computed check of Hypothesis 17.5 therefore tests a statement from which the load-bearing term has been deleted. This is the same blindness recorded in Proposition 16.7, appearing here in a form that would otherwise have passed as supporting evidence.

Accordingly, the large sieve does apply and Lemma 17.4 is correct, but the resulting statement is a *reformulation*: CRUX-A9 is equivalent to a kernel-collision bound, with the equivalence carried by the $k = 1$ term. The new attack surface is genuine, since the reformulated statement is unsigned and about products, but it is not a reduction and is not counted as progress on the crux. The unconditional record is Proposition A.7.

*The hypothesis measures, and its constant identifies itself.* $P(D, r)$ against $D^2/r + D$ has ratio 0.3688, 0.3688, 0.3689 at $r = 1009$; 0.3705, 0.3696, 0.3695 at $r = 10007$; and 0.3867, 0.3727, 0.3702 at $r = 100003$, for $D = 10^6, 5 \times 10^6, 2 \times 10^7$ (`code/sqsieve_bilinear.rs`). The ratio is not merely bounded but constant, and the constant is $(6/\pi^2)^2 = 0.3696$, the square of the squarefree density, which is exactly what equidistribution of $\sigma_r$ predicts once both $d$ and $d'$ are restricted to squarefree integers.

Compare Hypothesis A.14. That one asks for signed square-root cancellation, and Proposition A.17 showed the only available route to it fails. Theorem 17.3 instead bounds an *unsigned* second moment, and does so unconditionally; what it does not do is bound $T_r$, since the passage from bilinear forms to $T_r$ is exactly the gap recorded in Remark A.20.

*What made this work.* Every previous attack on A9 counted members of the minus layer directly, and each ran into the same obstruction: the layer is a union of lines $m' = 2m - k$, each thin by Proposition 15.18 but not summably so, and the peeling reorganisations that would combine them are all conjugate under the groupoid of Proposition 15.20. The square sieve never counts a member. It asks only whether $m'-2m$ is a square, and answers by averaging a Jacobi symbol, so the groupoid has nothing to act on. The one obstacle, that $m'$ is not a polynomial, is removed by Lemma 17.1: the symbol splits into a character in $m$ times a character in $\sigma_r(m)$, and $\sigma_r$ is additive, which is exactly the hypothesis Halász needs. The analytic inputs are all standard: Halász's theorem, the Gauss sum evaluation $|\tau(\chi)| = \sqrt{r}$, the classical zero–free region, the $\log L$ bound for non–principal characters, and Siegel–Walfisz. No Salié or Weil bound is used: the sum in question is a Gauss sum, as Lemma A.1 records. Note also that the theorem does not decide #307, and cannot: it is a statement about density, and a single two-cycle is a density-zero event.

**Proposition 17.7** (Slot stratification of the divisibility set)**.** *Fix $c$ and let $D(c) = \{m$ squarefree : $(2m - m') \mid (m - c),\ m \neq c\}$. For $m \in D(c)$ put $p = (m - c)/(2m - m')$, the* slot*. Then*

$$p\,m' \;=\; (2p-1)\,m + c, \qquad \textit{equivalently} \qquad \sigma(m) = 2 - \frac{m-c}{p\,m},$$

*so $D(c) = \bigsqcup_{p\neq 0} L(p, 2p-1, c)$ is a disjoint union of generalized lines in the sense of Proposition* 15.20*, the stratum $p = 1$ being the slope-*one *line $m' = m + c$. The identity was verified with* 0 *violations on all* 2,294 *members with $m \le 2 \times 10^7$, $|c| \le 400$* (`code/slot_stratify.rs`)*.*

*Proof.* $m - c = p(2m - m')$ rearranges to $pm' = (2p-1)m + c$; dividing by $pm$ gives the mass form. Distinct $p$ give disjoint strata since $p$ is a function of $m$. □

**Theorem 17.8** (The residual difficulty sits entirely at the mass-two wall)**.** *For every $\varepsilon > 0$ and every $c$,*

$$\#\big(D(c) \cap [1, Z]\big) \;\le\; \big(\delta(2-\varepsilon) + o_\varepsilon(1)\big)\, Z + O(|c|),$$

*where $\delta(x)$ is the density of $\{m : \sigma(m) \ge x\}$. Moreover*

$$\delta(x) \;\le\; \inf_{t>0}\; e^{-tx} \prod_{p} \Big(1 + \frac{e^{t/p} - 1}{p}\Big),$$

*which evaluates to $\delta(1) \le 10^{-0.8}$, $\delta(1.5) \le 10^{-7.2}$ and $\delta(2) \le 10^{-105.6}$* (`code/chernoff_mass.rs`)*.*

*Proof.* Split $D(c)$ at $|p| \le P$ and $|p| > P$. The first part is a union of at most $2P$ generalized lines, each of size $\ll Z/(\log Z)^{(1+o(1))\log_3 Z}$ by Proposition 15.18 (*caveat*: Proposition 15.18 is stated for *integer* slope $e \ge 1$, while the strata $L(p, 2p-1, c)$ of Proposition 17.7 have slope $(2p-1)/p$, which is non–integral for $p \neq 1$. The bound is therefore invoked outside the range in which it is proved, and the first part of this proof is CONJECTURE pending a version of Proposition 15.18 uniform over rational slopes); taking $P = P(Z) \to \infty$ slowly makes the total $o(Z)$. For the second, $|p| > P$ forces $|2 - \sigma(m)| < (1 + |c|/m)/P$ by the mass form, so for $m > |c|$ we get $\sigma(m) > 2 - 2/P$, and the count is at most $\delta(2-2/P)Z(1+o(1))$. For the Chernoff bound, $f(m) = \prod_{p|m} e^{t/p}$ is multiplicative with $f(p) \ge 1$, so $\sum_{m\le Z} f(m) = \sum_{d\le Z} \mu^2(d) \prod_{p|d} (f(p)-1)\lfloor Z/d \rfloor \le Z \prod_p (1 + (f(p)-1)/p)$, and $\#\{\sigma \ge x\} \le e^{-tx} \sum_{m\le Z} f(m)$. The product converges because $(e^{t/p} - 1)/p \sim t/p^2$. □

**Remark 17.9** (What this settles and what it does not)**.** Theorem 17.8 replaces the constant $\delta = 0.0420$ of Proposition 17.10 (ii) by one below $10^{-105}$, an unconditional gain of more than a hundred orders of magnitude, and it does so by *using* the divisibility rather than merely the mass

condition it forces. It also localises the residue exactly: away from the mass-two wall the set is a bounded union of thin lines, and *all* the remaining difficulty lies in $\{\sigma(m) \geq 2-\varepsilon\}$. It does not prove A9. The set $\{\sigma \geq 2\}$ has *positive* density, at least $1/\Pi_{59} = 10^{-112.9}$ since every multiple of $\Pi_{59}$ lies in it, so the method cannot reach $o(Z)$: the limiting constant is positive, not zero. The bound $10^{-105.6}$ sits within seven orders of that truth, so little is lost in the Chernoff step, and the obstruction is genuine rather than an artefact of the estimate. The residual question is therefore sharp: is $D(c)$ thin *within* the mass-two wall, the one place where the barrier (Theorem 4.3) also lives?

**Proposition 17.10** (The divisibility set: slope one is dense, slope two is not)**.** *Let $e \geq 1$ and $c$ be integers and let $m$ be squarefree with $(em - m') \mid (m - c)$ and $m \neq c$. Then*

$$\sigma(m) \;\geq\; e - 1 - \frac{|c|}{m}.$$

*Consequently the two slopes behave oppositely.*

(i) *For $e = 1$ the condition is vacuous and the set has positive density: every prime $m$ satisfies it when $c = 1$, since then $m' = 1$ and $em - m' = m - 1 = m - c$. Over $m \leq 2 \times 10^7$ this accounts for* 1,270,651 *of the members, exactly $\pi(2 \times 10^7)$.*

(ii) *For $e = 2$ the condition reads $\sigma(m) \geq 1 - |c|/m$, so by Proposition 6.5*

$$\#\{m \leq Z :\; (2m - m') \mid (m - c),\; m \neq c\} \;\leq\; (\delta + o(1))\, Z + O(|c|),$$

*with $\delta = 0.0420\ldots$ the abundance density, certified in Proposition 6.5 to lie in* $[0.04202, 0.04223]$.

*Proof.* If $|em - m'| > |m - c|$ the divisibility forces $m = c$. So $|em - m'| \leq |m - c| \leq m + |c|$; below the barrier $\sigma(m) < e$ for $e \geq 2$, so $em - m' = m(e - \sigma(m)) > 0$ and $m(e - \sigma(m)) \leq m + |c|$, which is the display. Part (i) is the computation just given, and (ii) is Proposition 6.5 applied to $\{\sigma \geq 1 - \varepsilon\}$. The mass condition was checked with 0 violations over all squarefree $m \leq 2 \times 10^7$ and all $|c| \leq 400$ at $e = 1, 2, 3$, where the nontrivial member counts are 2,197,230, 28 and 8 respectively (`code/a9prime_falsify.rs`). □

*Consequence for the residual question.* Proposition 17.10 settles the shape of the residual bound. A uniform statement over all slopes is false, and false for a trivial reason: at $e = 1$ the primes sit on the set by an identity. The correct target is confined to $e \geq 2$, where a genuine mass condition appears, and there Proposition 17.10(ii) gives the first unconditional bound, $\delta Z$ with $\delta < 0.043$. The gap to the $Z^{1/2}$ needed for Remark 15.26 is therefore now a gap between two *proved* quantities rather than between a proof and a wish, and the observed counts at $e = 2$ (28 over the whole range $|c| \leq 400$, against $Z^{1/2} = 4472$) sit far below both.

**Remark 17.11** (The two open routes are one question, asked in opposite directions)**.** The existence route of Remark 5.28 and the counting route of Remark 15.26 look unrelated (one is a search for a single solution, the other a density estimate) but they are the same question about the same kind of object. In each case an index determines a single integer, and everything turns on whether that integer is prime: in Remark 5.28 the Pell index $n$ determines $q_n = (x_n^2 - D_S)/A_S$, a term of a ternary linear recurrence, and #307 needs it prime *once*; in Remark 15.26 the cofactor $m$ determines the slot $(c - am)/(m' - em)$ of Proposition 15.20, and the density needs it prime *rarely*. Existence asks for a lower bound on the number of prime values, density for an upper bound, on determined values along a thin deterministic sequence, and neither estimate is available: no unconditional infinitude

of prime terms is known for any non–degenerate linear recurrence of order $\geq 2$, and no worst–case upper bound is known for single–class prime counts summed over all moduli. #307 sits exactly between two missing estimates on one object, which is why the heuristics can be read either way and why progress on either side would be informative about the other. The remaining walls are, in this sense, one wall seen from two directions.

### 17.3 Slot structure

**Corollary 17.12** (Quadratic–residue filter for #307 solutions)**.** *Let $(P, Q)$ solve Erdős #307 and set $a = D_P$, $b = D_Q$, $N = ab$. Since $N' = a^2+b^2$ and hence $N'+2N = (a+b)^2$ automatically, every prime $\ell \in P \cup Q$ satisfies $(N/\ell \mid \ell) = +1$: the cofactor $N/\ell$ is a nonzero quadratic residue modulo $\ell$. Indeed $\ell$ divides exactly one of $a, b$ (by $P \cap Q = \varnothing$), so $\ell \nmid a+b$ and $(a+b)^2 \not\equiv 0 \pmod{\ell}$; combined with the identity $N'+2N \equiv N/\ell \pmod{\ell}$ of* (v)*, this forces $N/\ell$ to be a nonzero square mod $\ell$. As a search filter the condition prunes candidate unions by a factor $\leq 2^{-|P\cup Q|} \leq 2^{-59}$ (heuristically each of the $\geq 59$ primes imposes an independent QR constraint); for genuine cycles it is automatic and does not alter the existence heuristics. We note that Corollary 17.12 is* not *independent of Bado: it is the squared form of his congruence $D_Q \equiv D_P/\ell \pmod{\ell}$ (itself $N_P$ reduced mod $\ell$ by cofactor survival), giving $N/\ell \equiv D_Q^2 \pmod{\ell}$. Squaring is lossy ($r$ and $-r$ share a square) so Bado's linear congruence is strictly stronger, pinning $D_Q \bmod \ell$ exactly where the residue filter determines it only up to sign; the filter's one added value is being* split–free *(the residue $N/\ell \bmod \ell$ is computable from the candidate union alone, with no knowledge of the $P \mid Q$ partition). The general law of Proposition 15.14(v), by contrast, holds for* any *squarefree plus–hit with no $P \mid Q$ split and is not subsumed by Bado.*

The plus– and minus–hits both organise into one–prime "slot" families, which makes the structure of Proposition 15.12 explicit at every $\omega$.

**Proposition 17.13** (Slot calculus for the discriminant layers)**.** *Let $N_0$ be squarefree, $r \nmid N_0$ a prime, $N = N_0 r$, and $E_0 := N_0' + 2N_0$. By Leibniz ($N' = N_0' r + N_0$),*

$$N' + 2N = rE_0 + N_0, \qquad N' - 2N = r(N_0' - 2N_0) + N_0,$$

*so $N$ is a plus–hit iff $r = (s^2 - N_0)/E_0$ for some integer $s$ (equivalently $s^2 \equiv N_0 \pmod{E_0}$).*

(a) Every $\omega$–stratum. *A plus–hit $N$ ($\omega \geq 2$) factors, through any prime divisor $r$, as $N = N_0 r$ with $N_0 = N/r$ squarefree, and lies in the quadratic family $\{N_0(s^2 - N_0)/E_0\}_s$ of its base; generalising the $p = 5$ family of Proposition 15.14(iii) beyond semiprimes (each step adds one prime, $\omega(N) = \omega(N_0) + 1$). The congruence $s^2 \equiv N_0 \pmod{E_0}$ is solvable for* 669 *of the* 3041 *squarefree $N_0 \in [2, 5000]$ (*22.0%*); solvability is necessary but not sufficient for the family to contain a hit.*

(b) Recursive towers: a trichotomy. *When the base is itself a plus–hit, $E_0 = s_0^2$ is a square; below $2 \times 10^5$ the* 114 *plus–hit bases split into three kinds.* No residue *(*84*): $s^2 \equiv N_0 \pmod{E_0}$ is unsolvable and the family is empty; a quadratic–residue obstruction, occurring for both even $E_0$ ($N_0 = 305$, with* 305 *a non–residue mod* 13*) and odd $E_0$ ($N_0 = 1870$, $E_0 = 73^2$).* Fixed divisor *(*13*, each with fixed divisor* 2*): the family is nonempty but every slot is even, so only $r = 2$ could serve and never gives a square; sterile (e.g. $N_0 = 145$, $E_0 = 18^2$).* Live *(*17*): fixed divisor* 1 *and a prime slot is exhibited (e.g. $N_0 = 130$, first prime slot $r =$ 83399, giving the plus–hit $N =$ 10,841,870 with $N' + 2N = 5487^2$).*

(c) Minus layer; the barrier in slot form. *For $\sigma(N_0) < 2$ the minus identity gives $N' - 2N \geq 0$ iff $r \leq 1/(2 - \sigma(N_0))$, i.e. iff $\sigma(N) \geq 2$: this is the slot–variable form of Corollary 14.3, not an independent proof of it. The bound $1/(2 - \sigma(N_0))$ is unbounded as $\sigma(N_0) \to 2^-$; over squarefree $N_0 < 10^5$ its maximum is only* 1.524 *(at $N_0 =$* 30030*), so no such base admits a prime minus–slot, yet for the* 58*–prime primorial the bound exceeds* 793 *and the next prime* 277 *fits, producing the* 59*–prime primorial, the smallest minus–hit base. A prime minus–slot thus first becomes feasible* exactly *at the* 59*–prime barrier; the minus–empty–below–barrier statement rests on the primorial growth of Corollary 14.3, not on the figure* 1.524*.*

(The two identities are checked symbolically and on $2 \times 10^4$ random pairs; the base trichotomy, slot lists, and the 1.524 maximum by exhaustive enumeration.)

The two square conditions over a base combine into a single binary–form representation, which places the obstruction inside Gauss's genus theory.

**Proposition 17.14** (The Pythagorean condition as a binary–form representation)**.** *For squarefree $N_0$ and prime $r \nmid N_0$, $N = N_0 r$, the identity*

$$(2N_0 - N_0')(N' + 2N) + (2N_0 + N_0')(N' - 2N) = 4N_0^2$$

*holds unconditionally (Leibniz; checked on $2 \times 10^4$ random pairs). Hence if $N$ is a Pythagorean pair product over $N_0$ (both $N' + 2N = s^2$ and $N' - 2N = d^2$ squares) then $(s, d)$ represents $4N_0^2$ by the binary form $Q_{N_0}(s, d) = A\,s^2 + B\,d^2$ with $A = 2N_0 - N_0' = N_0(2 - \sigma(N_0))$, $B = 2N_0 + N_0' = N_0(2 + \sigma(N_0))$, of discriminant $-4AB = 4(N_0'^2 - 4N_0^2)$; conversely such a representation yields a Pythagorean pair over $N_0$ exactly when the recovered slot $r = (s^2 - N_0)/E_0$ is a positive integer, prime, and coprime to $N_0$ (the integrality $s^2 \equiv N_0 \pmod{E_0}$ holds on every representation we found, $N_0 < 2 \times 10^4$). For $\sigma(N_0) < 2$ the form is positive–definite ($A > 0$), and below the barrier representations are rare and all "ghosts": the only squarefree $N_0 \leq 3000$ representing $4N_0^2$ are $N_0 = 30$ (with $(s, d) = (11, 1)$) and $N_0 = 1722$ ($(83, 1)$), each carrying the non–admissible slot $r = 1$, and no live prime slot occurs. As $\sigma(N_0) \to 2^-$ the leading coefficient $A = N_0(2 - \sigma(N_0)) \to 0$ and the ellipse $A\,s^2 + B\,d^2 = 4N_0^2$ degenerates to an arbitrarily thin sliver: the geometric face of the mass–2 barrier. Consequently #307 is the existence of a base with $\sigma(N_0) \geq 2 - 1/r$ whose form $Q_{N_0}$ represents $4N_0^2$ with an admissible, integral,* prime *slot and deviation $k = 0$.*

(We make no class–number claim: each base yields a form of a *different* discriminant, and the question is single–value representation, not a class count; only the degeneration $A \to 0$ is used.)

**Proposition 17.15** (An explicit Pythagorean family, and why it stops at rung $-1$)**.** *Let $N_0$ be squarefree with $N_0' = 2N_0 - 1$ (the slope–two line $c = -1$, i.e. $k_0 = 1$) and suppose $N_0 - 1$ is prime. Then $(a, b) = (N_0 - 1,\ N_0)$ is a Pythagorean pair:*

$$a^2 + b^2 \;=\; 2N_0^2 - 2N_0 + 1 \;=\; (ab)'.$$

*It arises from the form equation of Proposition 17.14 at its degenerate end: with $k_0 = 1$, $E_0 = 4N_0 - 1$, the pair $(s, d) = (2N_0 - 1,\ 1)$ represents $4N_0^2$ exactly, and the slot it yields is $r = (s^2 - N_0)/E_0 = N_0 - 1$. Every member of the family sits on rung $k = -1$, and no member can sit on rung 0: $a$ is prime, so $a' = 1$, and $a' = b$ would force $N_0 = 1$.*

*Proof.* $(ab)' = N_0'(N_0 - 1) + N_0(N_0 - 1)' = (2N_0 - 1)(N_0 - 1) + N_0 = 2N_0^2 - 2N_0 + 1 = (N_0 - 1)^2 + N_0^2$, using $(N_0 - 1)' = 1$ for $N_0 - 1$ prime. The form identity is $(2N_0 - 1)^2 + (4N_0 - 1) = 4N_0^2$, and $(s^2 - N_0)/E_0 = (4N_0^2 - 5N_0 + 1)/(4N_0 - 1) = N_0 - 1$. The rung is $k = (a' - b)/a = (1 - N_0)/(N_0 - 1) = -1$. □

*What the family shows.* This is the first explicit construction of Pythagorean pairs in this note: everything earlier established that they are rare (Corollary 14.3: none below $10^{112}$) without exhibiting a mechanism. The mechanism is a clean reduction (the pair exists as soon as the line $n' = 2n-1$ carries a member whose predecessor is prime) and it is Bunyakovsky–shaped, matching the observation in Remark 5.28 that a single square condition is of that class. It is also conditional in a way the barrier controls: $\sigma(N_0) = 2 - 1/N_0$ forces $\omega(N_0) \geq 58$, so the family is empty below the scale where the critical set of Remark 15.26 switches on.

Two further consequences. First, combining with Corollary 15.5: for odd $N_0$ the pair has mixed parity with odd member $N_0$, so $k \equiv \omega(N_0) \pmod 2$; since $k = -1$ here, *any odd member of the line $n' = 2n-1$ whose predecessor is prime has $\omega(N_0)$ odd*; a necessary condition on that line, derived rather than observed. Second, and this is the honest limitation, the construction is *permanently* off the cycle rung: primality of $a$ is exactly what makes it work and exactly what forces $a' = 1 \neq b$. A $k = 0$ analogue would need $a$ composite with $a' = b$, which is the original problem. So the family exhibits the Pythagorean locus without approaching #307; useful as the first worked mechanism, not as a route.

**Proposition 17.16** (The composite mechanism: $|P| = 2$ is one line and one primality)**.** *Write $a = 2m$ with $m$ odd squarefree, so that $b = a' = m + 2m'$ (Theorem 4.14). The two–cycle condition is then the single equation*

$$b' \;=\; 2m, \qquad b = m + 2m',$$

*in the one unknown $m$. For $m = q$ prime it reads $b = q+2$ and*

$$\boxed{b' = 2b - 4} \qquad \text{with } b-2 \text{ prime},$$

*and any such $b$ is a solution: $a = 2q$, $b = q+2$ satisfy $a' = b$, $b' = a$ and $\sigma(a)\sigma(b) = \left(\frac{1}{2} + \frac{1}{q}\right)\left(2 - \frac{4}{q+2}\right) = 1$ identically. Hence*

$$\#307 \text{ with } |P| = 2 \iff \exists\, b \text{ squarefree}:\ b' = 2b-4 \text{ and } b-2 \text{ prime}.$$

*Proof.* $(2m)' = m + 2m'$ by Leibniz, so $a' = b$ by definition of $b$, and the cycle needs only $b' = a = 2m$. For $m = q$ prime, $b = q+2$ and $b' = 2q = 2(b-2) = 2b-4$; the mass identity is the displayed product. □

*Two lines, two rungs.* Comparing with Proposition 17.15 isolates exactly what separates the cycle from its neighbour. Both are "a member of a slope–two line whose shifted predecessor is prime":

$$\begin{aligned} &\text{line } n' = 2n-1, \quad N_0 - 1 \text{ prime} \;\implies \text{Pythagorean pair, rung } k = -1;\\ &\text{line } n' = 2n-4, \quad b-2 \text{ prime} \;\implies \text{genuine two–cycle, rung } k = 0. \end{aligned}$$

The difference between the open problem and its solved neighbour is the constant in the line. And the obstruction changes character with it: Proposition 17.15 is barred from rung 0 *structurally* (its $a$ is prime, so $a' = 1 \neq b$) whereas Proposition 17.16 is barred only by *scale*, since $\sigma(b) = 2 - 4/b$ forces $\omega(b) \geq 58$ and $b \geq \Pi_{58}$, putting the whole family above $10^{108}$ and the resulting $N = 2qb$ at the barrier. A direct search of the reduction over odd squarefree $m \leq 2 \times 10^6$ (810,590 admissible $m$, of which 111,464 prime) finds none, with nearest miss $|b' - 2m| = 5$ at $m = 3$ (`code/composite_a.rs`); exactly as Corollary 14.3 demands. This is the composite mechanism the rung–0 case requires; what it does not do is evade the barrier, which reappears as the mass of the line.

**Proposition 17.17** (The barrier is a function of the split)**.** *For a solution with $|P| = j$ write $a = \prod P$, $b = \prod Q$. Since $\sigma(a)\sigma(b) = 1$ and $P \cap Q = \varnothing$, fixing $j$ forces $\sigma(b) = 1/\sigma(a)$ and leaves $Q$ only the odd primes outside $P$, which bounds $N$ from below split by split. Taking $a = 2 \cdot$ (the $j-1$ smallest odd primes), the most favourable configuration at that size:*

| $\lvert P\rvert$ | $\sigma(a)$ | $\sigma(b)$ | $\lvert Q\rvert \geq$ | $\log_{10} N \geq$ |
|---|---|---|---|---|
| 2 | 0.8333 | 1.2000 | 67 | 137.9 |
| 3 | 1.0333 | 0.9677 | 56 | 112.9 |
| 4 | 1.1762 | 0.8502 | 63 | 132.8 |
| 5 | 1.2671 | 0.7892 | 73 | 161.0 |
| 6 | 1.3440 | 0.7440 | 85 | 195.3 |
| 12 | 1.5927 | 0.6279 | 181 | 491.1 |

*The minimum over splits is $10^{112.9}$ at $|P| = 3$, recovering the constant of Theorem 4.3 by an independent route, and every other split is strictly dearer.*

*Proof.* $\sigma(b) = 1/\sigma(a)$ is the cross–match. $Q$ is disjoint from $P$, so its mass must be assembled from odd primes outside $P$; the least number of them reaching $\sigma(b)$, and the product of that many, are finite computations (`code/sector_barrier.rs`). Enlarging any prime of $a$ lowers $\sigma(a)$, raises $\sigma(b)$ and hence $|Q|$, so the displayed configuration is extremal for each $j$. □

*Why the composite mechanism is out of reach.* Proposition 17.17 settles the standing of the clean reduction of Proposition 17.16. There $|P| = 2$, and the mechanism is forced through the dearest small sector: $b$ is odd, so $Q$ must assemble mass $\sigma(b) = 2q/(q+2)$ from odd primes alone, and the sector cannot begin below $10^{137.9}$; some twenty–five orders of magnitude *above* the general barrier $10^{112.9}$. The line $n' = 2n - 4$ is therefore a correct and complete description of the $|P| = 2$ case and simultaneously the most expensive place to look.

This also explains the balance the barrier strikes. The cheapest configuration is the near–balanced one, $\sigma(a) \approx \sigma(b) \approx 1$ at $|P| = 3$; pushing the split either way (towards a small $P$ with $\sigma(b) \to 2$, or towards a large $P$ with $\sigma(b) \to 0$) costs orders rapidly, because mass near 2 needs the odd primes to $\approx 10^4$ and mass near 0 needs many large ones. The 59–prime barrier is thus not merely a statement about $|P \cup Q|$ but the *minimum of a convex profile over splits*, and Theorem 4.3's constant is the value at that minimum. (Rows with $|P| + |Q| < 60$ are further excluded by Proposition 4.5, which sharpens the $|P| = 3$ entry without moving the minimum's location.)

**Proposition 17.18** (The cycle condition does not improve the barrier)**.** *A two–cycle satisfies more than the mass inequality: it forces $b = a'$, that is $b = a\,\sigma(a)$ exactly. Since $\sigma(b) = 1/\sigma(a)$ and $P \cap Q = \varnothing$, writing $S$ for the small primes of $a$ and $\Pi_{\min}(S)$ for the least product of primes outside $S$ with reciprocal sum $\geq 1/\sigma(S)$, one gets*

$$a \;\geq\; \max\Big(\prod S,\; \Pi_{\min}(S)/\sigma(S)\Big), \qquad N \;=\; a^2\sigma(a) \;\geq\; \max\Big(\prod S,\; \Pi_{\min}(S)/\sigma(S)\Big)^2 \sigma(S).$$

*Minimised over $S$, this yields $\log_{10} N \geq 113.2$, attained at $S = \{\text{odd } p \leq 151\} \setminus \{3, 97\}$ with $\sigma(S) = 1.0495$ and $|Q| \geq 26$. The barrier of Theorem 4.3 is $\log_{10} N \geq 112.9$: the cycle condition buys essentially nothing beyond the mass condition, and Theorem 4.3 is already close to optimal for arguments of this class.*

Formalised. *The deductive content is `Erdos307.cycle_bound_max`: two independent lower bounds on $a$ combine through the maximum, and $N = a^2\sigma(a)$ is monotone in both arguments,*

*so the constraints give a bound on $N$. The conclusion drawn, that the gain over Theorem 4.3 is under one order, is an* upper *bound on a minimum, and an upper bound on a minimum needs one witness, not the search. Since $\Pi_{\min}(S)$ is itself a minimum, bounding the expression at $S$ from above needs a witness for it too: any $T$ disjoint from $S$ with $\sigma(T) \geq 1/\sigma(S)$ has $\Pi_{\min}(S) \leq \prod T$. Taking $S$ to be the odd primes to* 151 *less* $\{3, 97\}$ *and $T$ its greedy completion,* 26 *primes ending at* 277*, one checks $\prod T \leq \prod S \cdot \sigma(S)$, so the maximum is $\prod S$, and*

$$\Big(\big(\textstyle\prod S\big)^2 \sigma(S)\Big)^2 \;<\; 10^{227},$$

*squared so the exponent is an integer: the bound on $\log_{10} N$ at this $S$ is at most* 113.5*, and the gain over* 112.9 *is under* 0.6 *of one order. Note $\big(\prod S\big)^2\sigma(S)$ is an integer, $\sigma(S)$ having denominator exactly $\prod S$, so the whole check is exact integer arithmetic.* 0 `sorry` (`lean/Erdos307/NoGain.lean`, `lean/Erdos307/NoGainWitness.lean`)*. The exhaustive minimisation certifies the sharper claim that* 113.2 *is exactly the minimum; that is not needed here, and it was the only reason this proposition was carried as a formalisation refusal.*

**Remark 17.19** (Why restricted families overstated the gain)**.** Compare the conclusion of Proposition 17.18 with what restricted families suggest, because the discrepancy is large and instructive. Confining $S$ to an initial segment $\{2, 3, 5, \dots\}$ gives a minimum of $10^{222.9}$; enumerating all subsets of the first 14, 16, 18 primes gives $10^{195.9}$, $10^{188.9}$, $10^{181.7}$; still falling, hence not converged. Each of these looks like a large improvement of the barrier and none of them is one. The optimum sits outside all of them: it hands 2 to $Q$ (half of $Q$'s mass budget in a single prime), keeps the odd primes to 151 in $S$, and drops two of them, a configuration no initial segment and no small subset can express. Only after searching that structure does the bound settle at 113.2.

The lesson is about the shape of the search rather than the arithmetic. A bound minimised over a family that cannot express the optimum reports the family's best, not the problem's, and here that error was worth eighty orders of magnitude. We record the negative result as the substantive one: imposing $b = a'$, the strongest structural constraint available beyond the masses, does not move the barrier, which is evidence that $10^{112.9}$ is not an artefact of the AM–GM route but the true cost of the mass condition (`code/full_minimisation.rs`, `code/minimise_structured.rs`).

**Remark 17.20** (Decoupling the two squares: what it buys, and what it costs)**.** Remark 5.28 locates the exponential difficulty in the *conjunction* of the two squares, so it is worth asking for a family on which one of them is free. The slot calculus supplies one: on

$$\mathcal{F}(N_0) \;=\; \big\{\, N_0 r \;:\; r = (s^2 - N_0)/E_0 \,\big\}$$

the plus–square $N' + 2N = s^2$ holds *identically*, by construction. The minus–square then costs exactly the form equation of Proposition 17.14, $k_0 s^2 + E_0 d^2 = 4N_0^2$, and the class of the residual question genuinely changes. On a coupled tail family the solutions run along a Pell orbit and the candidate slots grow like $\varepsilon^n$; here, for $\sigma(N_0) < 2$ the form is positive definite, so $4N_0^2$ has only *finitely* many representations and each base carries finitely many candidate slots. Decoupling therefore trades a Lehmer sequence for a finite per–base check.

What it does not do is help. Over the 30,397 squarefree bases $N_0 \leq 5 \times 10^4$ with $\sigma(N_0) < 2$ the form represents $4N_0^2$ exactly *twice* (at most once per base, largest $s = 83$) and both representations are the ghost slot $r = 1$; there is no live prime slot at all (`code/plus_automatic.rs`, reproducing the sub–barrier census of Proposition 17.14 from an independent enumeration). The reason is the degeneration already noted: the barrier forces $\sigma(N_0) \uparrow 2$, so $k_0 = N_0(2 - \sigma)$ collapses and the form becomes ever more lopsided, while the target stays at $4N_0^2$. So freeing one square does not free the

problem; it relocates the difficulty from the primality of an exponential sequence to the solubility of a degenerating quadratic form. The pattern is the circularity of Section 8 once more, but the relocation is informative, since a finite, degenerating representation problem is a different kind of object from a Lehmer sequence, and no no–go here forbids attacking it.

The emptiness of that census is structural, and the reason fixes the range in which the question lives.

**Proposition 17.21** (The live–slot threshold)**.** *Let $N_0$ be squarefree and suppose $Q_{N_0}(s,d) = A\,s^2 + B\,d^2$ represents $4N_0^2$ with a slot $r = (s^2 - N_0)/B$ satisfying $r \geq 2$. Then*

$$\sigma(N_0) \;\geq\; \tfrac{3}{2},$$

*hence $\omega(N_0) \geq 10$ and $N_0 \geq \Pi_{10} = 6{,}469{,}693{,}230$.*

*Proof.* Assume first $\sigma = \sigma(N_0) < 2$, so $A = N_0(2-\sigma) > 0$ and the form is positive definite. From $r \geq 2$ and $B = N_0(2+\sigma)$,

$$s^2 \;\geq\; N_0 + 2B \;=\; N_0(5+2\sigma),$$

while $A\,s^2 \leq A\,s^2 + B\,d^2 = 4N_0^2$ gives $s^2 \leq 4N_0^2/A = 4N_0/(2-\sigma)$. The two are compatible only if $(5+2\sigma)(2-\sigma) \leq 4$, that is $2\sigma^2 + \sigma - 6 \geq 0$, whose positive root is exactly $\frac{3}{2}$. Since among $k$–element prime sets the reciprocal sum is maximised by the first $k$ primes (Lemma 4.2) and $\sigma(\Pi_9) = 1.498956\ldots < \frac{3}{2} \leq 1.533439\ldots = \sigma(\Pi_{10})$, the mass bound forces $\omega(N_0) \geq 10$ and $N_0 \geq \Pi_{10}$. If instead $\sigma(N_0) \geq 2$ then $N_0 \geq \Pi_{59} > \Pi_{10}$ by Theorem 4.3, so the conclusion holds a fortiori. □

The threshold is formalised. `Erdos307.live_slot_threshold` proves the analytic core, that a representation of $4N_0^2$ with $s^2 \geq N_0 + 2B$ forces $\sigma \geq \frac{3}{2}$, by dividing out $N_0^2$ and factoring $4 - (2-\sigma)(5+2\sigma) = (2\sigma-3)(\sigma+2)$ against $\sigma + 2 > 0$; and `threshold_bracket` checks the two numerals $\sigma(\Pi_9) < \frac{3}{2} \leq \sigma(\Pi_{10})$ that place the entry point at ten primes. Both carry 0 `sorry` and the three standard axioms (`lean/Erdos307/LiveSlot.lean`). The passage from the mass bound to $\omega(N_0) \geq 10$ is the step already formalised as `card_ge_59_of_recipSum_ge_two` for the barrier, and is cited rather than repeated.

The census reported above ran to $N_0 \leq 5 \times 10^4$, which is below $\Pi_{10}$ by a factor $1.3 \times 10^5$: it could not have met a live slot, and its emptiness is therefore not evidence about the family. The two representations it did find, at $N_0 = 30$ and $N_0 = 1722$, have $\sigma = 1.0333$ and $\sigma = 1.0006$, both far below $\frac{3}{2}$, which is why both are ghosts.

Three mass thresholds now organise the problem, and each is a Mertens count: $\sigma = 1$ at three primes, $\sigma = \frac{3}{2}$ at ten, and $\sigma = 2$ at fifty–nine. The first bounds the trivial regime, the third is Theorem 4.3, and the second is the entry point of the decoupled family. The entry region has now been swept: over the 105,104,561 squarefree $N_0 \leq 10^{18}$ with $\sigma(N_0) \geq \frac{3}{2}$, the form $Q_{N_0}$ represents $4N_0^2$ not once, so there is no live slot and no ghost either (`code/live_slot_census2.rs`). Two checks accompany it. The same program without the mass gate returns the two representations of Proposition 17.14 exactly, at $N_0 = 30$ with $(s,d) = (11,1)$ and $N_0 = 1722$ with $(83,1)$, both carrying $r = 1$; and the primorial bases $\Pi_{10}, \ldots, \Pi_{13}$ were re-checked by an independent implementation, each admitting no representation. The sweep uses the reduction $A \mid N_0'(N_0' - 2d^2)$, which follows from $2N_0 \equiv N_0' \pmod{A}$ and lets a single quadratic non-residue retire a base without any search; it retires 45% of them. The same program run without the mass gate returns the two representations of Proposition 17.14 exactly, at $N_0 = 30$ with $(s,d) = (11,1)$ and $N_0 = 1722$ with $(83,1)$, both with $r = 1$, which is the control that the search sees what it should.

### 17.4 Classical layers: genus, Giuga, parity

**Proposition 17.22** (A principal–genus necessary condition, and a cascade on solutions)**.** *As* $\gcd(N_0, N_0') = 1$*, the value* $4N_0^2$ *is a square coprime to the odd part of* $\operatorname{disc} Q_{N_0}$*, so every genus character is* $+1$ *on it and a primitive form represents it only from the principal genus [36]. Reading the odd assigned characters off* $Q_{N_0}(1,1) = 4N_0$ *gives the computable necessary condition*

$$(N_0 \mid \ell) = +1 \quad \textit{for every odd prime } \ell \mid (N_0'^2 - 4N_0^2) \textit{ to odd multiplicity}$$

*(a* 2*–adic condition accompanies it; for odd* $N_0$ *of even* $\omega$ *the form is imprimitive and must be primitivised first; so we state the criterion on the odd part, with a* 2*–adic caveat). Both ghost bases* $30, 1722$ *satisfy it, and about* 16% *of squarefree bases pass in the ranges tested (an* upper bound*, using only the odd characters, and mildly range–dependent).* Cascade. *In a solution of #307 with* $N = D_P D_Q$*, each* $\ell \in P \cup Q$ *gives a cofactor* $N/\ell$ *realising the Pythagorean structure of* $N$ *over base* $N/\ell$*; every cofactor with* $\sigma(N/\ell) < 2$ *(all of them for a minimal first–*59*–primes solution) must then satisfy the principal–genus criterion, a cascade of* $\geq 59$ *computable genus conditions complementing the quadratic–residue filter of Corollary 17.12.* (The further class–group and slot–admissibility stages of the fertility gap we observe only numerically at toy scale.)

Below the barrier the form has exactly one source of solutions, and it is entirely classical.

**Proposition 17.23** (The sub–barrier ghosts are pronic Giuga numbers)**.** *Every representation of* $4N_0^2$ *by* $Q_{N_0}$ *over squarefree* $N_0 \leq 3 \times 10^4$ *is a ghost* $(s, d) = (\sqrt{4N_0 + 1}, 1)$*, occurring exactly when* $N_0' = N_0 + 1$ *with* $N_0$ *pronic; in range only* $30 = 5 \cdot 6$ *and* $1722 = 41 \cdot 42$*. The two lines through such bases are classical: since* $n' = n\,\sigma(n)$ *for squarefree* $n$*,*

$\{n : n' = n-1\}$ *= primary pseudoperfect numbers* $(\sum_{p|n} \frac{1}{p} + \frac{1}{n} = 1)$*,* $\{n : n' = n+1\}$ *= Giuga numbers (quotien*

*The derivative characterisations are due to Lava (OEIS [37, 38], 2009; the Giuga form* $n' = n + 1$ *being Lava's* conjecture*, with the generalisation* $n' = an + 1$*,* $a \in \mathbb{N}$*, the theorem of Grau–Oller-Marcén [33]); the underlying number classes are Butske–Jaje–Mayernik [32] (exactly one primary pseudoperfect number per* $\omega \leq 8$*, eight known) and Giuga [29, 30], and the oblong chain (*$N$ *with* $N + 1$ *prime breeds the next term* $N(N + 1)$*) is classical in both directions (Forgues and Sondow, OEIS [37]). To* $2 \times 10^6$ *the lines are* $\{n' = n - 1\} = \{2, 6, 42, 1806, 47058\}$ *and* $\{n' = n + 1\} = \{30, 858, 1722, 66198\}$*, and the chain correctly predicts the next terms:* 47059 *is prime, so* $47058 \cdot 47059 = 2{,}214{,}502{,}422$ *is the sixth primary pseudoperfect number, and* 47057 *is prime, so the pronic* $47057 \cdot 47058 = 2{,}214{,}408{,}306$ *is the fifth Giuga number; both already known, recovered here by structure. Our contribution is the reframing: the chain step is the one–line Leibniz identity* $N' = N - 1 \Rightarrow (N(N+1))' = N'(N+1) + N^2 = N(N+1) - 1$*, exhibiting the chain as orbits of the Artin–Schreier map* $x \mapsto x(x+1) = x^2 + x$*; the same map, with the same halt (*$2 \to 6 \to 42 \to 1806$*, stopping at* $1807 = 13 \cdot 139$*), as the* $\mathbb{F}_2[t]$ *tower of Proposition 15.7: characteristic* 2 *and* $\mathbb{Z}$ *run identical breeders on the two lines.*

**Remark 17.24** (Converse, and the bridge)**.** Every pronic solution of $n' = n + 1$ is $m(m + 1)$ with $m$ prime and $m + 1$ primary pseudoperfect: Leibniz gives $m'(m + 1) + m(m + 1)' = m^2 + m + 1$, and writing $m' = km + 1$ forces $(m + 1)' = (1 - k)m - k$, which is $\geq 0$ only for $k = 0$, whence $m' = 1$ ($m$ prime) and $(m + 1)' = m$. (The congruence $m' \equiv 1 \pmod m$ alone does *not* force $m$ prime; its composite solutions are precisely the Giuga numbers $30, 858, 1722, 66198, \ldots$, so primality comes from the sign elimination $k = 0$.) Thus the only sub–barrier life in the form equation is the classical Giuga/primary–pseudoperfect world, known to sit in a web with Sylvester's sequence,

Znám's problem and the Erdős–Moser equation [34] and with the arithmetic derivative [33, 35]; we claim none of this web, only that #307 meets it through $n' = n\,\sigma(n)$, and that its $a \geq 2$ (equivalently $\sigma > 2$) branch lands on the same $\geq 59$–prime barrier [31] that bars genuine cycles (Corollary 14.3).

Writing both members of a cycle as members of these lines stratifies #307 by the gap.

**Proposition 17.25** (Gap stratification: #307 as adjacency of $\mu$–Sondow lines)**.** *For a squarefree two–cycle $a' = b$, $b' = a$ with $a > b$ and gap $d = a - b$, the smaller member lies on the line $n' = n + d$ and the larger on $n' = n - d$ (as $b' - b = d$ and $a' - a = -d$), with* $\gcd(d, ab) = 1$ *(a prime dividing $d$ and $a$ would divide $b = a - d$); conversely $x' = x + d$ with $(x + d)' = x$ is a two–cycle. These opposite–sign lines are the quotient–one $\mu$–Sondow numbers of parameter $\mp d$ [35]. Hence #307 holds iff for some $d \geq 1$ the lines $n' = n + d$ and $n' = n - d$ carry members exactly $d$ apart; for $d = 1$ this is* a Giuga number adjacent to a primary pseudoperfect number*. By the parity dichotomy (Theorem 4.14) the mixed–parity case forces $d$ odd and the both–odd case ($d$ even) carries the $10^{2519}$ bound, so every cycle below that bound has odd gap and $d = 1$ is the lead stratum. A $d = 1$ pair $(k, k + 1)$ has exactly one odd member, hence requires either an odd Giuga number (known to have at least* 9 *prime factors [30, 31]) or an odd primary pseudoperfect number (none is known): the lead stratum inherits two classical open problems. Tabulating cycle–eligible members (squarefree, coprime to $d$) to $2 \times 10^6$ for all $d \leq 50$ gives* 21 *on the plus lines (none of even gap, consistent with the parity remark) and* 81 *on the minus lines, with* no *adjacent pair at any $d \leq 50$; the barrier in gap–stratified form. (For coprime $M$ on $n' = n - \alpha$ and $N$ on $n' = n + \beta$, Leibniz gives $(MN)' - 2MN = \beta M - \alpha N$, so $\beta M - \alpha N = \square$ would manufacture a minus–side square in the sense of Proposition 15.12; the case $(\alpha, \beta) = (1, 1)$, $M - N = 1$ is the $d = 1$ cycle itself, and the known finite Giuga/primary–pseudoperfect lists contain no coprime pair, all members being even.) To our knowledge this gap–stratified equivalence and the pairing of opposite–sign lines at fixed additive distance are new; the lines themselves are [35].*

The line parameter $\delta(n) = n' - n$ obeys an affine recursion that organises the classical members into a genealogy.

**Proposition 17.26** (The $\delta$–recursion and the genealogy of the classical lines)**.** *With $\delta(n) = n' - n$, for squarefree $M$ and a prime $p \nmid M$,*

$$\delta(Mp) = p\,\delta(M) + M \qquad (\textit{Leibniz; checked on } 10^3 \textit{ random pairs}).$$

*Solving $p\,\delta(M) + M = \pm d$ gives the child rule $p = (M \mp d)/e$ from a parent $M$ on the minus line $\delta(M) = -e$. Every known Giuga number arises this way from a small–parameter parent:* $30 = 6 \cdot 5$ *($e = 1$),* $858 = 66 \cdot 13$ *($e = 5$),* $1722 = 42 \cdot 41$ *($e = 1$),* $66198 = 1122 \cdot 59$ *($e = 19$),* $2{,}214{,}408{,}306 = 47058 \cdot 47057$ *($e = 1$); the seed* 47058 *itself resolves as* $6 \to 66 \to 1518 \to 47058$ *(with $\delta = -1, -5, -49, -1$), so the previously unexplained members* 858 *and* 66198 *acquire genealogies. Two members of a cycle can never share a parent: siblings $M(M \mp d)/e$ differ by $2Md/e > d$, since $e = M - M' < 2M$. The oblong case $e = 1$ is the classical chain of Proposition 17.23 (its $\mu$–Sondow form is [35]); the recursion, the general–$e$ genealogy, and the resulting small–$|\delta|$ tree do not appear there.*

**Corollary 17.27** (Nonemptiness families for the $\mu$–Sondow conjecture)**.** *In the notation of [35] (their $S_\mu$; the quotient–one slice is the line $n' = n - \mu$), the recursion produces explicit members exceeding $|\mu|$ in three infinite parameter families: $2p \in S_{p-2}$ for every odd prime $p$; $Mp \in S_{p-M}$ for every primary pseudoperfect number $M$ and prime $p \nmid M$, and $Gp \in S_{-(p+G)}$ for every Giuga*

*number $G$ and prime $p \nmid G$; in each case $\mu/n + \sum_{q|n} 1/q = 1$ exactly (checked symbolically and on* 151 *instances). This settles their nonemptiness conjecture ($S_\mu \cap (|\mu|, \infty) \neq \varnothing$, their Conjecture 1(ii)) for infinitely many parameters, including infinitely many outside their verified range $-145 < \mu < 673$; these families do not appear among their constructions, which scale by* $\mathrm{rad}(\mu)$ *rather than extend by a new prime. Their two open instances resist precisely because every quotient–one parent route hits a composite: for $\mu = -145$, $M - 145$ is composite for all eight known primary pseudoperfect numbers (*$1661 = 11 \cdot 151$, $46913 = 43 \cdot 1091$*, and the three larger cases, including the recently found eighth member) while $p + G = 145$ has no prime solution; for $\mu = 673$, $M + 673$ is composite for all eight (*$679 = 7 \cdot 97$, $2479 = 37 \cdot 67$, $47731 = 59 \cdot 809$*, the rest ending in* 5*) and $p = 675$ fails. A prime in either family at the next generation would settle a published open instance; an exhaustive search found none. Concretely: no quotient–one member of $S_{-145}$ or $S_{673}$ exists with* any *prime–factor cofactor $\leq 10^{11}$ (every squarefree parent $M \leq 10^{11}$ was scanned against the child rules, a region whose members reach $\sim 10^{22}$, together with a depth–two descent), the quotient–zero slice of $S_{-145}$ is empty outright ($n' = 145$ forces $n \leq 5256$ by $n' \geq 2\sqrt{n}$, and the finite check is empty), and quotients $\geq 2$ force $\sigma \gtrsim 2$, hence at least* 59 *prime factors. Both instances therefore remain open, with the quotient–one frontier moved from the $10^{10}$ of [35] to cofactor scale $10^{11}$.*

**Remark 17.28** (The look–back)**.** Along small–$|\delta|$ paths the forced prime is $p \approx M/e$, so members grow doubly exponentially and the $10^{56}$–$10^{113}$ wall zone lies only a few generations from the seeds: a sparse, recursively enumerable tree (a child needs $(M - j)/e$ prime for small $j$). The adjacency of two such trees at a fixed additive distance places the final condition in the Pillai/$S$–unit genre (two multiplicatively generated sequences meeting at a fixed additive distance) where fixed–support finiteness is classical but growing–support statements are conjectural; a lens, not a transfer. Re–expressed in $\delta$–coordinates the obstruction is once more $\sigma > 1$ (positivity of $\delta$) plus integer reachability $\delta = \pm d$ under forced primes: the same additive–multiplicative exactness reappears in every coordinate tried (mass, the deviation $k$, the form $Q_{N_0}$, genus theory, $\mu$–Sondow lines, $\delta$–dynamics). The reframings buy structure, not a reduction.

**Proposition 17.29** (Even–gap plus lines are empty below $3.23 \times 10^9$)**.** *An even squarefree $x = 2u$ has $\delta(x) = 2u' - u$ odd, so every line member of even $\delta$ is odd (with $\omega$ odd). An odd member of a plus line has $\sigma > 1$ over odd primes, and the least odd squarefree integer of mass exceeding* 1 *is $3 \cdot 5 \cdots 29 =$* 3,234,846,615 *(the* 8*–prime product $3 \cdots 23 =$* 111,546,435 *has mass* 0.99896*). Hence every plus line of even gap $d$ is empty below* 3,234,846,615*, proving the census observation (no even–gap plus member to $2 \times 10^7$) and placing an explicit floor under the smaller member of any both–odd configuration.*

The lines also admit parity, infinitude, and congruence refinements.

**Proposition 17.30** (Parity floor for odd line members)**.** *For odd squarefree $n$ every cofactor $n/p$ is odd, so $n' \equiv \omega(n) \pmod 2$. Hence an odd primary pseudoperfect number ($n' = n - 1$, even) and an odd Giuga number ($n' = n + 1$, even) both have $\omega(n)$ even; combined with the classical bound $\omega \geq 9$ [30, 31] this forces $\omega \geq 10$, a parity sharpening (for the primary pseudoperfect side, the eight smallest odd primes give $\sum 1/p = 0.99896 < 1$, so $\omega \geq 9$ directly). The floor cascades when small primes are avoided: for an odd squarefree member of any plus line, or of a quotient–one line (where $\sigma(n) \geq 1 - 1/n$), exact greedy/exchange computation gives $\omega \geq 27$ if $3 \nmid n$; $\omega \geq 66$ if $3, 5 \nmid n$; $\omega \geq 144$ if $3, 5, 7 \nmid n$; and $\omega \geq 16$ if $3 \mid n$, $5 \nmid n$; each avoided small prime multiplies the required complexity, and for odd Giuga or primary pseudoperfect numbers the parity constraint rounds odd floors up (e.g. $3 \nmid n$ forces $\omega \geq 28$). (The restriction to plus or quotient–one lines is essential: on a general minus line $\sigma$ can be small (every prime $p$ trivially satisfies $p' = p - (p - 1)$) so no such floor exists there.)*

**Proposition 17.31** (Infinitely many gaps carry a nonempty plus line)**.** *By the recursion of Proposition 17.26,* $\delta(30p) = p\,\delta(30) + 30 = p + 30$ *for every prime* $p > 5$ *(as* $\delta(30) = 1$*), so the line* $n' = n + (p + 30)$ *is nonempty; by Euclid infinitely many distinct gaps* $d$ *carry a nonempty plus line, and the same runs from any Giuga number* $G$ *(*$\delta(Gp) = p + G$*). This fixes the census tail:* $330, 390, 510, 570 = 30{\cdot}\{11, 13, 17, 19\}$ *lie on the gaps* $d = 41, 43, 47, 49$*. (Per–gap infinitude remains open, of Bunyakovsky type.)*

**Proposition 17.32** (A difference–36 congruence modulo 72)**.** *Let* $n = 6m$ *be a Giuga or primary pseudoperfect number with* $\gcd(m, 6) = 1$*,* $n' = n + \varepsilon$ *(*$\varepsilon = +1$ *Giuga,* $-1$ *primary pseudoperfect). Then* $6m' = m + \varepsilon$*, and with* $m' \equiv \omega(m) \pmod 2$ *this yields* $n \equiv 36\,\omega(n) - 78 \pmod{72}$ *on the Giuga side and* $n \equiv 36\,\omega(n) - 66 \pmod{72}$ *on the primary pseudoperfect side; in particular equal* $\omega$ *forces equal residue. The congruence is verified on all twelve Giuga numbers and all seven* 6*–divisible primary pseudoperfect numbers listed in the OEIS (we recomputed the factorisations of the five largest Giuga members; all are squarefree, with* $\omega = 7, 7, 8, 8, 8$*), and gives, at the modulus* 72*, the difference–*36 *arithmetic progression observed by Sondow–MacMillan [34]; their finer modulus–*288 *pattern remains empirical.*

The modulus–72 argument is the $D = 6$ case of a general transport of the line equation across a fixed divisor.

**Proposition 17.33** (Divisor transport)**.** *Let* $n = Dm$ *with* $D, m$ *coprime squarefree and* $n' = n + \varepsilon$ *(*$\varepsilon = \pm 1$*) a quotient–one member. Leibniz transports the line equation to*

$$D\,m' = (D - D')\,m + \varepsilon.$$

(a) *If* $\sigma(D) > 1$ *(so* $D' > D$*; least case* $D = 30$*), the right side is negative for* $m > 1$ *while the left is nonnegative; the only escape is* $m = 1$*,* $\varepsilon = +1$*. Hence* no primary pseudoperfect number is divisible by 30, and the only Giuga number divisible by 30 is 30 itself *(elementary; we have not located it in the literature), consistent with the full known corpus, of which only* 30 *carries the factor* 5*.*

(b) *At a primary pseudoperfect block* $D$ *(where* $D - D' = 1$*) the slice* $m' = 1$ *reads* $m = D - \varepsilon$ *prime: the two classical chain rules (*$D(D + 1)$ *primary pseudoperfect when* $D + 1$ *is prime,* $D(D - 1)$ *Giuga when* $D - 1$ *is prime [35, 34]) are the two signs of one transported equation (*$1806 = 42 \cdot 43$ *and* $1722 = 42 \cdot 41$ *sit on the same slice* $D = 42$*).*

(c) *For any gap* $d$ *the same transport stratifies the line* $n' = n + d$ *by divisor blocks* $D$ *into the steeper lines* $D\,m' = (D - D')m + d$*, a further enumeration coordinate for the genealogy of Proposition 17.26.*

Finally, the form obstruction can be averaged: it eliminates almost every integer, though not the thin stratum where a solution could live.

**Proposition 17.34** (Average rarity of form–barrier survivors)**.** *Call a squarefree* $N$ blocked *at an odd prime* $\ell$ *if* $\ell \nmid N$*,* $N' \equiv \pm 2N \pmod \ell$*, and* $(N \mid \ell) = -1$*, and a* survivor *if it is blocked at no odd prime. A blocked* $N$ *admits no representation* $E\,s^2 + D\,d^2 = 4N^2$ *(reducing mod* $\ell$ *forces* $N$ *to be a quadratic residue, against* $(N \mid \ell) = -1$*; this is the genus fragment of Proposition 17.22), so every union of a #307 solution is a survivor. Now* $N' \equiv N\,S_\ell(N) \pmod \ell$ *with* $S_\ell(N) = \sum_{p \mid N} p^{-1}$ *an* additive *function, so blocking at* $\ell$ *is an additive–function event (*$S_\ell \equiv \pm 2$*) twisted by the quadratic character, of limiting density* $1/(\ell + 1)$ [HEURISTIC*: this constant is asserted, not derived. Equidistribution of* $S_\ell$ *over* $\mathbb{Z}/\ell$ *together with density* $\frac{1}{2}$ *for* $(N \mid \ell) = -1$ *would give* $2/\ell \cdot \frac{1}{2} = 1/\ell$

*instead. The two agree to leading order and $\sum_\ell 1/\ell$ diverges as well, so the shape of the conclusion is unaffected; the exact constant is not established here]. As $\sum_\ell 1/(\ell+1)$ diverges (Mertens), the Selberg–Delange/Halász theory of additive functions in residue classes [19] gives that the survivors have natural density zero among the squarefree integers (unconditional), with $\#\{\text{survivors} \le X\} = O\big(X/\log\log\log X\big)$ under a standard, but here unproved, uniformity estimate. Thus the single–prime form obstruction removes* almost every *squarefree integer. We stress the scope: survivors remain* infinite*, and a #307 solution; one integer beyond $2.09 \times 10^{56}$, in the density–zero stratum $\sigma(N) \ge 2$; is untouched by such an average statement, which speaks only to the typical integer, not to existence.*

# A The minus–layer density theorem, and the square sieve that does not reach its rate

Relegated from §17.1. This appendix is the historical route to Theorem A.9: a square sieve at $P \asymp (\log N)^{1/4}$ whose analytic input was never supplied, since the uniformity it needs is exactly what Remark A.3 forecloses at that range. The theorem is now proved instead by Proposition A.7, at the much smaller $P = \log\log N/(\log\log\log N)^{1+\varepsilon}$, where the same uniformity *is* available (Lemma A.4) (at the cost of a far weaker rate. The sieve is kept in full because its algebraic layer is correct and reusable, because the located gap is what identified the workable range, and because Theorem A.12) now proved, via Lemma A.5, is what this route gives once its input is supplied.

**Lemma A.1** (Cancellation in the character sums)**.** *Fix an odd squarefree $r > 1$. Then*

$$T_r(N) \;:=\; \sum_{\substack{m \le N,\ \mu^2(m)=1 \\ \gcd(m,r)=1}} \Big(\frac{m' - 2m}{r}\Big) \;=\; o_r(N) \qquad (N \to \infty).$$

*The statement is qualitative and per fixed $r$: the implied constant depends on $r$, and no uniformity is claimed. Remark A.3 records why the uniform version is not available; Remark A.2 records what the constant is and is not.*

*Proof.* Five standard facts are used, all at fixed $r$: Halász's theorem; $|\tau(\chi)| = \sqrt{r}$ for a primitive character; the classical zero–free region; $|\log L(s,\psi)| \ll_r \log\log(3+|\tau|)$ for $\psi$ non–principal at $s = 1 + 1/\log N + i\tau$; and Siegel–Walfisz. None is formalised.

*The expansion.* Write $\chi = (\cdot \mid r)$. For $r$ odd squarefree the Jacobi symbol is a primitive real character modulo $r$, so $\chi(x) = \tau(\chi)^{-1} \sum_{t \bmod r} \chi(t) e_r(tx)$ holds for *every* $x$, including $\gcd(x,r) > 1$, where both sides vanish. That case occurs: at $r = 15$ already $m = 2$ gives $\sigma_r(m) - 2 = 6$. By Lemma 17.1, $(m' - 2m \mid r) = \chi(m)\,\chi(\sigma_r(m) - 2)$, and expanding the second factor,

$$T_r(N) = \tau(\chi)^{-1} \sum_{t \bmod r} \chi(t)\, e_r(-2t)\, S_t(N), \qquad S_t(N) = \sum_{\substack{m \le N,\ \mu^2(m)=1 \\ \gcd(m,r)=1}} h_t(m),$$

where $h_t(m) = \chi(m) \prod_{p \mid m} e_r(tp^{-1})$, using that $\sigma_r$ is a sum over $p \mid m$ and $e_r$ carries sums to products. Extend $h_t$ to $\mathbb{N}$ by $h_t(p^k) = 0$ for $k \ge 2$ and $h_t(p) = 0$ for $p \mid r$, the only extension preserving $S_t$. It is multiplicative with $|h_t| \le 1$, which is Halász's hypothesis; $|h_t| = 1$ on the support only.

*The distance.* Halász needs $M = \min_{|\tau| \le \log N} \sum_{p \le N} \big(1 - \operatorname{Re} h_t(p) p^{-i\tau}\big)/p \to \infty$. Now $h_t$ is not a function of $m \bmod r$, being built from the factorisation: at $r = 7$, $h_1(3) = 0.2225 + 0.9749i$

while $h_1(10) = -1$, though $3 \equiv 10$. Its restriction to primes is, which is all that is needed. Put $g_t(a) = \chi(a)e_r(ta^{-1})$ on $(\mathbb{Z}/r)^\times$, so $h_t(p) = g_t(p \bmod r)$ for $p \nmid r$, and expand $g_t = \sum_\psi c_\psi \psi$. Reindexing by the involution $a \mapsto ta^{-1}$ of $(\mathbb{Z}/r)^\times$ and using $\chi(a^{-1}) = \chi(a)$,

$$c_{\psi_0} = \frac{1}{\varphi(r)} \sum_{a \in (\mathbb{Z}/r)^\times} \chi(a) e_r(ta^{-1}) = \frac{\chi(t)\,\tau(\chi)}{\varphi(r)}, \qquad |c_{\psi_0}| = \frac{\sqrt{r}}{\varphi(r)}.$$

This identity uses no primitivity: it holds for any quadratic character on any finite abelian group, which is what the formalisation records. Primitivity enters twice elsewhere, for $|\tau(\chi)| = \sqrt{r}$ and for the inversion at $\gcd(x, r) > 1$ above.

Splitting the prime sum by $\psi$,

$$\sum_{p \le N} h_t(p) p^{-1-i\tau} = c_{\psi_0} \sum_{p \le N,\, p \nmid r} p^{-1-i\tau} + \sum_{\psi \ne \psi_0} c_\psi \sum_{p \le N} \psi(p) p^{-1-i\tau}.$$

The first sum is at most $\sum_{p \le N} 1/p = \log\log N + O(1)$ in modulus, trivially and uniformly in $\tau$. For the coefficients, $|g_t| = 1$ gives $\sum_\psi |c_\psi|^2 = 1$ by Parseval, hence $\sum_{\psi \ne \psi_0} |c_\psi| \le \sqrt{\varphi(r)}$ by Cauchy–Schwarz; they are *not* small individually. Indeed $c_\psi = \varphi(r)^{-1} \sum_b (\chi\psi)(b) e_r(tb)$, so $|c_\psi| = \sqrt{\mathrm{cond}(\chi\psi)}/\varphi(r)$, and the maximum $\sqrt{r}/\varphi(r)$ is attained exactly on the $\prod_{p \mid r}(p-2)$ characters with $\chi\psi$ primitive: three of the eight at $r = 15$, five of the twelve at $r = 21$. Their contribution is small because the $L$–functions have no pole, and reaching that needs care: the cited bound governs $\log L$, an Euler product over all $p$, while what is required is the truncation. Both routes go by partial summation against Siegel–Walfisz and both shed the same factor $1 + |\tau|$; what differs is the *range*, and it is the range and not any shift that does the work. On $[2, N]$ one is left with $\int_{\log 2}^{\log N} u^{-A} du = O_A(1)$, so $(1 + |\tau|) \cdot O(1)$ is $\ll \log N$, worse than trivial. On $[N, \infty)$, writing $A(v) = \sum_{p \le v} \psi(p) \ll_{A,r} v(\log v)^{-A}$,

$$\sum_{p > N} \psi(p) p^{-s} \;=\; -A(N)N^{-s} + s\int_N^\infty A(v) v^{-s-1}\, dv \;\ll\; (\log N)^{-A} + (1+|\tau|)\frac{(\log N)^{1-A}}{A-1} \;\ll\; (\log N)^{2-A},$$

which is $o(1)$ for $A > 2$ and $|\tau| \le \log N$. Note what is *not* used: the damping $p^{-1/\log N}$ attached to the shifted point $s = 1 + 1/\log N + i\tau$ is bounded by 1 and is simply discarded above. It contributes nothing to the tail, and indeed the shift alone leaves the tail of size $\int_1^\infty e^{-w} w^{-1} dw = E_1(1) = 0.2194$. The shift costs $O(1)$ by Mertens, since $\sum_{p \le N} \log p/(p \log N) \ll 1$, and it is convenient for keeping the Euler logarithm absolutely convergent, but it is not what makes the tail small. The zero–free region then gives $|\log L(s, \psi)| \ll_r \log\log(3 + |\tau|)$, the argument of the double logarithm being taken above $e$ so that the bound is positive at $\tau = 0$; the non–principal terms are $O_r(\log\log(3 + |\tau|)) = O_r(\log\log\log N)$ uniformly for $|\tau| \le \log N$, since at the top of that range the cited input gives $\log\log(3 + \log N) \asymp \log\log\log N$ and not $O(1)$. Hence

$$M \;\ge\; c_r \log\log N - O_r(\log\log\log N), \qquad c_r := 1 - \frac{\sqrt{r}}{\varphi(r)} \;\ge\; 1 - \frac{\sqrt{3}}{2} \;=\; 0.1339\ldots,$$

the extremum being at $r = 3$: each local factor of $\sqrt{r}/\varphi(r)$ is $\sqrt{p}/(p-1)$, and $(p-1)/\sqrt{p} \ge 2/\sqrt{3}$ for every odd prime, so the supremum is the single factor at $r = 3$ (`code/charcancel_check.gp`).

*Conclusion.* With $T = \log N$, Halász gives $S_t(N) = \sum_{m \le N} h_t(m) \ll N\big((1 + M)e^{-M} + (\log N)^{-1/2}\big)$. Note $M \to \infty$ alone would not suffice: at fixed $T$ the bound is only $\ll NT^{-1/2}$. Both limits are in play, and $S_t(N) \ll_r N(\log N)^{-\min(c_r, 1/2) + o(1)} = o(N)$, the $\frac{1}{2}$ being the floor the

$(\log N)^{-1/2}$ term imposes however large $M$ becomes. Only $r = 3$ and $r = 5$ have $c_r \le \frac{1}{2}$, so for every other modulus, and in particular for every $r = pq$ arising in Theorem A.9, the floor is the binding constraint. Only the $\varphi(r)$ values of $t$ coprime to $r$ contribute, since $\chi(t) = 0$ otherwise, so

$$|T_r(N)| \;\le\; r^{-1/2} \sum_{\gcd(t,r)=1} |S_t(N)| \;\le\; \frac{\varphi(r)}{\sqrt{r}} \max_t |S_t(N)| \;\le\; \sqrt{r} \max_t |S_t(N)| \;=\; o_r(N).$$

The net factor is a *loss* of $\sqrt{r}$: the $\varphi(r)$ summands outweigh $|\tau(\chi)|^{-1}$. At fixed $r$ that is a constant. □

**Remark A.2** (What the constant $c_r$ is and is not)**.** Three things about $c_r = 1 - \sqrt{r}/\varphi(r)$ are worth recording, since each is easily overstated.

*Sharpness is conditional.* The bound $M \ge c_r \log\log N - O_r(1)$ is attained only when the main term $\chi(t)\tau(\chi)$ is real and positive. Since $\overline{\tau(\chi)} = \chi(-1)\tau(\chi)$ with $\chi(-1) = (-1)^{(r-1)/2}$, the Gauss sum is purely imaginary for $r \equiv 3 \pmod 4$, and there the true $M$ has constant 1 rather than $c_r$. For $\chi(t) = -1$ the main term has the wrong sign, and no admissible $\tau$ rotates it back, since $|\sum_{p \le X} p^{-1-i\tau}|$ collapses to $O(\log\log(3 + |\tau|))$ away from $\tau = 0$; there the constant is $1 + \sqrt{r}/\varphi(r)$. Measured at $r = 5$: the growth rate $\Delta M / \Delta \sum_p 1/p$ tends to $0.440887 \to 1 - \sqrt{5}/4 = 0.440983$ for $t \equiv \pm 1$, and to $1.558921 \to 1 + \sqrt{5}/4 = 1.559017$ for the other two residues, whose $M(10^7)$ is 4.378 against the leading term 1.341. So the constant is attained for $r \equiv 1 \pmod 4$ and $\chi(t) = +1$, and is not sharp otherwise.

*The finite-X deficit is a constant, though the proved error term is not.* The bound carries $-O_r(\log\log\log N)$, which is what the $L$–bound gives at the top of the $\tau$–range; what is *measured* at a single modulus is smaller and flat. At $r = 5$, $t = 1$ the deficit $c_r \sum_{p \le X} 1/p - M$ is $0.1399, 0.1401, 0.1402, 0.1402$ at $X = 10^4, \dots, 10^7$, flat to three places and still rising in the fourth, while $\log\log\log X$ rises by 28% over the same range. Both sums here run over *all* $p \le X$: $h_t$ is extended by $h_t(p) = 0$ at $p \mid r$, so those primes contribute $1/p$ to $M$, and omitting them shifts every number in this remark (`code/deficit_check.gp`). The $\log\log\log$ enters only through uniformity in $\tau$.

*The rate that follows, and the one that does not.* Halász gives $T_r(N) \ll_r N(\log N)^{-c_r + o(1)}$. Since $c_3 = 0.134$ and $c_5 = 0.441$ are below $\frac{1}{2}$, this does *not* yield $T_r(N) \ll \sqrt{r}\, N(\log N)^{-1/2}$ at those moduli. Quantitatively the lemma is asymptotic in the strong sense: at $r = 5$, with the measured constants and an implied Halász constant of 1, the bound first beats the trivial $|T_r(N)| \le N$ at $\log N = 143.0$, that is $N = 10^{62.1}$. Measured against the honest trivial bound, which is $0.5066N$ because $T_r$ has only that many terms, the crossing moves to $\log N = 1013.7$, that is $N = 10^{440}$.

**Remark A.3** (No uniform bound on the character sum near the threshold)**.** For $|\tau| \ge 1/\log N$ the bound $\left|\sum_{p \le N} p^{-1-i\tau}\right| \ll \log\log(3 + |\tau|)$ fails in the range that matters. Near the threshold the sum has size about $\log(1/|\tau|)$, which is of order $\log\log N$, not $\log\log(3 + |\tau|)$, which is of order 1. Measured at $N = 2 \times 10^7$ with $\tau = 2/\log N = 0.11897$: the left side is 2.6797 against a claimed bound of 0.12883, a violation by a factor 20.8; at $\tau = 1/\log N = 0.05948$ the left side is 2.9773 against 0.11176, a factor 26.6 (`code/charcancel_check.gp`). A second defect is that the inequality bounds the untwisted sum $\sum p^{-1-i\tau}$, whereas the Halász distance $M$ requires the $h_t$–twisted sum.

Consequences, stated exactly. Neither defect touches the lemma as now proved, because that proof does not use the offending step: it bounds the *twisted* sum through the character expansion of $h_t$, where the principal coefficient contributes at most $\sqrt{r}/\varphi(r) \le 0.8661$ of $\sum_{p \le N} 1/p$ and the non–principal ones contribute $O_r(\log\log\log N)$, both uniformly in $\tau$. The trivial bound $\left|\sum_{p \le N} p^{-1-i\tau}\right| \le \sum_{p \le N} 1/p$ is all that is needed of the untwisted sum, so the false inequality is not merely repaired but unnecessary.

Two claims must be distinguished. What is shown false above is the *r–uniform form of one input*, the $\log L$ bound, whose lower half needs $L(1,\psi)$ away from 0 and has only $L(1,\psi) \gg r^{-1/2}$ effectively or $\gg_\varepsilon r^{-\varepsilon}$ by Siegel, ineffectively. That is *not* the same as the uniform form of the lemma being false, and the distinction turned out to be the whole opening: the obstruction is *range–dependent*. At $r \le (\log\log N)^2$ one has $\log r \le 2\log\log\log N$, so even a Siegel–worst–case $|\log L(1,\psi)|$ costs $\eta\log r + O_\eta(1)$, which is negligible against the main term $\log\log N$; that is Lemma A.4, and with Lemma A.6 it proves Theorem A.9 with a rate. One caveat on how much this explains: at the far larger moduli of Theorem A.12 the $L$–function bound does also fail, but it is not what binds there. Even an $O(1)$ per–character bound would leave the requirement $\sqrt{\varphi(r)} = o(\log\log N)$, from the *coefficient mass* rather than from any $L$–function, a limitation of the character expansion, not of the arithmetic; see the blocker recorded at that corollary and the question posed in Remark A.11. What this forecloses is uniformity at the far larger $r$ that Theorem A.12 needs, where $\log r$ is comparable to the main term, and that foreclosure binds only the route that expands in characters. The rate is proved in Theorem A.12 by a route that does not expand (Lemma A.5).

**Lemma A.4** (Cancellation, uniform at polyloglog moduli)**.** (Label discipline, as for Theorem A.9: this statement went through six rounds of adversarial review, each by a session that did not write the text under review. Rounds one to five each found defects, twice in a repair made by the round before, which were repaired; round six, auditing statement and proof line by line and recomputing every printed constant, found nothing, which is what the label records. Proposition A.7 passed clean at round three.) *Fix $\varepsilon > 0$ and write $L = \log\log N$. For every odd squarefree $r$ with $7 \le r \le L^2/(\log L)^{2+\varepsilon}$,*

$$T_r(N) \;\ll\; N\,(\log N)^{-2/5},$$

*with an implied constant that depends only on $\varepsilon$ and is ineffective; the same bound, by the same proof with $e_r(+2t)$ in place of $e_r(-2t)$, holds for the sum with $(m'+2m \mid r)$ in place of $(m'-2m \mid r)$. The moduli $r = 3, 5$ are excluded: there $c_r < 1/2$, and they are covered instead at fixed $r$ by Lemma A.1 with the exponents of Remark A.2.*

*Proof.* The proof of Lemma A.1 is followed step by step; what is new is that every implied constant is tracked in $r$, and each is absolute in the stated range. Here $r \le L^2/(\log L)^{2+\varepsilon} \le L^2$, so $\log r \le 2\log L$. The range is exactly the one the argument supports: the binding requirement below is $\sqrt{r}\,\log L = o(L)$, that is $r = o\big((L/\log L)^2\big)$.

*Expansion.* Unchanged: $|T_r(N)| \le r^{-1/2}\sum_t |S_t(N)| \le r^{1/2}\max_t |S_t(N)|$, the sum over the $\varphi(r)$ values of $t$ with $\gcd(t,r)=1$, since $\chi(t)=0$ otherwise; the loss $r^{1/2} \le L$ is absorbed below.

*The distance.* As before, $\sum_{p\le N} h_t(p)p^{-1-i\tau} = c_{\psi_0}\sum_{p\le N,\,p\nmid r} p^{-1-i\tau} + \sum_{\psi\ne\psi_0} c_\psi S_\psi(\tau)$ with $S_\psi(\tau) = \sum_{p\le N}\psi(p)p^{-1-i\tau}$ and $|c_{\psi_0}| = \sqrt{r}/\varphi(r)$. The principal part is at most $(\sqrt{r}/\varphi(r))\log\log N$ in modulus, trivially and uniformly in $\tau$; dropping the primes $p \mid r$ from the untwisted sum costs $\sum_{p\mid r} 1/p \le \omega(r)/3 + O(1) = O(\log L)$, absorbed into the error below.

*The non–principal sums, uniformly.* Fix $\psi \ne \psi_0 \bmod r$ and $|\tau| \le \log N$, and put $s = 1 + 1/\log N + i\tau$. For composite $r$ the character $\psi$ need not be primitive; let $\psi^*$ of conductor $d \mid r$, $d > 1$, induce it. Then $S_\psi$ and $S_{\psi^*}$ differ only at the primes $p \mid r$, by at most $\sum_{p\mid r} 1/p = O(\log L)$, absorbed into the bound below, and Siegel–Walfisz, the zero–free region and Siegel's theorem are applied to $\psi^*$. Partial summation against Siegel–Walfisz, exactly as in Lemma A.1 with $A = 3$, replaces the truncated sum by $\log L(s,\psi) + O(1)$ at a cost $O((\log N)^{-1})$; Siegel–Walfisz is uniform, with an absolute ineffective constant, for conductors up to $(\log N)^3$, and $r \le L^2 \le (\log N)^3$ holds for all $N \ge 16$. For $\log L(s,\psi)$ two cases arise. If $L(\cdot,\psi)$ has no exceptional zero, the classical zero–free region gives $|\log L(s,\psi)| \ll \log\log\big(r(3+|\tau|)\big) \ll \log L$ with an absolute constant. If

$\psi$ is real with an exceptional zero $\beta$, split the $\tau$–range. For $|\tau| \le 1$ the local decomposition $\log L(s,\psi) = \log(s-\beta) + O\big(\log\log(r(3+|\tau|))\big)$ applies, and

$$|s-\beta| \;\ge\; \mathrm{Re}(s)-\beta \;>\; 1-\beta \;\ge\; c(\eta)\, r^{-\eta}$$

by Siegel's theorem in zero–location form, applied with any fixed $\eta > 0$, a parameter unrelated to the $\varepsilon$ of the range, so $|\log(s-\beta)| \le \log\big(1/(1-\beta)\big) + O(1) \le \eta \log r + O_\eta(1) \ll \eta \log L + O_\eta(1)$; the decomposition is *not* asserted beyond $|\tau| \le 1$, where $|s-\beta|$ can be as large as $\log N$ and $\log|s-\beta|$ as large as $L$. For $|\tau| > 1$ the zero $\beta$ lies at distance greater than 1 from $s$, outside the local zero–density disc, and the unconditional bound $|\log L(s,\psi)| \ll \log\log(r(3+|\tau|))$ applies with no exceptional case. In both regimes $|S_\psi(\tau)| \ll \log L$, absolutely and ineffectively, uniformly in $|\tau| \le \log N$.

*Assembly.* By Cauchy–Schwarz on the coefficients, $\sum_{\psi \ne \psi_0} |c_\psi| \le \sqrt{\varphi(r)} \le \sqrt{r} \le L/(\log L)^{1+\varepsilon/2}$, so the non–principal contribution to the distance is $\ll L(\log L)^{-\varepsilon/2} = o(L)$, and

$$M \;\ge\; \Big(1 - \frac{\sqrt{r}}{\varphi(r)}\Big)L \;-\; O\big(L(\log L)^{-\varepsilon/2}\big) \;\ge\; \Big(1 - \frac{\sqrt{r}}{\varphi(r)} - o(1)\Big)L.$$

The constant $1 - \sqrt{r}/\varphi(r)$ does *not* tend to 1 over the whole range: it is $0.133\ldots$ at $r = 3$ and $0.441\ldots$ at $r = 5$, and exceeds $1/2$ for every odd squarefree $r \ge 7$, with minimum $1 - \sqrt{15}/8 = 0.515\ldots$ at $r = 15$. Indeed $\sqrt{r}/\varphi(r) = \prod_{p \mid r} \sqrt{p}/(p-1)$, every factor is $< 1$, and $p \mapsto \sqrt{p}/(p-1)$ is decreasing on $p \ge 3$; so adjoining a prime only shrinks the product, and over $\omega(r) \ge 2$ the supremum is attained at $r = 15$, giving $\sqrt{r}/\varphi(r) \le \sqrt{15}/8 = 0.4841\ldots$, while for $\omega(r) = 1$ with $p \ge 7$ it is at most $\sqrt{7}/6 = 0.4409\ldots$. Only $r = 3, 5$ exceed $1/2$ (values confirmed for $r \le 20001$ in `code/uniform_min_sweep.gp`). Halász (Montgomery's form; the constants are absolute and the function enters only through $|h_t| \le 1$) then gives $\max_t |S_t(N)| \ll N(1+M)e^{-M} + N(\log N)^{-1/2} \ll N(\log N)^{-1/2+o(1)}$ for $r \ge 7$, and the expansion loss $r^{1/2} \le L$ leaves $T_r(N) \ll N(\log N)^{-1/2+o(1)} \ll N(\log N)^{-2/5}$.

Two honesty notes, as in Remark A.2. The $o(1)$ terms above are vacuous at every computable $N$: the crux inequality $\sqrt{r}\log L \ll L$ first bites at the top of the range around $\log\log N \approx e^{36}$, and the widening over the earlier $r \le L^{2-\delta}$ is asymptotic only, $L^\delta$ overtaking $(\log L)^{2+\varepsilon}$ far beyond even that. And the constant is ineffective twice over, through Siegel–Walfisz and through Siegel's theorem; no effective form is claimed. □

**Lemma A.5** (The distance, without expanding: uniform to power-of-log moduli)**.** *Fix $A > 0$ and write $L = \log\log N$. Let $r > 1$, let $g : (\mathbb{Z}/r)^\times \to \mathbb{C}$ satisfy $|g| \le 1$, and let $h$ be the multiplicative function with $h(\ell) = g(\ell \bmod r)$ for primes $\ell \nmid r$, $h(\ell) = 0$ for $\ell \mid r$, and $h(\ell^k) = 0$ for $k \ge 2$. Put $\widehat{g}_0 = \varphi(r)^{-1}\sum_a g(a)$, the mean of $g$, and let*

$$M \;=\; \min_{|\tau| \le \log N} \sum_{\ell \le N} \frac{1 - \mathrm{Re}\, h(\ell)\ell^{-i\tau}}{\ell}$$

*be Halász's distance for $h$. Then*

$$M \;\ge\; \big(1 - |\widehat{g}_0|\big)\log\log N \;-\; o(\log\log N),$$

*uniformly for $r \le (\log N)^A$ and for all such $g$, the $o(\cdot)$ being ineffective and of unspecified rate.*

*Proof.* Only the values of $h$ at primes are used. Fix $\varepsilon \in (0,1)$ and put $T_0 = \exp\big((\log N)^\varepsilon\big)$; $\varepsilon$ is an auxiliary of the proof and is removed at the end. Since $\sum_{\ell \le N} 1/\ell = L + O(1)$, it suffices to bound $\big|\sum_{\ell \le N} h(\ell)\ell^{-1-i\tau}\big|$ uniformly in $|\tau| \le \log N$.

*Small primes.* Trivially $\left|\sum_{\ell\le T_0} h(\ell)\ell^{-1-i\tau}\right| \le \sum_{\ell\le T_0} 1/\ell = \log\log T_0 + O(1) = \varepsilon L + O(1)$, with no $\tau$–dependence.

*Large primes.* Every $\ell > T_0$ is coprime to $r \le (\log N)^A$, so $h(\ell) = g(\ell \bmod r)$ there. For $a \in (\mathbb{Z}/r)^\times$ write $E_a(x) = \pi(x;r,a) - \operatorname{li} x/\varphi(r)$ and $E(x) = \sum_a g(a)E_a(x)$, and set

$$S(x) = \sum_{T_0<\ell\le x} h(\ell) = \widehat{g}_0\big(\operatorname{li} x - \operatorname{li} T_0\big) + R(x), \qquad R(x) = E(x) - E(T_0),$$

using $\sum_a g(a) = \varphi(r)\widehat{g}_0$; in particular $R(T_0) = 0$ identically. Siegel–Walfisz with the exponent $B = A/\varepsilon$ gives $|E_a(x)| \ll_B x\exp(-c_B\sqrt{\log x})$ uniformly in $a$, legitimately for every $x \ge T_0$, since then $(\log x)^{A/\varepsilon} \ge (\log T_0)^{A/\varepsilon} = (\log N)^A \ge r$. Summing over the $\varphi(r) \le r$ classes with $|g| \le 1$,

$$|R(x)| \;\ll\; r\Big(x\,e^{-c\sqrt{\log x}} + T_0\,e^{-c\sqrt{\log T_0}}\Big) \;\ll\; r\,x\,e^{-c\sqrt{\log x}},$$

the second term being dominated because $y \mapsto y\,e^{-c\sqrt{\log y}}$ increases once $\log y > c^2/4$. The factor $r$ comes from that class sum, not from Siegel–Walfisz, whose error carries no modulus.

Now integrate. Writing the large–prime sum as a Stieltjes integral against $dS$,

$$\sum_{T_0<\ell\le N} h(\ell)\ell^{-1-i\tau} = \widehat{g}_0\int_{T_0}^{N} \frac{x^{-i\tau}}{x\log x}\,dx + \Big[x^{-1-i\tau}R(x)\Big]_{T_0}^{N} + (1+i\tau)\int_{T_0}^{N} R(x)\,x^{-2-i\tau}dx.$$

The first term has modulus at most $|\widehat{g}_0|\int_{T_0}^{N} dx/(x\log x) = |\widehat{g}_0|(1-\varepsilon)L$. The bracket vanishes at $T_0$ and is $\ll r\,e^{-c\sqrt{\log N}} = o(1)$ at $N$. For the last, substituting $u = \log x$ and using $\int_U^\infty e^{-c\sqrt{u}}\,du = 2c^{-2}(1+c\sqrt{U})e^{-c\sqrt{U}}$,

$$(1+|\tau|)\int_{T_0}^{N} |R(x)|x^{-2}dx \;\ll\; (1+\log N)\,r\int_{(\log N)^\varepsilon}^{\log N} e^{-c\sqrt{u}}du \;\ll\; (1+\log N)(\log N)^A(\log N)^{\varepsilon/2}e^{-c(\log N)^{\varepsilon/2}},$$

which is $o(1)$ uniformly in $|\tau| \le \log N$, since $c(\log N)^{\varepsilon/2}$ eventually exceeds $(A+1+\frac{\varepsilon}{2})\log\log N$. Collecting,

$$\Big|\sum_{\ell\le N} h(\ell)\ell^{-1-i\tau}\Big| \;\le\; \varepsilon L + |\widehat{g}_0|(1-\varepsilon)L + O(1)$$

uniformly in $|\tau| \le \log N$, whence $M \ge L - \varepsilon L - |\widehat{g}_0|(1-\varepsilon)L - O(1) = (1-\varepsilon)\big(1-|\widehat{g}_0|\big)L - O(1)$.

*Removing $\varepsilon$.* For each $j \ge 1$ take $\varepsilon_j = 1/(2j)$ and $B_j = 2Aj$, both *fixed*, so that $c_{B_j}$ and the implied constant $C_j$ of Siegel–Walfisz are constants (ineffective, but constants). The display above then holds beyond a threshold $N_j$ depending only on $(A,j)$, and

$$\Big(1-\tfrac{1}{2j}\Big)\big(1-|\widehat{g}_0|\big)L - C_j \;\ge\; \big(1-|\widehat{g}_0|\big)L - \tfrac{1}{j}L \iff \tfrac{L}{2j}\big(1+|\widehat{g}_0|\big) \ge C_j,$$

which holds once $L \ge 2jC_j$; enlarge $N_j$ so that it does, and so that $N_j$ increases. Setting $j(N) = \max\{j : N \ge N_j\} \to \infty$ gives $M \ge (1-|\widehat{g}_0|)L - L/j(N) = (1-|\widehat{g}_0|)L - o(L)$, uniformly in $r$ and $g$ as claimed. The ineffectivity is exactly Siegel's, entering through Siegel–Walfisz; no exceptional–zero case split is needed, because no $L$–function is bounded anywhere above. □

*What this changes, and what it does not.* The constant is *not* an improvement: at $g_t(a) = \chi(a)e_r(ta^{-1})$ one has $\widehat{g}_0 = \chi(t)\tau(\chi)/\varphi(r)$, so $|\widehat{g}_0| = \sqrt{r}/\varphi(r)$ and Lemma A.5 returns the same $1-\sqrt{r}/\varphi(r)$ that Lemma A.4 already gives; and in the error term it is worse, an $o(L)$ of unspecified

rate against that lemma's explicit $O\big(L(\log L)^{-\varepsilon/2}\big)$. Both are ineffective, so ineffectivity is not the difference; the rate is.

What changes is the *range*, and by an exponential. Lemma A.4 is capped at $r \le L^2/(\log L)^{2+\varepsilon}$ because expanding $g$ in multiplicative characters pays $\sum_{\psi\neq\psi_0}|c_\psi| \asymp \sqrt{r}$ against a per–character bound $O(\log L)$. The proof above never expands, so never pays it; the price is instead a factor $r$ against $\exp(-c\sqrt{\log x})$, which is free at $r \le (\log N)^A$. Remark A.11 posed exactly this substitution as its open question, and the measurement recorded there, that $\max_s|\sum_{\ell\le 10^6} e_p(s\ell^{-1})/\ell|$ stays near 1.3 while $\sqrt{p}$ grows from 3.6 to 44.8, is the numerical shadow of the same fact.

Two honesty notes. The $o(L)$ is ineffective and of unspecified rate, so nothing here is quantitative at any particular $N$; and the error estimate is vacuous at every computable $N$, the bound $(1+\log N)(\log N)^A(\log N)^{\varepsilon/2}e^{-c(\log N)^{\varepsilon/2}}$, at $A=1,\varepsilon=0.2,c=\frac12$, *peaking* at $\log N = 10^{16.23}$ with value $9.29\times10^{24}$, and falling below 1 only past $\log N = 10^{23.58}$ (`code/swdirect_check.gp`). A reading of "turning over near $10^{18}$" is contradicted by the certificate itself in both directions. The uniform Mertens estimate underlying the untwisted case is classical (Norton, *Illinois J. Math.* **20** (1976), Lemma 6.3, gives $\sum_{p\le x,\,p\equiv\ell\,(k)}1/p = \log\log x/\varphi(k)+\delta_\ell/\ell+O(\log 3k/\varphi(k))$ with an absolute but ineffective constant) but we are not aware of a reference carrying the $\tau$–uniformity used here, and deduce it from Siegel–Walfisz (Montgomery–Vaughan, *Multiplicative Number Theory I*, §11.3) by the partial summation above.

**Lemma A.6** (Two uniform deficit bounds)**.** (Label discipline, as for Lemma A.4: three rounds of adversarial review, each by a session that did not write the text. The first two found five defects (including a principal–coefficient value false in one of three cases, and a numerical certificate that tested the wrong congruence with a silently inoperative coprimality filter) all repaired; the third found nothing, which is what the label records.) *Let $c\in\{2,-2\}$, $L=\log\log N$, and $P = L/(\log L\,w(L))$ with $w(L)\to\infty$. Uniformly for primes $p\neq q$ in $(\frac12P,P]$, with implied constants that depend only on $w$ and are ineffective,*

$$\#\{m\le N:\ \mu^2(m)=1,\ \gcd(m,p)=1,\ \sigma_p(m)\equiv c\ (\mathrm{mod}\ p)\}\ \ll\ N/p,$$

$$\#\{m\le N:\ \mu^2(m)=1,\ \gcd(m,pq)=1,\ \sigma_p(m)\equiv c\ (\mathrm{mod}\ p)\ \textit{and}\ \sigma_q(m)\equiv c\ (\mathrm{mod}\ q)\}\ \ll\ N/(pq).$$

*Proof.* Here $p,q\in(P/2,P]$ and $pq\le P^2 = L^2/(\log L\,w(L))^2$, inside Lemma A.4's range. Recall $\sigma_p(m)=\sum_{\ell\mid m}\ell^{-1}$ mod $p$, so that $e_p\big(t\sigma_p(m)\big)=\prod_{\ell\mid m}e_p(t\ell^{-1})$ is multiplicative in squarefree $m$.

*The second bound; the first is the same argument with one prime.* Detect both congruences additively:

$$\#\{\cdots\}\ =\ \frac{1}{pq}\sum_{s\bmod p}\sum_{t\bmod q}e_p(-sc)\,e_q(-tc)\,S_{s,t},\qquad S_{s,t}=\sum_{\substack{m\le N,\ \mu^2(m)=1\\ \gcd(m,pq)=1}}h_{s,t}(m),$$

where $h_{s,t}(m)=\prod_{\ell\mid m}e_p(s\ell^{-1})e_q(t\ell^{-1})$, extended by $h_{s,t}(\ell^k)=0$ for $k\ge2$ and $h_{s,t}(\ell)=0$ for $\ell\mid pq$. It is multiplicative with $|h_{s,t}|\le1$, which is Halász's hypothesis. The term $(s,t)=(0,0)$ contributes $(pq)^{-1}\#\{m\le N:\mu^2(m)=1,\gcd(m,pq)=1\}\le N/(pq)$, which is the asserted bound; the phases $e_p(-sc)e_q(-tc)$ are unimodular and play no further role, so the two values of $c$ are on the same footing. It therefore suffices to prove

$$\max_{(s,t)\neq(0,0)}|S_{s,t}(N)|\ \ll\ N/(pq),\qquad(*)$$

since the $pq-1$ remaining terms then contribute $\ll N/(pq)$ in total.

*The distance.* Put $g_{s,t}(a) = e_p(sa^{-1})e_q(ta^{-1})$ on $(\mathbb{Z}/pq)^\times$, so $h_{s,t}(\ell) = g_{s,t}(\ell \bmod pq)$ for $\ell \nmid pq$, and expand $g_{s,t} = \sum_\psi c_\psi \psi$ over the characters mod $pq$. By the Chinese remainder theorem $g_{s,t}$ factors and so do its coefficients, $c_\psi = c^{(p)}_{\psi_1} c^{(q)}_{\psi_2}$ with $\psi = \psi_1\psi_2$. At a single prime $p$ and $s \not\equiv 0$, reindexing by $b = sa^{-1}$,

$$c^{(p)}_{\psi_1} = \frac{1}{p-1}\sum_a e_p(sa^{-1})\overline{\psi_1}(a) = \frac{\overline{\psi_1}(s)}{p-1}\sum_{b\neq 0} e_p(b)\psi_1(b),$$

so $c^{(p)}_{\psi_{1,0}} = -1/(p-1)$ by the Ramanujan sum $\sum_{b\neq 0} e_p(b) = -1$, while $|c^{(p)}_{\psi_1}| = \sqrt{p}/(p-1)$ for $\psi_1 \neq \psi_{1,0}$ by $|\tau(\psi_1)| = \sqrt{p}$. For $s \equiv 0$ the factor $g$ is identically 1 and $c^{(p)}_{\psi_{1,0}} = 1$, the other coefficients vanishing. Hence for $(s,t) \neq (0,0)$ at least one factor is a Ramanujan sum: the three cases $s,t \neq 0$; $s \neq 0, t = 0$; $s = 0, t \neq 0$ give $|c_{\psi_0}|$ equal to $1/((p-1)(q-1))$, $1/(p-1)$ and $1/(q-1)$ respectively, so in every case, both $p$ and $q$ exceeding $P/2$,

$$|c_{\psi_0}| \;\le\; \max\Big(\frac{1}{p-1}, \frac{1}{q-1}\Big) \;\le\; \frac{2}{P-2} \;=\; o(1),$$

whereas Parseval $\sum_\psi |c_\psi|^2 = 1$ gives, by Cauchy–Schwarz, $\sum_{\psi\neq\psi_0}|c_\psi| \le \sqrt{\varphi(pq)} \le \sqrt{pq} \le L/(\log L\, w(L))$.

*The non–principal sums.* This is verbatim the corresponding step of Lemma A.4, at modulus $pq \le L^2/(\log L\, w(L))^2$: for $\psi \neq \psi_0$ and $|\tau| \le \log N$, partial summation against Siegel–Walfisz (legitimate since $pq \le L^2 \le (\log N)^3$) replaces the truncated sum by $\log L(s,\psi^*) + O(1)$ at the inducing character, and the classical zero–free region gives $|\log L(s,\psi^*)| \ll \log\log(pq(3+|\tau|)) \ll \log L$; in the exceptional case the same $\tau$–split applies, the branch $|\tau| \le 1$ using $|s-\beta| > 1-\beta \ge c(\eta)(pq)^{-\eta}$ for any fixed $\eta > 0$, unrelated to the $w$ of the range, and hence $|\log(s-\beta)| \le \eta\log(pq) + O_\eta(1) \ll \eta\log L + O_\eta(1)$, and the branch $|\tau| > 1$ needing no exceptional case. So $|S_\psi(\tau)| \ll \log L$ uniformly.

*Assembly and Halász.* Therefore

$$M \;=\; \min_{|\tau|\le\log N}\sum_{\ell\le N}\frac{1-\operatorname{Re} h_{s,t}(\ell)\ell^{-i\tau}}{\ell} \;\ge\; \big(1-o(1)\big)L \;-\; O\big(L/w(L)\big) \;=\; \big(1-o(1)\big)L,$$

the principal part costing at most $|c_{\psi_0}|L = o(L)$ and the non–principal part $O\big(L/w(L)\big) = o(L)$. Halász's theorem (Montgomery's form; absolute constants, and $h_{s,t}$ enters only through $|h_{s,t}| \le 1$) with $T = \log N$ gives

$$|S_{s,t}(N)| \;\ll\; N(1+M)e^{-M} + N(\log N)^{-1/2} \;\ll\; N(\log N)^{-1/2},$$

the floor dominating since $(1+M)e^{-M} \ll (\log N)^{-1+o(1)}$. Finally $pq \le L^2$, so $pq = o\big((\log N)^{1/2}\big)$ and hence $N(\log N)^{-1/2} = o\big(N/(pq)\big)$, which is $(*)$. The one–prime bound is the same computation with the single detector $p^{-1}\sum_s e_p(-sc)S_s$, principal coefficient $-1/(p-1)$, $\sum_{\psi\neq\psi_0}|c_\psi| \le \sqrt{p} \le \sqrt{P}$, and the requirement $\max_{s\neq 0}|S_s| \ll N/p$, which is weaker than $(*)$.

Two notes, as for Lemma A.4. The constants are ineffective, through Siegel–Walfisz and Siegel. And the distance constant here is $1 - 1/(p-1) \to 1$, larger than the $1 - \sqrt{r}/\varphi(r)$ of Lemma A.4: the deficit bounds have more room than the cancellation bound they are used beside, which is why no analogue of the $r \ge 7$ restriction is needed. Numerically, at $N = 2\times 10^6$ and $c = 2$, the ratio of the counted set to $N_{\mathrm{sf}}(p)/p$, where $N_{\mathrm{sf}}(p)$ counts the squarefree $m \le N$ coprime to $p$, lies in $[0.988, 1.023]$ for $3 \le p \le 31$, and the two–prime ratio against $N_{\mathrm{sf}}(pq)/(pq)$ lies in $[0.997, 1.057]$ for

pairs up to $(23, 29)$: the densities are not merely bounded but asymptotically exact, with no growth in $p$ (`code/deficit_numeric.gp`). The condition measured is $\sigma_p(m) \equiv 2$, not $m' \equiv 2$; these differ by the unit $m$. □

**Proposition A.7** (A rate on the minus layer)**.** *Write $L = \log\log N$. For every function $w$ with $w(L) \to \infty$, however slowly,*

$$\#\{m \le N :\ \mu^2(m) = 1,\ m' - 2m \text{ a positive perfect square}\} \ll_w N\,\frac{(\log\log\log N)^2\, w(L)}{\log\log N},$$

*which is the ceiling of the method (Remark A.11); in particular the bound $\ll_\varepsilon N(\log\log\log N)^{2+\varepsilon}/\log\log N$ holds for every fixed $\varepsilon > 0$, and the same for $m' + 2m$. In particular Theorem A.9 follows with a rate, by an argument with no diagonalisation over $r$.*

*Proof.* Let $P = L/(\log L\, w(L))$, $\mathcal{P}$ the primes in $(P/2, P]$, $\Pi = |\mathcal{P}| \gg P/\log P$, and $a_m = m' - 2m$. First, $a_m \neq 0$ for every squarefree $m$: $a_m = 0$ forces $\sigma(m) = 2$ exactly, impossible for $m > 1$ since $\gcd(D(m), m) = 1$ (Theorem 2.3) and $a_1 = -2$. If $a_m$ is a positive square then $(a_m \mid p) \neq -1$ for every $p$, and $(a_m \mid p) = 0$ exactly when $p \mid a_m$, so $\sum_{p\in\mathcal{P}}(a_m \mid p) \ \ge\ \Pi - \#\{p \in \mathcal{P} : p \mid a_m\}$. Hence $\Pi^2\,\#\{\text{squares}\} \le \sum_m \big(\sum_p (a_m \mid p) + \#\{p \in \mathcal{P} : p \mid a_m\}\big)^2$, the sum over all squarefree $m \le N$; expanding the square, the cross term is bounded through $\big|\sum_p (a_m \mid p)\big| \le \Pi$, which gives

$$\Pi^2\,\#\{\text{squares}\} \ \le\ \sum_{p\neq q}\big(T_{pq}(N) + E_{pq}\big) + \Pi N + 2\Pi\sum_p \#\{m : p \mid a_m\} + \sum_{p\neq q}\#\{m :\ p \mid a_m,\ q \mid a_m\} + \sum_p \#\{m : p \mid a$$

where $|E_{pq}| \le N/p + N/q \le 4N/P$ collects the $m$ with $\gcd(m, pq) > 1$, dropped from $T_{pq}$'s summation range exactly as in the argument for Theorem A.9. By Lemma A.4 at $r = pq \le L^2/(\log L\, w(L))^2$, inside its range once $w(L) \ge (\log L)^{\varepsilon/2}$ or by enlarging $w$ (each $r$ odd squarefree, $r \ge 7$), $\sum_{p\neq q}|T_{pq}| \ll \Pi^2 N(\log N)^{-2/5}$. By Lemma A.6 (applicable because $p \mid m$ forces $a_m \equiv m' \equiv m/p \not\equiv 0 \pmod p$ for squarefree $m$, so the $m$ counted here are automatically coprime to $p$, and likewise to $pq$ in the two–prime count) $\sum_p \#\{m : p \mid a_m\} \ll N\Pi/P \ll N/\log P$ and the two–prime sum is $\ll \Pi^2 N/P^2 \ll N/(\log P)^2$. Dividing by $\Pi^2$: the diagonal $\Pi N$ gives the main term $N/\Pi \ll N\log P/P$, and every other term is smaller. Since $\log P \asymp \log\log\log N$, the bound follows: $\log P \asymp \log L$, so $N\log P/P \asymp N(\log L)^2 w(L)/L$. The fixed–$\varepsilon$ form is the case $w = (\log L)^\varepsilon$. Note $\mu^2(m)$ and the coprimality conditions are carried inside every count; no Möbius expansion is performed. □

Measured at $r = 1009$, $N = 3\times 10^7$: $|\sum_m h_t(m)|/N$ equals $3.6\times10^{-6}$, $1.3\times10^{-3}$, $2.1\times10^{-3}$, $5.7\times10^{-4}$, $1.5\times10^{-3}$ at $t = 0, 1, 2, 17, 500$, all of size $\sqrt{N}$ rather than merely $o(N)$. The sums $T_r(N)$ themselves were measured at $N = 5\times10^7$ for $r = 101, 1009, 10007, 100003$ and $r = pq = 101\cdot1009$, $1009\cdot10007$, giving $|T_r|/\sqrt{N} = 8.1$, $0.1$, $1.2$, $1.0$, $1.0$, $0.2$ (`code/sqsieve_test.rs`).

**Lemma A.8** (Local densities at $p^2$)**.** *For each odd prime $p$, $\limsup_{N\to\infty} N^{-1}\#\{m \le N : \mu^2(m) = 1,\ p^2 \mid (m' - 2m)\} \le p^{-2}$, with absolute constant. (The density form is the one Theorem A.9 needs, because it sends $P \to \infty$; the form "$\ll N/p^2$ for each fixed $p$" advertises a $p$–dependent implied constant and would not license that limit.)*

*Proof.* For $p \nmid m$ the condition is $\sigma_{p^2}(m) \equiv 2 \pmod{p^2}$ by Lemma 17.1 applied with modulus $p^2$. Detecting it by additive characters, the non-trivial frequencies give multiplicative sums $\prod_{\ell\mid m} e_{p^2}(t/\ell)$ whose values at primes depend only on $\ell \bmod p^2$ and which are not Dirichlet characters, because $u \mapsto e_{p^2}(t/u)$ is not multiplicative; the Halász argument of Lemma A.1 applies, but *not* verbatim,

and the difference is worth naming. That lemma uses primitivity of $\chi = (\cdot \mid r)$ twice, for $|\tau(\chi)| = \sqrt{r}$ and for the principal coefficient $\chi(t)\tau(\chi)/\varphi(r)$; here $(\cdot \mid p^2)$ is trivial on $(\mathbb{Z}/p^2)^\times$, so there is no character twist at all. The principal coefficient is instead a Ramanujan sum $c_{p^2}(t)/\varphi(p^2)$, which vanishes for $\gcd(t, p^2) = 1$ and equals $-1/(p-1)$ when $p \,\|\, t$; the distance bound follows more easily than in Lemma A.1, not by the same computation. Hence the density is $\frac{6}{\pi^2} \cdot \frac{p}{p+1} \cdot p^{-2} + o(1) \le p^{-2}$. The two factors in front are not optional: the count runs over *squarefree* $m$ coprime to $p$, so it carries the squarefree density $6/\pi^2$ and the coprimality factor $p/(p+1)$. The error was in the safe direction, the stated $p^{-2}$ being an over-estimate, so the inequality the lemma asserts survives with room to spare. The case $p \mid m$ does not arise at all: writing $m = pn$ with $p \nmid n$ gives $m' = pn' + n$, so $m' - 2m = n + p(n' - 2n) \equiv n \not\equiv 0 \pmod p$, and $p$ does not even divide $m' - 2m$, let alone $p^2$. □

Measured at $N = 3 \times 10^7$ for $p = 5, 7, 11, 13, 101$, the count divided by $\frac{6}{\pi^2}\frac{p}{p+1}N/p^2$ is 0.842, 0.887, 0.930, 0.925, 0.993, approaching 1 as the corrected density predicts. Against the bare $N/p^2$ the same data read 0.512, 0.539, 0.566, 0.562, 0.604, approaching $6/\pi^2 = 0.6079$.

**Theorem A.9** (The minus layer has density zero on the squarefree stratum)**.** Label: PROVED. Two independent routes reach this statement, and the label rests on the second. The first is the square sieve at fixed $r$ recorded in Remark A.10 below: it needs no uniformity, since it sends $P \to \infty$ only after $N$, but it yields no rate, and its *text* failed three consecutive reviews, a load–bearing display silently shedding a hypothesis each time, so it has never had a clean round and the label does not rest on it. The second is Proposition A.7, which proves the statement *with a rate* from Lemma A.4 and Lemma A.6; each of those, and the proposition itself, earned its label under this project's standard, a review by a session that did not write it, finding nothing.

*$\#\{m \le N : \ \mu^2(m) = 1, \ m' - 2m$ is a positive perfect square$\} = o(N)$; indeed, by Proposition A.7, the count is $\ll_\varepsilon N(\log\log\log N)^{2+\varepsilon}/\log\log N$ for every $\varepsilon > 0$. Equivalently, atom* A9 *holds on the squarefree stratum: the squarefree minus layer has density zero. The same argument with $m' + 2m$ gives the squarefree plus–analogue, so the squarefree Pythagorean layer has density zero unconditionally. That is* not *a reproof of Proposition* 15.15*, which is a sharper $O(\sqrt{X}(\log X)^2)$ on the narrower semiprime stratum; the two are incomparable, and that proposition rests on its own elementary divisor–sum proof.*

*The squarefree restriction is not cosmetic. Every tool in the proof, Rigidity, the cofactor identity, and Lemma A.1, is stated for squarefree $m$; the non-squarefree stratum, of density $1 - 6/\pi^2 \approx 0.392$, is untreated. Since a #307 solution is squarefree by Rigidity, the restriction costs nothing for the application, but the unrestricted statement is not proved here.*

*Proof.* Immediate from Proposition A.7, which gives the minus–layer count $\ll_\varepsilon N(\log\log\log N)^{2+\varepsilon}/\log\log N = o(N)$ on the squarefree stratum for any fixed $\varepsilon > 0$, and the plus–layer count by the same statement. Remark A.10 records an independent route to the qualitative form. □

**Remark A.10** (The original route, and what it does and does not give)**.** This is the argument that carried Theorem A.9 before Proposition A.7 existed. It is recorded because it is independent of that proposition, because it needs no uniformity in $r$ whatsoever, and because the errata inside it are the ones that cost the most to find. It is *not* what the label rests on: its text failed three consecutive reviews and has never had a clean one, and it yields no rate, since $P \to \infty$ only after $N$.

Write $a_m = m' - 2m$ and fix $P$, letting $\mathcal{P}$ be the primes in $(P, 2P]$ and $\Pi = |\mathcal{P}| \sim P/\log P$. If $a_m$ is a perfect square coprime to every $p \in \mathcal{P}$ then $(a_m/p) = 1$ for all such $p$, so $\sum_{p\in\mathcal{P}}(a_m/p) = \Pi$.

Hence

$$\Pi^2\,\#\{m\le N:\ \mu^2(m)=1,\ a_m=\square,\ \gcd(a_m,\textstyle\prod\mathcal{P})=1\}\ \le\sum_{\substack{m\le N\\ \mu^2(m)=1}}\Big(\sum_{p\in\mathcal{P}}\Big(\frac{a_m}{p}\Big)\Big)^2\ =\ \sum_{p,q\in\mathcal{P}}\ \sum_{\substack{m\le N\\ \mu^2(m)=1}}\Big(\frac{a_m}{pq}\Big).$$

Both $m$–sums carry $\mu^2(m)=1$ throughout, and so does the counting set on the left. This is not a convenience: every tool below is squarefree–only, and writing these sums unrestricted makes the identity with $T_{pq}+E_{pq}$ false by the non–squarefree terms. Those are not negligible: measured at $N=3\times10^7$ the omitted sum is 119,473 against a $T_{pq}$ of 2,699 at $r=101\cdot1009$, and 169,949 against 314 at $r=1009\cdot10007$, exceeding the term the lemma controls by a factor 44 to 540. The paper bounds it by nothing better than the trivial $(1-6/\pi^2)N$. The diagonal $p=q$ contributes $(a_m/p^2)=(a_m/p)^2\in\{0,1\}$, hence at most $\Pi N$. The off-diagonal is $\sum_{p\neq q}\big(T_{pq}(N)+E_{pq}(N)\big)$, where $T_{pq}$ carries the coprimality condition $\gcd(m,pq)=1$ of Lemma A.1 and $E_{pq}(N)=\sum_{m\le N,\ \mu^2(m)=1,\ \gcd(m,pq)>1}(a_m/pq)$ collects the terms it omits. That correction is harmless but is not empty and must be carried: $|E_{pq}(N)|\le\#\{m\le N:\ p\mid m\text{ or }q\mid m\}\le N(1/p+1/q)\le 2N/P$ (the primes here lie in $(P,2P]$, whereas Proposition A.7 takes them in $(P/2,P]$ and so carries $4N/P$) so $\Pi^{-2}\sum_{p\neq q}|E_{pq}(N)|\le 2N/P$, which is dominated by the diagonal's own $N/\Pi\asymp N\log P/P$ and changes nothing below. Each $T_{pq}(N)$ is $o(N)$, for each of the finitely many pairs, by Lemma A.1. A square divisible by $p$ is divisible by $p^2$, so the excluded $m$ are counted by Lemma A.8 in its *density* form: $\limsup_N N^{-1}\#\{m\le N:\mu^2(m)=1,\ p^2\mid a_m\}\le p^{-2}$, with absolute constant 1. That form is what the endgame needs, since sending $P\to\infty$ against a $p$–dependent implied constant would not be legitimate; the statement "$\ll N/p^2$ for each fixed $p$" advertises exactly such a constant. Summing, $\sum_{p\in\mathcal{P}}p^{-2}\le\Pi/P^2\asymp 1/(P\log P)$. Here $p\mid m$ is impossible, since $m=pn$ with $p\nmid n$ gives $a_m=n+p(n'-2n)\equiv n\not\equiv 0$, a computation that itself needs $\mu^2(m)=1$. The case $a_m=0$ does not arise: the theorem counts $a_m=\square>0$, and in any case $m'=2m$ has *no* squarefree solution, since $\sigma(m)=2\in\mathbb{Z}$ forces $D_m=1$ by rigidity. Altogether

$$\limsup_{N\to\infty}\frac{1}{N}\#\{m\le N:\ \mu^2(m)=1,\ a_m=\square>0\}\ \ll\ \frac{1}{\Pi}+\frac{1}{P\log P}.$$

The left side does not depend on $P$; letting $P\to\infty$ gives 0. The restriction $\mu^2(m)=1$ must be carried here as everywhere above: without it the display would assert density zero for the *full* minus layer, which is exactly the statement this theorem disclaims, and three successive revisions of this proof dropped it from one display or another.

The proof of Theorem A.9 sends $P\to\infty$ only after $N$, which costs all quantitative information. Keeping $P$ a function of $N$ would turn it into an explicit rate, and what that requires is uniformity in $r$ as $P\to\infty$.

*That uniformity is exactly what is not available.* Lemma A.1 gives $M\ge\big(1-\sqrt{r}/\varphi(r)\big)\log\log N-O_r(\log\log\log N)$, with the error depending on $r$ through the non–principal $L$–functions, and its statement and the remark following it both say so. The constant is $\sqrt{r}/\varphi(r)$, not the $3/\sqrt{r}$ printed here previously; the two are not interchangeable, since $\varphi(r)\ge r/3$ fails for odd squarefree $r$, first at $r=3\cdot5\cdots23=111{,}546{,}435$ with $\varphi(r)/r=0.3272$. The statement below was therefore stated as a CONJECTURE when this remark was written, and its argument here is a sketch rather than a proof. The uniform input it wanted is Lemma A.5, and the rate itself is Theorem A.12, proved by a different route.

**Remark A.11** (The ceiling of the method, and the one input that would break it)**.** Two things are worth recording about how far Proposition A.7 can be pushed, since the same accounting error was made twice in this paper's history.

*The ceiling.* The binding constraint there is again the coefficient mass: $\sqrt{pq}\cdot O(\log\log\log N) = o(\log\log N)$, that is $P = o(L/\log L)$ with $L = \log\log N$. The proposition takes $P = L/(\log L)^{1+\varepsilon}$, which is short of that ceiling by $(\log L)^{\varepsilon}$, Taking $P = L/(\log L\, w(L))$ for any $w \to \infty$ gives

$$\#\{m \le N :\ \mu^2(m) = 1,\ m' - 2m = \square > 0\} \ \ll\ N\,\frac{(\log\log\log N)^2\, w(\log\log N)}{\log\log N},$$

so the true ceiling of the method is $N(\log\log\log N)^{2+o(1)}/\log\log N$ with the $o(1)$ positive, unremovable, and arbitrarily slow. We state the proposition with a fixed $\varepsilon$ because it is the usable form; the sharpening is asymptotic and worth nothing numerically. What matters is the shape: no choice of $P$ puts a power of $\log N$ in the denominator *so long as the distance is estimated through the character expansion.* That proviso is not decoration: see the next paragraph, where it is exactly what is in question.

*Where the loss actually is, and what it is not.* Both walls, this one and the corollary's, are the same $\sqrt{\varphi(r)}$ paid when the phase $\ell \mapsto e_r(t\ell^{-1})$ is expanded in multiplicative characters. It is tempting to ask for a Weil–type bound for that phase over primes, and an earlier draft of this remark did. That framing is *wrong*, and the error is worth recording. The phase depends only on $\ell \bmod p$, so

$$\sum_{\ell\le X} e_p(s\ell^{-1})\ell^{-1-i\tau} \ =\ \sum_{a\not\equiv 0} e_p(sa^{-1}) \sum_{\substack{\ell\le X\\ \ell\equiv a\ (p)}} \ell^{-1-i\tau},$$

an exact reorganisation with no loss; its discrete Fourier transform is a Gauss sum, so a "direct" estimate is the character expansion restated rather than an independent route. The Kloosterman literature over primes (Fouvry–Michel, Fouvry–Kowalski–Michel, Garaev, Korolev, Irving) is *not* the relevant technology: every bound there is nontrivial only for modulus at least a fixed power of $X$, and ours is at most a power of $\log X$. Note also that the sum is not $o(\log\log X)$ at fixed $p$: since $\sum_{a\not\equiv 0} e_p(sa^{-1}) = -1$, Mertens in progressions gives $-\log\log X/(p-1)+O(1)$, measured as $-0.2502$, $-0.1667$, $-0.0830$, $-0.0105$ at $p = 5, 7, 13, 101$ against $-1/(p-1)$ (`code/inverse_phase_check.gp`). That failure is confined to bounded $p$ and is harmless, $1/(p-1)$ being itself $o(1)$.

What the same computation does show is that the *accounting* is lossy: measured, $\max_s |\sum_{\ell\le 10^6} e_p(s\ell^{-1})/\ell|$ is $1.04, 1.17, 1.35, 1.32, 1.33$ at $p = 13, 101, 503, 1009, 2003$, flat, while $\sqrt{p}$ grows from 3.6 to 44.8. So the triangle inequality over characters discards almost everything, and the wall above is an artefact of that step rather than an arithmetic obstruction. It can be avoided, and that is Lemma A.5: applying Siegel–Walfisz to the residue classes directly pays a factor $r$ against an error $\exp(-c\sqrt{\log x})$ rather than $\sqrt{r}$ against a bound on $\log L$, which is free at $r \le (\log N)^A$. The ceiling computed in this remark is therefore the ceiling of the *character–expansion* route only; Theorem A.12 passes it by a quarter–power of $\log N$. The measurement above, $\max_s|\cdot|$ flat near 1.3 while $\sqrt{p}$ grows to 44.8, was the numerical shadow of exactly this.

**Theorem A.12** (A rate for A9)**.** (Label: PROVED. The input that makes it unconditional is Lemma A.5.) *For every $\delta > 0$,*

$$\#\{m \le N :\ \mu^2(m) = 1,\ m' - 2m \text{ is a positive perfect square}\} \ \ll_\delta\ N\,\frac{(\log\log N)^{1/2+\delta}}{(\log N)^{1/4}},$$

*and the same for $m' + 2m$. The squarefree restriction is inherited from Theorem A.9 and from Lemma A.1. The implied constant is ineffective, through Siegel.*

*Proof.* Run the sieve of Proposition A.7 at $P = (\log N)^{1/4}(\log\log N)^{1/2}$, with $\mathcal{P}$ the primes in $(P/2, P]$ and $\Pi = |\mathcal{P}| \gg P/\log P$; here $\log P \asymp \log\log N$. The expanded square and its five terms are exactly as there, and we bound each after dividing by $\Pi^2$.

*The distance, at these moduli.* Every modulus arising is $r = pq \le P^2 \ll (\log N)^{1/2} \log\log N$, so Lemma A.5 applies with $A = 1$. For the character sums it gives, at $g_t(a) = \chi(a)e_r(ta^{-1})$ and hence $|\widehat{g}_0| = \sqrt{r}/\varphi(r) = o(1)$, $M \ge (1 - o(1)) \log\log N$; for the deficit sums, at $g_{s,t}(a) = e_p(sa^{-1})e_q(ta^{-1})$ and hence $|\widehat{g}_0| \le \max\big(\frac{1}{p-1}, \frac{1}{q-1}\big) \le 2/(P-2) = o(1)$, the same. In both cases Halász (Montgomery's form, $T = \log N$) gives $N(1 + M)e^{-M} + N(\log N)^{-1/2} \ll N(\log N)^{-1+o(1)} + N(\log N)^{-1/2} \ll N(\log N)^{-1/2}$: the floor binds.

*The five terms.* The diagonal contributes the main term $N/\Pi \asymp N \log P/P \asymp N(\log\log N)^{1/2}(\log N)^{-1/4}$, which is the stated bound. The character term is $\Pi^{-2} \sum_{p\ne q} |T_{pq}| \le \max_{p\ne q} \sqrt{pq} \cdot \max_t |S_t| \ll 2P \cdot N(\log N)^{-1/2} \asymp N(\log\log N)^{1/2}(\log N)^{-1/4}$, the same order, this is what fixes $P$, the balance $N \log P/P \asymp PN(\log N)^{-1/2}$ giving $P \asymp (\log N)^{1/4}(\log\log N)^{1/2}$. The correction $|E_{pq}| \le 4N/P$ contributes $\ll N(\log N)^{-1/4}(\log\log N)^{-1/2}$, smaller by $\log\log N$. For the two divisibility terms, note first that no $p^2$–count is needed and Lemma A.8 does not enter: what the sieve requires is $\#\{m : p \mid a_m\} \ll N/\Pi$, weaker than $N/p$ by the factor $\log P$, and the two–prime count needs nothing at all, since $\#\{m : p \mid a_m,\ q \mid a_m\} \le \#\{m : p \mid a_m\}$ makes $\sum_{p\ne q} \#\{\cdots\} \le \Pi \sum_p \#\{\cdots\}$. By the detector identity of Lemma A.6 (whose $1/(pq)$ prefactor cancels the $pq - 1$ non–principal terms exactly, with no $\sqrt{r}$ amplification) and the distance bound above, $\#\{m : p \mid a_m\} \le N/p + \max_{s\ne 0} |S_s| \ll N/p + N(\log N)^{-1/2} \ll N/\Pi$, as required. Every non–diagonal term is therefore at most the main term, and dividing by $\Pi^2$ gives the bound. The proof in fact delivers the sharper exponent $\frac{1}{2}$ on $\log\log N$; the $\delta$ in the statement is a deliberate weakening, so that no reader has to track which constants are absolute. The plus case is identical with $c = -2$. □

*Why the exponent is $-1/2$ and not $-1/4$.* A rate of $\ll N \log\log N(\log N)^{-1/4}$, balancing at $P \asymp (\log N)^{1/4}$. Carrying the balance properly against the Halász floor gives $P \asymp (\log N)^{1/4}(\log\log N)^{1/2}$ and a bound better by $(\log\log N)^{1/2}$. Compared with Proposition A.7, which remains the statement proved without Lemma A.5, the gain is a full quarter–power of $\log N$: that proposition's ceiling is $N(\log\log\log N)^{2+o(1)}/\log\log N$, and no choice of $P$ improves it while the distance is estimated through the character expansion.

*Sketch, conditional on a uniform-in-r form of Lemma A.1.* Take $\mathcal{P}$ to be the primes in $(P, 2P]$ with $P = P(N)$ to be chosen, so $\Pi \sim P/\log P$. The three terms of Theorem A.9 are then

$$\frac{N}{\Pi} \ll \frac{N \log P}{P}, \qquad \frac{1}{\Pi^2} \sum_{p\ne q} |T_{pq}(N)| \le \max_{p\ne q} |T_{pq}(N)|, \qquad \sum_{p\in\mathcal{P}} O\Big(\frac{N}{p^2}\Big) \ll \frac{N}{P \log P}.$$

Granting a uniform-in-$r$ distance bound, which Lemma A.1 does *not* supply, Halász would give $T_r(N) \ll \sqrt{r}\, N(\log N)^{-1/2}$, and $r = pq \le 4P^2$ gives $\sqrt{r} \le 2P$, so the middle term is $\ll PN(\log N)^{-1/2}$. Balancing it against the first, $N \log P/P = PN(\log N)^{-1/2}$, gives $P \asymp (\log N)^{1/4}$, at which both equal $N \log\log N(\log N)^{-1/4}$ up to constants; the third term is then $O\big(N(\log N)^{-1/4}/\log\log N\big)$ and is absorbed. □

**Remark A.13** (Where the exponent comes from)**.** The exponent $\frac{1}{4}$ is not intrinsic to the arithmetic. It is what remains after balancing against the floor $N(\log N)^{-1/2}$ in the Montgomery form of Halász's theorem, which holds however large the pretentious distance $M$ becomes. Any improvement of that floor transfers here without change: a floor of $N(\log N)^{-\alpha}$ yields the exponent $\alpha/2$. The measured character sums are of size $\sqrt{N}$ rather than $N(\log N)^{-1/2}$ (Section 17), so the truth is far stronger than what the general theorem can deliver, and the gap is entirely in the analytic input rather than in the sieve.

The exponent $\frac{1}{4}$ is an artefact of the general theorem, not of the arithmetic, and the gap can be closed off exactly. Everything in the sieve except the character-sum input is elementary, so the reachable exponent is a function of one measurable quantity: how $T_r(N)$ grows with $r$.

**Hypothesis A.14** (Square-root cancellation, uniform in the modulus)**.** There is an absolute $A \geq 0$ with $|T_r(N)| \ll r^A N^{1/2+o(1)}$ for every odd squarefree $r$.

**Theorem A.15** (Conditional power saving, and the crux exactly)**.** *Assume Hypothesis* A.14*. Then*

$$\#\{m \leq N : \ \mu^2(m) = 1, \ m' - 2m \text{ is a positive perfect square}\} \ \ll \ N^{1-\delta}, \qquad \delta = \frac{1}{2(2A+1)}.$$

*In particular $A = 0$ gives $N^{1/2+o(1)}$, which is exactly the bound* CRUX-A9 *asks for.*

*Proof.* Run the sieve of Theorem A.9 with $\mathcal{P}$ the primes in $(P, 2P]$, $\Pi \sim P/\log P$, and dispose of the $m$ with $p \mid a_m$ by the divisor bound rather than Lemma A.8: a perfect square $a_m$ has at most $\log(2N)/\log P$ prime factors in $\mathcal{P}$, so $\sum_{p\in\mathcal{P}}(a_m/p) \geq \Pi/2$ once $\Pi \geq 2\log(2N)/\log P$. Then

$$\#\{\,\cdot\,\} \ \ll \ \frac{N}{\Pi} + \frac{1}{\Pi^2}\sum_{p\neq q} |T_{pq}(N)| \ \ll \ \frac{N\log P}{P} + P^{2A}N^{1/2+o(1)},$$

since $pq \leq 4P^2$. Balancing the two terms gives $P = N^{1/(2(2A+1))}$ and the stated exponent; the side condition $\Pi \geq 2\log(2N)/\log P$ holds comfortably at that $P$. For $A = 0$ take $P = N^{1/2}$, whence both terms are $N^{1/2+o(1)}$. □

**Remark A.16** (The hypothesis is measured, and $A$ reads as zero)**.** Hypothesis A.14 is not a wish. At $N = 4 \times 10^7$, so $\sqrt{N} = 6.325 \times 10^3$, the ratio $|T_r(N)|/\sqrt{N}$ across twelve moduli spanning ten orders of magnitude is

| | | | | | | |
|---|---|---|---|---|---|---|
| $r$ prime : | 101 | 1009 | 10007 | 100003 | $10^6$+3 | $10^7$+19 |
| $\lvert T_r\rvert/\sqrt{N}$ : | 7.38 | 0.17 | 0.84 | 1.07 | 0.05 | 0.15 |
| $r = pq$ : | 101909 | 10097063 | 1000730021 | 100003300009 | | |
| $\lvert T_r\rvert/\sqrt{N}$ : | 0.95 | 0.21 | 1.30 | 0.64 | | |

together with 2.34 and 1.00 at the squarefree 1155 and 1113121. The implied exponent $A = \log(|T_r|/\sqrt{N})/\log r$ is $+0.006$, $-0.004$, $+0.013$, $-0.018$ and $-0.000$ at the five largest moduli; the outlying $+0.433$ occurs at $r = 101$, where $\log r$ is too small for the ratio to be meaningful. Note that the last two moduli *exceed* $N$, and the sums are still of size $\sqrt{N}$ (`code/sqsieve_uniform.rs`).

So the sharp form of A9 is not blocked by the sieve, which is elementary, nor by the arithmetic, which cooperates: it is blocked by the absence of a Weil-type bound for a character sum whose argument is not a polynomial. That is a single, isolated, and precisely stated analytic deficit, and Lemma 17.1 has already reduced it to a sum over a multiplicative function of $\sigma_r$.

The obvious route to Hypothesis A.14 is the one already used in Lemma A.1: expand by Gauss sums and bound each multiplicative sum $\sum_{m\leq N} h_t(m)$ separately, where square-root cancellation would follow from a Weil-type input or, in the analytic form, from GRH applied to $\prod_\chi L(s,\chi)^{\widehat{\alpha}(\chi)}$. That route is closed. The reason is not the one a first look suggests.

**Proposition A.17** (The decomposition destroys the cancellation)**.** *Write $\alpha(u) = (u/r)e_r(tu^{-1})$, so that $h_t(p) = \alpha(p \bmod r)$. The principal Fourier coefficient $\widehat{\alpha}(\chi_0) = \varphi(r)^{-1}\sum_u \alpha(u)$ is a Salié sum; measured, it has modulus exactly $\sqrt{r}$ before normalisation, at $r = 1009$ and $r = 10007$ giving*

31.8 *and* 100.0 *against* $\sqrt{r} = 31.76$ *and* 100.03. *It is therefore nonzero, and* $\prod_\chi L(s,\chi)^{\widehat{\alpha}(\chi)}$ *retains a* $\zeta$*-type branch point at* $s = 1$. *This does* not *by itself decide the matter, since the measured sums fall* 50 *to* 350 *times below what that branch point predicts. What decides it is the growth in* $N$*: at* $r = 10007$, $t = 1$,

| $N$: | $10^6$ | $3\times10^6$ | $10^7$ | $3\times10^7$ | $6\times10^7$ |
|---|---|---|---|---|---|
| $\lvert\sum h_t\rvert/\sqrt{N}$: | 0.67 | 0.48 | 0.69 | 2.08 | 2.29 |

*a fitted exponent of* 0.800 *over that range, with* $|\sum h_t|/(N/\log N)$ *flat near* 0.005. *The individual multiplicative sums thus grow strictly faster than* $\sqrt{N}$ (`code/sqsieve_route.rs`).

**Remark A.18** (Where the cancellation actually lives)**.** Since $T_r(N)$ itself measures at $\sqrt{N}$ across ten orders of magnitude in $r$ (Remark A.16) while each $\sum_m h_t(m)$ measured here grows like $N^{0.8}$, the cancellation in $T_r$ is not present frequency by frequency: it lies in the sum over $t$. The magnitude is not marginal. At $r = 1009$, $N = 4\times10^7$, bounding each frequency separately and summing trivially gives $\sqrt{r}\,N/\log N \approx 7\times10^7$, against a measured $|T_r| = 1050$: a factor of about $7\times10^4$ is lost the moment the Gauss expansion is taken.

The deficit behind Hypothesis A.14 is therefore not a missing Weil bound for a single character sum, which is how Remark A.16 first framed it. It is that the only available decomposition of $T_r$ scatters the cancellation across frequencies, and no current technique grips cancellation in that position. Stating it this way is more useful than the Weil framing, because it says what would have to be new: an estimate for $T_r$ that never passes through the multiplicative decomposition at all.

Remark A.18 asked for an estimate for $T_r$ that never passes through the multiplicative decomposition. The following is the mechanism such an estimate would use, isolated and proved; the reduction of $T_r$ to it is *not* established here, and we say precisely what is missing.

**Proposition A.19** (Kernel norm)**.** *Since* $\sigma_r$ *is additive, for coprime* $d, e$ *Lemma 17.1 gives*

$$\Big(\frac{(de)' - 2de}{r}\Big) = \Big(\frac{d}{r}\Big)\Big(\frac{e}{r}\Big)\,K\big(\sigma_r(d),\sigma_r(e)\big), \qquad K(u,v) = \Big(\frac{u+v-2}{r}\Big).$$

*The matrix* $K$ *is Toeplitz in* $u+v$, *so* $K(u,v) = \sum_\xi \widehat{f}(\xi)e_r(\xi u)e_r(\xi v)$ *diagonalises it against the orthogonal system* $(e_r(\xi u))_u$ *of norm* $\sqrt{r}$. *As* $\widehat{f}(\xi)$ *is a normalised Gauss sum,* $|\widehat{f}(\xi)| = r^{-1/2}$ *for* $\xi \neq 0$ *and* $\widehat{f}(0) = 0$, *so every singular value of* $K$ *equals* $\sqrt{r}$ *and*

$$\|K\|_{\mathrm{op}} = \sqrt{r}.$$

*Consequently* any *bilinear form* $\sum_{d\sim D}\sum_{e\sim E}\alpha_d\beta_e\,K(\sigma_r(d),\sigma_r(e))$ *is at most* $\sqrt{r}\,\|A\|_2\|B\|_2$, *where* $A(u) = \sum_{d\sim D,\ \sigma_r(d)\equiv u}\alpha_d$ *and* $B$ *is defined likewise. No multiplicative decomposition is used; the only input is the modulus of a Gauss sum.*

**Remark A.20** (What is missing, stated exactly)**.** Proposition A.19 bounds bilinear forms, not $T_r$. Passing from one to the other requires a decomposition of $\sum_{m\le N}$ into bilinear pieces, and two obstacles are real rather than cosmetic.

(i) Summing over coprime pairs $(d,e)$ with $de = m$ weights each $m$ by $2^{\omega(m)}$: at $N = 2\times10^6$ the coprime-pair count is 10,014,190 against 1,215,876 squarefree $m$, a factor 8.24. Since the summand is *signed*, no inequality relates $\sum_m F(m)$ to $\sum_m 2^{\omega(m)}F(m)$ in either direction.

(ii) A single dyadic block $d \sim D$ sees only a minority of $m$: for $D = \sqrt{N}$ at $N = 2\times10^6$, 24.7% (`code/sqsieve_bilinear.rs`).

The natural repair is the largest-prime-factor split $m = pn$ with $p = P^+(m)$, which is unique and so avoids (i); the $A$-side is then supported on primes, where $\|A\|_2^2 \ll \pi(D)^2/\varphi(r) + \pi(D)$ follows from Brun–Titchmarsh uniformly in $r$, and the $B$-side is governed by Theorem 17.3. What remains is the coupling $P^+(n) < p$ and the dyadic bookkeeping, a Type I/II analysis that we have not carried out. Until it is, the power saving below is **CONJECTURAL**. The best unconditional rate proved here is Theorem A.12's $N(\log\log N)^{1/2+\delta}(\log N)^{-1/4}$; Proposition A.7 is what the character–expansion route gives without Lemma A.5.

**Conjecture A.21** (Power saving, pending the decomposition)**.** *If the decomposition of Remark* A.20 *is carried out with the stated bounds, then combining Proposition A.19 with Theorem 17.3 and Theorem A.15 at $P = N^{1/4}$ gives*

$$\#\{m \le N : \ \mu^2(m) = 1, \ m' - 2m \text{ is a positive perfect square}\} \ \ll \ N^{3/4+o(1)}.$$

The two ingredients that would feed it are themselves unconditional, and are recorded separately because they do not depend on the missing step.

## References


[1] E. J. Barbeau, *Remarks on an arithmetic derivative*, Canad. Math. Bull. **4** (1961), 117–122.

[2] E. J. Barbeau, *Computer challenge corner: Problem 477*, J. Recreational Math. **9** (1976/77), 30.

[3] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monogr. Enseign. Math. **28**, Geneva, 1980, p. 38.

[4] R. L. Graham, *Paul Erdős and Egyptian fractions*, in *Erdős Centennial*, Bolyai Soc. Math. Stud. **25**, Springer, 2013, pp. 289–309.

[5] T. Watanabe, *New examples of the representation of* 1 *by the sum of reciprocals of semiprime numbers*, arXiv:2009.03275 (2020).

[6] I.-O. Bado, *Primary-pseudoperfect sectors in two-cycles of the arithmetic derivative*, preprint (2026), communicated to the author.

[7] T. F. Bloom, *Unit fractions with shifted prime denominators*, Proc. Roy. Soc. Edinburgh Sect. A (2024); arXiv:2305.02689.

[8] V. Ufnarovski and B. Åhlander, *How to differentiate a number*, J. Integer Seq. **6** (2003), Article 03.3.4.

[9] J. Kovič, *The arithmetic derivative and antiderivative*, J. Integer Seq. **15** (2012), Article 12.3.8.

[10] MathOverflow, *Can the product of two sums of prime reciprocals be* 1*?*, question 320838 (comments by R. Israel and J. Rosen, answer by "Woett"), 2019. `https://mathoverflow.net/questions/320838`.

[11] T. F. Bloom, *Erdős problem #307*, `https://www.erdosproblems.com/307`.

[12] OEIS Foundation, *Sequence A003415 (arithmetic derivative)*, `https://oeis.org/A003415`.

[13] P. Pollack, *On amicable tuples*, Illinois J. Math. **62** (2018), no. 1–4, 225–240; arXiv:1711.06847. (Explicit bounds for relatively prime amicable pairs, refining the finiteness of Artjuhov and Borho.)

[14] H. J. J. te Riele, *Computation of all the amicable pairs below* $10^{10}$, Math. Comp. **47** (1986), 361–368.

[15] R. C. Baker, G. Harman, and J. Pintz, *The difference between consecutive primes, II*, Proc. London Math. Soc. (3) **83** (2001), 532–562.

[16] H. Iwaniec, *Almost-primes represented by quadratic polynomials*, Invent. Math. **47** (1978), 171–188.

[17] R. J. Lemke Oliver, *Almost-primes represented by quadratic polynomials*, Acta Arith. **151** (2012), 241–261.

[18] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford Univ. Press, 2008.

[19] G. Tenenbaum, *Introduction to Analytic and Probabilistic Number Theory*, 3rd ed., Grad. Stud. Math. **163**, Amer. Math. Soc., 2015 (Erdős–Wintner theorem).

[20] W. W. Stothers, *Polynomial identities and Hauptmoduln*, Quart. J. Math. Oxford (2) **32** (1981), 349–370; R. C. Mason, *Diophantine Equations over Function Fields*, Cambridge Univ. Press, 1984.

[21] Seva (`mathoverflow.net` user), *A recursion with a number–theoretic function*, MathOverflow question 323194 (2019). `https://mathoverflow.net/questions/323194`.

[22] *Egyptian fractions for few primes*, arXiv:2507.03727 (July 2025).

[23] S. Steinerberger, *On an iterated arithmetic function problem of Erdős and Graham*, arXiv:2504.08023 (April 2025).

[24] *A stretched–exponential bound for an Erdős–Graham unit–fraction problem*, arXiv:2607.04157 (July 2026).

[25] *Port fillings for primary pseudoperfect numbers*, arXiv:2605.21518 (May 2026).

[26] *Every natural number is a sum of distinct semiprime unit fractions*, arXiv:2606.15159 (June 2026).

[27] W. Butske, L. M. Jaje and D. R. Mayernik, *On the equation* $\sum_{p|N} 1/p+1/N = 1$*, pseudoperfect numbers, and perfectly weighted graphs*, Math. Comp. **69** (2000), 407–420.

[28] I. O. Bado, *Structural constraints for an Erdős unit–fraction problem over primes*, preprint, Aix–Marseille University (May 2026). `https://doi.org/10.13140/RG.2.2.32245.54245`.

[29] G. Giuga, *Su una presumibile proprietà caratteristica dei numeri primi*, Ist. Lombardo Sci. Lett. Rend. Cl. Sci. Mat. Nat. (3) **14** (1950), 511–528.

[30] D. Borwein, J. M. Borwein, P. B. Borwein, and R. Girgensohn, *Giuga's conjecture on primality*, Amer. Math. Monthly **103** (1996), no. 1, 40–50.

[31] J. M. Borwein and E. Wong, *A survey of results relating to Giuga's conjecture on primality*, CRM Proc. Lecture Notes **11** (1997), 13–27.

[32] W. Butske, L. M. Jaje, and D. R. Mayernik, *On the equation* $\sum_{p|N} 1/p+1/N = 1$*, pseudoperfect numbers, and perfectly weighted graphs*, Math. Comp. **69** (2000), no. 229, 407–420.

[33] J. M. Grau and A. M. Oller-Marcén, *Giuga numbers and the arithmetic derivative*, J. Integer Seq. **15** (2012), Article 12.4.1; arXiv:1103.2298.

[34] J. Sondow and K. MacMillan, *Primary pseudoperfect numbers, arithmetic progressions, and the Erdős–Moser equation*, Amer. Math. Monthly **124** (2017), no. 3, 232–240; arXiv:1812.06566.

[35] J. M. Grau, A. M. Oller-Marcén, and D. Sadornil, *On* $\mu$*-Sondow numbers*, arXiv:2111.14211 (2021); Acta Math. Hungar. (2022).

[36] D. A. Cox, *Primes of the Form* $x^2 + ny^2$*: Fermat, Class Field Theory, and Complex Multiplication*, 2nd ed., Wiley, 2013.

[37] OEIS Foundation, *Sequence A054377 (primary pseudoperfect numbers)*, `https://oeis.org/A054377`.

[38] OEIS Foundation, *Sequence A007850 (Giuga numbers)*, `https://oeis.org/A007850`.

[39] OEIS Foundation, *Sequence A396915 (composite squarefree* $k$ *with* $k' + 2k$ *a perfect square)*, `https://oeis.org/A396915`.

[40] J.-H. Evertse, *On sums of* $S$*-units and linear recurrences*, Compositio Math. **53** (1984), 225–244.

[41] J. Friedlander and H. Iwaniec, *The polynomial* $X^2 + Y^4$ *captures its primes*, Ann. of Math. **148** (1998), 945–1040.

[42] D. R. Heath-Brown, *Primes represented by* $x^3 + 2y^3$, Acta Math. **186** (2001), 1–84.

[43] C. L. Stewart, *On divisors of Lucas and Lehmer numbers*, Acta Math. **211** (2013), 291–314.

[44] Y. Bugeaud, P. Corvaja, and U. Zannier, *An upper bound for the G.C.D. of* $a^n - 1$ *and* $b^n - 1$, Math. Z. **243** (2003), 79–84.

[45] J. Merikoski, *On the largest prime factor of* $n^2 + 1$, J. Eur. Math. Soc. **25** (2023), 1253–1284.

[46] J. Bourgain, A. Gamburd, and P. Sarnak, *Affine linear sieve, expanders, and sundry applications*, Invent. Math. **179** (2010), 559–644.

[47] J.-H. Evertse, H. P. Schlickewei, and W. M. Schmidt, *Linear equations in variables which lie in a multiplicative group*, Ann. of Math. (2) **155** (2002), 807–836.

[48] P. Erdős, *On the sum* $\sum_{k=1}^{x} d(f(k))$, J. London Math. Soc. **27** (1952), 7–15.

[49] Y. Bilu, G. Hanrot, and P. M. Voutier, *Existence of primitive divisors of Lucas and Lehmer numbers* (with an appendix by M. Mignotte), J. Reine Angew. Math. **539** (2001), 75–122.